

Chapter I

Transformers

LEARNING OBJECTIVES

After reading this chapter, you will be able to understand:

- Single-phase transformer
- Windings
- Bushings
- Breather
- Voltage transformation ratio
- Ideal two-winding transformer
- Phasor diagram of transformer under load
- Equivalent circuit
- Transformer tests
- Sumpner's test
- Losses and efficiency
- Voltage regulation of a transformer
- Auto-transformer
- Parallel operation of single-phase transformers

INTRODUCTION

Single-Phase Transformer

The transformer is a device that

1. Transfers electrical energy from one electrical circuit to another
2. Does so without a change in the frequency
3. Works on the principle of mutual induction
4. Has electric circuits that are linked by a common magnetic circuit

The energy transfer usually takes place with a change of voltage. If the energy transfer occurs at the same voltage, the purpose of transformer is merely to isolate the two electric circuits

- Transformer is a static device
- No rotating or moving parts
- Single-phase transformer has two circuits which are electrically isolated but magnetically coupled
- Transformer is an electromagnetic energy conversion device, since the energy received by the primary is first converted to magnetic energy and it is then reconverted to useful electrical energy in the secondary winding circuit
- Voltage and current levels change depending on transformation ratio
- Due to the absence of moving parts, transformers require very little maintenance and it has the highest possible efficiency out of all electrical machines

- Transformer is the main reason for the widespread popularity of AC systems over DC systems

Constructional Details

- Core: Sheet steel or silicon steel with 4% silicon is used
- The core is laminated and the laminations are insulated from each other by a coat of varnish or by an oxide layer on the surface to minimize eddy current losses
- The thickness of laminations varies from 0.35 mm for a frequency of 50 Hz to 0.5 mm for a frequency of 25 Hz
- The core provides a continuous magnetic path of low reluctance. The relative permeability of the core material is of the order of 1000
- Core stepping gives high space factor and also results in reduced length of mean turn and the consequent I^2R loss

Types of Transformers

1. **Core type:** In this type of transformers, the windings surround a considerable part of the core
2. **Shell type:** In this type of transformers, the core surrounds a considerable portion of the windings
3. **Spiral-core or wound-core (trade name—Spirakore):** In this type of transformers, the core is assembled of a continuous strip or ribbon of transformer steel wound in the form of a circular or elliptical cylinder

Advantages of Spiral—Core Construction

- High flux densities are used which reduces the weight/kVA
- Lower iron losses, lesser weight and size per kVA, lower cost of manufacture and a relatively more rigid core

Windings

Conventional transformer has two windings, which are made of high-grade copper or aluminium.

The winding in which electrical energy is fed from the AC supply mains is called primary winding and the other which delivers electrical energy is called secondary winding.

- Depending on voltage level, the windings can also be classified into LV and HV
- Low voltage winding is of higher cross section as it carries higher current compared to high voltage winding
- The winding is insulated from core and other parts of transformers

During the operation of transformer, two types of losses occur, namely core (iron) and Cu losses. These losses result in increase in temperature of both core and winding. To keep the temperature of core and winding within limits, it is necessary to radiate the heat produced due to losses in the transformer. Different cooling methods are used, depending upon the rating of the transformer.

Conservator

In an oil-filled transformer, due to variations in load and change in climatic conditions the oil expands or contracts. To allow room for expansion of the oil, conservator is used. In the absence of a conservator tank, high pressures may develop in transformer tank which may lead to bursting of tank.

Bushings

The purpose of bushings is to provide proper insulation to the output leads of the transformer tank. Porcelain bushings are used up to 33 kV. For higher voltages beyond 33 kV, condenser type or oil-filled type bushings are used.

Breather

Absorption of moisture and dust by the transformer oil will deteriorate the dielectric strength of transformer oil. To prevent moisture and dust from entering the transformer tank, breather is provided

E.m.f. Equation

- Transformer works on the principle of mutual induction. Voltage applied to the primary and the magnetic flux set up in the core assumed to be sinusoidal

Core flux $\phi = \phi_m \sin \omega t$

The primary induced e.m.f.

$$\begin{aligned} e_1 &= -N_1 \frac{d\phi}{dt} = -N_1 \phi_m \omega \cos \omega t \\ &= -N_1 \omega \phi_m \sin \left(\omega t - \frac{\pi}{2} \right) \end{aligned}$$

e_1 is maximum when $\sin \left(\omega t - \frac{\pi}{2} \right) = 1$

$$E_{1_{\max}} = N_1 \omega \phi_m$$

The RMS value of induced e.m.f.

$$\begin{aligned} E_1 &= \frac{1}{\sqrt{2}} E_{\max} = \frac{1}{\sqrt{2}} N_1 \cdot 2\pi f \phi_m \\ &= \sqrt{2} \pi N_1 f \phi_m = 4.44 N_1 f \phi_m \end{aligned}$$

Similarly $E_2 = \sqrt{2} \pi N_2 f \phi_m = 4.44 N_2 f \phi_m$

$$\therefore \frac{E_1}{E_2} = \frac{N_1}{N_2}$$

- The voltage per turn of the primary and secondary windings is the same, since the same mutual flux is linking with both the windings
- E_1 and E_2 are in phase and lag behind the core flux by an angle of 90°
- In a loaded transformer, the primary draws a current to compensate for the secondary ampere turns so that core flux is maintained constant
- Core flux is constant and is independent of the load

If no load, primary ampere turns are neglected, then

$$\begin{aligned} N_1 I_1 &= N_2 I_2 \\ \frac{I_1}{I_2} &= \frac{N_2}{N_1} = \frac{V_2}{V_1} \end{aligned}$$

Voltage Transformation Ratio (K)

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$

- If $N_2 > N_1$, i.e., $K > 1$, then the transformer is called step-up transformer
- If $N_2 < N_1$, i.e., $K < 1$, then the transformer is known as step down transformer

For an ideal transformer, input VA = output VA

$$V_1 I_1 = V_2 I_2 \quad \text{or} \quad \frac{I_2}{I_1} = \frac{V_1}{V_2} = \frac{1}{K}$$

Hence, currents are in the inverse ratio of the transformation ratio

Solved Examples

Example 1: If the applied voltage of a certain transformer is increased by 50% and the frequency is reduced to 50% (assuming that the magnetic circuit remains unsaturated), the maximum core flux density will. _____

Solution: Induced e.m.f. $E \propto B_m f$
neglecting the impedance drops

$$\frac{V_2}{V_1} = \frac{E_2}{E_1}; \quad \frac{V_2}{V_1} = \frac{B_{m_2} f_2}{B_{m_1} f_1}$$

$$B_{m_2} = \frac{V_2}{V_1} \times \frac{f_1}{f_2} \times B_{m_1}$$

$$= 1.5 \times \frac{1}{0.5} \times B_{m_1} = 3B_{m_1}$$

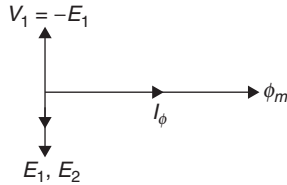
Ans: change to three times the original value.

IDEAL TWO-WINDING TRANSFORMER

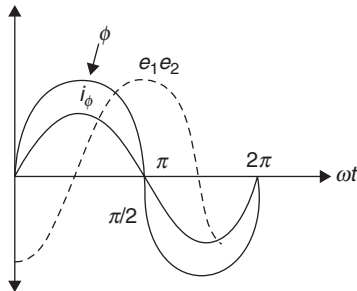
When the transformer is assumed to be ideal,

1. Winding resistances are negligible
2. No flux leakage. All the flux is confined to the magnetic core
3. The core losses are negligible
4. The core has constant permeability, i.e., the magnetization curve for the core is linear

(a) Phasor diagram



(b) Waveforms



When secondary of transformer is loaded, the load will draw secondary current I_2 , which sets up $N_2 I_2$, the secondary MMF and it opposes the main flux ϕ_m . The secondary MMF F_2 weakens the main flux ϕ_m . But any reduction in ϕ_m reduces E_1 and in case of ideal transformer $V_1 = -E_1$. Hence, an extra load component of current I_1' is drawn by primary to compensate the secondary ampere-turns to maintain constant core flux.

$$I_1' N_1 = N_2 I_2 \Rightarrow \frac{I_1'}{I_2} = \frac{N_2}{N_1}$$

As I_ϕ is negligible

$$\bar{I}_1 = \bar{I}_1' + \bar{I}_\phi \Rightarrow \frac{I_1}{I_2} = \frac{N_2}{N_1}$$

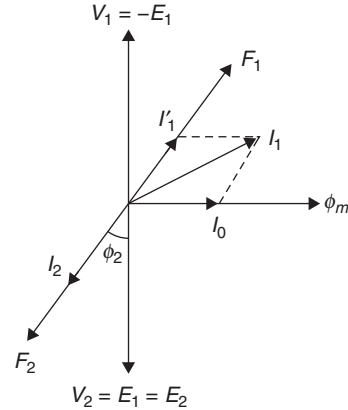
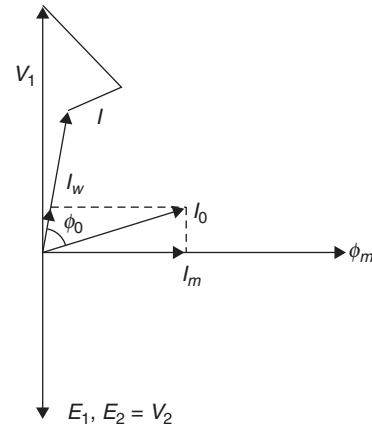


Figure 1 Phasor diagram of ideal transformer for inductive load

Phasor Diagram at no Load



The no-load primary current I_0 is called the exciting current of the transformer and can be resolved into two components.

1. One, in phase with V_1 , known as active or working or iron loss component I_w because it mainly supplies the iron loss plus small quantity of primary Cu loss

$$I_w = I_0 \cos \phi_0$$

2. The other component is in quadrature with V_1 and is known as magnetizing component I_μ because its function is to sustain the alternating flux in the core. It is wattless.

$$I_\mu = I_0 \sin \phi_0$$

Even when the transformer is on no-load, the primary input current is not wholly reactive. The no-load primary current lags V_1 by an angle ϕ_0

$$W_0 = V_1 I_0 \cos \phi_0$$

This no-load power supplies the losses in transformer, i.e., a small amount of Cu loss and iron losses in the magnetic core. As it is principally the core-loss which is responsible for the shift in current vector from 90° position, angle ϕ_0 is known as hysteresis angle of advance

$$I_0 = \sqrt{I_w^2 + I_\mu^2}$$

Phasor Diagram of Transformer Under Load

The transformer is said to be loaded, when its secondary circuit is completed through an impedance or load. The magnitude and phase of secondary current I_2 with respect to secondary terminals depends on the characteristics of the load, i.e., I_2 will be in phase, lags behind and lead the terminal voltage V_2 , respectively, when load is resistive, inductive and capacitive. The net flux in the core remains constant irrespective of the load.

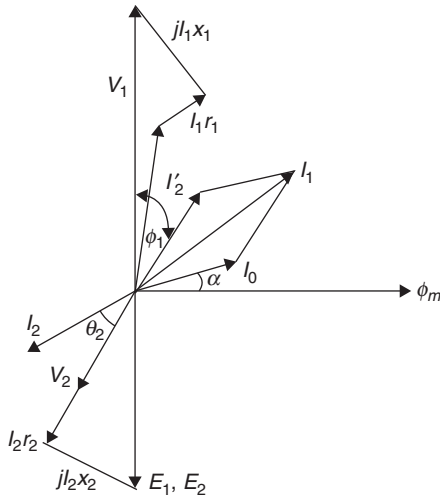


Figure 2 Phasor diagram for inductive load

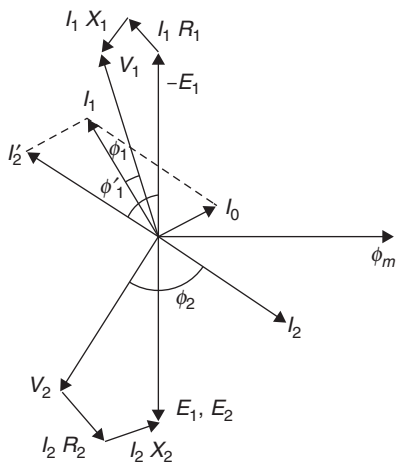


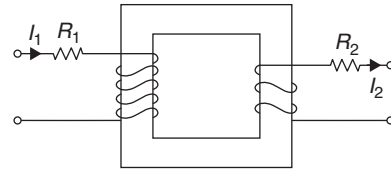
Figure 3 Phasor diagram for capacitive load

$$\overline{E_2} = \overline{V_2} + \overline{I_2} (r_2 + jx_2) = \overline{V_2} + \overline{I_2} Z_2$$

$$\overline{V_1} = \overline{V_1'} + \overline{I_1} (r_1 + jx_1) = \overline{V_1'} + \overline{I_1} Z_1$$

Impedance Transformation

1. Equivalent resistance



In the above figure, a transformer is shown whose primary and secondary windings have resistances of R_1 and R_2 , respectively.

The copper loss in secondary is $I_2^2 R_2$. This loss is supplied by primary which takes a current I_1 . Hence if R_2' is the equivalent resistance in primary which would have caused the same loss as R_2 in secondary, then

$$I_1^2 R_2' = I_2^2 R_2 \text{ or } R_2' = \left(\frac{I_2}{I_1} \right)^2 R_2$$

If we neglect no-load current I_0 , then $\frac{I_2}{I_1} = \frac{1}{K}$

$$\text{Hence } R_2' = \frac{R_2}{K^2}$$

Similarly equivalent primary resistance as referred to secondary is

$$R_1' = K^2 R_1$$

Total or effective resistance of the transformer as referred to primary is

$$R_{01} = R_1 + R_2' = R_1 + \frac{R_2}{K^2}$$

Similarly total transformer resistance as referred to secondary is

$$R_{02} = R_2 + R_1' = R_2 + K^2 R_1$$

The leakage reactance can also be transferred from one winding to the other in the same way as resistance

$$X_2' = \frac{X_2}{K^2} \text{ and } X_1' = K^2 X_1$$

$$\text{and } X_{01} = X_1 + X_2' = X_1 + \frac{X_2}{K^2}$$

$$\text{and } X_{02} = X_2 + X_1' = X_2 + K^2 X_1$$

The total impedance of the transformer referred to primary and secondary is given by

$$Z_{01} = \sqrt{R_{01}^2 + X_{01}^2} \text{ and } Z_{02} = \sqrt{R_{02}^2 + X_{02}^2}$$

Example 2: A 2400 V/400 V single-phase transformer takes a no-load current of 0.5 A and core loss = 400 W. Determine the values of the magnetizing and core loss components of no-load current. Draw the no load phasor diagram for the transformer.

Solution: $V_1 = 2400$; $V_2 = 400$ V and $I_o = 0.5$ A

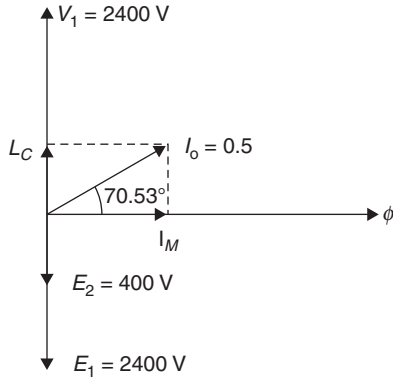
Core loss = Iron loss = 400

$$= V_1 I_o \cos \phi_o$$

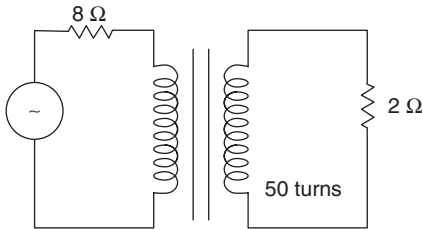
$$\Rightarrow 400 = 2400 \times 0.5 \cos \phi_o$$

$$\phi_o = 70.53^\circ$$

$\therefore$ Phasor



Example 3: If the secondary winding of the transformer of the circuit shown in the given figure has 50 turns, then for maximum power transfer to the 2-ohm resistor, the number of turns required in the primary winding will be



Solution: For maximum power transfer 2

$$= 8 \cdot \left(\frac{N_2}{N_1} \right)^2 \quad (\text{or}) \quad \frac{N_2}{N_1} = \frac{1}{2}$$

$$\text{Or } N_1 = 2N_2 = 2 \times 50 = 100 \text{ turns}$$

EQUIVALENT CIRCUIT

By making use of equivalent circuit, different performance indices such as efficiency, voltage regulation can be determined.

The equivalent circuit of a transformer having voltage transformation ratio $K = \frac{E_2}{E_1}$ is given in the following figure

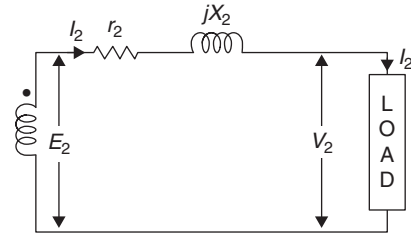
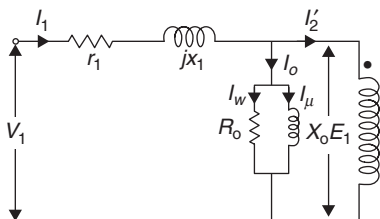


Figure 4(a)

To make calculations simpler, it is preferable to transfer voltage, current and impedance either to the primary or to the secondary. In that case we would have to work in one winding only which is more convenient.

The secondary circuit is shown in figure 4(a) and its equivalent primary values are shown in figure 4(b).

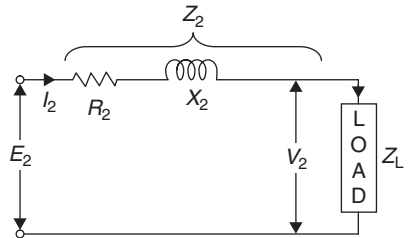
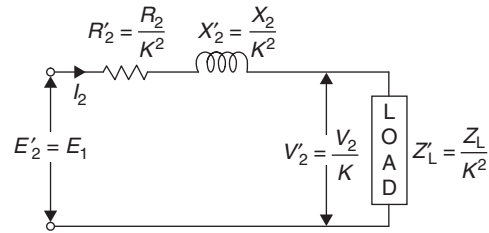
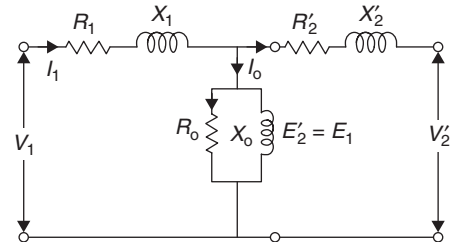


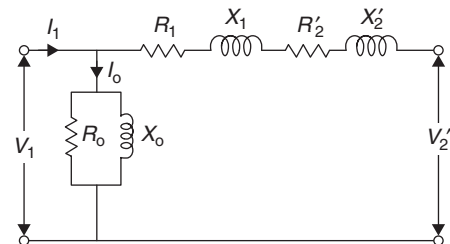
Figure 4(b)



The total equivalent circuit of the transformer is obtained by adding in the primary impedance as shown in the following figure. This is known as exact equivalent circuit. A simplification can be made by transferring the existing circuit across the supply terminals as shown in below figure.

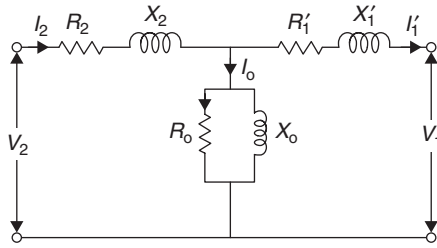


Approximate equivalent ckt referred to primary

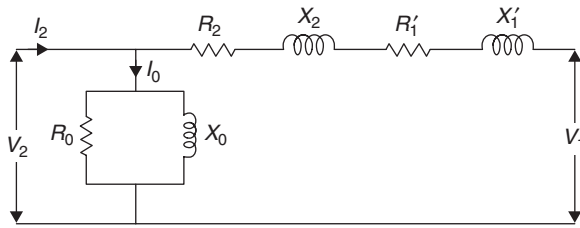


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In a similar manner, the equivalent circuit referred to the secondary can be obtained. Which is shown in the following figure?



Approximate equivalent circuit referred to the secondary is shown in figure.



Example 4: A 15 kVA, 2300/230 V, 50 Hz single-phase transformer gave the following test data.

Open-circuit test (H. V. side)

$$V_0 = 2300 \text{ V}, I_0 = 0.21 \text{ A}$$

$$W_0 = 50 \text{ W}$$

Short-circuit test (H. V. side)

$$V_{sc} = 47 \text{ V}, I_{sc} = 6.0 \text{ A}$$

$$W_{sc} = 160 \text{ W}$$

Obtain the equivalent circuit referred to high voltage side.

Solution: As both O.C. test and S.C. test carried out on high voltage side the results will directly lead to equivalent circuit referred to high voltage side.

Shunt branch elements

$$V_0 = 2300 \text{ V}, I_0 = 0.21 \text{ A}, W_0 = 50 \text{ W}$$

$$R_0 = \frac{V_0^2}{W_0} = \frac{2300^2}{50} = 105.8 \times 10^3 \Omega$$

$$I_w = I_0 \cos \phi_0 = \frac{W_0}{V_0} = \frac{50}{2300} = 0.02174 \text{ A}$$

$$I_\mu = \sqrt{I_0^2 - I_w^2} = \sqrt{0.21^2 - (0.02174)^2} = 0.20887 \text{ A}$$

$$\therefore X_0 = \frac{V_0}{I_\mu} = \frac{2300}{0.20887} = 11.011 \times 10^3 \Omega$$

Series branch elements

$$V_{sc} = 47 \text{ V}, I_{sc} = 6.0 \text{ A},$$

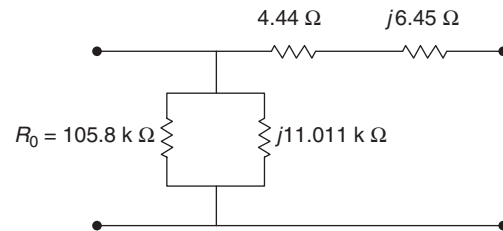
$$W_{sc} = 160 \text{ W}$$

$$R = \frac{160}{6^2} = 4.44 \Omega$$

$$Z = \frac{47}{6.0} = 7.833 \Omega$$

$$X = \sqrt{Z^2 - R^2} = \sqrt{41.608} = 6.45 \Omega$$

Equivalent circuit referred to h.v. side.



Example 5: A 3300/230 V, 50-kVA transformer is found to have impedance of 4% and a Cu. loss of 1.8% at full load. Find its percentage reactance and also the ohmic values of resistance, reactance and impedance as referred to primary. What would be the value of primary short-circuit current if primary voltage is assumed constant?

$$\text{Solution: } \% X = \sqrt{(\% Z^2 - \% R^2)}$$

$$= \sqrt{4^2 - 1.8^2} = 3.57\%$$

$$(\therefore \% \text{ Cu loss} = \% R)$$

$$I_1, \text{ full load} = \frac{50 \times 10^3}{3300} = 15.2 \text{ A}$$

Considering primary winding, we have $\% R = 1.8$

$$\% R = \frac{R_{01} \times I_1 \times 100}{V_L}$$

$$R_{01} = \frac{\% R \times V_L}{I_1 \times 100} = 3.91$$

$$\text{Similarly } \% X = \frac{X_{01} \times I_1 \times 100}{V_1} = 3.57$$

$$\therefore X_{01} = \frac{3.57 \times 3300}{100 \times 15.2} = 7.76 \Omega$$

$$\text{Similarly } Z_{01} = \frac{4 \times 3300}{100 \times 15.2} = 8.7 \Omega$$

Short-circuit current

$$I_{sc} = \frac{V_1}{Z_{01}}$$

$$\text{Now } Z_{01} = \frac{V_1 \times \%Z}{100 \times I_1}$$

$$I_{sc} = \frac{V_1 \times 100 \times I_1}{V_1 \times \%Z} = \frac{100 \times I_1}{\%Z}$$

∴ Short-circuit current

$$= \frac{100 \times 15.2}{4} = 380 \text{ A}$$

Example 6: Calculate the regulation of a transformer in which the ohmic loss is 1% of the output and the reactance drop is 5% of the output voltage, when the power factor is 0.8 lead.

Solution: Per unit regulation at 0.8 leading
 p.f. = $0.01 \times 0.8 - 0.05 \times 0.6$
 = -0.022 or -2.2%

TRANSFORMER TESTS

The usual tests conducted on the transformer are

1. Open-circuit or no-load test
2. Short-circuit or impedance test
3. Sumpner's or back-to-back test

Open-circuit Test

The purpose of this test is to determine no load loss or core loss and no-load current I_0 , which is helpful in finding shunt branch parameters of the equivalent circuit, i.e., R_0 and X_0 .

Also this test is helpful in determining the magnetizing current I_μ and core loss component (working component) I_w .

- It is preferable to apply the rated voltage to the LV winding keeping the HV winding open. This is from the point of view of the safety of the testing personnel and also for reading the meters with great accuracy.

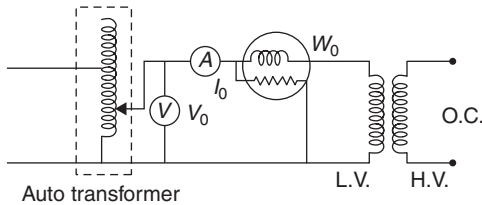


Figure 5 O.C. test circuit diagram

If W_0 is the no-load power input, (V_0 rated voltage), and I_0 the no-load current we have

$$\text{No-load p.f., } \cos \phi_0 = \frac{W_0}{V_0 I_0}$$

$$\text{Active component of no-load current, } I_w = I_0 \cos \phi_0 = \frac{W_0}{V_0}$$

$$\text{The magnetizing component, } I_\mu = \sqrt{I_0^2 - I_w^2}$$

Shunt branch parameters of the equivalent circuit are

$$R_0 = \frac{V_0}{I_w} \text{ and } X_0 = \frac{V_0}{I_\mu}$$

Note: Since I_0 is only of the order of 5% of the rated current, the no-load Cu loss of the primary winding will be $\left(\frac{1}{20}\right)^2$ times the full-load copper loss. Hence copper loss can be neglected. So, the no load power input W_0 is equal to the iron or core loss P_i .

S.C. Test

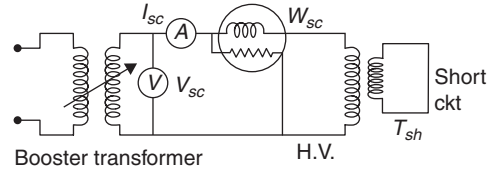


Figure 6 S.C. test circuit diagram

- Since the H.V. rated current is less, short-circuit test is conducted preferably on H.V. side keeping L.V. short circuited.
- For conducting short-circuit test only a small percentage (of the order 5% to 10%) of the rated voltage is sufficient to cause rated current to flow through the windings. As the voltage is less, the core flux density is very much less compared to normal value, so the iron loss which depends on flux density (voltage) can be neglected. The input is therefore equal to total copper loss.
- Through this test, the full-load Cu loss and the equivalent resistance and leakage reactance are determined.
- Let I_{sc} be the current, V_{sc} be the voltage and W_{sc} be the power input, respectively, on short circuit.

$$\text{Then, } R_{01} = \frac{W_{sc}}{I_{sc}^2} \text{ and } Z_{01} = \frac{V_{sc}}{I_{sc}} \text{ and } X_{01} = \sqrt{Z_{01}^2 - R_{01}^2}$$

If the H.V. winding is energized and L.V. is short circuited as is the usual case, the values of R and X obtained above are the values referred to the H.V. side.

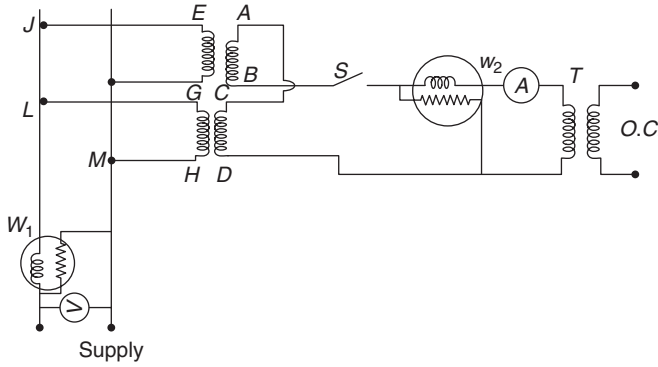
- For drawing the equivalent circuit referred to H.V. or L.V. side, one set of parameters obtained from either O.C. side or S.C. side are to be referred to the other side.
- Limitations of O.C. and S.C. tests

In the O.C. test, though the rated voltage is applied to one of the windings, the other winding carries no current, so that the effect of temperature rise is not taken care of. In the short-circuit test, though the windings are made to carry the rated currents, yet the voltage applied is only a small fraction of a rated voltage, with the result, the flux density is not at its usual value.

- When the transformer is operating under load conditions, both core loss and ohmic loss exists. But in O.C. and S.C. tests, each of them exist separately. Hence from O.C.

and S.C. tests the effect of temperature rises because the losses in the temperature are not taken care of, which is possible with Sumpner's or back-to-back test.

Sumpner's Test



This test is mainly performed to determine temperature rise and the drawbacks of the O.C. and S.C. tests are overcome. For conducting this test two identical transformers are required. The primaries are connected in parallel across the supply voltage and rated voltage of the winding is impressed. The secondaries are connected in series opposition, so that the resultant voltage acting around the closed circuit formed by the secondaries is zero. To circulate the current through the secondaries (and hence in primaries), a voltage is injected into the circuit.

By suitably adjusting the value of the injected voltage, both the secondaries can be made to carry the rated (or any desired value) of the current.

Before closing the switch 'S' to complete the local circuit formed by the secondaries, the correctness of the polarities is to be ensured. If zero reading of the voltmeter connected across the switch ensures that the polarities are correct.

- The wattmeter connected on the primary side reads twice the iron loss of each transformer. The watt meter connected in the secondary circuit reads twice the copper loss corresponding to the short-circuit current I_{sc} and V_{sc} equals twice the impedance drop of each transformer.

All-day Efficiency

The all-day efficiency is defined as the ratio of output power in kWh to the input in kWh during the day.

Transformers used for distribution have their primaries energized all 24 hours, although their secondaries supply little or no-load much of the time during the day except during the house lighting period. Hence, it is considered a good practice to design such transformers so that core losses are very low.

The Cu losses are relatively less important, because they depend on the load. The performance of distribution transformer is compared on the basis of energy consumed during a certain time period, usually a day of 24 hours.

$$\eta_{\text{all-day}} = \frac{\text{Output in kWh}}{\text{Input in kWh}} \text{ (for 24 hours)}$$

This efficiency is always less than the commercial efficiency of a transformer.

The Per Unit (p.u.) System

Per unit quantity

$$= \frac{\text{Actual value of the quantity in any unit}}{\text{Base value of the quantity in the same unit}}$$

Percentage Resistance, Reactance and Impedance

These quantities are usually measured by the voltage drop at full-load current expressed as a percentage of the normal voltage of the winding on which calculations are made.

1. Percentage resistance at full load

$$\begin{aligned} \% R &= \frac{I_1 R_{01}}{V_1} \times 100 \\ &= \frac{I_1^2 R_{01}}{V_1 I_1} \times 100 \\ &= \frac{I_2^2 R_{02}}{V_2 I_2} \times 100 = \% \text{ Cu loss at full load} \\ \% R &= \% \text{ Cu loss} = V_r \end{aligned}$$

2. Percentage reactance at full load

$$\begin{aligned} \% X &= \frac{I_1 X_{01}}{V_1} \times 100 \\ &= \frac{I_2 X_{02}}{V_2} \times 100 = V_x \end{aligned}$$

3. Percentage impedance at full-load

$$\begin{aligned} \% Z &= \frac{I_1 Z_{01}}{V_1} \times 100 \\ &= \frac{I_2 Z_{02}}{V_2} \times 100 \end{aligned}$$

4. $\% Z = \sqrt{\% R^2 + \% X^2}$

The ohmic values of resistance and reactance can be obtained from % values as given below.

$$R_{01} = \frac{\% R \times V_1}{100 \times I_1} = \frac{\% \text{ Cu loss} \times V_1}{100 \times I_1}$$

Similarly

$$R_{02} = \frac{\% R \times V_2}{100 \times I_2} = \frac{\% \text{ Cu loss} \times V_2}{100 \times I_2}$$

$$X_{01} = \frac{\% X \times V_1}{100 \times I_1} = \frac{V_x \times V_1}{100 \times I_1}$$

Similarly

$$X_{02} = \frac{\%X \times V_2}{100 \times I_2} = \frac{V_x \times V_2}{100 \times I_2}$$

Note: The percentage resistance and impedance are the same value whether referred to primary or secondary.

Example 7: A 3300/230 V, 50-kVA transformer is found to have impedance of 4% and a Cu loss of 1.8% at full load. Find its percentage reactance and also the ohmic values of resistance, reactance and impedance as referred to primary. What would be the value of primary short-circuit current if primary voltage is assumed constant?

Solution: $\%X = \sqrt{(\%Z^2 - \%R^2)}$

$$= \sqrt{4^2 - 1.8^2} = 3.57\%$$

($\therefore$ % Cu loss = %R)

$$I_1, \text{ full load} = \frac{50 \times 10^3}{3300} = 15.2 \text{ A}$$

Considering primary winding, we have

$$\%R = 1.8$$

$$\%R = \frac{R_{01} \times I_1}{V_1} \times 100$$

$$R_{01} = \frac{\%R \times V_1}{I_1 \times 100} = 3.91$$

Similarly

$$\%X = \frac{X_{01} \times I_1}{V_1} = 3.57$$

$$\therefore X_{01} = \frac{3.57 \times 3300}{100 \times 15.2} = 7.76 \Omega$$

$$\text{Similarly } Z_{01} = \frac{4 \times 3300}{100 \times 15.2} = 8.7 \Omega$$

Short-circuit current

$$I_{sc} = \frac{V_1}{Z_{01}}$$

Now

$$Z_{01} = \frac{V_1 \times \%Z}{100 \times I_1}$$

$$I_{sc} = \frac{V_1 \times 100 \times I_1}{V_1 \times \%Z} = \frac{100 \times I_1}{\%Z}$$

$\therefore$ Short-circuit current

$$= \frac{100 \times 15.2}{4} = 380 \text{ A}$$

LOSSES AND EFFICIENCY

Transformer is a static device; hence there are no mechanical losses. Losses in the transformer are

1. Iron loss or core loss, i.e., hysteresis and eddy current losses, which occur in magnetic core.
2. Copper losses or I^2R loss which occurs in the windings.

The core loss of a transformer depends upon the frequency and the maximum flux density when the volume and the thickness of the core laminations are given. The core loss is made up of two parts.

1. Hysteresis loss, $W_h = PB_{\max}^{1.6} f$ as given by Steinmetz empirical relation and

2. Eddy current loss, $W_e = QB_{\max}^2 f^2$,

where P and Q are constants.

The total core loss is given by

$$W_i = W_h + W_e = PB_{\max}^{1.6} f + QB_{\max}^2 f^2$$

Separation of Core Losses

- If we carry out two experiments using two different frequencies but the same maximum flux density, we should be able to find the constants P and Q and hence calculate hysteresis and eddy current losses separately.
- The flow of current through the windings gives rise to the copper losses, i.e., $I_1^2 r_1$ and $I_2^2 r_2$.
- The magnetic losses are present as long as primary is energized.
- The no-load current is only of the order of 5% of the rated or full-load current. Hence no load Cu loss can be neglected. So, the no-load input to a transformer is considered as Iron or core or magnetic loss. The iron loss is assumed to be the same under all operating conditions, i.e., from no load to full load. It is denoted by P_i .
- The Cu loss varies with the secondary (and hence primary) current. The copper loss corresponding to the rated value of current is called full-load Cu loss. It is denoted by P_{cu} .
- The efficiency (or commercial efficiency) of a transformer is given by

$$\eta = \frac{\text{Output power}}{\text{Input power}} = \frac{V_2 I_2 \cos \phi_2}{V_2 I_2 \cos \phi_2 + P_i + I_2^2 R_{02}}$$

where P_i = iron loss

$I_2^2 R_{02}$ = ohmic loss or Cu loss.

$V_2 I_2$ = Output kVA and $\cos \phi_2$ = load pf.

Condition for Maximum Efficiency

Maximum efficiency occurs when variable losses (i.e., Cu losses) are equal to the fixed loss (i.e., core loss).

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$$I_2^2 R_{02} = P_i$$

$$\text{or } I_2 = \sqrt{\frac{P_i}{R_{02}}}$$

$$= I_{FL} \sqrt{\frac{P_i}{\text{Full-load Cu loss}}}$$

kVA corresponding to maximum efficiency.

$$(\text{kVA})_{\max \eta} = \text{Full-load kVA} \sqrt{\frac{P_i}{I_{FL}^2 R_{02}}}$$

Example 8: A 50 kVA transformer has a core loss of 400 W and a full-load copper loss of 800 W. what is the load corresponding to maximum efficiency.

Solution: Iron loss $P_i = 400$ W
Full-load copper loss $P_{Cu} = 800$ W
kVA corresponding to max efficiency

$$= \text{full-load kVA} \times \sqrt{\frac{P_i}{P_{Cu}}}$$

$$= 50 \times \sqrt{\frac{400}{800}} = 35.35 \text{ kVA}$$

Example 9: In a transformer, the core loss is found to be 52 W at 40 Hz and 90 W at 60 Hz measured at same flux density. Calculate the hysteresis and eddy current losses at 50 Hz.

Solution: Since the flux density is the same in both cases, we can use the relation

Total core loss, $W_i = Af + Bf^2$ or $W_i/f = A + Bf$,

$$\therefore \frac{52}{40} = A + 40 B$$

$$\text{And } \frac{90}{60} = A + 60 B$$

$$\therefore A = 0.9 \text{ and } B = 0.01$$

At 50 Hz,

$$\text{The hysteresis loss } W_h = Af$$

$$= 0.9 \times 50 = 45 \text{ W}$$

The eddy current loss

$$W_e = Bf^2 = 0.01 \times 50^2 = 25 \text{ W}$$

Common Data for Examples 10 to 12

A transformer with normal voltage impressed has a flux density of 1.4 wb/m² and a core loss comprising of 1000 W eddy current loss and 3000 W hysteresis loss. What do these losses become under the following conditions?

Example 10: Increasing the applied voltage by 10% at rated frequency

Solution: Hysteresis loss $W_h \propto B_{\max}^{1.6}$

$$= PB_{\max}^{1.6} f$$

Eddy current loss $W_e \propto B_{\max}^2 f^2$

$$= QB_{\max}^2 f^2$$

From the relation

$$E = 4.44fNB_{\max}A.$$

We get $B_{\max} \propto E/f$

Putting this value of $B_{\max}$ in the above equations, we have

$$W_h = P \left(\frac{E}{f} \right)^2 f = PE^{1.6} f^{-0.6}$$

$$\text{And } W_e = Q \left[\frac{E}{f} \right]^2 f^2 = QE^2$$

From the given data, we have

$$3000 = PE^{1.6} f^{-0.6} \quad (1)$$

$$\text{And } 1000 = QE^2 \quad (2)$$

where E and f are the normal values of primary voltage and frequency, respectively.

If the voltage is increased by 10% at rated frequency, the new hysteresis loss is

$$W_h = P(1.1E)^{1.6} f^{-0.6} \quad (3)$$

Dividing Eq. (3) by (1),

$$\text{We get } \frac{W_h}{3000} = 1.1^{1.6}$$

$$W_h = 3000 \times 1.165 = 3495 \text{ W}$$

The new eddy current loss is

$$W_e = Q(1.1E)^2 \Rightarrow \therefore \frac{W_e}{1000} = 1.1^2$$

$$W_e = 1000 \times 1.21 = 1210 \text{ W}$$

Example 11: Reducing the frequency by 10% with normal voltage impressed.

Solution: Since $W_e = QE^2$, eddy current loss is unaffected with change in the supply frequency. The new hysteresis loss is,

$$W_h = PE^{1.6} (0.9f)^{-0.6} \quad (4)$$

From Eq. (1) and (4), we get

$$\frac{W_h}{3000} = 0.9^{-0.6}$$

$$W_h = 3000 \times 1.065 = 3196 \text{ W}$$

Example 12: Increasing both impressed voltage and frequency by 10 per cent

Solution: In this case, both E and f are increased by 10%. The new losses are as under;

$$W_h = P(1.1E)^{1.6} (1.1f)^{-0.6}$$

$$\begin{aligned}\frac{W_h}{3000} &= 1.1^{1.6} \times 1.1^{-0.6} \\ &= 1.165 \times 0.944 \\ W_h &= 3000 \times 1.165 \times 0.944 \\ &= 3299 \text{ W}\end{aligned}$$

The eddy current loss, W_e depends only on applied voltage and is unaffected by changes in f ,

$$W_e = Q(1.1)E^2$$

$$\begin{aligned}\therefore \frac{W_e}{1000} &= 1.1^2 \\ &= 1000 \times 1.21 = 1210 \text{ W.}\end{aligned}$$

VOLTAGE REGULATION OF A TRANSFORMER

When a transformer is supplied with a constant primary voltage, the secondary terminal voltage varies as the load current and p.f. of the load vary. The voltage regulation is defined as the change in secondary voltage, expressed as a percentage (or p.u.) of the secondary rated voltage, when load at a given power factor is reduced to zero, with primary applied voltage held constant.

$$\begin{aligned}\text{The per unit voltage regulation} &= \\ \frac{(\text{No load secondary voltage} - \text{Secondary voltage at rated load})}{\text{Secondary voltage at rated load}}\end{aligned}$$

$$\% \text{Regulation} = (\text{p.u. regulation} \times 100)$$

The magnitude of change in secondary voltage depends on the load power factor, load current total resistance and total reactance of the transformer.

$$\begin{aligned}\text{Voltage regulation} &= \frac{E_2 - V_2}{E_2} \text{ in p.u.} \\ &= \frac{E_2 - V_2}{E_2} \times 100 \text{ in percentage}\end{aligned}$$

From the approximate equivalent circuit, the expression for voltage regulation can be given as follows.

$$E_2 - V_2 = I_2 R_{02} \cos \phi \pm I_2 X_{02} \sin \phi$$

[+ve and -ve signs are taken for lagging and leading p.f.'s, respectively]

where $\cos \phi$ is load power factor.

Therefore per unit voltage regulation

$$\varepsilon = \varepsilon_r \cos \phi \pm \varepsilon_x \sin \phi$$

where ε - p.u. voltage regulation

ε_r - p.u. resistance

ε_x - p.u. reactance

+ for lagging p.f. and

- for leading p.f.

- condition for maximum voltage regulation:

$$\text{Since, } |I| R_{02} \cos \theta - |I| X_{02} \sin \theta = 0$$

$$\phi = \tan^{-1} \left[\frac{X_{01}}{R_{01}} \right] = \tan^{-1} \left[\frac{X_{02}}{R_{02}} \right]$$

Maximum regulation occurs at a lagging p.f.

Condition for Zero Voltage Regulation

$$\phi = \tan^{-1} \left[\frac{R_{01}}{X_{01}} \right] = \tan^{-1} \left[\frac{R_{02}}{X_{02}} \right]$$

Zero voltage regulation occurs at a leading p.f.

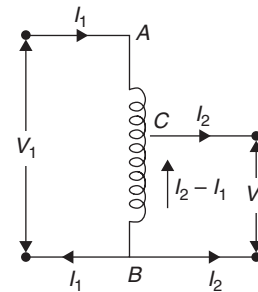
Example 13: The equivalent resistance and reactance of a transformer are, respectively, 0.6Ω and 0.8Ω . At what power factor the voltage regulation will be zero?

Solution: % Regulation is zero at a p.f. of

$$\begin{aligned}\phi &= \tan^{-1} \left[\frac{R}{X} \right] \text{ leading} \Rightarrow \phi = \tan^{-1} \left[\frac{0.6}{0.8} \right] \\ \cos \phi &= \cos \left(\tan^{-1} \left(\frac{6}{8} \right) \right)_{\text{lead}}\end{aligned}$$

Auto-transformer

It is a transformer with one winding only, part of this being common to both primary and secondary. The primary and secondary are connected electrically as well as coupled magnetically.



- There is a superimposition of the input and output currents in the part of the winding common to primary and secondary.
- As an ordinary transformer, the transformer ratio,

$$\frac{V_2}{V_1} = \frac{E_{BC}}{E_{AB}} = \frac{N_2}{N_1} = \frac{I_1}{I_2} = K$$

Power is transferred from the primary to the secondary both inductively and conductively.

$$\text{Power delivered to load } P = V_2 I_2$$

$$\begin{aligned}\text{Power transformed or power transferred inductively} \\ = V_2 I_2 (1 - K)\end{aligned}$$

$$\text{Power transferred conductively} = K V_2 I_2$$

- As compared to an ordinary two-winding transformer of same output, an auto-transformer uses less copper

because of one winding, has higher efficiency, smaller in size and cheaper.

- Since the HV and LV windings are electrically connected, a fault on the HV side may cause a high voltage to appear on LV side.

- Weight of Cu in auto-transformer

$$= (1 - K) \times \text{weight of Cu in two-winding transformer}$$

Saving = $K \times$ weight of Cu in two-winding transformer.

$$\frac{\text{kVA rating as an auto-transformer}}{\text{kVA rating as two-winding transformer}} = \frac{1}{1 - K}$$

$$\frac{\text{Full load losses as an auto-transformer}}{\text{Full load losses as a two-winding transformer}} = (1 - K)$$

$$\frac{\text{p.u. impedance drop in auto-transformer}}{\text{p.u. impedance drop in two-winding transformer}} = 1 - K$$

$$\frac{\text{Voltage regulation as an auto-transformer}}{\text{Voltage regulation in two-winding transformer}} = 1 - K$$

Applications

1. It is used as an auto starter for induction motor.
2. It is used to give small boost to a distribution line.
3. It is used in testing laboratories.

CONVERSION OF TWO-WINDING TRANSFORMER INTO AUTO-TRANSFORMER

Any two-winding transformer can be converted into an auto-transformer either step-down or step-up.

There are two possibilities of such conversion.

1. Connecting two windings with additive polarity to get a step-up auto-transformer.
2. Connecting two windings with subtractive polarity to get a step-down auto-transformer.

For example consider a 20-kVA, 2400/240 V two-winding transformer with its polarity markings shown in figure 7(a).

(a) Additive polarity

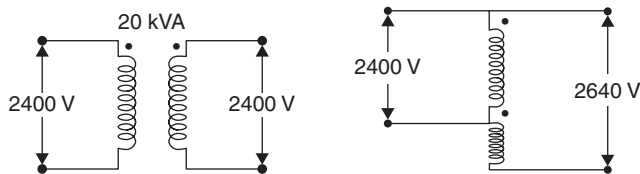


Figure 7(a)

Figure 7(b)

Because of additive polarity

$$V_2 = 2400 + 240 = 2640 \text{ and}$$

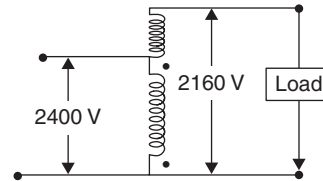
$$V_1 = 2400 \text{ V}$$

There is a marked increase in the kVA rating of the auto-transformer.

kVA rating of auto-transformer

$$= \frac{V_2 I_2}{1000} = \frac{2640 \times 83.3}{1000} = 220 \text{ kVA}$$

(b) Subtractive polarity



$$\text{Here } V_2 = 2400 - 240 = 2160 \text{ V}$$

In this case also there is a large increase in the kVA rating of auto-transformer.

kVA rating of auto-transformer

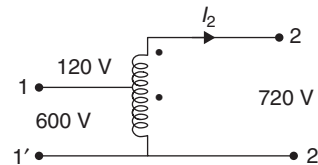
$$= \frac{V_2 I_2}{1000} = \frac{2160 \times 83.3}{1000} = 180 \text{ kVA.}$$

When two-winding transformer is converted to an auto-transformer some portion (or common portion) of winding is overloaded. The increase in kVA capacity is due to power transfer both by conduction and induction.

Example 14: A single-phase transformer has rating of 15 kVA, 600/120 V. It is recommended as an auto-transformer to supply at 720 V from a 600 V primary source. The maximum load it can supply is _____

Solution: Current rating of L.V. winding

$$= \frac{15 \times 10^3}{120} = 125 \text{ A}$$



Maximum load can be supplied, if the two windings are connected in additive polarity as shown in above figure.

$$\text{Output kVA} = \frac{720 \times 125}{1000} = 90 \text{ kVA}$$

PARALLEL OPERATION OF SINGLE-PHASE TRANSFORMERS

For supplying a load in excess of the rating of an existing transformer, a second transformer may be connected in parallel with it. If the demand is more than present supply, two or more transformers are placed in parallel instead of a single large unit, due to which the power system becomes more

reliable. Transformers are switched off or added as per load requirement.

There are certain definite conditions which must be satisfied in order to avoid any local circulating currents and to ensure that transformers share the common load in proportion to their kVA ratings. The conditions are

1. Primary windings of the transformers should be suitable for the supply system voltage and frequency
2. The transformers should be properly connected with regard to polarity.
3. The voltage ratings of both primaries and secondaries should be identical. In other words, the transformers should have the same turn ratio, i.e., transformation ratio.
4. The percentage impedance should be equal in magnitude and have the same X/R ratio in order to avoid circulating currents and operation at different power factors.
5. With transformers having different kVA ratings, the equivalent impedances should be inversely proportional to the individual kVA rating if circulating currents are to be avoided.

Of these conditions, condition (2) is absolutely essential (otherwise paralleling with incorrect polarities will result in dead short-circuit).

If condition (3) is not exactly satisfied, i.e., the two transformers have slightly different transformation or voltage ratios, even then parallel operation is possible. But due to inequality of induced e.m.f.'s in secondaries, there will be some circulating current between the secondaries even on no-load. When secondaries are loaded, this localized circulating current will tend to produce unequal loading condition. Hence, it may be impossible to take full kVA output from the parallel connected group without one of the transformers becoming overheated.

If condition (4) is not exactly satisfied, i.e., impedance triangles are not identical in shape and size, parallel operation will still be possible, but the power factors at which the transformers operate will be different from the power factor of common load. Therefore in this case, the two transformers will not share the load in proportion to their kVA ratings.

The currents carried by the two transformers are proportional to their ratings, provided the numerical impedances are inversely proportional to these ratings and their percentage impedances are identical.

If E_A and E_B are the no-load induced e.m.f.'s of the two transformers connected in parallel, ' I_A ' and ' I_B ' are the currents delivered and ' Z_A ' and ' Z_B ' are the ohmic impedances referred to the L.V side of the two transformers and Z_L is the Load impedance then

$$I_A = \frac{[E_A Z_B + (E_A - E_B) Z_L]}{(Z_A + Z_B) Z_L + Z_A Z_B} \quad I_B = \frac{[E_B Z_A + (E_A - E_B) Z_L]}{(Z_A + Z_B) Z_L + Z_A Z_B}$$

Let ' S ', ' S_A ' and ' S_B ' be the total kVA, kVA delivered by the transformer ' A ' and that delivered by the transformer ' B ', respectively, then we have

$$S_A = S \frac{Z_B}{Z_A + Z_B} \quad S_B = S \frac{Z_A}{Z_A + Z_B}$$

Example 15: Two 1- ϕ transformers are connected in parallel at no-load. One has a turn ratio of 5000/440 and a rating of 200 kVA, the other has a ratio of 5000/480 and a rating of 350 kVA. The leakage reactance of each is 3.5%. What is the no-load circulation current expressed as a percentage of the nominal current of the 200 kVA transformer.

Solution: Assume that per turn voltage is 1V. the normal currents are

$$\frac{200 \times 10^3}{440} = 455 \text{ A}$$

And $\frac{350 \times 10^3}{480} = 730 \text{ A}$

Reactances seen from the secondary side are

$$\frac{3.5}{100} \times \frac{440}{455} = 0.034 \Omega$$

$$\frac{3.5}{100} \times \frac{480}{730} = 0.023 \Omega$$

The difference of induced voltages is 40 V. The circulating current is

$$I_c = \frac{40}{(0.034 + 0.023)} = 704 \text{ A}$$

$$= 1.55 \text{ times or } 155\% \text{ the normal current of } 200 \text{ kVA unit.}$$

Example 16: A 500-kVA single-phase transformer with 0.012 p.u resistance and 0.06 p.u. reactance, is connected in parallel with a 250 kVA single-phase transformer with 0.014 p.u. resistance and 0.045 p.u. reactance to share a load of 600 kVA at 0.8 p.f. lagging. Find the load shared by each transformer.

Solution: The p.u. resistance and reactance given above refer to different ratings. These should be adjusted to some basic kVA.

Let Base kVA = 500

$$Z_A \text{ p.u.} = 0.012 + j0.06 \\ = 0.061 \angle 78.69^\circ$$

$$Z_B \text{ p.u.} = (0.014 + j0.045) \times \frac{500}{250} \\ = 0.094 \angle 72.7^\circ$$

Load shared by transformer A

$$S_A = \frac{Z_B}{Z_A + Z_B} \cdot S$$

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$$= \frac{0.094 \angle 72.7^\circ \times 600 \angle -36.87^\circ}{0.1552 \angle 75.07^\circ}$$

$$= 364.37 \angle -39.22^\circ$$

364 kVA at a p.f. of 0.774 lag

Load shared by transformer B

$$S_B = \frac{Z_A \cdot S}{Z_A + Z_B}$$

$$= \frac{0.0612 \angle 78.69^\circ \times 600 \angle -36.87^\circ}{0.1552 \angle 75.07^\circ}$$

$$= 236.60 \angle -33.25^\circ$$

236.6 kVA at a p.f. of 0.836 lag.

Example 17: Two transformers A and B are connected in parallel to a load of $2 + j1.5$ ohms. Their impedances in secondary terms are $Z_A = (0.15 + j0.5)$ ohm and $Z_B = (0.1 + j0.6)$ ohm. Their no-load terminal voltages are $E_A = 207 \angle 0^\circ$ V and $E_B = 205 \angle 0^\circ$ V. Find the power output and power factor of each transformer.

Solution: $I_A = \frac{E_A Z_B + (E_A - E_B) Z_L}{Z_A Z_B + Z_L (Z_A + Z_B)}$

$$Z_A = (0.15 + j0.5) \Omega$$

$$Z_B = (0.1 + j0.6) \Omega;$$

$$Z_L = (2 + j1.5) = 2.5 \angle 36.9^\circ$$

$$I_A = \frac{207(0.1 + j0.6) + (207 - 205)(2 + j1.5)}{(0.15 + j0.5) \times (0.1 + j0.6) + (2 + j1.5)(0.25 + j1.1)}$$

$$= \frac{24.7 + j127.2}{-1.435 + j2.715} = \frac{129.7 \angle 79^\circ}{3.07 \angle 117.9^\circ}$$

$$= 42.26 \angle -38.9^\circ$$

$$= (32.89 - j26.55) \text{ A}$$

$$I_B = \frac{E_B Z_A - (E_A - E_B) Z_L}{Z_A Z_B + Z_L (Z_A + Z_B)}$$

$$= \frac{205(0.15 + j0.5) - 2(2 + j1.5)}{-1.435 + j2.715}$$

$$= \frac{103 \angle 75^\circ}{3.07 \angle 117.9^\circ} = 33.56 \angle -42.9^\circ$$

$$= (24.58 - j22.84) \text{ A}$$

$$V_2 = I Z_L = (I_A + I_B) Z_L$$

$$= (57.47 - j49.39)(2 + j1.5)$$

$$= 189 - j12.58 = 189.4 \angle -3.9^\circ$$

Power factor angle of transformer A

$$= -3.9^\circ - (-38.9^\circ) = 35^\circ$$

$$\therefore \text{p.f. of A} = \cos 35^\circ = 0.818 \text{ lag}$$

$$\text{p.f. of B} = \cos(-3.9^\circ) - (-42.9^\circ)$$

$$= 0.776 \text{ lag}$$

Power output of transformer

$$A \text{ is } P_A = 189.4 \times 42.26 \times 0.818 = 6548 \text{ W}$$

$$\text{Similarly } P_B = 189.4 \times 33.56 \times 0.776 = 4900 \text{ W}$$

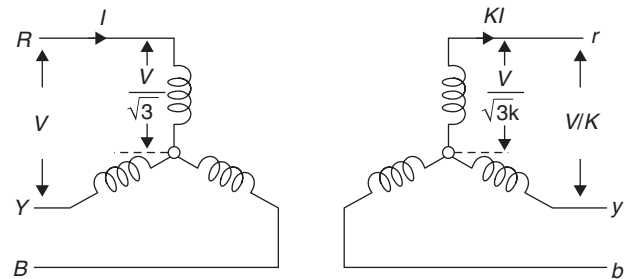
THREE-PHASE TRANSFORMERS

Three-phase transformation can be obtained by having a single, three-phase transformer or by suitable inter connection of three single-phase transformers.

As compared to a bank of single-phase transformers, the main advantages of a three-phase transformer are that it occupies less floor space for equal rating, weight less, costs about 15% less and further, that only one unit is to be handled and connected. However, if a fault occurs, all the loads connected to the three-phase transformer will be interrupted till the repair works are carried out. But, in the case of 3-phase bank of single-phase transformers if one transformer goes out of order, the system can still be run open- Δ at reduced capacity or the faulty transformer can be readily replaced by a single spare.

Three-phase Transformer Connections

Star/Star: 1:1



This connection is most economical for small, high-voltage transformers because the number of turns/phase and the amount of insulation required is minimum (as phase voltage is only $1/\sqrt{3}$ of line voltage). The ratio of line voltages on the primary and secondary is the same as the transformation ratio of each transformer.

There is a phase shift of 30° between the phase voltages and line voltages both on primary and secondary sides.

This connection works only if the load is balanced. With the unbalanced load, the neutral point shifts thereby making the three line-to-neutral voltages unequal. This difficulty of shifting (or floating) neutral can be obviated by connecting the primary neutral to neutral of generator. Star-Star transformers are operated with grounded neutrals.

With an isolated neutral, any unbalanced load causes the shifting of the neutral.

For delivering a sine wave of voltage, it is necessary to have a sine wave of flux in the core but on account of the characteristics of iron, a sine wave of flux requires a third harmonic component in the exciting current. At any instant the third harmonic component tends to flow towards or

away from neutral. If primary neutral is isolated, the third harmonic components of the magnetizing current do not have a path. Hence, the flux in the core cannot be a sine wave and so the phase voltages become non-sinusoidal, though the line voltages are sinusoidal.

Another way of avoiding this trouble of oscillating neutral is to provide each of the transformers with a third or tertiary winding of relatively low kVA rating. This tertiary winding is connected in Δ and provides a circuit in which the third harmonic component can flow.

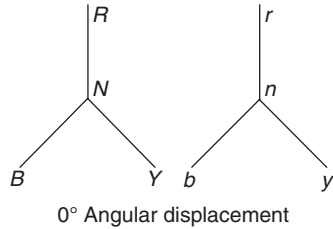
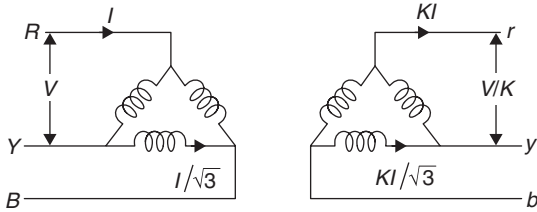


Figure 8 Phasor diagram

Delta/Delta



This connection is economical for large, low voltage transformers in which insulation problem is not so urgent, because it increases number of turns/phase.

The ratio of transformation between primary and secondary line voltage is exactly same as that of each transformer and there is no angular displacement between them. This connection has the following advantages:

1. In order that the output voltage be sinusoidal, it is necessary that the magnetizing current of the transformer must contain a third order harmonic component. In this case, the third harmonic component of the magnetizing current can flow in the Δ -connected transformer primaries without flowing in the line wires. The three phases are 120° apart, which is $3 \times 120^\circ = 360^\circ$ with respect to the third harmonic, hence it merely circulates in the delta. Therefore the flux is sinusoidal, which results in sinusoidal voltages.
2. No difficulty is experienced from unbalanced loading as was the case in star-star connection. The three-phase voltages remain practically constant regardless of the load impedance.
3. An added advantage of this connection is that if one of the transformer phases becomes disabled, the system can continue to operate in open-delta or in $V-V$

although with reduced available capacity. The reduced capacity is 58% and not 66.7% of the normal capacity.

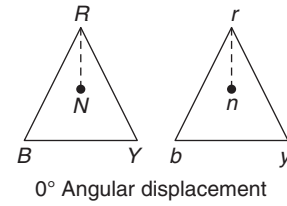
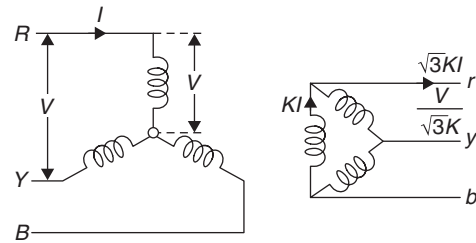


Figure 9 Phasor diagram

Star/Delta



The main use of this connection is at the substation end of the transmission line where the voltage is to be stepped down. The primary winding is star connected with grounded neutral.

The ratio between the secondary and primary line voltage is $1/\sqrt{3}$ times the transformation ratio of each transformer. There is a 30° shift between the primary and secondary voltages which means that a star-delta transformer bank cannot be paralleled with either a star-star or a delta-delta bank.

The third harmonic currents flow in the delta to provide a sinusoidal flux.

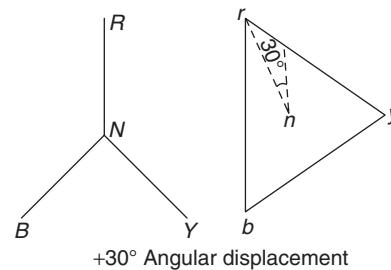
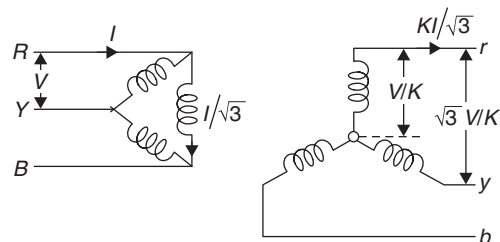


Figure 10 Phasor diagram

Delta/Star



This connection is generally employed where it is necessary to step up the voltage as for example, at the beginning of high tension transmission system. The neutral of the secondary is grounded for providing 3-phase, 4-wire service.

- This is the most popular connection as it can be used to serve both the 3-phase power equipment and single-phase lighting circuits.
- This connection is not open to the objection of a floating neutral and voltage distortion because the existence of a Δ -connection allows a path for the third-harmonic currents.
- The primary and secondary line voltages and line currents are out of phase with each other by 30° . Because of this 30° shift, it is impossible to parallel such a bank of transformers even though the voltage ratios are correctly adjusted. The ratio of secondary to primary voltage is $\sqrt{3}$ times the transformation ratio of each transformer.

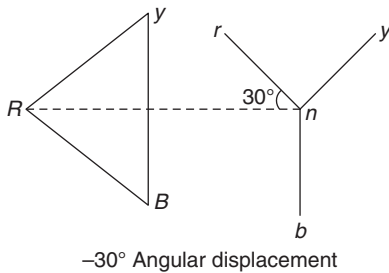


Figure 11 Phasor diagram

Example 18: The percentage impedance of a 100 kVA, 11 kV /400 V, delta/bye, 50 Hz transformer is 4.55 for the circulation of half the full-load current during short-circuit test, with low voltage terminals shorted, what should be the applied voltage on high voltage side?

Solution: Full-load primary current

$$I_1 = \frac{100 \times 10^3}{11 \times 10^3} = 9.09 \text{ A}$$

$$Z_{01} = \% \frac{Z \times V_1}{100 I_1}$$

$$= \frac{4.5 \times 11 \times 10^3}{100 \times 9.09} = 55 \Omega$$

For half full-load current during short-circuit test

$$Z_{01} = \% \frac{Z \times V_1}{100 \times I_1 / 2}$$

$$Z_{01} = \frac{4.5 \times 11 \times 10^3}{100 \times \frac{9.09}{2}} = 110 \Omega$$

$$\text{Half full-load current} = \frac{9.09}{2} = 4.5$$

$$\text{Applied voltage} = 4.5 \times 55$$

$$= 247.5 \text{ V}$$

Open-Delta or V-V Connection

If one of the transformers of Δ - Δ is removed and 3-phase supply is connected to the primaries as shown in fig, then three equal 3-phase voltages will be available at the secondary terminals on no-load. This method of transforming 3-phase power by means of only two transformers is called the open- Δ or V - V connection.

The total load that can be carried by a V - V bank is not two-third of the capacity of a Δ - Δ bank but it is only 57.7% of it.

$$\frac{V-V \text{ capacity}}{\Delta-\Delta \text{ capacity}} = \frac{\sqrt{3} V_L I_s}{3 V_L I_s}$$

$$= \frac{1}{\sqrt{3}} = 0.577 \text{ or } 58\%$$

i.e., only 86.6% of the rated capacity of the two remaining transformers is available. In other words, ratio of operating capacity to available capacity of an open Δ is 0.866 or utility factor is 0.866.

Each transformer will supply 57.7% of load and not 50% when operating in V - V .

If three transformers in a Δ - Δ bank are delivering their rated load and one transformer is removed, the overload on each of the two remaining transformers is 73.2% because

$$\frac{\text{Total load in } V-V}{\text{VA of transformer}} = \frac{\sqrt{3} V_L I_s}{V_L I_s} = \sqrt{3} = 1.732$$

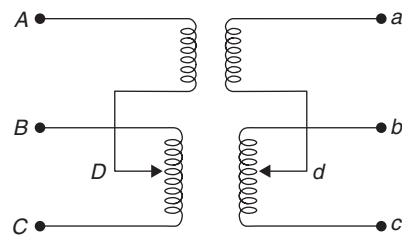
The disadvantages of this connection are

1. The average power factor is 86.6% of the balanced load power factor. One of two transformers operates at a p.f. of $\cos(30 + \theta)$ and the other at $\cos(30 - \theta)$.
2. The secondary terminal voltages tend to become unbalanced to a great extent when the load is increased, this happens even when the load is perfectly balanced.

SCOTT Connection or T-T Connection

This is a connection by which 3-phase to 3-phase transformation is accomplished using two transformers as shown in below figure.

This connection can also be used for 3-phase to 2-phase transformation.



If the two transformers are of identical voltage ratings, one of the transformers has centre taps both on the primary and

secondary windings and is known as the main transformer. The other has 0.866 tap and is known as teaser transformer.

- Alternatively teaser may be designed for 86.6% voltage of the main transformer.
- If the load power factor is $\cos\theta$ lag then
 - (a) Teaser transformer operates at a lagging p.f. of $\cos\theta$
 - (b) one half of the main transformer operates a p.f. of $\cos(30 + \theta)$ lag
- If V is the line voltage and I is the rated line current, the combined VA rating of the two transformers = $VI(1 + 0.866)$ if both the teaser primary and secondary windings are designed for 0.866 times the voltage on the main transformer.

$$\text{Output} = \sqrt{3} VI$$

$$\text{Utilization factor} = \frac{\sqrt{3}VI}{(1 + 0.866)VI} = 0.928$$

If the two transformers are of equal voltage rating (and with taps at 86.6% points), 13.4% of the winding is unutilized.

$$\begin{aligned}\text{Then utility factor of the winding} &= \frac{\sqrt{3}VI}{2 VI} \\ &= 0.866\end{aligned}$$

Example 19: A 5000 kVA, three-phase transformer 6.6/33-kV, Δ/Y , has a no-load loss of 15 kW and a full-load loss of 50 kW. The impedance drop at full load is 7%. Calculate the primary voltage when a load of 3,200 kW at 0.8 p.f. is delivered at 33 kV.

Solution: Full-load $I_2 = 5 \times 10^6 / \sqrt{3} \times 33,000$
 $= 87.5 \text{ A}$

Impedance drop /phase

$$\begin{aligned}&= 7\% \text{ of } (33 / \sqrt{3}) \text{ kV} \\ &= 1330 \text{ V}\end{aligned}$$

$$\therefore Z_{02} = \frac{1330}{87.5} = 15.3 \Omega/\text{phase}$$

F. L. Cu loss = 50 – 15 = 35 kW

$$\therefore 3I_2^2 R_{02} = 35,000$$

$$\begin{aligned}R_{02} &= 35,000 / 3 \times (87.5)^2 \\ &= 1.53 \Omega/\text{phase}\end{aligned}$$

$$X_{02} = \sqrt{15.3^2 - 1.53^2} = 15.23 \Omega$$

When the load is 3200 kW at 0.8 p.f.

$$I_2 = 3,200 / \sqrt{3} \times 33 \times 0.8 = 70 \text{ A}$$

$$\begin{aligned}\text{Voltage drop} &= 70(1.53 \times 0.8 + 15.23 \times 0.6) \\ &= 725 \text{ V/phase}\end{aligned}$$

$$\therefore \% \text{ regulation} = \frac{725 \times 100}{19,000} = 3.8\%$$

Primary voltage will have to be increased by 3.8%

$$\begin{aligned}\therefore \text{Primary voltage} &= 6.6 + 3.8\% \text{ of } 6.6 \\ &= 6.58 \text{ kV} = 6580 \text{ V}\end{aligned}$$

Common data for Examples 20 and 21

A $\Delta-\Delta$ bank consisting of three 30-kVA 2300/230-V transformer supplies a load of 60 kVA. If one transformer is removed, find for the resulting $V-V$ connecting.

Example 20: (a) KVA load carried by each transformer
 (b) Per cent of rated load carried by each transformer

$$\text{Solution: (a) } \frac{\text{Total kVA load in } V-V \text{ bank}}{\text{VA/transformer}} = \sqrt{3}$$

$$\begin{aligned}\therefore \text{ kVA load supplied by each of the two transformers} &= \frac{60}{\sqrt{3}} \\ &= 34.64 \text{ kVA}\end{aligned}$$

Obviously, each transformer in $V-V$ bank does not carry 50% of the original load but 57.7%

(b) Percentage of rated load

$$\begin{aligned}&= \frac{\text{KVA load/transformer}}{\text{kVA rating/transformer}} \\ &= \frac{34.64}{30} = 115.5\%\end{aligned}$$

Carried by each transformer obviously, in this case, each transformer is overloaded to the extent of 15.5 per cent.

Example 21: (a) Total kVA rating of the $V-V$ bank
 (b) Ratio of the $V-V$ bank to $\Delta-\Delta$ bank transformer ratings
 (c) Percentage increase in load on each transformer

Solution: (a) kVA rating of the $V-V$ bank = $(2 \times 30) \times 0.8666 = 51.96 \text{ kVA}$

$$\begin{aligned}\text{(b) } \frac{V-V \text{ rating}}{\Delta-\Delta \text{ rating}} &= \frac{51.96}{90} \\ &= 0.577 = 57.7\%\end{aligned}$$

As seen, the rating is reduced to 57.7% of the original rating.

(c) Load supplied by each transformer in $\Delta-\Delta$ bank = 60/3 = 20 kVA

$\therefore$ Percentage increase in load supplied by each transformer

$$\begin{aligned}&= \frac{\text{kVA load/transformer in } V-V \text{ bank}}{\text{kVA load/transformer in } \Delta-\Delta \text{ bank}} \\ &= \frac{34.64}{20} = 1.732 = 173.2\%\end{aligned}$$

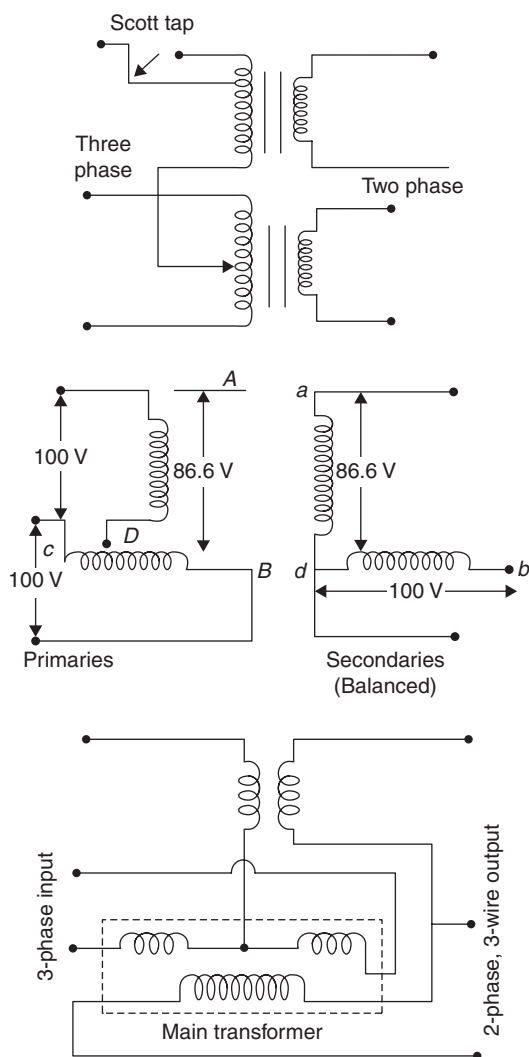
It is obvious that each transformer in the $\Delta-\Delta$ bank supplying 60 kVA was running underloaded (20 vs. 30 kVA) but runs over loaded (34.64 vs. 30) in $V-V$ connection.

THREE-PHASE TO TWO-PHASE CONVERSION

This conversion is required (i) to supply 2-phase furnaces. (ii) to interconnect 2-phase systems with 3-phase systems. (iii) to supply a 3-phase apparatus from 2-phase supply source.

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For this purpose, Scott connection as shown in following fig is employed.



- Teaser transformer primary has $\sqrt{3}/2$ times the turns of main primary. But volt /turn is the same. Their secondaries have the same turns which results in equal secondary terminal voltages.

- If main primary has N_1 turns and main secondary has N_2 turns then main transformation ratio is N_2/N_1 . However the transformation ratio of teaser is

$$\frac{N_2}{(\sqrt{3}N_1/2)} = 1.15N_2/N_1 = 1.15 K$$

- If the load is balanced on one side, it is balanced on the other side as well.
- Under balanced condition, main transformer rating is 15% greater than that of the teaser.
- The currents in either of the two halves of main primary are the vector sum of KI_{2M} and $0.58 KI_{2T}$ (or) $\left(\frac{1}{2} I_{1T}\right)$.

PARALLEL OPERATIONS OF THREE-PHASE TRANSFORMERS

All the conditions which apply to the parallel operation of single-phase transformer also apply to the parallel operation of three-phase transformer but with following conditions.

1. The voltage ratio must refer to the terminal voltage of primary and secondary. It is obvious that this ratio may not be equal to the ratio of the number of turns per phase. For example if V_1 , V_2 are the primary and secondary terminal voltages then for star/delta connection, this turns ratio is $V_2/(V_1/\sqrt{3})$

$$= \sqrt{3} V_2/V_1$$
2. The phase displacement between primary and secondary voltages must be the same for all transformers which are to be connected for parallel operation.
3. The phase sequence must be the same.
4. All the three transformers in the three-phase transformer bank will be of the same construction either core or shell.

EXERCISES

Practice Problems I

Directions for questions 1 to 45: Select the correct alternative from the given choices.

1. In a transformer ratio of the copper losses at half load to that at full load will be
 (A) 2 times (B) one fourth
 (C) half (D) 4 times
2. The eddy current loss of a transformer at 120 V, 60 Hz is 150 W. The eddy current loss at 120 V, 30 Hz will be
 (A) 150 W (B) 75 W
 (C) 37.5 W (D) 300 W

3. The ratio of primary to secondary turns of a transformer is 4. The primary resistance is 1Ω and the secondary resistance is 0.5Ω . The resistance of the transformer referred to primary side is
 (A) 9Ω (B) $\frac{1}{9} \Omega$
 (C) 3Ω (D) $\frac{1}{3} \Omega$
4. At 50 Hz operation, a single-phase transformer has hysteresis and eddy current losses of 300 W and 150 W, respectively. Its core loss at 60 Hz will be
 (A) 450 W (B) 576 W
 (C) 648 W (D) 540 W

5. When the iron and full-load copper losses in a transformer are 900 W and 1600 W, respectively, the maximum efficiency occurs at _____ of full load.
(A) 25% (B) 50%
(C) 75% (D) 100%
6. A transformer has 600 primary turns and 120 secondary turns. If the load current is 10 A, its primary load component of current is _____.
(A) 20 A (B) 10 A
(C) 2 A (D) 5 A
7. A transformer possesses a percentage resistance and percentage reactance of 2.2% and 4%, respectively. Its voltage regulation at power factor of 0.6 leading and 0.6 lagging would be, respectively,
(A) -2% and -2% (B) 4.4% and -2%
(C) -2% and 4.4% (D) 4.4% and -4.4%
8. If the applied voltage of a certain transformer is increased by 100% and the frequency is reduced to 25% (assuming that the magnetic circuit remains unsaturated) the maximum flux density will change to _____ times the original value
(A) 2 (B) 4
(C) 6 (D) 8
9. A 50 kVA transformer has a core loss of 500 W and a full-load copper loss of 1000 W. Maximum efficiency occurs at _____ per cent of full load
(A) 50 (B) 37.5
(C) 70.7 (D) 100
10. A single-phase transformer has rating of 10 kVA, 400/100 V. It is recommended as an auto-transformer to supply at 500 V from a 400 V primary source. The maximum load which it can supply is
(A) 50 kVA (B) 25 kVA
(C) 100 kVA (D) 12.5 kVA
11. An auto-transformer having a transformation ratio of 0.5, supplies a load of 20 kW. The power transferred inductively from the primary to secondary is
(A) 20 kW (B) 10 kW
(C) 5 kW (D) zero
12. The exciting current in p.u. on the LV as well as HV side, respectively _____.
(A) 0.04, 0.02 (B) 0.02, 0.04
(C) 0.04, 0.04 (D) 0.08, 0.08
13. The equivalent impedance in p.u. on the HV side is
(A) $0.105 + j0.085$ (B) $0.21 + j0.17$
(C) $0.085 + j0.105$ (D) None of these
14. Two transformers of the same type, using the same grade of iron and conductor materials, are designed to work at the same flux and current densities, but the linear dimensions, of one are two times those of the other in all respects. The ratio of kVA of the two transformers closely equals
(A) 2 (B) 4
(C) 8 (D) 16
15. Two transformers of different kVA ratings working in parallel share the load in proportion to their ratings when their
(A) ohmic values of the leakage impedances are inversely proportional to their ratings.
(B) ohmic values of the leakage magnetizing reactances are the same.
(C) per unit leakage impedances on the same kVA base are the same.
(D) per unit leakage impedances on their respective ratings are equal.
16. The percentage impedance of a 100 kVA, 11 kV/400 V, delta/wye, 50 Hz transformer is 5%. For the circulation of half the full-load current during short-circuit test, with low voltage terminals, shorted, the applied voltage on the high voltage side will be
(A) 245 V (B) 275 V
(C) 315 V (D) 550 V
17. The laws of electromagnetic induction are summarized in which of the following equation?
(A) $e = iR$ (B) $e = \frac{d\psi}{dt}$
(C) $e = -\frac{Ldi}{dt}$ (D) $e = \frac{Nd\phi}{dt}$
18. Keeping in view the requirement of parallel operation, which of the 3-phase connections given below are possible?
(A) star-star to delta-star
(B) delta-delta to star-delta
(C) delta-delta to delta-star
(D) delta-star to star-delta
19. If an AC voltage wave is corrupted with an arbitrary number of harmonics, then the overall voltage wave form differs from its fundamental frequency component in terms of
(A) only the peak values.
(B) only the average values.

Common Data for Questions 12 and 13:

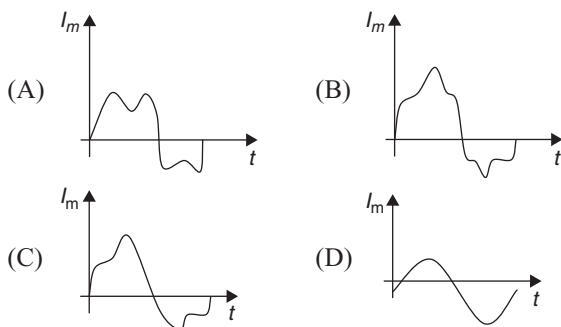
A single-phase transformer on full load has an impedance drop of 25 V and resistance drop of 15 V.

12. The reactance drop is ____ V
(A) 15 (B) 20
(C) 25 (D) 30
13. The p.f. at which regulation will be zero is _____.
(A) zero (B) 0.6
(C) 0.8 (D) 0.707

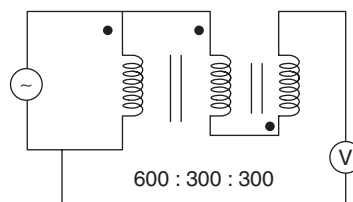
Common Data for Questions 14 and 15:

The exciting current was found to be 4A when measured on the LV side of a 10 – kVA, 1000/100 V transformer. Its equivalent impedance referred to the HV side is $8.5 + j12.5$ (Assume that the transformer rating as the base.)

- (C) only the rms values.
(D) all the three measures (Peak, rms and average values).
22. A single-phase transformer is to be switched to the supply to have minimum inrush current. The switch should be closed at
(A) zero supply voltage.
(B) $\frac{1}{2}$ maximum supply voltage.
(C) $\frac{1}{\sqrt{2}}$ maximum supply voltage.
(D) maximum supply voltage.
23. In the protection of transformers harmonic restraint is used to guard against
(A) switching.
(B) lightning.
(C) magnetizing inrush current.
(D) unbalanced operation.
24. The efficiency of a 100 kVA transformer is 0.98 at full as well as at half full-load. For this transformer at full load, the copper loss
(A) is equal to core loss.
(B) is less than core loss.
(C) is more than core loss.
(D) None of the above
25. The core flux of a practical transformer with a resistive load
(A) increases linearly with load.
(B) increases as the square root of the load.
(C) is strictly constant with load changes.
(D) decreases with increased load.
26. The single-phase, 50 Hz iron core transformer in the circuit has both the vertical arms of cross sectional area 20 cm^2 and both the horizontal arms of cross sectional area 10 cm^2 . If the windings were wound instead on opposite horizontal arms, the mutual inductances will
(A) remain same.
(B) become one quarter.
(C) be halved.
(D) doubled.
27. A single-phase air core transformer, fed from a rated sinusoidal supply, is operating at no load. The steady state magnetizing current drawn by the transformers from the supply will have the wave form

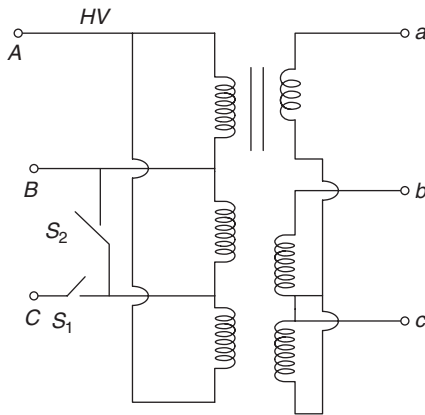


28. In transformer, which of the following statements is valid?
(A) In an oc test, current is drawn at high power factor.
(B) In an oc test, copper losses are obtained, while in short-circuit test core losses are obtained.
(C) In a sc test, current is drawn at zero power factor.
(D) In an oc test, current is drawn at low power factor.
29. In a transformer, zero voltage regulation at full load is
(A) possible at unity power factor load.
(B) possible at leading power factor load.
(C) possible at lagging power factor load.
(D) Not possible.
30. A 50 Hz transformer having equal hysteresis and eddy current losses at rated excitation is operated at 45 Hz at 90% of its rated voltage. Compared to rated operating point the core losses under this condition
(A) increase by 10%.
(B) reduce by 10%.
(C) reduce by 14.5%.
(D) remain unchanged.
31. The hysteresis loop of a magnetic material has an area of 10 cm^2 with the scales given as $1 \text{ cm} = 2 \text{ AT}$ and $1 \text{ cm} = 50 \text{ mwb}$. At 50 Hz the total hysteresis loss is
(A) 10 W. (B) 25 W.
(C) 50 W. (D) 100 W.
32. The windings of a Q kVA, V_1/V_2 volt, three phase delta connected core type transformer are reconnected to work as a single-phase transformer. The maximum voltage and the power ratings are
(A) $\frac{V_1}{V_2}$, $Q/3$. (B) $\frac{\sqrt{3}V_1}{\sqrt{3}V_2}$, $3Q$.
(C) $\frac{2V_1}{2V_2}$, $\frac{2Q}{3}$. (D) $\frac{\sqrt{3}V_1}{\sqrt{3}V_2}$, Q .
33. A 10 kVA 400 V/200V single-phase transformer with 5% impedance draws short-circuit line current of
(A) 50 A. (B) 100 A.
(C) 250 A. (D) 500 A.
34. A 600 V / 300 V/ 300 V, 50 Hz three-winding transformer is connected as shown in figure. The reading of the voltmeter 'V' will be



- (A) 0 V. (B) 300 V.
(C) 600 V. (D) 1200 V.
35. Figure shows Δ -Y connected 3-phase distribution transformer used to step down the voltage from 11 kV to 415 V line-to-line. It has two switches S_1 and S_2 .

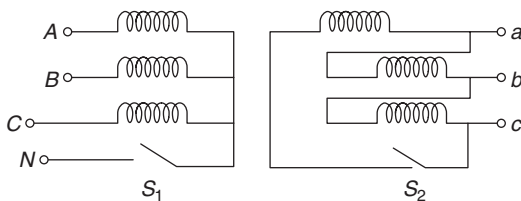
Under normal conditions S_1 is closed and S_2 is open. The magnitude of the voltage across LV terminals a and c with S_1 open and S_2 closed is



- (A) 240 V. (B) 415 V.
(C) 830 V. (D) 0 V.

36. Two transformers are to be operated in parallel such that they share the load in proportion to their kVA ratings. The rating of the first transformer is 200 kVA and its p.u. leakage impedance is 0.05 p.u. If the second transformer is 100 kVA, its p.u. leakage impedance is
(A) 0.10. (B) 0.20.
(C) 0.05. (D) 0.025.
37. A single-phase 50 kVA, 250 V/500 V two-winding transformer has an efficiency of 95% at full load, unity power factor. It is reconfigured as a 500/750 V auto-transformer, its efficiency at its new rated load at unity power factor will be
(A) 93.275%. (B) 95.752%.
(C) 98.276%. (D) 99.241%.

Common Data for Questions 38 and 39:

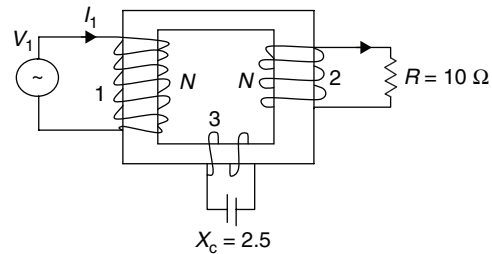


The star-delta transformer shown above is excited on the star side with balanced, 4-wire, 3-phase, sinusoidal voltage supply of rated magnitude. The transformer is under no load condition.

38. With both S_1 and S_2 open, the core flux waveform will be
(A) flat-topped with third harmonic.
(B) peaky with third harmonic.
(C) A sinusoid at fundamental frequency.
(D) flat-topped with fifth harmonic.
39. With S_2 closed and S_1 open, the current wave form in the delta winding will be

- (A) flat-topped with fifth harmonic.
(B) a sinusoid with third harmonic.
(C) only third harmonic.
(D) flat-topped with third harmonic.

40. Figure shows an ideal three-winding transformer. The three windings 1, 2, and 3 of the transformer are wound as shown in fig. the turns ratio $N_1:N_2:N_3$ is 4:2:1. A resistor of $10\ \Omega$ is connected across winding-2. A capacitor of reactance $2.5\ \Omega$ is connected across winding - 3. Winding -1 is connected to a 200 V supply. If the supply voltage phasor $V_1 = 200 \angle 0^\circ$, the supply current phasor I_1 is given by

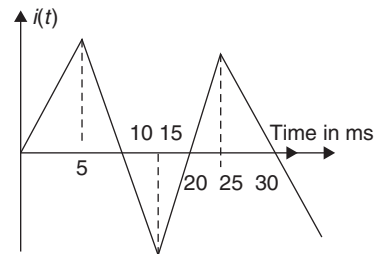
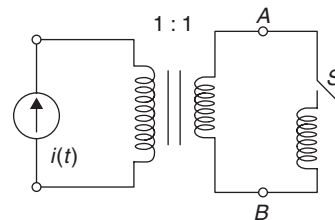


- (A) $(-5 + j5)$ A. (B) $(5 + j5)$ A.
(C) $(5 - j5)$ A. (D) $(-5 - j5)$ A.

41. The resistance and reactance of a 100 kVA 11000/400 V, Δ -y distribution transformer are 0.02 and 0.07 p.u., respectively. The phase impedance of the transformer referred to the primary is
(A) $(0.55 + j1.925)\ \Omega$. (B) $(2.15 + j7.35)\ \Omega$.
(C) $(15.125 + j52.94)\ \Omega$. (D) $(72.6 + j254.1)\ \Omega$.

Common Data for Questions 42 and 43:

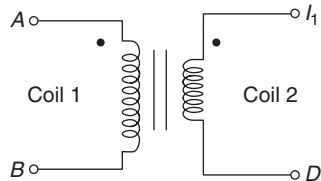
The circuit diagram shows a two-winding loss less transformer with no leakage flux, excited from a current source, $i(t)$, whose wave form is also shown. The transformer has a magnetizing inductance of $200/\pi$ mH.



42. The peak voltage across A and B with S open is
(A) $200/\pi$ V. (B) 400 V.
(C) $\frac{2000}{\pi}$ V. (D) $\frac{400}{\pi}$ V.

43. If the wave form of $i(t) = 10\sin(100\pi t)$ A, the peak voltage across A and B with 'S' closed is
 (A) 200 V. (B) $200 \times \sqrt{2}$ V.
 (C) 400 V. (D) $400 \times \sqrt{2}$ V.

Common Data for Questions 44 and 45:



The figure above shows coils 1 and 2, with dot markings as shown, having 2000 and 3000 turns, respectively. Both the coils have a rated current of 25 A. Coil-1 is excited with single phase, 400 V, 50 Hz supply.

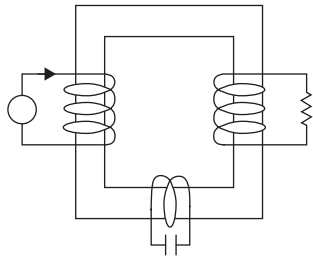
44. The coils are to be connected as a single phase, 400/1000 V auto-transformer to drive a load of 10 kVA. Which of the options given should be exercised to realize the required auto-transformer?
 (A) connect A and C; common B.
 (B) connect A and C; common D.
 (C) connect A and D; common B.
 (D) connect B and D; common C.
45. In the auto-transformer obtained in above question the current in each coil is
 (A) coil-1 is 15 A and coil-2 is 10 A.
 (B) coil-1 is 10 A and coil-2 is 25 A.
 (C) coil-1 is 10 A and coil-2 is 15 A.
 (D) coil-1 is 25 A and coil-2 is 10 A.

Practice Problems 2

Directions for questions 1 to 50: Select the correct alternative from the given choices.

- The purpose of transformer oil is
 (A) to provide lubrication
 (B) to provide insulation and cooling
 (C) to provide protection against short circuit
 (D) to provide protection against lightning
- A voltage $V = 400 \sin 314t$ is applied to a single-phase transformer on no load. If the no-load current of the transformer is $2\sin(314.16t - \angle 85^\circ)$, then magnetization branch impedance will be approximately equal to
 (A) $200 \angle -85^\circ \Omega$ (B) $200 \angle 85^\circ \Omega$
 (C) $200 \angle 5^\circ \Omega$ (D) $200 \angle -5^\circ \Omega$
- The efficiency of a 1000 kVA transformer is 95% at full load as well as at half the full-load both at u.p.f. The Cu loss at full load
 (A) is less than core loss
 (B) is equal to core loss
 (C) is more than core loss
 (D) none of these
- The dominant harmonic in the magnetizing current of a transformer is
 (A) 3rd harmonic (B) 5th harmonic
 (C) 7th harmonic (D) 13th harmonic
- A 60 Hz transformer having equal hysteresis and eddy current losses at rated excitation is operated at 48 Hz and 80% rated voltage. The core losses under this condition
 (A) reduce by 20%
 (B) reduce by 38%
 (C) reduce by 28%
 (D) remain unchanged
- A 400 / 200 V 20 kVA two-winding transformer is reconnected as an auto-transformer. The maximum possible rating of auto-transformer could be
 (A) 50 kVA (B) 15 kVA
 (C) 40 kVA (D) 60 kVA
- In a constant voltage transformer (CVT) output voltage remains constant because of
 (A) saturation
 (B) tapped windings
 (C) capacitor across the terminals
 (D) a DC voltage source across the terminals
- Two transformers of different kVA rating working in parallel share the load in proportion to their rating when
 (A) ohmic values of leakage impedances are inversely proportional to their ratings
 (B) ohmic values of magnetizing reactances are same
 (C) per unit leakage reactance on their respective kVA bases are equal
 (D) per unit leakage impedance on same kVA base are equal
- A transformer is subjected to a short circuit. The turns of the winding experience
 (A) attractive force
 (B) repulsive force
 (C) no force
 (D) can be attractive or repulsive
- Maximum efficiency of a single-phase transformer is 90% at full load and u.p.f. Efficiency at half load and u.p.f. is
 (A) 89.1% (B) 87.8%
 (C) 90% (D) 86.8%

11.



$N_1:N_2:N_3 = 4:2:1$, $V_1 = 400 \angle 0$. The current phasor in N_1 , I_1 is given by

- (A) $-10 - 10j$ (B) $-10 + 10j$
(C) $10 - 10j$ (D) $10 + 10j$

12. Building steel core out of stampings reduces eddy current loss because it

- (A) increases the effective length of eddy current paths thereby increasing effective resistance to flow of eddy currents
(B) reduces effective length of eddy current path, thereby reducing effective resistance to flow of eddy currents
(C) increases core resistivity
(D) increases core permeability

13. Assuming P_i is the iron loss and P_c is the copper loss, the maximum kVA delivered to the load corresponding to maximum efficiency is

- (A) $\text{rated kVA} \times \sqrt{\frac{P_c}{P_i}}$ (B) $\text{rated kVA} \times \sqrt{\frac{P_i}{P_c}}$
(C) $\text{rated kVA} \times \left(\frac{P_c}{P_i}\right)^2$ (D) $\text{rated kVA} \times \left(\frac{P_i}{P_c}\right)^2$

14. A transformer of turns ratio 200/400, 50 Hz is to be excited from 40 Hz from 200 V side. For the exciting current to remain same, the applied voltage should be

- (A) 300 V (B) 250 V
(C) 200 V (D) 60 V

15. The leakage reactance of a transformer can be found out using

- (A) OC test only (B) SC test only
(C) both OC and SC (D) cannot be found

16. A transformer on no load, when switched on draws a current

- (A) twice the steady-state magnetizing current provided core has no residual flux
(B) several times the steady-state magnetizing current independent of initial state of residual flux in the core
(C) Several times steady-state magnetizing current dependent on initial state of residual flux in core
(D) same as steady-state magnetizing current

17. If a two-winding 2/1 transformer is reconnected as auto-transformer, its kVA

- (A) remains same (B) becomes 2 times
(C) becomes 3 times (D) becomes 1.5 times

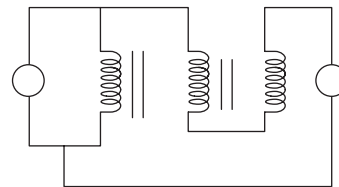
18. A transformer has 95% efficiency at full load and 0.85 p.f. lag. Efficiency at full load 0.85 pf lead will be

- (A) $> 95\%$ (B) $< 95\%$
(C) $= 95\%$ (D) 100%

19. A single-phase transformer is to be switched to have minimum inrush current. Switch should be closed at

- (A) $\frac{1}{\sqrt{2}}$ times maximum value of V
(B) $\frac{1}{2}$ times maximum value of V
(C) at maximum value of V
(D) at zero value of V

20. A 400 V/200 V/200 V is connected as shown. Reading in voltmeter V is



- (A) 0 V (B) 200 V
(C) 400 V (D) 800 V

21. Auto-transformer is used in transmission and distribution when

- (A) operator is not available
(B) iron losses are to be reduced
(C) efficiency consideration can be ignored
(D) when transformation ratio is small

22. Which of the following 3-phase connection can possibly be operated in parallel

- (A) $\Delta - \Delta$ to $\Delta - Y$ (B) $\Delta - \Delta$ to $Y - \Delta$
(C) $Y - Y$ to $\Delta - \Delta$ (D) $\Delta - Y$ to $Y - \Delta$

23. A 2400/240 V, 200 kVA single-phase transformer has a core loss of 1.8 kW at rated voltage. $R = 1.1\%$. Transformer efficiency at 0.9 p.f. and full load is

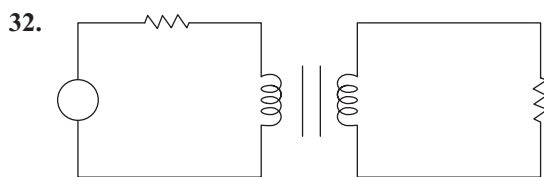
- (A) 97.57% (B) 98.05%
(C) 97.82% (D) 96.56%

24. Hysteresis loss and eddy current loss of 340 W and 120 W, respectively, for a 220 V, 60 Hz single-phase transformer. If the transformer is operated from 230 V, 50 Hz supply mains then core loss assuming (Steinmetz) constant equal to 1.6 is

- (A) 539.3 W (B) 408 W
(C) 349.4 W (D) 308 W

25. Full-load voltage regulation of a power transformer is zero when p.f. of load is

- (A) unity and leading
(B) zero and leading
(C) zero and lagging
(D) unity and lagging
26. Three single-phase transformers each with a 10 kVA rating are connected in a closed delta arrangement. If one transformer is taken out, o/p capacity of system will be
(A) 20 kVA (B) 8.66 kVA
(C) 17.39 kVA (D) 10 kVA
27. A three-phase transformer has rating at 20 mVA, 220 kV (Y), 33 kV (Δ) with leakage reactance at 12%. The transformer reactance (in Ω) referred to each phase of L.V delta is
(A) 23.5 (B) 19.6
(C) 18.5 (D) 8
28. While performing the O.C. and S.C. tests on a transformer
(A) in O.C. LV is open, in S.C LV is short
(B) in O.C. hv is open in SC LV is short
(C) in O.C. hv is open and in S.C hv is short
(D) in O.C. lv is open and in S.C hv is short
29. Iron loss and full-load copper loss in a transformer are 40.5 kW and 50 kW. Efficiency is maximum at load fraction equal to
(A) 0.57 (B) 0.8
(C) 0.9 (D) 0.75
30. In a transformer, fed from a fundamental frequency voltage source, the source of harmonics is
(A) poor insulation
(B) iron loss
(C) overload
(D) saturation of core
31. In a 3 phase Δ -Y transformer phase displacement of secondary line voltages with primary line voltage will be
(A) 180° (B) zero
(C) 30° lead (D) 30° lag

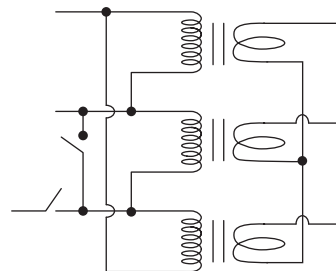


In the figure, transformer is ideal. For what number of turns in primary, maximum power is transferred.

- (A) 40 (B) 80
(C) 160 (D) 320
33. A 5 kVA transformer has iron loss of 200 W and full-load copper loss of 300 W. At maximum efficiency, total loss is
(A) 200 W (B) 400 W
(C) 600 W (D) 800 W

34. A transformer has turns ratio 4:1. Reactance referred to secondary side is 0.05 p.u. Reactance referred to primary is
(A) 0.2 ohm (B) 0.05 ohm
(C) 0.2 p.u. (D) 0.05 p.u.

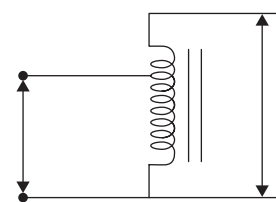
35.



Transformer is used to step down voltage from 11000 V to 415 V line to line. Normally S_1 is closed and S_2 is open. If S_1 is open and S_2 is closed, what is the voltage across a and c?

- (A) 415 V (B) 240 V
(C) 480 V (D) 0 V
36. A 500 kVA, three-phase transformer has iron losses of 300 W and full-load copper loss of 600 W. Percentage load at which transformer is expected to have maximum efficiency?
(A) 50.0% (B) 70.7%
(C) 141.4% (D) 200%

37.



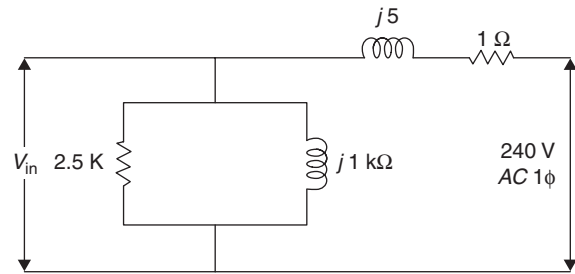
50 kVA 3300/230 V transformer is reconnected as an auto-transformer. The nominal rating of auto-transformer would be

- (A) 50 kVA (B) 53.5 kVA
(C) 717.4 kVA (D) 767.4 kVA
38. A 100 kVA, 11000/400 V, Δ -Y distribution transformer has resistance and reactance 0.02 and 0.07 p.u., respectively, at the secondary side. The phase impedance at primary side is
(A) $24.2 + j84.7 \Omega$ (B) $0.02 + j0.07 \Omega$
(C) $41.91 + j146.70 \Omega$ (D) $72.6 + j254.1 \Omega$
39. The equivalent circuit of a transformer has leakage reactance X_1 and magnetizing reactance X_m , their magnitudes satisfy
(A) $X_1 \ll X_2' \ll X_m$
(B) $X_1 \gg X_2' \gg X_m$
(C) $X_1 \cong X_2 \ll X_m$
(D) $X_1 \cong X_2 \gg X_m$

40. To introduce a phase difference of 30° between input and output line voltages, which of the 3 phase connections can be used
 (A) Y-Y (B) Y- Δ
 (C) Δ - Δ (D) Δ -zigzag
41. Which of the following statements are valid with respect to transformers?
 (A) In O.C. Cu losses are obtained, while in S.C. core losses are obtained
 (B) In O.C. current is drawn at high power factor
 (C) In O.C. current is drawn at low p.f.
 (D) In O.C. current is drawn at zero p.f.
42. Two transformers are operated in parallel such that they share load in proportion to their kVA ratings. The rating of first transformer is 500 kVA and its p.u. leakage impedance is 0.05 p.u. If rating of second transformer is 250 kVA leakage impedance is
 (A) 0.05 (B) 0.025
 (C) 0.1 (D) 0.2
43. A bank of three identical transformers of 11 kV /230 V is used to provide 400 V from a 11 kV 3 phase substation. The kVA rating of the bank will be
 (A) $250\sqrt{3}$ (B) 250
 (C) 500 (D) 750
44. The frequency of power supply of a transformer designed for 60 Hz is changed to 50 Hz with voltage and current ratings remaining the same. The change in efficiency
 (A) increases marginally
 (B) increases by a factor 1.2
 (C) remains unaltered
 (D) decreases marginally
45. If the height-to-width ratio of the window core type transformer is increased then
 (A) leakage reactance and % voltage regulation decreases
 (B) leakage reactance and % voltage regulation increases

- (C) leakage reactance will be unaffected
 (D) efficiency will increase temperature rise decreases

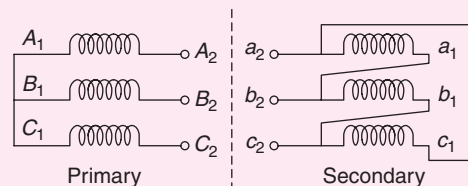
Common Data for Questions 46 and 47:



46. Equivalent circuit of 240/110 V 1.2 kVA single-phase transformer referred to primary. Delivers full load at rated voltage and u.p.f. Efficiency will be
 (A) 98% (B) 96%
 (C) 94% (D) 92%
47. Regulation will be
 (A) 1.546% (B) 2.546%
 (C) 3.546% (D) 4.546%
48. A 250 kVA, 230/115 V 50 Hz transformer has $r_1 = 0.12 \Omega$ $r_2 = 0.04 \Omega$, $X_1 = 0.2 \Omega$ $X_2 = 0.05 \Omega$
 At what loading will the primary induced emf be equal to primary terminal voltage when transformer is carrying full-load current? Neglect magnetizing current
 (A) 15.57 kW (B) 16.57 kW
 (C) 17.57 kW (D) 19.57 kW
49. The leakage flux in a transformer depends on
 (A) applied voltage (B) load current
 (C) frequency (D) mutual flux
50. 15 kVA 600/120 V transformer is reconnected as an auto-transformer to supply 720 V from a 600 V source. Its new rating is
 (A) 90 kVA (B) 18 kVA
 (C) 15 kVA (D) 12 kVA

PREVIOUS YEARS' QUESTIONS

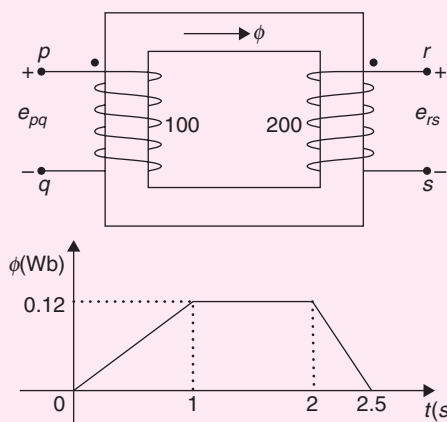
1. The equivalent circuit of a transformer has leakage reactance X_1 , X'_2 and magnetizing reactance X_M . Their magnitudes satisfy [2005]
 (A) $X_1 \gg X'_2 \gg X_M$ (B) $X_1 \ll X'_2 \ll X_M$
 (C) $X_1 = X'_2 \gg X_M$ (D) $X_1 = X'_2 \ll X_M$
2. Three single-phase transformers are connected to form a three-phase transformer bank. The transformers are connected in the following manner



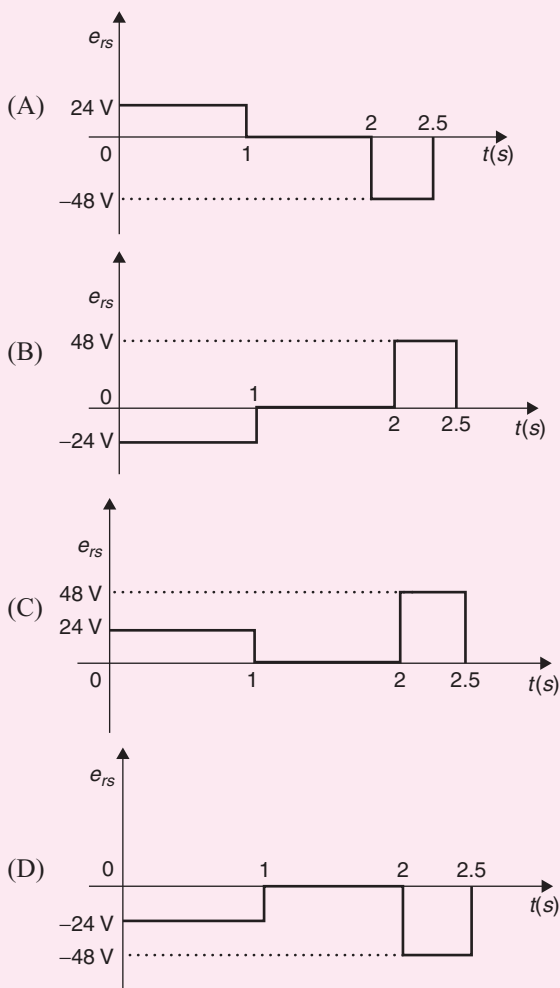
The transformer connection will be represented by [2008]

- (A) Y d0 (B) Y d1
 (C) Y d6 (D) Y d11

3. The core of a two-winding transformer is subjected to a magnetic flux variation as indicated in the figure. [2008]

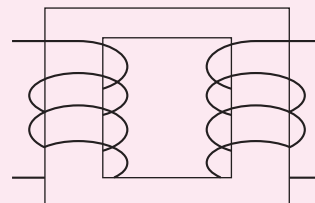


The induced emf (e_{rs}) in the secondary winding as a function of time will be of the form



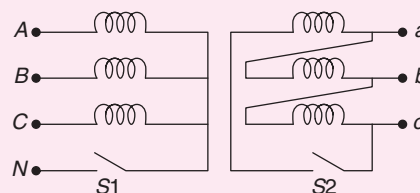
4. The single phase, 50 Hz, iron core transformer in the circuit has both the vertical arms of cross-sectional area 20 cm^2 and both the horizontal arms of cross-sectional area 10 cm^2 . If the two windings shown

were wound instead on opposite horizontal arms, the mutual inductance will [2009]



- (A) double (B) remain same
(C) be halved (D) become one quarter

Common Data for Questions 5 and 6

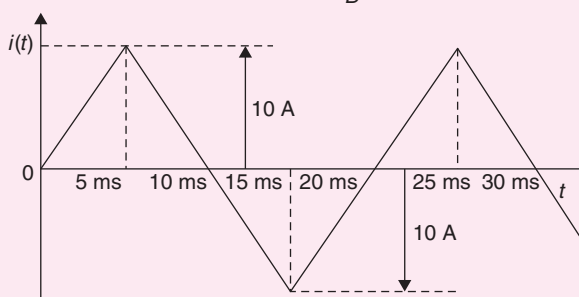
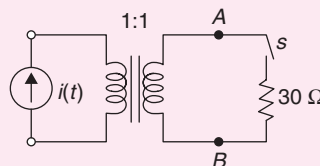


The star-delta transformer shown above is excited on the star side with a balanced, 4-wire, 3-phase, sinusoidal voltage supply of rated magnitude. The transformer is under no load condition.

5. With both S1 and S2 open, the core flux waveform will be [2009]
(A) a sinusoid at fundamental frequency
(B) flat topped with third harmonic
(C) peaky with third harmonic
(D) none of these
6. With S2 closed and S1 open, the current waveform in the delta winding will be [2009]
(A) a sinusoid at fundamental frequency
(B) flat topped with third harmonic
(C) only third harmonic
(D) None of these

Common Data for Questions 7 and 8

The circuit diagram shows a two-winding, lossless transformer with no leakage flux, excited from a current source, $i(t)$, whose waveform is also shown. The transformer has a magnetizing inductance of $400/\pi \text{ mH}$.



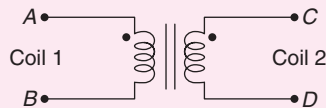
7. The peak voltage across A and B, with S open is [2009]

(A) $400/\pi$ V (B) 800 V
(C) $4000/\pi$ V (D) $800/\pi$ V

8. If the waveform of $i(t)$ is changed to $i(t) = 10\sin(100\pi t)$ A, the peak voltage across A and B with S closed is [2009]

(A) 400 V (B) 240 V
(C) 320 V (D) 160 V

Common Data for Questions 9 and 10:



The figure above shows coils 1 and 2, with dot markings as shown, having 4000 and 6000 turns, respectively. Both coils have a rated current of 25 A. Coil 1 is excited with single phase, 400 V, 50 Hz supply

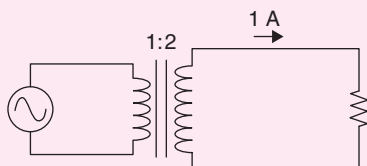
9. The coils are to be connected to obtain a single phase, 400/1000 V, auto-transformer to drive a load of 10 kVA. Which of the options given should be exercised to realize the required auto-transformer? [2009]

(A) Connect A and D; Common B
(B) Connect B and D; Common C
(C) Connect A and C; Common B
(D) Connect A and C; Common D

10. In the auto-transformer obtained in Question 57, the current in each coil is [2009]

(A) Coil-1 is 25 A and Coil-2 is 10 A
(B) Coil-1 is 10 A and Coil-2 is 25 A
(C) Coil-1 is 10 A and Coil-2 is 15 A
(D) Coil-1 is 15 A and Coil-2 is 10 A

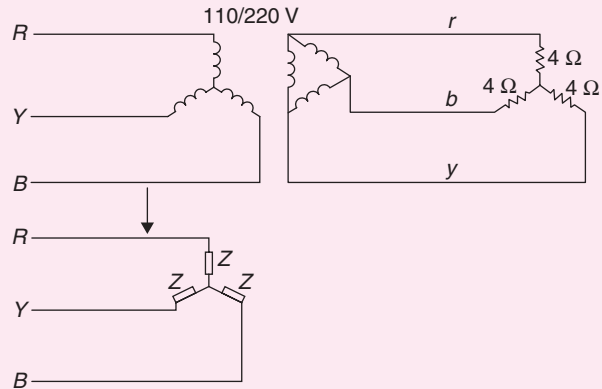
11. A single-phase transformer has a turns ratio of 1:2, and is connected to a purely resistive load as shown in the figure. The magnetizing current drawn is 1 A, and the secondary current is 1 A. If core losses and leakage reactances are neglected, the primary current is [2010]



(A) 1.41 A (B) 2 A
(C) 2.24 A (D) 3 A

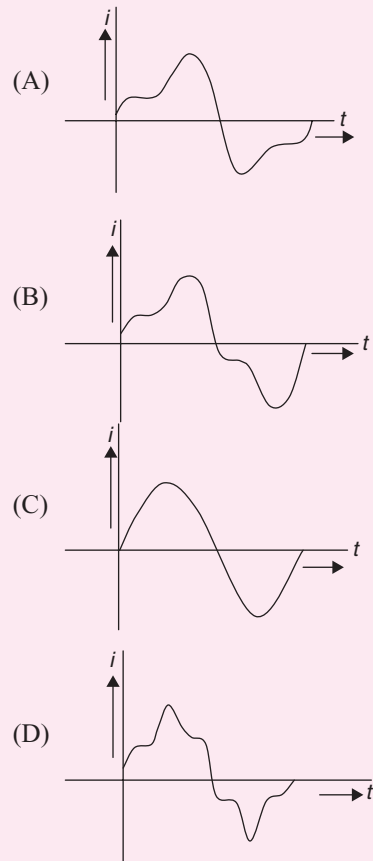
12. A balanced star-connected and purely resistive load is connected at the secondary of a star-delta transformer as shown in the figure. The line-to-line voltage rating of the transformer is 110 V/220 V. Neglecting the non-idealities of the transformer, the impedance 'Z'

of the equivalent star-connected load, referred to the primary side of the transformer, is [2010]



(A) $(3 + j0) \Omega$ (B) $(0.866 - j0.5) \Omega$
(C) $(0.866 + j0.5) \Omega$ (D) $(1 + j0) \Omega$

13. A single-phase air core transformer, fed from a rated sinusoidal supply, is operating at no load. The steady state magnetizing current drawn by the transformer from the supply will have the waveform. [2011]



14. A single-phase 10 kVA, 50 Hz transformer with 1 kV primary winding draws 0.5 A and 55 W, at rated voltage and frequency, on no load. A second transformer has a core with all its linear dimensions $\sqrt{2}$ times the corresponding dimensions of the first transformer.

The core material and lamination thickness are the same in both transformers. The primary windings of both the transformers have the same number of turns. If a rated voltage of 2 kV at 50 Hz is applied to the primary of the second transformer, then the no-load current and power, respectively, are [2012]

- (A) 0.7 A, 77.8 W (B) 0.7 A, 155.6 W
(C) 1 A, 110 W (D) 1 A, 220 W

15. A single-phase transformer has no-load loss of 64 W, as obtained from an open-circuit test. When a short-circuit test is performed on it with 90% of the rated currents flowing in its both LV and HV windings, the measured loss is 81 W. The transformer has maximum efficiency when operated at [2013]

- (A) 50.0% of the rated current
(B) 64.0% of the rated current
(C) 80.0% of the rated current
(D) 88.8% of the rated current

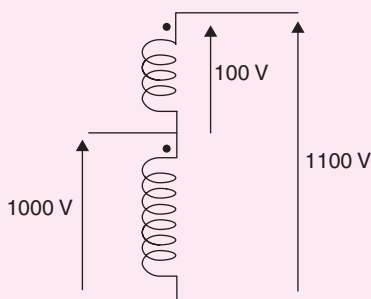
16. For a specified input voltage and frequency, if the equivalent radius of the core of a transformer is reduced by half, the factor by which the number of turns in the primary should change to maintain the same no-load current is [2014]

- (A) $1/4$ (B) $1/2$
(C) 2 (D) 4

17. The core loss of a single phase, 230/115 V, 50 Hz power transformer is measured from 230 V side by feeding the primary (230 V side) from a variable voltage variable frequency source while keeping the secondary open circuited. The core loss is measured to be 1050 W for 230 V, 50 Hz input. The core loss is again measured to be 500 W for 138 V, 30 Hz input. The hysteresis and eddy current losses of the transformer for 230 V, 50 Hz input are, respectively, [2014]

- (A) 508 W and 542 W (B) 468 W and 582 W
(C) 498 W and 552 W (D) 488 W and 562 W

18. A single-phase, 50 kVA, 1000 V/100 V two-winding transformer is connected as an auto-transformer as shown in the figure.

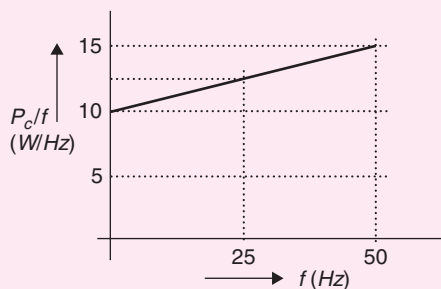


The kVA rating of the auto-transformer is _____. [2014]

19. For a single phase, two-winding transformer, the supply frequency and voltage are both increased by 10%. The percentage changes in the hysteresis loss and eddy current loss, respectively, are [2014]

- (A) 10 and 21 (B) -10 and 21
(C) 21 and 10 (D) -21 and 10

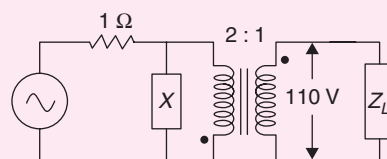
20. An open-circuit test is performed on 50 Hz transformer, using variable frequency source and keeping V/f ratio constant, to separate its eddy current and hysteresis losses. The variation of core loss/frequency as function of frequency is shown in the figure



The hysteresis and eddy current losses of the transformer at 25 Hz, respectively, are [2014]

- (A) 250 W and 2.5 W
(B) 250 W and 62.5 W
(C) 312.5 W and 62.5 W
(D) 312.5 W and 250 W

21. The load shown in the figure absorbs 4 kW at a power factor of 0.89 lagging.



Assuming the transformer to be ideal, the value of the reactance X to improve the input power factor to unity is _____. [2014]

22. The parameters measured for a 220 V/110 V, 50 Hz, single-phase transformer are:

Self-inductance of primary winding = 45 mH

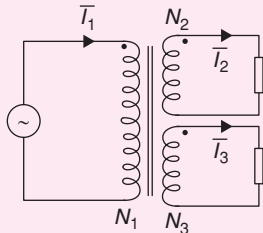
Self-inductance of secondary winding = 30 mH

Mutual inductance between primary and secondary windings = 20 mH

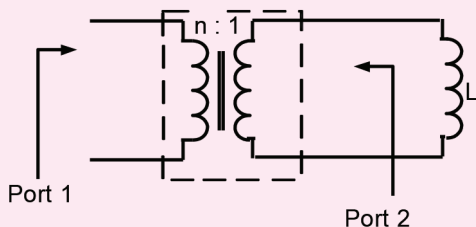
Using the above parameters, the leakage (L_{l1} , L_{l2}) and magnetizing (L_m) inductances as referred to primary side in the equivalent circuit, respectively, are [2014]

- (A) 5 mH, 20 mH and 40 mH
(B) 5 mH, 80 mH and 40 mH
(C) 25 mH, 10 mH and 20 mH
(D) 45 mH, 30 mH and 20 mH

23. The primary mmf is least affected by the secondary terminal conditions in a [2015]
 (A) power transformer
 (B) potential transformer
 (C) current transformer
 (D) distribution transformer
24. A 200/400 V, 50 Hz, two-winding transformer is rated at 20 kVA. Its windings are connected as an auto transformer of rating 200/600 V. A resistive load of 12 Ω is connected to the high voltage (600 V) side of the auto-transformer. The value of equivalent load resistance (in Ohm) as seen from low voltage side is _____. [2015]
25. Two single-phase transformers T_1 and T_2 each rated at 500 kVA are operated in parallel. Percentage impedances of T_1 and T_2 are $(1 + j6)$ and $(0.8 + j4.8)$, respectively. To share a load of 1000 kVA at 0.8 lagging power factor, the contribution of T_2 (in kVA) is _____. [2015]
26. A three-winding transformer is connected to an AC voltage source as shown in the figure. The number of turns are as follows: $N_1 = 100$, $N_2 = 50$, $N_3 = 50$. If the magnetizing current is neglected, and the currents in two windings are $\underline{I_2} = 2 \angle 30^\circ$ A and $\underline{I_3} = 2 \angle 150^\circ$ A, then what is the value of the current $\underline{I_1}$ in Ampere? [2015]

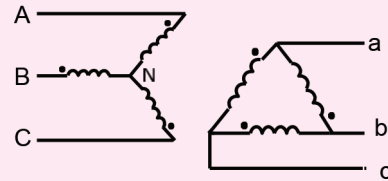


- (A) $1 \angle 90^\circ$
 (B) $1 \angle 270^\circ$
 (C) $4 \angle 90^\circ$
 (D) $4 \angle 270^\circ$
27. In a constant V/f induction motor drive, the slip at the maximum torque [2016]
 (A) is directly proportional to the synchronous speed.
 (B) remains constant with respect to the synchronous speed.
 (C) has an inverse relation with the synchronous speed.
 (D) has no relation with the synchronous speed.
28. If an ideal transformer has an inductive load element at port 2 as shown in the figure below, the equivalent inductance at port 1 is [2016]

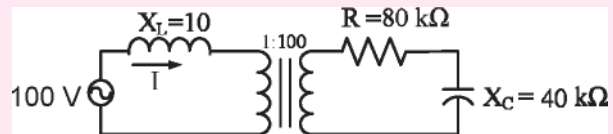


- (A) nL
 (B) $\frac{n^2 L}{L}$
 (C) $\frac{n}{L}$
 (D) $\frac{n^2}{L}$

29. If the star side of the star-delta transformer shown in the figure is excited by a negative sequence voltage, then [2016]



- (A) $\underline{V_{AB}}$ leads $\underline{V_{ab}}$ by 60°
 (B) $\underline{V_{AB}}$ lags $\underline{V_{ab}}$ by 60°
 (C) $\underline{V_{AB}}$ leads $\underline{V_{ab}}$ by 30°
 (D) $\underline{V_{AB}}$ lags $\underline{V_{ab}}$ by 30°
30. A single-phase 400 V, 50 Hz transformer has an iron loss of 5000 W at the rated condition. When operated at 200 V, 25 Hz, the iron loss is 2000 W. When operated at 416 V, 52 Hz, the value of the hysteresis loss divided by the eddy current loss is _____. [2016]
31. A single-phase, 22 kVA, 2200 V/ 220 V, 50 Hz, distribution transformer is to be connected as an auto-transformer to get an output voltage of 2420 V. Its maximum KVA rating as an auto-transformer is _____. [2016]
 (A) 22
 (B) 24.2
 (C) 242
 (D) 2420
32. The following figure shows the connection of an ideal transformer with primary to secondary turns ratio of 1: 100. The applied primary voltage is 100V (rms), 50Hz, AC. The rms value of the current I, in ampere, is _____. [2016]



33. A single-phase, 2kVA, 100/200V transformer is reconnected as an auto-transformer such that its kVA rating is maximum. The new rating, in kVA is _____. [2016]
34. Three single-phase transformers are connected to form a delta-star three-phase transformer of 110kV/11kV. The transformer supplies at 11kV a load of 8MW at 0.8p.f. lagging to a near by plant. Neglect the transformer losses. The ratio of phase currents in delta side to star side is _____. [2016]
 (A) $1:10\sqrt{3}$
 (B) $10\sqrt{3}:1$
 (C) $1:10$
 (D) $\sqrt{3}:10$

ANSWER KEYS

EXERCISES

Practice Problems 1

1. B	2. A	3. A	4. B	5. C	6. C	7. C	8. D	9. C	10. A
11. B	12. B	13. C	14. C	15. C	16. D	17. D	18. B	19. C	20. D
21. D	22. D	23. C	24. C	25. C	26. C	27. D	28. D	29. B	30. C
31. C	32. C	33. D	34. A	35. D	36. C	37. C	38. A	39. C	40. B
41. D	42. D	43. A	44. C	45. A					

Practice Problems 2

1. B	2. B	3. C	4. A	5. C	6. D	7. B	8. C	9. A	10. B
11. D	12. A	13. B	14. D	15. B	16. C	17. C	18. C	19. C	20. A
21. D	22. C	23. C	24. A	25. A	26. C	27. B	28. B	29. C	30. D
31. D	32. B	33. B	34. D	35. D	36. B	37. D	38. D	39. C	40. B
41. C	42. C	43. D	44. A	45. A	46. B	47. B	48. D	49. B	50. A

Previous Years' Questions

1. D	2. B	3. B	4. C	5. B	6. A	7. D	8. A	9. A	10. D
11. C	12. D	13. C	14. B	15. C	16. C	17. A	18. 550	19. A	20. B
21. 23.6	22. B	23. C	24. 1.3 Ω	25. 555.75 KVA	26. A	27. C	28. B	29. D	
30. 1.44	31. C	32. 10	33. 6	34. A					

Chapter 2

Principles of Electro Mechanical Energy Conversion

LEARNING OBJECTIVES

After reading this chapter, you will be able to understand:

- Single excited magnetic systems
- Electrical energy input
- Doubly excited magnetic systems
- MMF produced by polyphase machines
- Armature windings
- Distributed windings
- Pole-pitch
- Coil-pitch or coil-span
- Simplex and multiplex windings
- Closed-winding terms
- Simplex wave winding
- Multiplex lap and wave windings
- Open windings or AC armature windings

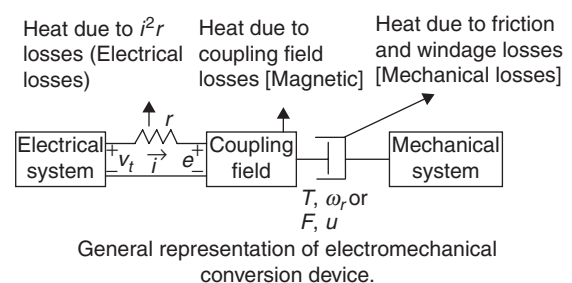
INTRODUCTION

Conversion of other forms of energy into electrical energy is a common practice owing to the advantage of electrical energy as it can be transmitted, utilized and controlled more easily, reliably and efficiently. Energy conversion devices are required at both ends of an electrical system, since energy is neither available and nor required in electrical form. An electromechanical energy conversion device is one which converts electrical energy into mechanical energy or vice versa. The operating principles of energy-conversion devices are similar, but their structural details differ depending upon their function. Energy conversion process is basically a reversible process. However in practice, the electromechanical energy conversion devices may be designed and constructed to suit one particular mode of conversion or the other. The electro-mechanical-energy conversion principles are developed with field energy as basis as the field energy concept is more general and broad-based, since it is applicable to all devices possessing rotational, linear or vibratory motions. The energy storing capacity of the magnetic field is much greater (about 25,000 times) than that of the electric field. In view of this fact, electromechanical energy conversion devices with magnetic field as the coupling medium between electrical and mechanical systems are more common in commercial applications.

Various electromechanical energy conversion devices may be categorized as under:

1. Low energy conversion devices such as telephone receivers, loud-speakers, microphones, low signal transducers, etc.
2. The second category consists of force, or torque-producing devices with limited mechanical motion such as electromagnets, relays, moving-iron instruments, moving-coil instruments, actuators, etc.
3. The third category includes bulk energy conversion and utilization devices such as generators and motors. Motors and generators are continuous energy conversion devices.

An electromechanical energy conversion device consists of three essential parts, namely, electrical system, coupling field and mechanical as shown in following figure. The coupling field may be magnetic or electric.



When energy is converted from one form to another, the law of conservation of energy can be invoked. According to this law,

energy can neither be created nor be destroyed, it can merely be converted from one form into another.

In an energy conversion device, out of the total input energy, some energy is converted into the required form, some energy is stored and the rest is dissipated. The energy balance equation includes these four terms. For a motor, it can be written as

$$\left(\begin{array}{l} \text{Total electrical} \\ \text{energy input} \end{array} \right) = \left(\begin{array}{l} \text{Mechanical energy output} \\ + (\text{Total energy stored}) \\ + (\text{Total energy dissipated}) \end{array} \right) \quad (1)$$

For generator action

$$\left(\begin{array}{l} \text{Total mechanical} \\ \text{energy input} \end{array} \right) = \left(\begin{array}{l} \text{Electrical energy output} \\ + (\text{Total energy stored}) \\ + (\text{Total energy dissipated}) \end{array} \right) \quad (2)$$

The energy balance equation, i.e., Eq. (1) can be written in more specific terms as

$$\begin{aligned} W_{ei} &= W_{mo} + (W_{es} + W_{ms}) \\ &+ (\text{Ohmic energy losses}) \\ &+ (\text{Coupling field energy losses}) \\ &+ (\text{Energy losses in mechanical system}) \end{aligned}$$

The subscripts e, m, I, s and o stand for electrical, mechanical, input, stored and output respectively.

The various forms of energies involved in Eq. (1) for an electromechanical energy conversion device are given below:

1. Total electrical energy input from the supply mains is W_{ei} .
2. The mechanical energy output is W_{mo} .
3. Total energy stored in any device = Energy stored in magnetic field, W_{es} + Energy stored in mechanical system, W_{ms} .
4. Total energy dissipated = Energy dissipated in electric circuit as ohmic losses + Energy dissipated as magnetic core loss (hysteresis and eddy-current losses) + Energy dissipated in mechanical system (friction and windage losses etc.).

Associated the losses with the corresponding systems Eq. (1) can be rewritten as

$$\left(\begin{array}{l} \text{Electric energy input} \\ - \text{ohmic losses} \end{array} \right) = \left(\begin{array}{l} \text{Mechanical energy} \\ \text{output} + \text{energy} \\ \text{stored in mechanical} \\ \text{system} + \text{mechanical} \\ \text{losses (friction \& windage)} \end{array} \right) + \left(\begin{array}{l} \text{Energy stored in the} \\ \text{coupling field} + \\ \text{coupling field} \\ \text{energy losses} \end{array} \right) \quad (2a)$$

or

$$W_{elec} = W_{mech} + W_{fld}$$

where

W_{elec} = Net energy input to the coupling field.

W_{mech} = The total energy converted to mechanical form.

[Mechanical energy output + Energy stored in the mechanical system + friction and windage losses] and W_{fld} = Total energy absorbed by the coupling field [which is equal to the sum of both the stored field energy w_{es} and the coupling field energy losses]

$$W_{elec} = W_{mech} + W_{fld} \quad (2b)$$

The electrical losses (ohmic and coupling – field losses) as well as mechanical losses (friction and windage losses), though always present, play no basic role in the energy conversion process.

Equation (2b) can be written in differential form as

$$dW_{elec} = dW_{mech} + dW_{fld} \quad (3)$$

where

dW_{elec} = differential electrical energy input to coupling field

dW_{mech} = differential mechanical energy output

dW_{fld} = differential change in energy stored in the coupling field.

From Eq. (1), the differential electrical energy input in time dt is

$$dW_{ei} = v_i i dt$$

ohmic loss in resistance r in time dt is $i^2 r dt$.

$\therefore$ Differential electrical energy input or net energy input to the coupling field,

$$\begin{aligned} dW_{elec} &= dW_{ei} - \text{ohmic loss} \\ &= (V_i - ir) i dt = ei dt \end{aligned} \quad (4)$$

Equation (3) now becomes

$$ei dt = dW_{mech} + dW_{fld} \quad (5)$$

Energy balance Eq. (5) is obtained by applying law of conservation of energy to the motoring mode. This Eq. (5) along with faradays laws of electromagnetic induction forms the fundamental basis for the analysis of energy-conversion devices.

The coupling field is the link between electrical and mechanical systems.

In order that a moving member can rotate, or move, with respect to the stationary member, an air gap must exist in between the stator and rotator. The energy stored in the coupling field acts as interaction medium between electrical and mechanical systems.

This energy stored in the coupling field must produce action and reaction on electrical and mechanical systems for the conversion of energy from electrical to mechanical

(motoring mode) or from mechanical to electrical (generating mode).

If the output is mechanical, the coupling field must react with the electrical system in order to absorb electrical energy from it. In a motor, this reaction is the back emf or counter emf 'e'. The coupling field extracts energy proportional to $e \cdot i$ from the electrical system, converts and delivers energy proportional to $T \cdot \omega_r$ (or F.u) to the mechanical system.

If the output is electrical, the coupling field must react with the mechanical system so as to absorb mechanical energy from it. In a generator this reaction is counter-torque T (or reaction torque), opposite to the applied mechanical torque of the prime mover.

Thus, the reaction of the coupling field on the electrical or mechanical system is essential for the process of electro-mechanical energy conversion.

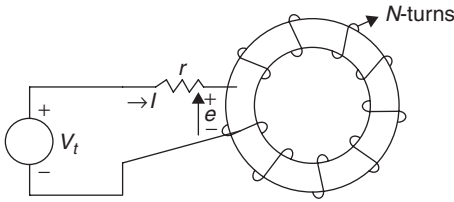
The induced emf 'e' and torque, T are called electromechanical coupling terms as they are associated with the coupling field.

Electromechanical energy conversion devices are slow-moving because of the inertia associated with mechanical components. Therefore the coupling field must be slowly varying and as such the nature of this field is quasi-static. Electromagnetic radiation from the coupling field is almost negligible.

SINGLE EXCITED MAGNETIC SYSTEMS

Electrical Energy Input

Consider a simple magnetic system of a toroid, excited by a single coil as shown in the figure.



Applying KVL for the electrical circuit, the instantaneous voltage equation can be obtained as $V_t = ir + e$

where e is the reaction emf (or back emf) taken as a voltage drop in the direction of current.

$$\therefore e = \frac{d\psi}{dt} \text{ and } V_t = ir + \frac{d\psi}{dt}$$

Where ψ denotes the instantaneous flux linkages with the circuit, multiplying both sides of Eq. (2) we have

$$V_t i dt = i^2 r dt + i d\psi$$

$$\text{or } (V_t - ir) i dt = i d\psi$$

$$\text{or } e i dt = i d\psi$$

$$dW_{elec} = v'_t i dt = dW_{ei} - \text{ohmic loss} = (v_t - ir) i dt = e i dt.$$

If the toroidal core is made of good magnetic material (ferro-magnetic material), most of the flux ϕ links all the N turns of the coil, we have flux linkages, $\psi = N\phi$

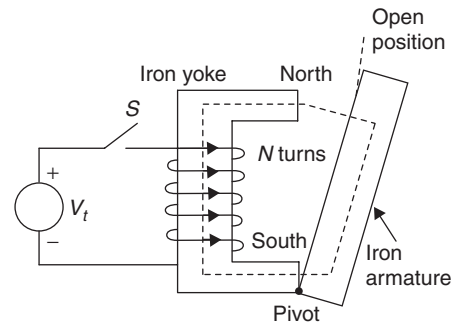
$$\begin{aligned} dW_{elec} &= i d\psi \\ &= iN d\phi = F d\phi \end{aligned} \quad (6)$$

where ϕ is the instantaneous value of the coil flux and $F = iN$ is the instantaneous coil mmf. From Eq. (6) it is evident that the flux linkages of the system must change for the toroid to extract energy from the supply system.

These changing flux linkages cause the generation of reaction emf e (or back emf or counter emf). The flow of current against the reaction emf e causes the extraction of energy from the electrical system or in other words e equals the extracted power from electrical system.

Magnetic Field Energy Stored

Consider a simple magnetic relay shown in following figure.



Initially the switch 'S' and the armature are in open position. When switch 'S' is closed, current i flows through the coil, setting up a flux. The magnetic field thus produced, creates north and south poles on yoke and armature respectively. As a result of the magnetic field there is established a magnetic force that acts on the armature tending to shorten the air-gap. If the armature is not allowed to move, the mechanical work done is zero. i.e. $dW_{mech} = 0$. Therefore from the energy balance equation,

$$dW_{elec} = 0 + dW_{fld}$$

The above relation infers that when the movable part of any physical system is kept fixed, the entire electrical energy input is stored in the magnetic field.

$$\therefore dw_{fld} = dW_{elec} = i d\psi = F \cdot d\phi$$

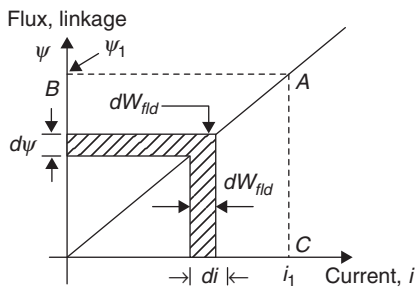
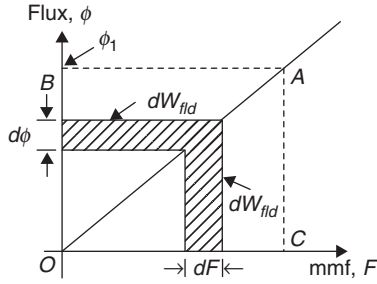
If the initial flux is zero, the magnetic field energy stored W_{fld} , in establishing a flux ϕ , or flux linkage ψ , is given by

$$W_{fld} = \int_0^{\psi_1} i \cdot d\psi = \int_0^{\phi} F \cdot d\phi$$

In the above equation, i must be expressed in terms of ψ and F in terms of ϕ .

3. 550 | Electrical Machines

When the armature is held open, then most of the mmf is consumed in the air gap and magnetic saturation may not occur. Then ψ and ϕ vary linearly with i and F , respectively.



For figure (a)

$$W_{fld} = \int_0^{\phi_1} dW_{fld} = \int_0^{\phi_1} F \cdot d\phi = \text{area OABO}$$

For figure (b),

$$W_{fld} = \int_0^{\psi_1} dW_{fld} = \int_0^{\psi_1} i d\psi = \text{area OABO}$$

and area OACO = $\int dW_{fld}$

$$= \int_0^{F_1} \phi dF = \int_0^{i_1} \psi di$$

This area OACO is called the co-energy W'_{fld} . Co-energy has no physical significance, it is however useful in calculating the magnetic forces.

With no magnetic saturation

$$\text{Area OABO} = \text{Area OACO} \text{ or } W_{fld} = W'_{fld}$$

$$\begin{aligned} \text{and } W_{fld} + W'_{fld} &= \text{Area OCABO} \\ &= \text{Area OABO} + \text{Area OACO} \\ &= \phi_1 F_1 = \psi_1 i_1 \end{aligned}$$

In general for a linear magnetic circuit,

$$W_{fld} = W'_{fld} = \frac{1}{2} \psi i = \frac{1}{2} F \phi$$

The magnetic stored energy and co-energy expressed in terms of reluctance and permeance is as follows.

$$\begin{aligned} \text{mmf} &= (\text{flux})(\text{Reluctance}) \\ &= \phi S \end{aligned}$$

$$= \frac{\phi}{\text{Permeance, } \Delta}$$

$$W_{fld} = W'_{fld} = \frac{1}{2} \phi^2 S = \frac{1}{2} \frac{\phi^2}{\Delta}$$

$$W_{fld} = W'_{fld} = \frac{1}{2} F^2 \Delta = \frac{1}{2} \frac{F^2}{S}$$

In terms of self-inductance, $L = \frac{\psi}{i}$

$$W_{fld} = W'_{fld} = \frac{1}{2} L i^2 = \frac{1}{2} \frac{\psi^2}{L}$$

The stored magnetic energy W_{fld} and co-energy W'_{fld} can be written as follows

$$\begin{aligned} W_{fld} = W'_{fld} &= \frac{1}{2} F \phi = \frac{1}{2} \psi i \\ &= \frac{1}{2} \phi^2 S = \frac{1}{2} \frac{\phi^2}{\Delta} = \frac{1}{2} \frac{F^2}{S} \\ &= \frac{1}{2} F^2 \Delta = \frac{1}{2} L i^2 = \frac{1}{2} \frac{\psi^2}{L} \end{aligned}$$

The magnetic stored energy density is given by

$$W_{fld} = \frac{1}{2} HB = \frac{1}{2} \mu H^2 = \frac{1}{2} \frac{B^2}{\mu} \text{ J/m}^3$$

For linear magnetic circuit

$$W'_{fld} = W_{fld}$$

Note: Here W'_{fld} is the co-energy density.

The field-energy approach serves as the physical basis for the generalized theory of electrical machines.

During the motion of armature the electromagnetic force or torque developed in any physical system acts in such a direction as to tend to

1. Decrease the magnetic stored energy at constant ψ or ϕ .
2. Increase both field stored and co-energy at constant current or mmf.
3. Decrease the reluctance.
4. Increase the permeance and inductance.
5. Decrease current i at constant flux linkages ψ or increase flux linkages ψ at constant i .

Examples of singly excited electromagnetic systems.

1. Moving iron instruments
2. Relays
3. Electromagnets
4. Reluctance motor, etc.

$$\text{Magnetic force, } f_e = \frac{-1}{2} \psi \frac{\partial i}{\partial x}(i, x)$$

If ψ is expressed in terms of i and x

$$f_e = \frac{1}{2} i \frac{\partial \psi}{\partial x}(i, x)$$

where ' x ' is the displacement of the armature in the direction of magnetic force.

The electromagnetic torque,

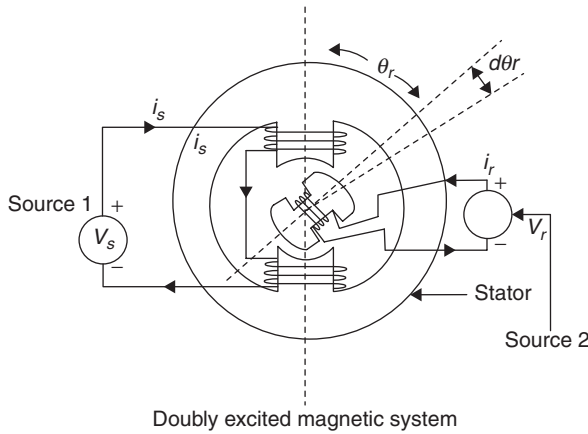
$$\begin{aligned} T_e &= \frac{-1}{2} \phi^2 \frac{dS}{d\theta} = \frac{1}{2} I^2 \frac{dL}{d\theta} \\ &= \frac{-1}{2} \psi \frac{\partial i}{\partial \theta}(\psi, \theta) \\ &= \frac{1}{2} i \frac{\partial \psi}{\partial \theta}(i, \theta) \end{aligned}$$

DOUBLY EXCITED MAGNETIC SYSTEMS

Most of the electromagnetic energy conversion devices belong to doubly excited or multiply excited magnetic systems. A doubly excited magnetic system is one which has two independent sources of excitations.

For example, Synchronous machines, DC shunt machines, loud speakers, tachometers, etc.

The following figure shows a simple model of doubly excited magnetic system.



The stator with N_s turns is energized from source 1 and the rotor with N_r turns is excited from source 2. The mmf's (or rmf's) produced by both the stator and rotor windings are in the same direction and the electromagnetic torque, T_e is in the counter clockwise direction as shown in fig. The magnetic saturation and hysteresis are neglected in the following developments. The differential electrical energy input dW_{elec} for the doubly excited system from two energy sources 1 and 2 is

$$dW_{elec} = i_s d\psi_s + i_r d\psi_r \quad (7)$$

where ψ_s and ψ_r are the instantaneous total flux linkages of stator and rotor windings respectively. Since the magnetic saturation is neglected, ψ_s and ψ_r can be expressed as

$$\psi_s = L_s i_s + M_{sr} i_r$$

$$\psi_r = L_r i_r + M_{rs} i_s$$

where L_s and L_r are the self-inductances of the stator and rotor windings, respectively, and

$M_{sr} = M_{rs}$ = mutual inductance between the stator and rotor windings.

Initially the space angle between stator and rotor axes is θ_r and both the currents i_s and i_r are assumed to be zero. On switching the stator and rotor windings to their respective energy sources, the currents rise from zero to i_s and i_r , respectively. If the rotor is not allowed to move, then $dW_{mech} = 0$.

$$\therefore dW_{elec} = 0 + dW_{fld}$$

Thus, with rotor held fixed, all the energy supplied by sources is stored in the magnetic field.

From Eq. (7)

$$\begin{aligned} dW_{fld} &= dW_{elec} = i_s d\psi_s + i_r d\psi_r \\ &= i_s d(L_s i_s + M_{sr} i_r) + i_r d(L_r i_r + M_{rs} i_s) \end{aligned} \quad (8)$$

L_s , L_r and $M_{sr}(=M_{rs})$ are given by

$$L_s = \frac{N_s^2}{S_s}, L_r = \frac{N_r^2}{S_r}$$

$$M_{sr} = M_{rs} = \frac{N_s N_r}{S_{sr}}$$

where S_s = reluctance seen by the stator flux
 S_r = reluctance seen by the rotor flux
 S_{sr} = reluctance seen by the resultant of stator and rotor fluxes.

Since the rotor is not allowed to move, the reluctances and therefore the inductances are constant.

$$\therefore dL_s = dL_r = dM_{sr} = dM_{rs} = 0$$

From Eq. (3), we can have

$$\begin{aligned} dW_{fld} &= i_s L_s di_s + i_s M_{sr} di_r + i_r L_r di_r + i_r M_{sr} di_s \\ &= i_s L_s di_s + i_r L_r di_r + M_{sr} d(i_s i_r) \end{aligned}$$

Therefore, the magnetic field energy stored in establishing the currents from zero to i_s and i_r is given by

$$\begin{aligned} W_{fld} &= L_s \int_0^{i_s} i_s di_s + L_r \int_0^{i_r} i_r di_r + M_{sr} \int_0^{i_s} \int_0^{i_r} d(i_s i_r) \\ &= \frac{1}{2} i_s^2 L_s + \frac{1}{2} i_r^2 L_r + M_{sr} i_s i_r \end{aligned}$$

For obtaining the magnetic torque T_e , assume the rotor to move through a virtual displacement $d\theta_r$ in the direction of T_e . With rotor movement, reluctances S_s , S_r , S_{sr} and therefore L_s , L_r , M_{sr} must vary.

Therefore, the differential electrical energy input dW_{elec} during virtual displacement $d\theta_r$ is given by

$$\begin{aligned} dW_{elec} &= i_s d[L_s i_s + M_{sr} i_r] + i_r d[L_r i_r + M_{sr} i_s] \\ &= i_s L_s di_s + i_s^2 dL_s + i_s M_{sr} di_r + i_s i_r dM_{sr} \\ &\quad + i_r L_r di_r + i_r^2 dL_r + i_r M_{sr} di_s + i_r i_s dM_{sr} \end{aligned}$$

and the differential magnetic energy stored dW_{fld} during the virtual displacement $d\theta_r$ is given by

$$\begin{aligned} dW_{fld} &= \frac{1}{2} i_s^2 dL_s + L_s i_s di_s + \frac{1}{2} i_r^2 dL_r + L_r i_r di_r \\ &\quad + M_{sr} i_s di_r + M_{sr} i_r di_s + i_s i_r dM_{sr} + i_s i_r dM_{sr} \end{aligned}$$

The differential mechanical work done dW_{mech} during the virtual displacement $d\theta_r$ is given by

$$dW_{mech} = T_e d\theta_r$$

From the energy balance equation,

$$dW_{elec} = dW_{fld} + dW_{mech}$$

substituting the value of dW_{elec} , dW_{mech} and dW_{mech} and on simplification, we get

$$T_e = \frac{1}{2} i_s^2 \frac{dL_s}{d\theta_r} + \frac{1}{2} i_r^2 \frac{dL_r}{d\theta_r} + i_s i_r \frac{dM_{sr}}{d\theta_r}$$

It can be seen from the above equation of T_e , that the differential changes di_s and di_r do not contribute to the production of magnetic torque T_e (In other words, T_e is independent of di_s and di_r)

Therefore torque depends on

1. Instantaneous values of currents i_s and i_r and
2. The angular rate of change of inductances,

If W_{fld} is differentiated with respect to the space angle θ_r (with constant i_s and i_r) then the electromagnetic torque T_e is given by

$$T_e = \frac{\partial W_{fld}}{\partial \theta_r}(i_s, i_r, \theta_r)$$

i.e., torque can be obtained from the space derivative of field energy expression, when it is expressed in terms of i_s , i_r and θ_r .

Integration of dW_{elec} , with constant currents, gives

$$\begin{aligned} W_{elec} &= \int dw_{elec} \\ &= \frac{1}{2} i_s^2 L_s + \frac{1}{2} i_r^2 L_r + 2 i_s i_r M_{sr} \end{aligned}$$

For a linear magnetic circuit, $W_{fld} = W'_{fld}$

$$\therefore T_e = \frac{\partial W'_{fld}}{\partial \theta_r}(i_s, i_r, \theta)$$

If the electrical energy input takes place at constant currents, then half of it is converted to mechanical energy, and the remaining half is stored in the magnetic field at constant currents.

In other words, the magnetic energy stored at constant currents is equal to the mechanical work done.

The linear magnetic force f_e , for a doubly excited system can be obtained as follows.

The differential electrical energy input dW_{elec} , from two energy sources is

$$dW_{elec} = i_1 d\psi_1 + i_2 d\psi_2$$

Where

$$\psi_1 = L_1 i_1 + M_{12} i_2$$

$$\psi_2 = L_2 i_2 + M_{21} i_1$$

Also $M_{12} = M_{21}$

$$f_e = \frac{1}{2} i_1^2 \frac{dL_1}{dx} + \frac{1}{2} i_2^2 \frac{dL_2}{dx} + i_1 i_2 \frac{dM_{12}}{dx}$$

$$f_e = \frac{\partial W_{fld}}{\partial x}(i_1, i_2, x)$$

$$= \frac{\partial W'_{fld}}{\partial x}(i_1, i_2, x)$$

MMF Produced by Polyphase Machines

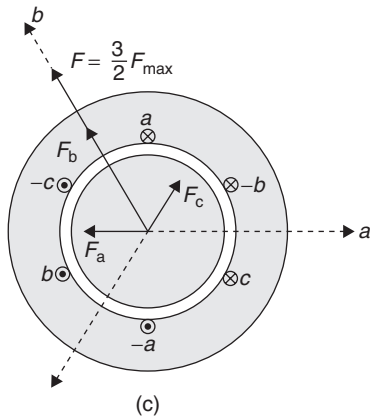
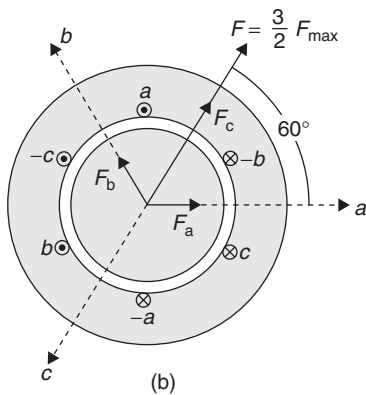
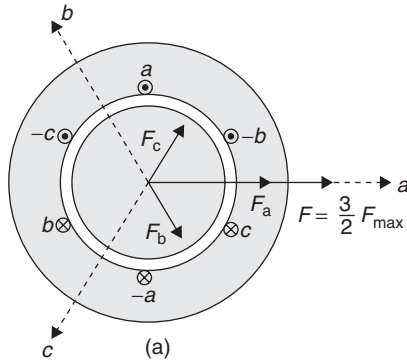
The production of a rotating mmf can be shown graphically. Consider the state of affairs at $t = 0$, the moment when the phase a — a current is at its maximum value I_m . The mmf of phase a then has its maximum value F_{max} , as shown by the vector $F_a = F_{max}$ drawn along the magnetic axis of phase a in the two — pole machine shown schematically in figure (a). At this moment, currents i_b and i_c are both $I_m/2$ in the negative direction, as shown by the dots and crosses in figure (a) indicating the actual instantaneous directions. The corresponding mmf's of phases b and c are shown by the vectors F_b and F_c , both of magnitude $F_{max}/2$ drawn in the negative direction along the magnetic axes of phases b and c , respectively. The resultant, obtained by adding the individual contributions of the three phases, is a vector of magnitude

$F = \frac{3}{2} F_{max}$ centred on the axis of phase α . It represents a sinusoidal space wave with its positive peak centred on the axis of phase α and having an amplitude $\frac{3}{2}$ times that of the phase- α contribution alone.

At a later time $\omega_e t = \pi/3$ the currents in phases α and β are a positive half maximum, and that in phase c is a negative maximum. The individual mmf components and their resultant are now shown in Figure (b). The resultant has the same amplitude as at $t = 0$, but it has now rotated counterclockwise 60 electrical degrees in space. Similarly, at $\omega_e t = 2\pi/3$ (when the phase- b current is a positive maximum and the phase- α and phase- c currents are a negative half maximum) the same resultant mmf distribution is again obtained, but it has rotated counterclockwise 60 electrical degrees still farther and is now aligned with the magnetic axis of phase b figure c. As time passes, then, the resultant mmf wave retains its sinusoidal form and amplitude but

rotates progressively around the air gap; the net result can be seen to be an mmf wave of constant amplitude rotating at a uniform angular velocity.

In one cycle the resultant mmf must be back in the position of Figure (a). The mmf wave therefore makes one revolution per electrical cycle in a two-pole machine. In a multipole machine the mmf wave travels one pole-pair per electrical cycle and hence one revolution in pole/2 electrical cycles.



Solved Examples

Example 1: A coil of 1000 turns on a core would create a flux of 2 m Wb when carrying a current of 1 A. Calculate the energy stored in the magnetic field.

Solution: Flux linkages $\Psi = N\phi$
 $= 1000 \times 2 \times 10^{-3}$
 $= 2 \text{ Weber-turns}$

Energy stored in the magnetic field

$$= \frac{1}{2} \Psi i$$

$$= \frac{1}{2} \times 2 \times 1 = 1 \text{ J}$$

Example 2: One 800-turn flat coil with an area of $5 \times 10^{-2} \text{ m}^2$ is rotating in a magnetic field of flux density $60 \times 10^{-3} \text{ Wb/m}^2$

Solution: emf induced, $e = N \phi \omega \sin \theta$

Here $\theta = 90^\circ$.

$$\phi = 5 \times 10^{-2} \times 60 \times 10^{-3}$$

$$= 3 \times 10^{-3} \text{ Wb}$$

$$\omega = \frac{2\pi \times 1500}{60} = 50\pi \text{ rad/s}$$

And $N = 800$ turns

$$e = 800 \times 3 \times 10^{-3} \times 50\pi$$

$$= 377 \text{ V}$$

ARMATURE WINDINGS

There are two types of windings that are used in rotating electrical machines, namely

1. Concentrated windings
2. Distributing windings

Concentrated Winding

All the turns of the winding are wound together in series to form one multi-turn coil. In concentrated windings, all the turns have the same magnetic axis.

Example: Field winding of salient-pole synchronous machines and DC machines, primary and secondary windings of a transformer.

Distributed Windings

All the winding turns are arranged in several full-pitch or fractional (short) – pitch coils which are housed in the slots spread around the air-gap periphery to form phase or commutator winding.

Example: Stator and rotor windings of induction machines, armature windings of both synchronous and DC machines.

The field (or exciting) windings produces the working flux. The armature winding is one in which working emf is induced by the working flux.

The armature windings, in general are classified as

1. Closed windings and
2. Open windings

The closed windings are used only for commutator machines, such as DC machines and AC commutator (e.g.: universal motor) machines. The open windings are used

only for AC machines like synchronous machines, induction machines etc.

In closed windings, there is a closed path, i.e., if one starts from any point on the winding and traverses it, one again reaches the starting point from where one had started, whereas the open windings terminate at a suitable number of slip-rings or terminals.

Note: Three phase delta connected winding used for AC machines forms a closed circuit. This delta can, however, be open-circuited and reconnected in star if desired. The closed circuit in a delta connected winding is a result of interconnection of the armature windings of different phases. The closed windings used for commutator machines can, in no case, be open-circuited.

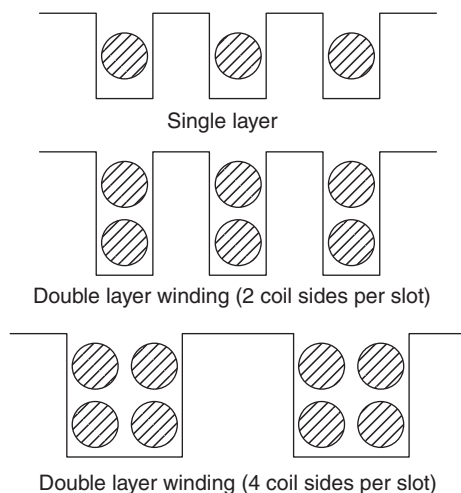
Some of the terms common to both the open and closed armature windings are described below.

1. **Conductor:** The active length of wire which takes part in energy-conversion process.
2. **Turn:** One turn consists of two conductors.
3. **Coil:** A coil may consist of any number of turns. E.g. a coil having one turn is called one-turn coil. A multi-turn coil is one which has more than one-turn.
4. **Coil-sides:** One coil with any number of turns has two coil-sides.
No. of coil sides = $2 \times \text{No. of coils}$
Each coil-side will have several conductors.

Single-layer and Double-layer Windings

In a single-layer winding, one coil-side occupies the total slot area.

In case the slot contains even number of coil-sides (may be 2, 4, 6, etc.) the winding is referred to as a two-layer winding.



The one occupying the upper portion of the slot is called top layer and the other bottom layer.

If one coil-side of a coil occupies the top position in a slot, the other occupies the bottom position in a slot which is one coil-pitch apart.

Advantages of double-layer windings:

1. Easier to house the winding in slots during repairs.
2. Leakage reactance is low which results in better performance.
3. Better induced emf waveform in case of generators.
4. In a single-layer winding, variety of coils that are of different sizes and shapes are used. In double-layer windings, identical diamond-shaped coils are used. So they are more economical.

Pole-pitch

The term 'pitch' indicates a particular method of measurement in terms of coil sides, teeth etc. A pole pitch is defined as the peripheral distance between identical points on two adjacent poles. Pole pitch is always equal to 180° electrical.

Coil-pitch or Coil-span

The distance between the two coil-sides of a coil measured in terms of teeth, slots or electrical degrees.

Chorded-coil

A coil with full-pitch will have the coil-span equal to the pole-pitch. If the coil-pitch is less than pole-pitch, then it is called chorded, short-pitch or fractional-pitch coil.

If there are S slots (or teeth) and P poles, then

$$\text{Pole pitch} = \frac{S}{P} \text{ slots per pole}$$

$$\text{For a full-pitched coil, coil pitch} = \frac{S}{P}$$

In case coil-pitch $< \frac{S}{P}$ it results in chorded, short pitched or fractional-pitch winding. The coil-pitch is rarely greater than pole pitch. In such a case, for a full-pitched coil $\frac{S}{P}$ must be a positive integer.

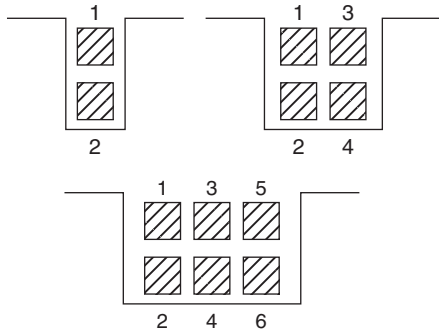
Simplex and Multiplex Windings

In case higher current ratings are to be obtained with given size of a conductor, multiplex windings are used.

In some cases, the number of parallel circuits may not be adequate enough from design considerations. For example in an 8-pole commutator machine, simplex wave winding gives only 2 parallel paths. In case the parallel paths required are in between 2 and 8 or more than 8, then multiplex windings are employed.

Closed-winding Terms

In commutator machines having closed windings, the coil sides for convenience are numbered as shown in the following figure.



The top coil-sides are numbered 1, 3, 5.... and bottom coil sides are numbered 2, 4, 6....

Back Pitch (Y_b)

The distance between the top and bottom coil sides of one coil, measured at the back of armature (or measured at the rear side of the commutator) is called back pitch.

Front Pitch (Y_f)

The distance between the two coil-sides connected to the same commutator segment, is called front pitch.

For convenience, Y_b and Y_f are expressed in terms of coil sides.

Winding Pitch (Y_w)

The distance between the two consecutive and similar top, or bottom coil sides as the winding progresses is called the winding pitch. It is expressed in terms of coil-sides. The winding pitch Y_w is always even, because it is equal to either the difference or the addition of two odd numbers Y_b and Y_f .

For simplex lap winding

$$Y_w = Y_b - Y_f \text{ (progressive)}$$

$$Y_w = Y_f - Y_b \text{ (retrogressive)}$$

For simplex wave winding

$$Y_w = Y_b + Y_f$$

Commutator Pitch (Y_c)

The distance between the two commutator segments, to which ends of one coil are joined, is called the commutator pitch. Y_c is always expressed in terms of commutator segments.

The armature windings are of two types, namely (i) Lap winding (ii) Wave winding.

In lap winding, the two coil-ends of a coil are connected to the two adjacent commutator segments. In other words, for simplex lap winding each commutator segment has two coil-ends connected to it one coil-end is from the top coil-side of the one coil and the other coil-end is from the bottom coil-side of the adjacent coil.

In simplex wave winding, the two coil-ends of a coil are bent in opposite directions and connected to the commutator

segments which are approximately two pole-pitches (i.e., 360° electrical) apart. In wave winding also, each commutator segment has two coil-ends connected to it one from the top coil-side and the other from bottom coil-side.

If lap coils are traversed, the movement is forward and backward alternately, whereas for wave winding, the movement is forward (or backward for retrogressive) only. In both lap and wave windings, all the coils in one parallel path are connected in series.

Simplex Lap Windings

Let C be the number of armature coils and P the number of poles. Since each coil has two coil-sides, total number of coil-sides is $2C$. The back pitch Y_b , almost equal to pole pitch, must be odd

$$\therefore Y_b = \text{coil-sides per pole} \pm K \text{ or } Y_b = \frac{2C}{P} \pm K$$

where K is a number (integer or fraction) added to $2C/P$ to make Y_b an odd integer. Hence, $2C/P$ is equal to pole pitch in terms of coil-sides per pole.

For the progressive simplex lap winding

$$Y_w = Y_b - Y_f = +2 \text{ and } Y_c = \pm 1$$

For simplex retrogressive lap winding

$$Y_w = Y_b - Y_f = -2 \text{ and } Y_c = -1$$

Split Coils

In the process of winding design, a situation may arise where same coils with their top coil-sides together in one slot, may not have their bottom coil-sides together in another slot. Such cases arise only when there are more than two coil-sides per slot. These coils with their top coil-sides in one slot and bottom coil-sides in two different slots are called split coils. The split coils are helpful in commutation but the labour charges increase significantly.

Simplex Wave Winding

In simplex wave winding, two ends of a coil are connected to the commutator segments which are approximately 360° electrical apart.

The number of commutator segments is equal to the number of coils.

For a P -pole machine there are $P/2$ commutator pitches,

$$\frac{P}{2} Y_c = C \pm 1 \text{ or } Y_c = \frac{C \pm 1}{P/2}$$

where C is the number of coils or commutator segments.

Winding pitch for a simplex wave winding is

$$Y_w = \frac{2C \pm 2}{P/2} = 2Y_c$$

Note: +sign for progressive and -sign for retrogressive windings.

Wave winding is also known as series winding. The cause of designating it as wave windings is that during the traverse of a parallel path, one moves through the winding in a wave like progression.

Dummy Coils

For simplex wave winding the equation $Y_c = \frac{C \pm 1}{P/2}$ must be satisfied. If this relation is not fulfilled, then two ends of one (or more) coil may be cut off, insulated and not connected to the commutator segments. This unused and inactive coil, though placed in the armature slots for mechanical balance is called a dummy or dead coil. In order to have symmetrical armature winding, a dummy coil should be avoided as far as possible. However, it may be used in low-rating commutator machines where armature core with existing number of slots have to be used in order to avoid extra expenditure on new punching tools etc.

Note: If dummy coils are to be avoided, then $\frac{S}{P/2}$ should not be an integer.

Multiplex Lap and Wave Windings

$Y_c = \pm m$ for lap winding and $Y_c = \frac{C \pm m}{P/2}$ for wave winding

Where m is multiplicity of winding.

In the above expressions for Y_c positive sign is used for progressive winding and negative sign for retrogressive winding.

The number of armature parallel paths, A . In lap winding, $A = mp$ and in wave winding $A = 2$ m.

Multiplex wave windings are rarely used in DC machines.

Double layer windings are used in DC machines invariably.

Example 3: A synchronous generator with a synchronous reactance of 1.3 p.u. is connected to an infinite bus whose voltage is 1 p.u., through an equivalent reactance of 0.2 p.u., for maximum output of 1.2 pu, the alternator emf must be

- (A) 1.5 p.u. (B) 1.56 p.u.
(C) 1.8 p.u. (D) 2.5 p.u.

Solution: (C)

$$P = \frac{EV_t}{X} \sin \delta$$

$$P_{\max} = \frac{EV_t}{X}$$

$$\Rightarrow E = \frac{X P_{\max}}{V_t}$$

$$\Rightarrow \frac{(0.2 + 1.3) \times 1.2}{1} = 1.8 \text{ p.u.}$$

Equalizer Rings

In order to avoid overloading and sparking at the brushes which is caused by unequal emf's of parallel paths it is essential to take some steps so that circulating currents are not handled by the brushes. An equalizer ring is a low resistance conductor wire which connects together the points in the armature winding which should be at same potential whose function is to cause the armature currents to flow within the armature itself, without letting them pass through the brushes. It is used to provide equalizer rings at the back of the armature, i.e., at the other side of the commutator.

Note: Equalizer rings are not required for simplex wave windings even if there is magnetic unbalance because the coil-sides in two parallel paths are distributed under all the poles. Therefore both the parallel circuits affected equally and there is no need of equalizer rings in simplex wave windings.

Advantages of Wave Windings Over Lap Windings

1. A wave winding required only two brushes (or two sets of brushes) whereas lap winding requires brushes equal to the number of poles.
A wave winding may be fitted with as many brushes as the number of poles. In such a case, if one or more brushes develop poor contact with the commutator, the commutator machine continues operating satisfactorily whereas this is not possible in case of lap-connected machines.
2. Equalizer rings are required for lap-connected machines for obtaining better commutation, whereas wave windings do not require any equalizer ring.
3. Lap-connected machines require more brushes and equalizer rings hence they are costly compared to wave-connected machines.

Lap windings are used on low-voltage (medium power (50 to 500 kW) and high power (above 500 kW) machines. Wave windings are used on high voltage low-power (less than 50 kW) and medium power machines.

Wave windings are preferred because of the fact that they do not require equalizer-rings and are less expensive. However for armature currents more than 400 A, lap winding is the only choice. It is because for armature currents above 400 A, current per path in wave winding would be more than 200 A and such a high current leads to poor commutation in wave-wound machines.

In addition to Lap and wave windings, there is another type of winding called composite winding or frog-leg winding which is a combination of simplex lap and multiplex wave windings and requires no equalizer rings. This type of winding is used mainly on high-power commutator machines.

Open Windings or AC Armature Windings

AC machines do not require a commutator and their armature windings are open at one or more points. In commutator machines, the turns per coil are limited for obtaining better commutation. But in machines without commutator, there is no such restriction on the choice of number of turns per coil.

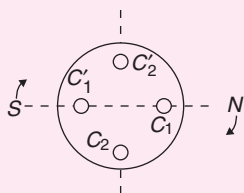
Closed or commutator windings are always double layer windings whereas AC armature windings may be single layer or double layer. Single layer windings are used for small AC machines, whereas double layer windings are used for machines above 5 kW.

PREVIOUS YEARS' QUESTIONS

1. A rotating electrical machine having its self-inductances of both the stator and the rotor windings, independent of the rotor position will be definitely not develop [2004]

(A) starting torque
(B) synchronizing torque
(C) hysteresis torque
(D) reluctance torque

2. Two magnetic poles revolve around a stationary armature carrying two coils ($c_1 - c'_1$, $c_2 - c'_2$) as shown in Figure. Consider the instant when the poles are in a position as



shown. Identify the correct statement regarding the polarity of the induced emf at this instant in coil sides c_1 and c_2 . [2005]

(A) $\odot$ in c_1 , no emf in c_2
(B) $\otimes$ in c_1 , no emf in c_2
(C) $\odot$ in c_2 , no emf in c_1
(D) $\otimes$ in c_2 , no emf in c_1

3. Distributed winding and short chording employed in AC machines will result in [2008]

(A) increase in emf and reduction in harmonics
(B) reduction in emf and increase in harmonics
(C) increase in both emf and harmonics
(D) reduction in both emf and harmonics.

ANSWER KEYS

Previous Years' Questions

1. D 2. A 3. D

Chapter 3

DC Machines

LEARNING OBJECTIVES

After reading this chapter, you will be able to understand:

- Armature windings
- Types of generators
- E.M.F. equation of a DC generator
- Performance equations of DC generators
- Losses in a DC generator
- Armature reaction
- Demagnetizing ampere-turns per pole
- Compensating windings
- Commutation
- Equalizing connections
- Characteristics of DC generator
- Parallel operation of DC generators
- DC motors
- Motor characteristics

FUNDAMENTALS OF DC MACHINES

Energy can neither be created nor be destroyed. We can change its forms, using appropriate energy -conversion processes. Change of the form of energy requires an energy conversion device.

- DC generator and DC motor are electromechanical energy conversion devices,
- A DC generator converts mechanical energy into electrical energy. It requires a prime mover such as a turbine diesel engine, etc.
- In a DC generator, the energy conversion is based on the principle of the production of dynamically induced e.m.f. This process can be explained by Faraday's laws of electromagnetic induction.
- The two basic essential parts of an electrical generator are
 1. A magnetic field, and
 2. A conductor or conductors which can so move as to cut the flux.
- Whenever there is relative motion between the conductor and the magnetic field, an emf is induced in the conductors which causes a current to flow if the conductor circuit is closed.
- In a DC generator, relative motion is produced by rotation of the armature. The armature winding is on the rotor and field winding is on the stator.

- In normal DC machines stator core is not laminated, armature core is laminated to reduce eddy current losses.

Constructional Details

The major parts of a DC machine are

1. Field system
2. Armature
3. Commutator
4. Brush and brush gear.

Constructionally there is no difference between a DC generator and DC motor.

1. The Field System

The purpose of the field system is to provide a uniform magnetic field within which armature rotates. The magnetic field is generated by electromagnets rather than permanent magnets on account of their greater magnetic strength and field strength regulation (i.e., field flux can be varied according to the requirement). The field system consists of four parts:

1. **Yoke:** This provides mechanical support for the poles and acts as a protecting cover for the whole machine and it carries the magnetic flux produced by the pole.

2. **Pole core:** The pole-core offers low reluctance path for magnetic flux. The pole core is of smaller cross-section than the pole shoe.
3. **Pole shoe:** The pole shoes serve two purposes.
 - (a) They spread out the flux in the air-gap and also, being of larger cross-section, reduce the reluctance of the magnetic path.
 - (b) They support the exciting coils (or field coils)
4. **Magnetizing coils (or) field winding:** These coils electromagnetize the poles which produce the necessary flux. The field coils which consist of copper wire or strip, are former wound for the correct dimension. Then, the former is removed and wound coil is put into place over the pole core.

2. Armature Core

It houses the armature conductors or coils and causes them to rotate and hence cut the magnetic flux of field magnets.

In addition to this, its most important function is to provide a path of very low reluctance to the flux through the armature from a N-pole to a S-pole. Armature core is made of high permeability material such as silicon steel. The use of high grade steel is made to keep hysteresis loss low. The core is laminated to reduce eddy current losses. These laminations are coated with insulating varnish.

3. Commutator

The function of commutator is to facilitate collection of current from the armature conductors. Inherently the voltage induced in the armature winding is alternating. Commutator acts as a mechanical rectifier to convert induced alternating voltages into DC voltage at brush terminals.

[In a DC Motor commutator acts as a mechanical inverter to convert DC applied voltage to AC Voltage in the armature winding].

- Commutator is of cylindrical structure and is built up of wedge-shaped hard-drawn copper segments. These segments are insulated from each other by thin layers of mica.

4. Brushes and Brush Gear

The function of brushes is to collect current from the commutator and supply it to the external-load circuit. Brushes are usually made of carbon or graphite. These brushes are housed in brush-holders. The brushes are made to bear down on the commutator by a spring to avoid any air-gap which may give rise to sparking.

Armature Windings

Two types of windings are mostly used for DC machine. They are lap winding and wave winding. For a given number of poles and armature conductors, the wave winding

gives more emf than lap winding. Conversely for the same emf lap winding would require more no. of conductors which will result in higher winding cost. Wave windings are suitable for low capacity generators with a voltage rating of 500–600 V. Lap windings are used for large capacity generators because lap winding have more parallel paths.

- Another advantage with the wave winding is that it requires no equalizing connections, where in a lap-wound machine they are necessary.
- Lap winding is suitable for comparatively low voltage but high current machines whereas wave winding is used for comparatively high voltage, low-current machines.

TYPES OF GENERATORS

DC generators are broadly classified as

1. Separately excited generators: Field is energized from an independent external source of DC current.
2. Self-excited generators – field current is provided by the generators themselves. Due to residual magnetism, there is always present some flux in the poles. When the armature is rotated some emf and hence some induced current is produced which is partly or fully passed through the field coils there by strengthening the residual pole flux.

- For the process of self-excitation to sustain,

1. There must be residual flux in the field poles.
2. The field winding flux must aid residual flux.

There are 3 types of self-excited generators named according to the manner in which their field coils are connected to the armature.

1. **Shunt wound:** Field winding is connected directly across the armature terminals.
2. **Series wound:** Field winding is connected in series with the armature winding.
3. **Compound wound:** It is a combination of a few series and few shunt windings.

There are two types of connections in a compound machine. They are (1) Short shunt (2) Long shunt.

In a compound generator, the shunt field is stronger, than the series field. When series field aids the shunt field, generator is said to be cumulatively compounded. On the other hand if series field opposes shunt field the generator is said to be differentially compounded.

- By reversing the connections of the series field winding, cumulative connection can be converted to differential connection and vice versa.

EMF EQUATION OF A DC GENERATOR

Let ϕ = Flux / pole in Wb
 Z = Total number of armature conductors
 P = No. of generator poles
 A = No. of armature parallel paths
 N = Armature speed in rpm
 E = emf induced in any parallel path in armature,

$$\text{The induced E.M.F.E.} = \frac{\phi Z N}{60} \times \frac{P}{A}$$

- The number of parallel paths A , depends upon the type of the armature winding, i.e., whether wave or lap-wound, For wave-wound armature $A = 2m$, where m is multiplicity of the winding.

$m = 1$ for simplex, $m = 2$ for duplex, $m = 3$ for triplex and so on for a lap-wound armature.

$A = Pm$, where m is the multiplicity of the winding
For a given DC machine Z , P and A are constants.

$$\text{Hence putting } K_a = \frac{ZP}{A}$$

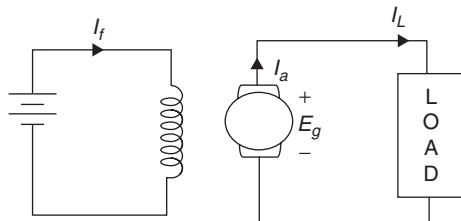
$$\text{We get } E_g = K_a \phi N$$

The induced emf can be increased by

- Increasing the flux / pole ϕ by increasing the field current (up to saturation point).
- Increasing the speed of the prime mover.

Performance Equations of DC Generators

Separately Excited



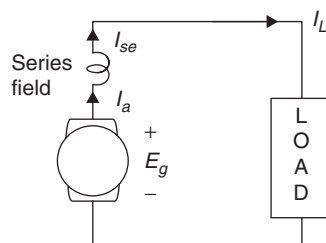
$$I_a = I_L = I$$

$$V = E_g - I_a r_a$$

$$\text{Power developed} = E_g I_a$$

$$\text{Power delivered to load} = V I_L$$

Series Wound



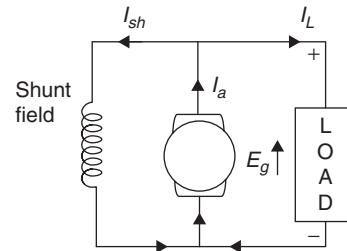
$$I_a = I_{se} = I_L = I$$

$$V = E_g - I(r_a + r_{se})$$

$$\text{Power developed} = E_g I_a$$

$$\text{Power delivered} = VI$$

Shunt Wound



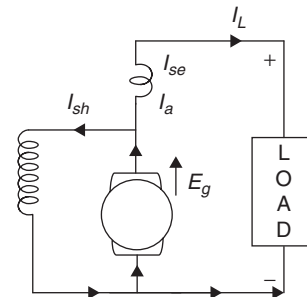
$$I_{sh} = \frac{V}{R_{sh}}$$

$$I_a = I_L + I_{sh}$$

$$V = E_g - I_a r_a$$

$$\text{Power developed} = E_g I_a$$

Short Shunt Compound



$$I_{se} = I_L$$

$$I_{sh} = \frac{V + I_{se} r_{se}}{R_{sh}} \text{ suffixes}$$

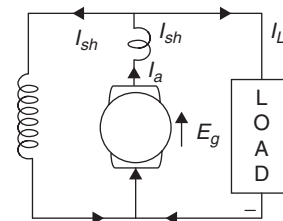
$$I_a = I_L + I_{sh}$$

$$V = E_g - I_a r_a - I_{se} r_{se}$$

$$\text{Power developed} = E_g I_a$$

$$\text{Power delivered} = VI_L$$

Long Shunt Compound



$$I_{sh} = \frac{V}{R_{sh}}$$

$$I_a = I_L + I_{sh}$$

$$V = E_g - I_a r_a - I_{se} r_{se}$$

$$\text{Power developed} = E_g I_a$$

$$\text{Power delivered} = VI_L$$

Solved Examples

Example 1: A 4-pole lap-connected DC generator with 480 conductors has an armature resistance 0.06Ω . If the conductors are reconnected to form a wave winding, other things remaining unchanged what is the value of armature resistance?

Solution: $\frac{Z_{lap}}{Z_{wave}} = \frac{4}{P^2}$ or

$$Z_{wave} = \frac{P^2}{4} \times Z_{Lap} \\ = \frac{P^2}{4} \times 0.06 = 0.24 \Omega$$

Example 2: Calculate commutator pitch for two-circuit simplex-wave winding having 4-poles and 129 commutator segments.

Solution: Commutator pitch for simplex wave winding is given by

$$Y_c = \frac{C \pm 1}{P/2} = \frac{129 \pm 1}{4/2} = \frac{128 \text{ or } 130}{2}$$

The commutator pitch may be 64 or 65.

Example 3: A 4-pole generator with 24 conductors has a two layer lap winding. The pole pitch is ____.

(A) 24 (B) 12 (C) 6 (D) 4

Solution: (C)

$$\text{Pole pitch} = Y_p = \frac{Z}{P}$$

$$Z = 24; P = 4$$

$$Y_p = \frac{Z}{P} = \frac{24}{4} = 6$$

Example 4: An 8-pole, lap-wound armature running at 350 rpm is required to generate 260 V. The useful flux/pole is 0.05 Wb. If the armature has 120 slots, the no. of conductors/slot is _____

(A) 6 (B) 8 (C) 10 (D) 4

Solution: (B)

$$E_g = \frac{\phi Z N P}{60 A} \Rightarrow 260 = \frac{0.05 \times Z \times 350 \times 8}{60 \times 8}$$

$$\Rightarrow Z = 890$$

$$\therefore \text{No of conductors/slot} = \frac{890}{120} = 7.14$$

Conductors/slot should be an integer

Hence, conductors/slot = 8

Example 5: A 4-pole lap-wound generator is run at 900 rpm. The reactance voltage if self inductance is $153 \times 10^{-6} \text{ H}$, armature current is 108 A, no of segments = 55, width of brush = 1.74 segments is _____

(A) 3.195 V
(C) 5.391 V

(B) 3.915 V
(D) 5.915 V

Solution: (B)

$$I_c = \frac{I_a}{A} = \frac{108}{4} = 27 \text{ A}$$

$$\text{Time taken by one commutator segment to cross brush axis} \\ = \frac{60}{900 \times 55}$$

$$\text{Time taken for 1.74 segments} = \frac{60 \times 1.74}{900 \times 55} = 2.11 \times 10^{-3} \text{ sec}$$

$$\therefore \text{Reactance voltage} = E = L \cdot \frac{2I_c}{t_c}$$

$$\therefore E = \frac{153 \times 10^{-6} \times 2 \times 27}{2.11 \times 10^{-3}} = 3.915 \text{ V}$$

LOSSES IN A DC GENERATOR

The various losses occurring in a generator can be subdivided as follows.

Copper Losses

1. Armature copper loss $= I_a^2 R_a$.

where R_a = resistance of armature and inter poles and series field winding etc.

This is about 30% to 40% of full-load losses.

2. Field copper loss:

In case of shunt generators, it is practically constant and $I_{sh}^2 R_{sh}$. In the case of series generator, it is $I_{se}^2 R_{se}$, where R_{se} is resistance of the series field winding.

3. The loss due to brush contact resistance. It is usually included in the armature copper loss.

Magnetic Losses (Iron or Core Losses)

Hysteresis Loss

This loss is due to the reversal of magnetization of the armature core. Every portion of the rotating core undergoes one complete cycle of magnetic reversal after passing under one pair of poles.

For normal flux densities (i.e., up to 1.5 Wb/m^2), hysteresis loss is given by Steinmetz formula.

$$W_h = \eta B_{\max}^{1.6} \times f V W.$$

where η = Steinmetz hysteresis coefficient

$B_{\max}$ = Maximum flux density

f = Frequency of magnetic reversals

V = Volume of the core in m^3

The hysteresis loss is reduced by choosing magnetic materials which have a low hysteresis coefficient such as silicon steel.

Eddy Current Loss

When the armature core rotates, it also Cuts The Magnetic Flux. Hence an emf is induced in the magnetic core which

sets up large current in the body of the core due to its small resistance. These currents circulate in the core and find no other path, hence they are known as eddy current. The power loss due to the flow of this current is known as eddy current loss. In order to reduce this loss and consequent heating of the core to a small value, the core is built up of thin laminations which are insulated from each other by a thin coating of varnish. These laminations are stacked and riveted at right angles to the path of the eddy current.

Eddy current loss is given by the following relation

$$W_e = KB_{\max}^2 f^2 t^2 V W.$$

Where

$B_{\max}$ = Maximum flux density

F = Frequency of magnetic reversals

T = Thickness of each lamination

V = Volume of the core.

The iron losses are practically constant for shunt and compound wound generators, because in their case, field current is approximately constant, the iron losses are about 20% to 30% of full-load losses.

Mechanical Losses

These losses consist of

1. Friction loss at bearings and commutators.
 2. Air friction or windage loss of rotating armature.
- These are about 10% to 20% of full load losses.

Stray Losses

Magnetic and mechanical losses are collectively known as stray losses. These are also known as rotational losses as they exist only when the rotor is rotating.

Constant or Standing Losses

Field Cu loss for shunt and compound generators is constant. Hence, stray losses and shunt Cu loss are constant in their case. These losses are together known as standing or constant losses W_c .

Example 6: A 10 kW, 230 V, DC shunt generator operation are 80% efficiency. Input power and total losses are

- (A) 5 kW, 2.5 kW (B) 12.5 kW, 1 kW
(C) 12.5 kW, 2.5 kW (D) 5 kW, 1 kW

Solution: (C)

$$\text{Output} = 10 \text{ kW}$$

$$\therefore \frac{80}{100} = \frac{10}{\text{input}}$$

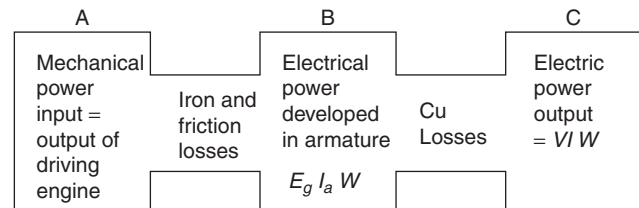
$$\Rightarrow 0.8x = 10$$

$$\Rightarrow x = \frac{10}{0.8} = 12.5 \text{ kW}$$

$$\therefore \text{Losses} = \text{input} - \text{output} = 2.5 \text{ kW}.$$

Power Stages in Generator

Various power stages in generator are as shown below



Following are the three generator efficiencies

1. Mechanical efficiency

$$\eta_M = \frac{B}{A} = \frac{\text{Total power generated}}{\text{Mech power input}}$$

2. Electrical efficiency

$$\eta_E = \frac{C}{B} = \frac{\text{Electric power output}}{\text{Electrical power developed}}$$

3. Overall or commercial efficiency

$$\eta_c = \frac{C}{A} = \frac{\text{Electric power output}}{\text{Mechanical power input}}$$

Condition for Maximum Efficiency

Generator efficiency is maximum, when

$$\text{Variable loss} = \text{Constant loss}$$

ARMATURE REACTION

The effect of the magnetic field set up by armature current on the distribution of flux under main poles of a generator is called armature reaction.

The armature magnetic field has two effects.

1. It demagnetizes or weakens the main flux and
2. It cross magnetizes or distorts it.

- The first effect leads to reduced generated voltage and the second to the sparking at the brushes.
- The flux is strengthened at the trailing pole tips but weakened at the leading pole tips.
- Because of armature reaction there is a shift in the magnetic neutral axis (M. N. A.) in the direction of rotation of the armature.
- Armature reaction gives rise to cross magnetization and de-magnetization.

Demagnetizing Ampere-turns Per Pole

The armature demagnetizing ampere-turns are neutralized by adding extra ampere-turns to the main field winding.

Demagnetizing ampere-turns pole,

$$AT_d / \text{pole} = ZI \times \frac{\theta_m}{360^\circ}$$

where

Z = Total number of armature conductors

I = Current in each armature conductor (or)
Current in a parallel path

$I_{a/2}$ for simplex wave winding

$I_{a/p}$ for simple lap winding

θ_m = Forward lead in mechanical or geometrical or
angular degrees

Cross-magnetizing Ampere-turns Per Pole

Cross magnetizing ampere-turns per pole,

$$AT_c / \text{pole} = ZI \left[\frac{1}{2P} - \frac{\theta_m}{360^\circ} \right]$$

To reduce the effects of armature reaction the following methods are normally used.

1. Provision of inter poles.
2. Provision of compensating windings.
3. Chamfering of the poles.

Compensating Windings

In large direct current machines which are subjected to large fluctuations in load (i.e., rolling mill motors and turbo-generators, etc.) compensating windings are used. Their function is to neutralize the cross-magnetizing effect of armature reaction.

In the absence of compensating winding, with sudden changes in load, the flux will be suddenly shifting backward and forward which will induce an emf in the armature coils. It may be so high as to strike an arc between the consecutive commutator segments which may lead to flash over around the whole commutator there by resulting in a short circuit of armature.

- These windings are embedded in slots in the pole shoes and are connected in series with armature in such a way that the current in them flows in opposite direction to that following in the armature conductors directly below the pole shoes.

Number of Compensating Windings

No. of armature amp-turns/pole for compensating winding

$$= \frac{ZI}{2P} \times \frac{\text{Pole arc}}{\text{Pole pitch}} = 0.7 \times \frac{ZI}{2P} \text{ (approximately)}$$

COMMUTATION

The process by which the current in the short-circuited armature coil is reversed while it crosses the magnetic neutral axis is called commutation.

During the commutation period the brush spans and hence short circuits that particular coil under going reversal

of current through it. The self-induced emf in the coil under-going commutation is known as reactance voltage.

Reactance voltage = (Coefficient of self-inductance)
× Rate of change of current

Time of commutation or commutation period,

$$T_c = \frac{W_b - W_m}{v} \text{ second}$$

where

W_b = Brush width in cm

W_m = Width of mica insulation in cm

v = Peripheral velocity of commutator segments in
cm/sec

Reactance voltage

$$L \times \frac{2I}{T_c}$$

If commutation is linear = $1.11 L \times \frac{2I}{T_c}$

If commutation is sinusoidal

Methods of Improving Commutation

There are two practical ways of improving commutation, i.e., making current reversal in the short-circuited coil as sparkless as possible.

Resistance Commutation

In this method, low resistance copper brushes are replaced by comparatively high resistance carbon brushes

EMF-commutation

In this method, arrangement is made to neutralize the reactance voltage by producing a reversing emf in the short-circuited coil under commutation. The reversing emf may be produced in two ways.

- (a) Either by giving the brushes a forward lead, sufficient enough to bring the short – circuited coil under the influence of next pole of opposite polarity. Shifting of the brushes is not a solution for reducing the effect of armature reaction. Though the cross-magnetization effect may be reduced somewhat, de-magnetization is also introduced.
- (b) By providing inter poles.

Inter poles: These are small poles fixed to the yoke and spaced in between the main poles. They are wound with comparatively few heavy gauge Cu wire turns and are connected in series with the armature so that they carry full armature current.

Their polarity in the case of a generator is the same as that of the main pole a head in the direction of rotation. The function of inter poles is two-fold:

1. As their polarity is the same as that of the main pole ahead, they induce an emf in the coil (under

commutation) which helps the reversal of current. The emf induced by inter-poles is known as commutating or reversing emf. The commutating emf neutralizes the reactance emf thereby making commutation sparkless. As inter poles carry armature current, their commutating emf is proportional to the armature current. This ensures automatic neutralization of reactance voltage which is also due to armature current.

2. Another function of inter poles is to neutralize the cross-magnetizing effect of armature reaction. Cancellation of cross magnetization is automatic and for all loads because both are produced by the same armature current.

Equalizing Connections

It is characteristic of lap-winding that all conductors in any parallel path lie under one pair of poles. If fluxes from all poles are exactly the same, then the emf induced in each parallel path is same and each path carries same current. But some inequalities in flux inevitably occur due to which there is always a slight imbalance of emf in the various parallel paths.

The function of equalizer rings is to avoid unequal distribution of current at the brushes there by helping to get sparkless commutation.

$$\text{Number of equalizer rings} = \frac{\text{No. of conductors}}{\text{No. of pair of poles}}$$

Equalizer rings are not used in wave-wound armature, because there is no imbalance in the emf's of the two parallel paths, this is due to the characteristic of wave-winding that armature conductors in either parallel path are not confined under one pair of poles as in lap winding but are distributed under all poles.

CHARACTERISTICS OF A DC GENERATOR

There are three important characteristics or curves of a DC generator.

1. **Open-circuit Characteristic:** O.C.C. (or No-load saturation characteristic). It shows the relation between the generated emf E_g and field current I_f at a given fixed speed. It is also known as magnetic characteristic.
2. **External Characteristic:** It gives relation between the terminal voltage V and the load current I_L .
3. **Internal or total characteristic:** This characteristic gives relation between the emf, E actually induced in the armature (after allowing for the demagnetizing effect of armature reaction) and the armature current I_a .

Characteristic Curves

Separately Excited Generator

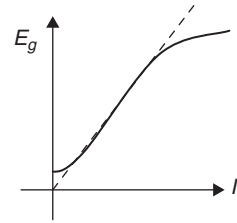
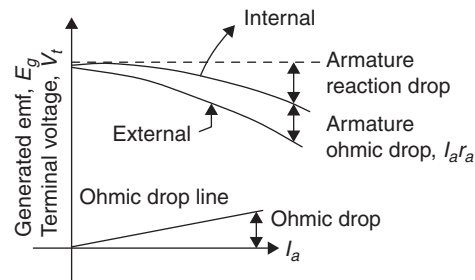


Figure 1 No-load characteristic (For all types of generators)



Series-wound Generators

For a series generator, as the field windings are in series with the armature, the field current (which is same as armature current) is zero, hence we do not have O.C.C. for a series wound generator. But it can be drawn from O.C.C. by considering extra current needed to neutralize the effect of armature reaction.

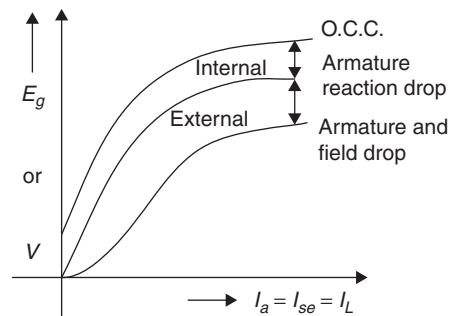
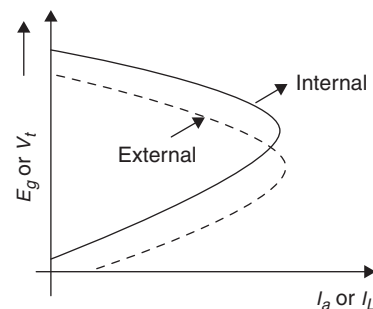
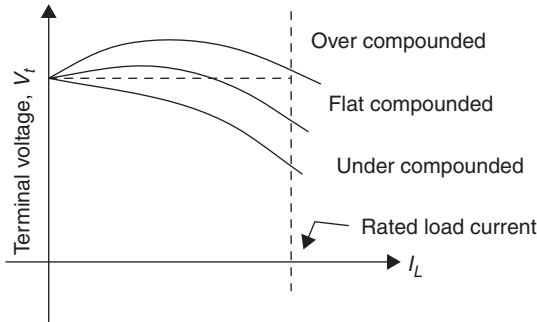


Figure 2 Series generator has rising voltage characteristic

Shunt-wound Generator



Compound Generator



Condition for build-up of a self excited (shunt) generator:
The conditions required to be fulfilled for the build up of a shunt generator are as follows.

1. There must be some residual magnetism in the generator poles.
 2. For the given direction of rotation the shunt field coils should be correctly connected to the armature i.e., the shunt field current produces a flux in the direction of residual flux.
 3. If excited on open circuit, its shunt field resistance should be less than the critical resistance.
 4. If excited on load, then its shunt field resistance should be more than certain minimum value of resistance whose value is given by internal characteristic.
- For a series wound generator the external (or load) circuit resistance should be less than the critical resistance and for a shunt generator load circuit resistance must be greater than critical load resistance.

A DC generator may fail to build up voltage because of failure of any of the above given cases.

PARALLEL OPERATION OF DC GENERATORS

Parallel operations is extremely desirable for the following reasons

1. Continuity of service
2. Efficiency
3. Maintenance and repair
4. Additions to plant

Condition to be Satisfied

- Polarity of the incoming generator should be same as that of existing generator.
- The no-load induced emf's are to be the same in order to avoid circulating currents on no load.

Shunt generators are best suited for parallel operation because of the drooping nature of their external characteristics

Because of the rising nature of the external characteristics of series generators, their parallel operations will pose some problems, which can be overcome by

1. Cross-connecting the fields.
2. Providing equalizing bar.

Example 7: Two DC generators having rectilinear external characteristics operate in parallel. One machine has the terminal voltage of 270 V on no load and 220 V at the load current of 30 A. The other has a voltage of 280 V at no load and 220 V at the load current of 30 A. Calculate the load shared and bus voltage of each machine when the total load current is 50 A.

Solution: Generator 1: Slope of external characteristic or voltage drop/ampere

$$= \frac{270 - 220}{30} = \frac{50}{30} = 5/3 \text{ V/A.}$$

Generator 2: Slope of external characteristic or voltage drop/ampere

$$= \frac{280 - 220}{30} = \frac{60}{30} = 2 \text{ V/A}$$

Let V be the common bus voltage

$$I_1 = \text{Load current shared by generator 1}$$

$$I_2 = \text{Load current shared by generator 2.}$$

Then

$$V = 270 - \frac{5}{3} I_1 \text{ for generator 1}$$

$$V = 280 - 2 I_2 \text{ for generator 2.}$$

Since the terminal voltage is same

$$\therefore 270 - \frac{5}{3} I_1 = 280 - 2 I_2$$

$$\Rightarrow \frac{-5}{3} I_1 + 2 I_2 = 10 \quad (1)$$

$$\text{Also } I_1 + I_2 = 50 \quad (2)$$

Solving Equations (1) and (2)

We get,

$$I_1 = 24.54 \text{ A and } I_2 = 25.46 \text{ A}$$

and bus bar voltage $V = 229 \text{ V}$.

DC MOTORS

An electric motor is a machine which converts electrical energy into mechanical energy. It works on the principle that when a current carrying conductor is placed in a magnetic field, it experiences a mechanical force. Whose direction is given by Fleming's left-hand rule and hence the conductor moves in the direction of force.

The magnitude of force is given by

$$F = BIL \sin \theta \text{ Newton}$$

where

$$B = \text{Flux density in Wb/m}^2$$

$$I = \text{Current flowing in the conductor}$$

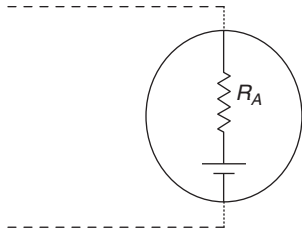
$$L = \text{Length of the conductor}$$

$$\theta = \text{Angle made by the tangent drawn at conductor with field direction}$$

Significance of the Back emf

As the armature of a DC motor rotates, all the necessary conditions for an emf to be induced in a conductor are satisfied, namely a conductor, a magnetic field and the relative motion between the conductor and magnetic field. Hence an emf is induced in the armature, the direction of which can be determined by Flemings right hand rule. This induced emf is in opposition to the applied voltage hence it is known as back or counter emf.

The applied voltage has to overcome the back emf in addition to the armature drop. The equivalent circuit of armature is shown in below figure.



The voltage equation of a DC motor is

$$V = E_b + I_a R_a$$

Where E_b is the back emf given by

$$E_b = \frac{\phi Z N}{60} \times \frac{P}{A}$$

From the voltage equation, on multiplying both the side by I_a , we get

$$VI_a = E_b I_a + I_a^2 R_a$$

where

VI_a = Electrical input to the armature.

$E_b I_a$ = Electrical equivalent of mechanical power developed in the armature.

$I_a^2 R_a$ = Cu loss in the armature.

Example 8: DC motor gives maximum power when

(A) $V = 2E_b$ (B) $V = E_b/2$

(C) $V = E_b/\sqrt{2}$ (D) $V = E_b$

Solution: (A)

$$V = E_b + I_a R_a$$

$$\therefore P_m = E_b I_a = VI_a - I_a^2 R_a$$

It is maximum when $\frac{dP_m}{dI_a} = 0$

$$V - 2I_a R_a = 0 \Rightarrow I_a R_a = \frac{V}{2}$$

$$\therefore \text{Back emf: } E_b = V - I_a R_a = V - \frac{V}{2} = \frac{V}{2}$$

$$\therefore E_b = \frac{V}{2}$$

Condition for Maximum Power

The gross mechanical power developed by a motor is $P_m = VI_a - I_a^2 R_a$.

Differentiating both sides w.r.t I_a and equating the result to zero, we get,

$$\frac{dP_m}{dI_a} = V - 2I_a R_a = 0$$

$$\therefore I_a R_a = \frac{V}{2}$$

$$\text{As } V = E_b + I_a R_a \text{ and } I_a R_a = V/2$$

$$\therefore E_b = V/2$$

Thus gross mechanical power developed by a motor is maximum when back emf is equal to half the applied voltage. This condition is however, not realized in practice, because in that case current would be much beyond the normal current of the motor. Moreover, half the input would be wasted in the form of heat and taking other losses into consideration, the motor efficiency will be below 50 percent.

Condition for Maximum Efficiency

Variable loss = Constant loss
(same as that of generator)

Armature Torque (T_a) or Torque developed (T_d):

Let T_a be the torque developed by the armature of a motor running at N rps. If T_a is in Nm, then

$$\text{Power developed} = T_a \times 2\pi N$$

$$T_a \times 2\pi N = E_b I_a$$

$$T_a = \frac{E_b I_a}{2\pi N} = \frac{\phi Z N \left(\frac{P}{A} \right) \times I_a}{2\pi N}$$

$$= \frac{1}{2\pi} \phi Z \left[\frac{P}{A} \right] I_a \text{ Nm}$$

$$T_a = 0.159 \phi Z I_a \left[\frac{P}{A} \right] \text{ Nm}$$

From the above equation for the torque, we find that $T_a \propto \phi I_a$

1. For series motor, ϕ is directly proportional to I_a (before saturation) because field windings carry full armature current

$$\therefore T_a \propto I_a^2$$

2. For shunt motors, ϕ is practically constant hence $T_a \propto I_a$

Alternate expression for torque developed.

If N is in rpm then,

$$T_a \times \frac{2\pi N}{60} = E_b I_a$$

$$T_a = \frac{60 E_b I_a}{2\pi N} = \frac{60}{2\pi} \frac{E_b I_a}{N} = 9.55 \frac{E_b I_a}{N} \text{ Nm}$$

Shaft Torque

The torque which is available for doing useful work is known as shaft torque, T_{sh} .

$$T_{sh} = \frac{\text{Output in watts}}{2\pi N / 60} N \text{ in rpm}$$

$$= \frac{60}{2\pi} \frac{\text{Output}}{N} = 9.55 \frac{\text{Output}}{N}$$

The difference ($T_a - T_{sh}$) is known as lost torque and is due to iron and friction losses of the motor.

Speed of a DC Motor

From the voltage equation of a DC motor, we get

$$E_b = V - I_a R_a$$

$$\text{Or } \frac{\phi Z N}{60} \frac{P}{A} = V - I_a r_a$$

$$\therefore N = \frac{V - I_a r_a}{\phi} \times \frac{60 A}{Z P} \text{ rpm}$$

$$\text{Or } N = \frac{E_b}{\phi} \times \left(\frac{60 A}{Z P} \right)$$

For a given DC motor A , Z and P are fixed.

$$\therefore N = K \frac{E_b}{\phi}$$

where K is a constant

$$N \propto \frac{E_b}{\phi} \quad \text{Or} \quad \frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_2}$$

1. For series motor,
 $\phi \propto I_a$ (prior to saturation of magnetic poles)

$$\therefore \frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{I_{a1}}{I_{a2}}$$

2. For shunt motor,

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_2}$$

If flux is constant i.e., $\phi_2 = \phi_1$

$$\text{Then } \frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}}$$

Speed Regulation

The speed regulation is defined as the change in speed when the load on the motor is reduced from rated value to zero, expressed as percent of the rated load speed.

$$\% \text{Speed regulation} = \frac{\text{No load speed} - \text{Full load speed}}{\text{Full load speed}} \times 100$$

$$\text{P. U. speed regulation} = \frac{\text{No load speed} - \text{Full load speed}}{\text{Full load speed}}$$

Example 9: Determine the full-load speed shunt field, current.

$$\text{Solution: } I_{sh} = \frac{230}{230} = 1 \text{ A}$$

$$I_{a1} = 3 - 1 = 2 \text{ A}$$

$$E_{b1} = 230 - 0.5 \times 2 = 229 \text{ V}$$

$$I_{a2} = 23 - 1 = 22 \text{ A}$$

$$E_{b2} = 230 - 22 \times 0.5 = 219 \text{ V}$$

$$N_2 = \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_2} \times N_1$$

$$= \frac{219}{229} \times \frac{\phi_1}{0.98\phi_1} \times 1000$$

$$(\because \phi_2 = 0.98\phi_1) = 975.84 \text{ rpm}$$

Example 10: Determine the full load torque

Solution:

$$\text{Power developed} = E_{b2} I_{a2} = 219 \times 22$$

$$= 4.818 \text{ kW}$$

$$\text{Angular speed, } \omega = \frac{2 \times 3.14 \times 975.84}{60}$$

$$= 102.14 \text{ rad/sec}$$

$$\text{Torque} = \frac{4818}{102.14} = 47.17 \text{ Nm.}$$

Common Data for Examples 11 and 12

A 220 V DC shunt motor takes 20 A at rated voltage and runs at 1000 rpm. Its field circuit resistance is 100 Ω and armature circuit resistance is 0.1 Ω . Compute the value of additional resistance required in the armature circuit to reduce the speed to 800 rpm when

Example 11: The load torque is proportional to the speed

Solution: Given the condition torque, $T \propto$ speed, N

$$\therefore \frac{T_1}{T_2} = \frac{N_1}{N_2}$$

$$\text{Shunt field current, } I_{sh} = \frac{220}{100} = 2.2 \text{ A}$$

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$$I_{a1} = I_L - I_{sh} = 20 - 2.2 = 17.8 \text{ A}$$

$$E_{b1} = V - I_{a1} R_a = 220 - 17.8 \times 0.1$$

$$= 218.22 \text{ V}$$

$$E_{b2} = E_{b1} \times \frac{N_2}{N_1}$$

$$218.22 \times \frac{800}{1000} = 174.57 \text{ V}$$

Also $T \propto \phi I_a$, where ϕ is constant

$$\therefore I_{a2} = \frac{I_{a1} N_2}{N_1} = \frac{17.8 \times 800}{1000}$$

Total armature circuit resistance

$$R = \frac{V - E_b}{I_a} = \frac{220 - 174.57}{14.24} = 3.19 \Omega$$

$$R_{ext} = 3.19 - 0.1 = 3.09 \Omega$$

Example 12: The load torque varies as the square of the speed

Solution: Given that $T \propto N^2$

$$\therefore I_{a2} = I_{a1} \times \left(\frac{N_2}{N_1} \right)^2$$

$$= 17.8 \times \left(\frac{800}{1000} \right)^2 = 11.4 \text{ A}$$

$$R = \frac{220 - 174.57}{11.40} = 3.988 \Omega$$

$$R_{ext} = 3.988 - 0.1 = 3.89 \Omega$$

Example 13: The full-load speed regulation is 8% when the speed at full load is 800 RPM. The no-load speed is _____ RPM.

- (A) 832 (B) 864
(C) 736 (D) 768

Solution: (B)

$$\text{Speed regulation} = \frac{N_0 - N}{N}$$

$$N_0 = \text{No-load speed}$$

$$N = \text{Speed in load}$$

$$\frac{8}{100} = \frac{N_0 - 800}{800} \Rightarrow 64 = N_0 - 800$$

$$N_0 = 864 \text{ rpm}$$

Example 14: A 230 V shunt motor develops a torque of 48 Nm at an armature current of 12 A. The torque produced when the armature current is 15 A, is

- (A) 48 Nm (B) 36 Nm
(C) 60 Nm (D) 72 Nm

Solution: (C)

$$T \propto \phi I_a$$

For a shunt motor, ϕ is constant

$$T \propto I_a$$

$$\frac{T_2}{T_1} = \frac{I_{a2}}{I_{a1}} \Rightarrow T_2 = T_1 \times \frac{I_{a2}}{I_{a1}}$$

$$= 48 \times \frac{15}{12} = 60 \text{ Nm}$$

Example 15: A DC motor develops a torque of 120 Nm at 20 rps. At 30 rps it will develop a torque of _____ Nm.

- (A) 160 (B) 120
(C) 80 (D) 40

Solution: (C)

$$T \propto \frac{1}{N}$$

$$N_1 T_1 = N_2 T_2$$

$$T_2 = \frac{N_1 T_1}{N_2} = \frac{20 \times 120}{30} = 80 \text{ Nm}$$

Example 16: A DC motor connected to a 230 V supply has an armature resistance of 0.15 Ω . The value of armature current when the back emf $E_b = 200 \text{ V}$ is _____ A.

- (A) 50 (B) 100
(C) 400 (D) 200

Solution: (D)

$$E_b = V - I_a R_a$$

$$I_a = \frac{V - E_b}{R_a}$$

$$= \frac{230 - 200}{0.15} = \frac{30}{0.15} = 200 \text{ A}$$

MOTOR CHARACTERISTICS

The characteristic curves of DC motors can be broadly classified as

1. Electrical characteristics: ' T ' vs. ' I_a ' and ' N ' vs. ' I_a '.
2. Mechanical characteristic: ' N ' vs. ' T '.

Characteristics of Shunt Motors

1. T_a / I_a characteristic

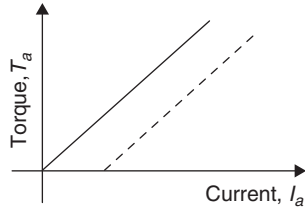
$$T = K \phi I_a$$

where K is a constant

In the case of shunt motors, the flux is assumed to be practically constant (though at heavy loads, ϕ decreases somewhat due to increased armature reaction)

$$\therefore T \propto I_a$$

The torque varies linearly with armature current.

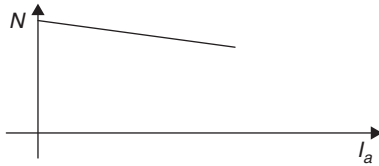


2. N / I_a characteristic

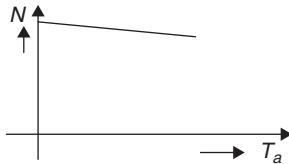
Variations of speed can be obtained from the relation

$$N \propto \frac{E_b}{\phi} \text{ or } \frac{V - I_a r_a}{\phi}$$

Since the flux is practically constant for a shunt motor, the speed decreases linearly with armature current. The speed curve is slightly dropping as shown in the figure but for all practical purposes, shunt motor is taken as a constant – speed motor.



3. N / T_a characteristic

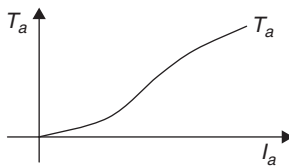


Characteristics of Series Motors

1. T_a / I_a characteristic

$$\text{Torque } T_a \propto \phi I_a$$

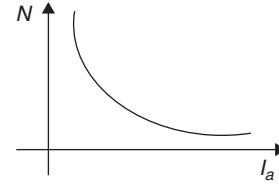
In the case of the series motors field windings also carry the armature current, $\phi \propto I_a$ up to the point of magnetic saturation. At light loads, I_a and hence ϕ are small. But as I_a increases, T_a increases as the square of the current, i.e., $T_a \propto I_a^2$. Hence T_a / I_a curve is a parabola as shown in below figure. After saturation ϕ is almost independent of I_a hence $T_a \propto I_a$ only. So the characteristic becomes a straight line



2. N / I_a characteristic

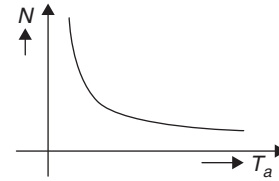
$$N \propto \frac{E_b}{\phi}$$

Change in E_b for various load currents is small and hence may be neglected with increased I_a . ϕ also increases hence the speed varies inversely as armature current as shown in the following figure.



- The series motor is a variable speed motor.
- On no-load and at light loads, a DC series motor has a tendency to run at dangerously high value of speed.

3. N / T_a or Mechanical characteristic.



When the speed is high, the torque is low and vice versa.

Characteristic of Compound Motors

These motors have both series and shunt windings. So the characteristics of such motors lie in between those of shunt and series motors. The characteristics approach those of a shunt or a series motor, depending upon the relative field strengths of the shunt and the series field windings.

If series excitation helps the shunt excitation, then the motor is said to be cumulatively compounded. If on the other hand, series field opposes the shunt field, then the motor is said to be differentially compounded.

Cumulative Compound Motors

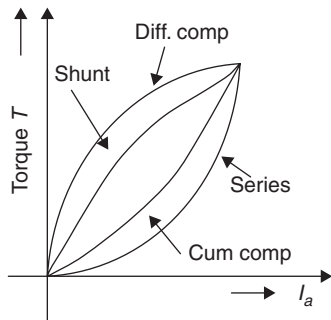
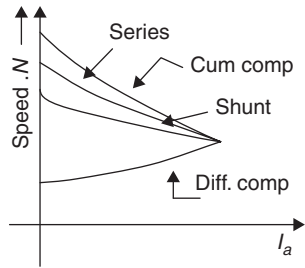
These machines are used where series characteristics are required and where, in addition, the load is likely to be removed totally as in some types of coal cutting machines or for driving heavy machine tools which have to take sudden cuts quite often. Such machines are used to drive electric shovels, metal stamping, machines, reciprocating pumps, hoists and compressors etc.

Differential Compound Motors

Since series field opposes the shunt field, the flux is decreased as load is applied to the motor. This results in the motor speed remaining almost constant or even increasing with the load ($\therefore N \propto E_b / \phi$). Due to this reason there is a decrease in the rate at which the motor torque increases with load. Such motors are not in common use. One of the biggest drawback of such a motor is that due to weakening of flux with increase in load, there is a tendency towards

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speed instability and motor running away unless designed properly.



Type of motor	Characteristics	Applications
Shunt	Approximately constant speed, adjustable speed	For driving constant speed line shafting. Lathes, centrifugal pumps, machine tools, blowers and fans, reciprocating pumps
Series	Variable speed, Adjustable varying speed, high starting torque	For traction work, i.e., electric locomotives rapid transit systems, trolley, cars, cranes and hoists, conveyors
Cumulative compound	Variable speed, adjustable varying speed, high starting torque	For intermittent high torque loads, for shears and punches, elevators, conveyors, heavy planers, rolling mills, ice machines, printing presses, air compressors.

SPEED CONTROL OF DC MOTORS

The speed of a DC motor is given by the relation

$$N = \frac{V - I_a r_a}{Z\phi} \left[\frac{A}{P} \right] = K \frac{V - I_a r_a}{Z\phi} \text{ rps}$$

where K is a constant.

Sometimes, it may become necessary to vary the speed of a DC motor at a given load condition. There are various methods of controlling the speed of a DC motor.

1. By varying the applied voltage
2. By varying the resistance in the armature circuit
3. By varying flux/pole

Example 17: The value of torque produced by armature of 4-pole motor having 774 conductors, two paths in parallel, 24 mWb flux per pole, when the total armature current is 50 A, is ____.

- (A) 350 Nm (B) 373 Nm
(C) 295.35 Nm (D) 420 Nm

Solution: (C)

$$\begin{aligned} T_a &= 0.159 \cdot \phi I_a \left(\frac{P}{A} \right) \\ &= 0.159 \times 774 \times 24 \times 10^{-3} \times 50 \times \frac{4}{2} \\ &= 295.35 \text{ Nm} \end{aligned}$$

Common Data for Example 18

A 230 V DC shunt motor takes a current of 40 A and runs at 1100 RPM. If armature and shunt field resistances are 0.25 Ω and 230 Ω respectively.

Example 18: The torque developed by the motor is ____.

- (A) 78.9 Nm (B) 74.66 Nm
(C) 98.7 Nm (D) 87.9 Nm

Solution: (B)

$$\begin{aligned} E_b &= 220.25 \text{ V} \\ T_a &= \frac{E_b I_a}{N} \times 9.55 \\ &= \frac{220.25 \times 39}{1100} \times 9.55 \\ \therefore T_a &= 74.66 \text{ Nm} \end{aligned}$$

Common Data for Examples 19 and 20:

The counter emf of a shunt motor is 227 V, the field resistance is 160 Ω and the field current is 1.5 A.

Example 19: If the line current is 39.5 A, the armature resistance is ____.

- (A) 0.342 Ω (B) 0.243 Ω
(C) 0.423 A (D) 0.432 Ω

Solution: (A)

Line current = $I_L = 39.5 \text{ A}$

$$I_a = I_L - I_{sh} = 39.5 - 1.5 = 38 \text{ A}$$

$$V = E_b + I_a R_a \Rightarrow R_a = \frac{V - E_b}{I_a}$$

$$\therefore R_a = \frac{240 - 227}{38} = 0.342 \Omega$$

$$[\because V = I_{sh} R_{sh} = 1.5 \times 160 = 240 \text{ V}]$$

Example 20: The starting current of the motor is _____.

- (A) 250 A (B) 107.57 A
(C) 255.5 A (D) 701.75 A

Solution: (D)

When the motor is starting, $E_b = 0$

$$I_a = \frac{V}{R_a} = \frac{240}{0.342} = 701.75 \text{ A}$$

Common Data for Examples 21 and 22:

A 450 V series motor runs at 500 rpm taking a current of 40 A. Total resistance of the armature and field circuits is 0.8Ω , assuming flux is proportional to field current and if the load is reduced so that motor draws a current of 30 A.

Example 21: The percentage change in torque is

- (A) 38.75% (B) 41.75%
(C) 39.75% (D) 43.75%

Solution: (D)

$$\phi \propto I_a$$

$$\Rightarrow T \propto \phi I_a$$

$$T \propto I_a^2 \quad [\because I_a = I_{sc} \text{ in series motor}]$$

$$\therefore T_1 \propto 40^2$$

$$T_2 \propto 30^2$$

$$\frac{T_2}{T_1} = \frac{9}{16}$$

$$\% \text{ change in torque} = \frac{T_1 - T_2}{T_1} \times 100$$

$$= \frac{7}{16} \times 100 = 43.75\%$$

Example 22: Speed is _____ rpm

- (A) 627.94 (B) 697.4
(C) 679.42 (D) 649.72

Solution: (C)

$$E_{b1} = 450 - (40 \times 0.8) = 450 - 32 = 418 \text{ V}$$

$$E_{b2} = 450 - (30 \times 0.8) \\ = 450 - 24 = 426 \text{ V}$$

$$E_b \propto \phi N$$

$$\propto I_a N$$

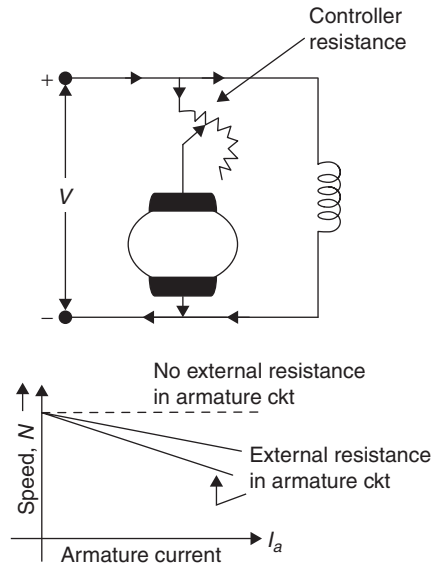
$$\Rightarrow \frac{E_{b2}}{E_{b1}} = \frac{I_{a2} N_2}{I_{a1} N_1}$$

$$\therefore N_2 = N_1 \frac{E_{b2}}{E_{b1}} \times \frac{I_{a1}}{I_{a2}} \\ = 500 \times \frac{426}{418} \times \frac{40}{30} = 679.42 \text{ rpm}$$

Speed Control of Shunt Motors

Armature or Rheostatic Control Method

This method is used when speeds below the no-load speed are required. As the supply voltage is constant, the voltage across the armature is varied by inserting an additional resistance in series with the armature circuit. Different speeds can be obtained by varying the additional series resistance.



Let I_{a1} = Armature current with no external resistance

I_{a2} = Armature current with external resistance in the armature ckt

N_1, N_2 = Corresponding speeds

V = Supply voltage

Then $N_1 \propto E_{b1}$

Or $\propto V - I_{a1} R_a$

Suppose the external resistance in series with the armature ckt be r , and $r + r_a = R_t$.

$$N_2 \propto E_{b2} \\ \propto V - I_{a2} R_t$$

$$\therefore \frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}}$$

If N_0 be the no-load speed (and $I_a r_a$ drop at no-load neglected)

$$\text{Then } \frac{N}{N_0} = \frac{V - I_a R_t}{V} = \left(1 - \frac{I_a R_t}{V} \right)$$

It is seen that for a given resistance R_t , speed is a linear function of armature current

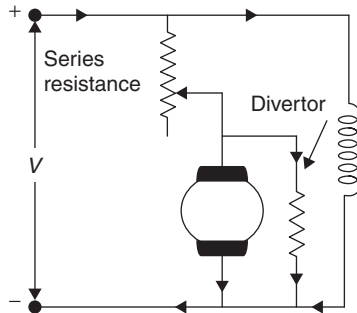
At $N = 0$

$I_a = \frac{V}{R_t}$ is the maximum current and is called stalling current

The main disadvantage of this method lies in the fact that a considerable amount of power is lost in the external resistance.

This method is expensive and unsuitable for rapidly changing loads because for a given value of R_p , speed will change with load.

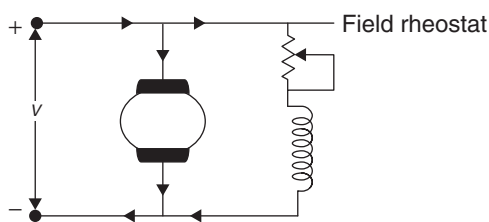
- A more stable operation can be obtained by using a diverter across the armature in addition to armature control resistance as shown in below figure.



Now, the changes in armature current (due to changes in the load torque) will not be so effective in changing the p.d., across the armature. (hence the armature speed).

Flux (or) Field Control

As $N \propto \frac{1}{\phi}$, by decreasing the flux, the speed can be increased and vice versa. The flux of a DC shunt motor can be changed by changing I_{sh} with help of a shunt field rheostat as shown in below figure.

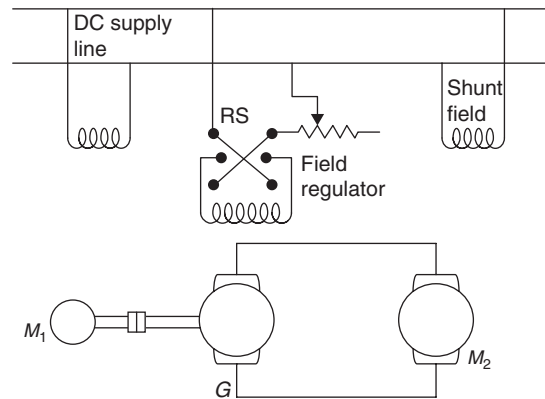


The external resistance connected in series with the shunt field winding is called a regulator.

Voltage Control Method

- Multiple voltage control:** In this method, the shunt field of the motor is connected permanently to a fixed exciting voltage, but the armature is supplied with different voltages by connecting it across one of several different voltages by means of suitable switchgear. The armature speed will be proportional to their different voltages. The intermediate speeds can be obtained by adjusting the shunt field regulator. This method is not much used.

2. Ward Leonard system:



This system is used where a usually wide (up to 10:1) and very sensitive speed control is required as for coillery winders, electric excavators, elevators and the main drives in steel mills and blooming and paper mills.

With the help of Ward Leonard set speed in both directions can be varied.

M_1 is the main motor whose speed is to be controlled. This motor is separately excited. It is fed from a variable voltage generator G , which is driven by an auxiliary motor M_2 which may be another DC or AC motor. By applying a variable voltage across armature terminals of M_1 , any desired speed can be obtained. This variable voltage is supplied by motor-generator set, i.e., $M_2 - G$. By controlling the speed of auxiliary motor M_2 , magnitude of voltage can be controlled. The polarity of voltage is changed by changing the direction of field current of generator G with the help of reversing switch RS .

This system has advantages of smooth variation in speed and the change of the sense of rotation.

The disadvantages of this system are

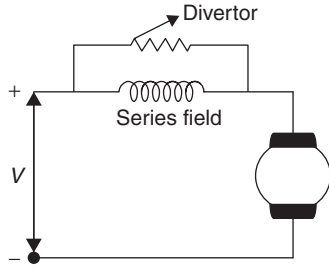
- It requires an auxiliary motor to drive the variable voltage generator which involves high capital outlay.
- Low efficiency especially at light loads.

Speed Control of Series Motors

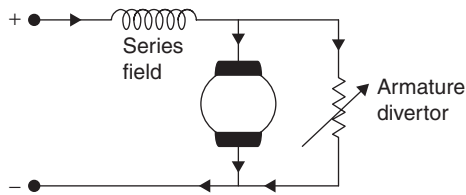
Flux Control Method

A series motor speed can be controlled by changing the flux with the help of any one of the following methods.

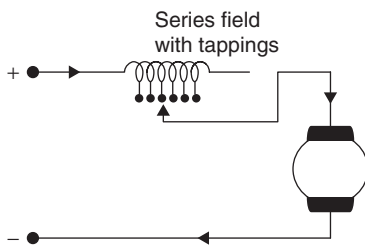
- Field divertors:** The series field winding is shunted by a variable resistance known as field diverter. Any desired amount of current can be passed through the diverter by adjusting its resistance. Hence speed is increased by decreasing flux.



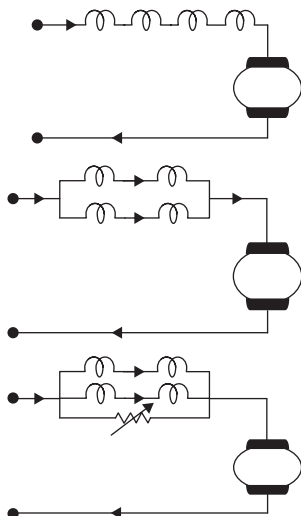
- (b) **Armature Divertor:** A divertor across the armature can be used for given speeds lower than the normal speed. For a given constant-load torque, if I_a is reduced due to armature divertor, then ϕ must ($\because T_a \propto \phi I_a$) an increase in current drawn from the supply $\left(N \propto \frac{1}{\phi} \right)$, and hence the speed is reduced.



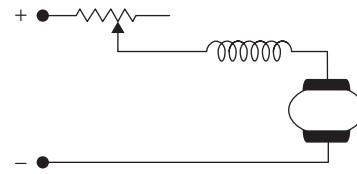
- (c) **Tapped field control:** The number of series field turns in the circuit can be changed as per the requirement. With full field, the motor runs at its minimum speed which can be raised in steps by cutting out some of the series turns.



- (d) **Paralleling field coils:** In this method, used for fan motors, several speeds can be obtained by regrouping the field coils as shown in below figure.



Variable Resistance in Series with Motor

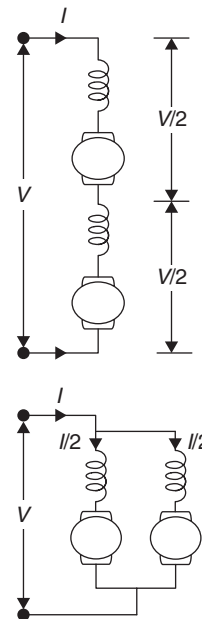


By increasing the resistance in series with the armature as shown in the figure, the voltage applied across the terminals can be decreased. With reduced voltage across the armature, the speed is reduced.

In this method, as full motor current passes through this series resistance there is a considerable loss of power in it.

Series Parallel Control

In this system of speed control, which is widely used in electric traction, two or more similar mechanically coupled series motors are employed. At low speeds, the motors are joined in series and for high speeds, are joined in parallel



When in parallel,

$$\text{Speed} \propto \frac{E_b}{\phi} \propto \frac{V}{I/2} \propto \frac{2V}{I}$$

As torque $\propto \phi I \propto I^2$

$$\therefore T \propto \left(\frac{I}{2} \right)^2 \propto \frac{I^2}{4}$$

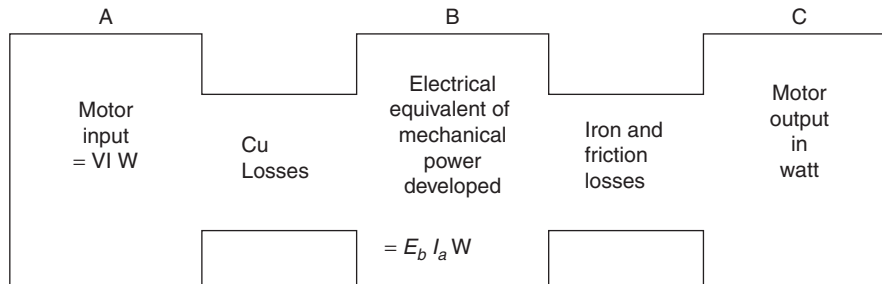
When in series

$$\text{Speed} \propto \frac{E_b}{\phi} \propto \frac{(V/2)}{I} \propto \frac{V}{2I}$$

$$T \propto \phi I \propto I^2$$

This speed is one-fourth of the speed of the motors when in parallel and the torque is four times that produced by two motors when in parallel.

Power Stages in a DC Motor



When the motor is at rest there is, as yet, obviously no back emf developed in the armature. If full supply voltage is applied across the stationary armature, it will draw a very large current because armature resistance is relatively small. This excessive current will blow out the fuses and prior to that it will damage the commutator, and brushes etc

- To limit the starting current to a safe predetermined value, every DC motor is to be provided with a starter or starting resistance. The starting resistance is gradually cut out as the motor gains speed and develops the back emf which then regulates its speed.
- There are two types of starters for DC shunt motors. They are as follows:
 - (i) Three-point starter
 - (ii) Four-point starter.
- Along with the limiting resistance protective devices are also mounted on the three-point starter of DC shunt motor, they are N.V.R. (no volt release) and the O.L.R. (over load relay)

$A - B$ = Copper losses and $B - C$ = Iron and friction losses.

$$\text{Overall or commercial efficiency } \eta_c = \frac{C}{A}$$

$$\text{Electrical efficiency } \eta_e = \frac{B}{A}$$

$$\text{Mechanical efficiency } \eta_m = \frac{C}{B}$$

Example 23: A 440 V shunt motor has an armature resistance of 0.8 Ω . The back emf when giving an output of 7.46 kW at 80% efficiency is _____ V.

- (A) 440 (B) 400
(C) 220 (D) 425

Solution: (C)

$$\begin{aligned} \text{Input power} &= \frac{7.46 \times 1000}{0.8} \\ &= 9325 \text{ W} \end{aligned}$$

$$\text{Input current} = \frac{9325}{440} = 21.193 \text{ A}$$

$$I_f = \frac{440}{220} = 2.2 \text{ A}$$

$$I_a = 21.193 - 2.2 = 18.99 \text{ A}$$

$$\begin{aligned} E_b &= V - I_a R_a \\ &= 440 - 18.99 \times 0.8 \\ &= 424.8 \text{ V} \end{aligned}$$

Starters for DC Motors

The current drawn by a motor armature is given by the relation

$$I_a = \frac{(V - E_b)}{R_a}$$

- The main drawback of the three-point starter is maloperation of the N.V.R. which makes it unsuitable for use with variable speed motors, if the speed is controlled by the field flux control method.

ELECTRIC BRAKING

A motor comes to rest by applying Brake to the motor. There are generally two breaking phenomenon (i) electro mechanical braking (or) friction braking (ii) electrical braking. In these electro mechanical Braking drawback is sudden application of braking force.

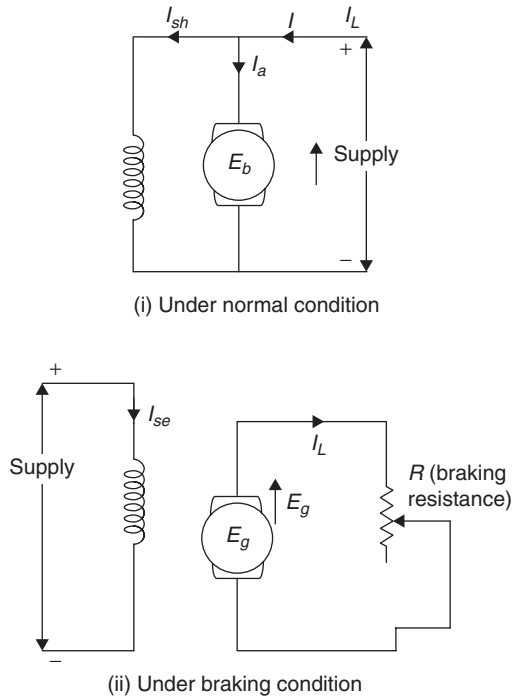
Electrical braking are generally three types.

1. Dynamic (or) rheostatic braking
2. Plugging (or) counter current braking
3. Regenerative braking

Dynamic (or) Rheostatic Braking

In this method the armature of DC motor is disconnected from the supply and is connected across a Braking (variable) resistance. The different DC motor connection as shown below.

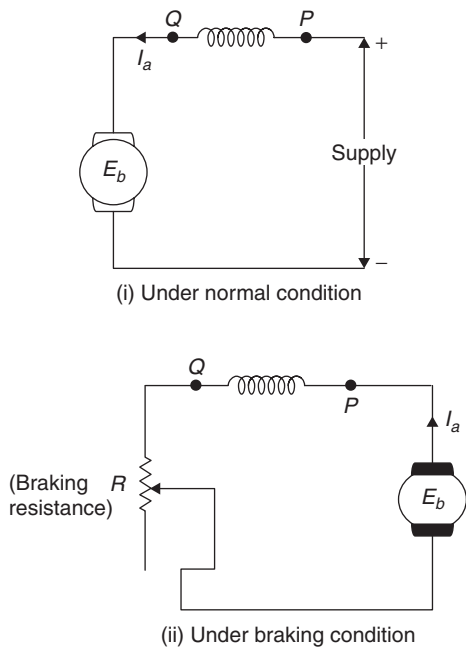
DC Shunt Motor



$$\text{Braking torque } T_B = \frac{1}{2\pi} \phi Z I_a \left(\frac{P}{A} \right) \text{ Nm}$$

DC Series Motor

In this motor, the field connection also reversed during braking to make the current flowing through the field winding in the same direction. The connection of the motor under normal condition and braking condition is as shown below.



Plugging (or) Counter Current Braking

In these method, the armature terminals is reversed without changing the supply terminals and a variable resistance is connected in series with the armature terminals to limit the armature current. The circuit connection of DC motors is as shown below.

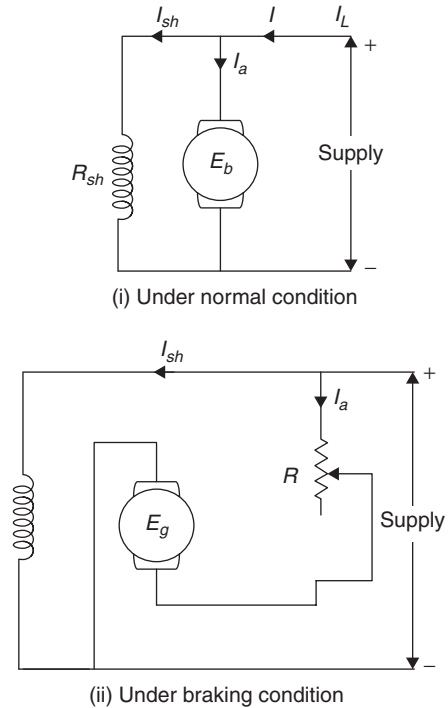


Figure 3 DC Shunt motor

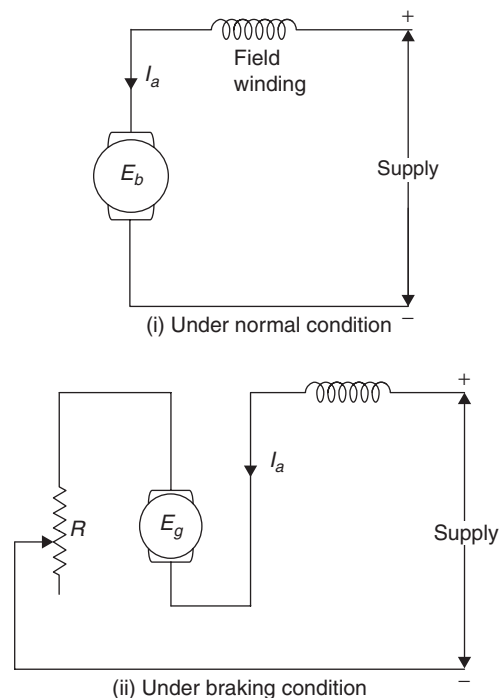


Figure 4 DC Series motor

Regenerative Braking

Regenerative braking takes place when E_b becomes greater than supply voltage. Regenerative braking can be easily applied to DC shunt motors without changing any circuit connections. But In DC series motor it is not possible without changing circuit connection because back emf more than the supply voltage and the current flowing through the field winding is in reverse direction. So this method is applied to DC series motor by special arrangements only.

TESTING OF DC MACHINES

DC machines are to be tested for determining the efficiency and other performance indices.

There are three types of testing DC machines. They are

1. Direct testing
2. Indirect testing
3. Regenerative or Back-to-Back method.

Brake Test

It is a direct method and consists of applying a brake to a water-cooled pulley mounted on the motor shaft.

Swinburne's Test

It is a simple indirect test in which losses are measured separately and from their knowledge, efficiency at any desired load can be predetermined. However this test is applicable to those machines which have practically flux, i.e., shunt and compound machines.

Advantages

1. It is simple and economical because power required to test a large machine is small, i.e., only no-load input power.
2. The efficiency can be predetermined at any load because constant losses are known.

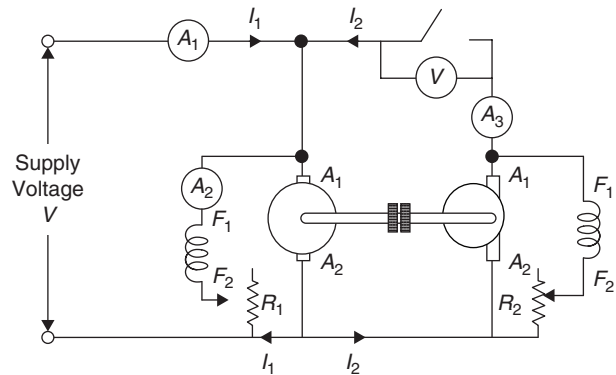
Disadvantages

1. No account is taken of the change in iron losses from no load to full load. At full-load due to armature, reaction flux is distorted which increases the iron losses in some cases as much as 50%.
2. As the test is on no load, it is impossible to know the effects of the temperature rise and commutation at full load.

Regenerative or Hopkinson's Test (Back-to-Back test)

It is performed on a pair of identical shunt machines, the two machines are mechanically coupled and field excitations are so adjusted that one of them runs as a motor and the other as a generator. By this method full-load test can

be carried out, without wasting their outputs and hence the name regenerative test. The mechanical output of motor drives the generator and the electrical output of generator is used in supplying the greater part of input to the motor. The power supplied in this test is to meet the losses in two machines.



If η is efficiency of both motor and generator

$$\text{Input to motor} = V(I_1 + I_2)$$

$$\text{Output of motor} = \eta \cdot V(I_1 + I_2)$$

Output of generator = $\eta \cdot \eta V(I_1 + I_2) = \eta^2 V(I_1 + I_2)$ but output of generator is $I_2 \cdot V$

$$\therefore \eta^2 V(I_1 + I_2) = I_2 \cdot V$$

$$\eta = \sqrt{\frac{I_2}{I_1 + I_2}}$$

Merits of Hopkinson's Test

1. Power required for the test is small as compared to the full-load powers of the two machines.
2. As machines are being tested under full load conditions, the temperature rise and the commutation qualities of the machines can be observed.
3. Because of full-load conditions, any change in iron loss due to flux distortion at full load, is being taken into account.

*The only disadvantage is with regard to the availability of two identical machines.

Retardation or Running Down Test

This method is applicable to shunt motors and generators and is used for finding stray losses. Then knowing the armature and shunt Cu losses at a given load current, efficiency can be calculated.

The machine under test is speeded up slightly beyond its normal speed and then supply is cut off from the armature while keeping the field excited.

Consequently the armature slows down and its kinetic energy is used to meet the rotational losses if friction, windage and iron losses.

Field Test for Series Motor

This test is applicable to two similar series motors. Series motors mainly used for traction work are easily available in pairs. The two machines are coupled mechanically. One machine runs normally as a motor and drives generator whose output is wasted in a variable load R .

Iron and frictional losses of two machines are made equal

1. By joining the series field winding of the generator in the motor armature circuit so that both machines are equally excited and
2. By running them at equal speed.

Field test is not a regenerative method although the two machines are mechanically coupled, because the generator output is wasted instead of being fed back into the motor as in Hopkinson's test.

EXERCISES

Practice Problems I

Directions for questions 1 to 29: Select the correct alternative from the given choices.

1. DC motor gives maximum power when
(A) $V = 2E_b$ (B) $V = E_b/2$
(C) $V = E_b/\sqrt{2}$ (D) $V = E_b$
2. The full-load speed regulation is 8% when the speed at full load is 800 rpm. The no-load speed is _____ rpm.
(A) 832 (B) 864
(C) 736 (D) 768
3. A 4-pole generator with 24 conductors has a two-layer lap winding. The pole pitch is _____.
(A) 24 (B) 12
(C) 6 (D) 4
4. A DC motor develops a torque of 120 Nm at 20 rps. At 30 rps, it will develop a torque of _____ Nm.
(A) 160 (B) 120
(C) 80 (D) 40
5. A DC motor connected to a 230 V supply has an armature resistance of $0.15\ \Omega$. The value of armature current when the back emf $E_b = 200\text{ V}$ is _____ A.
(A) 50 (B) 100
(C) 400 (D) 200
6. A 230 V shunt motor develops a torque of 48 Nm at an armature current of 12 A. The torque produced when the armature current is 15 A, is
(A) 48 Nm (B) 36 Nm
(C) 60 Nm (D) 72 Nm
7. An 8-pole, lap-wound armature running at 350 rpm is required to generate 260 V. The useful flux/pole is 0.05 Wb. If the armature has 120 slots, the no. of conductors/slot is _____.
(A) 6 (B) 8
(C) 10 (D) 4
8. The value of torque produced by armature of 4-pole motor having 774 conductors, two paths in parallel, 24 mWb flux per pole, when the total armature current is 50 A, is _____.
(A) 350 Nm (B) 373 Nm
(C) 295.35 Nm (D) 420 Nm

9. A 440 V shunt motor has an armature resistance of $0.8\ \Omega$. The back emf when giving an output of 7.46 kW at 80% efficiency is _____ V.
(A) 440 (B) 400
(C) 220 (D) 425

Common Data for Question 10:

A 230 V DC shunt motor takes a current of 40 A and runs at 1100 rpm. If armature and shunt field resistances are $0.25\ \Omega$ and $230\ \Omega$ respectively.

10. The torque developed by the motor is _____.
(A) 78.9 Nm (B) 74.66 Nm
(C) 98.7 Nm (D) 87.9 Nm

Common Data for Questions 11 and 12:

The counter emf of a shunt motor is 227 V, the field resistance is $160\ \Omega$ and the field current is 1.5 A.

11. If the line current is 39.5 A, the armature resistance is _____.
(A) $0.342\ \Omega$ (B) $0.243\ \Omega$
(C) $0.423\ \Omega$ (D) $0.432\ \Omega$
12. The starting current of the motor is _____.
(A) 250 A (B) 107.57 A
(C) 255.5 A (D) 701.75 A

Common Data for Question 13:

A 250 V DC shunt motor takes a total current of 20 A. Resistance of shunt field winding is $200\ \Omega$ and that of armature is $0.3\ \Omega$.

13. A 4-pole lap-wound generator is run at 900 rpm. The reactance voltage if self inductance is $153 \times 10^{-6}\text{ H}$, armature current is 108 A, no of segments = 55, width of brush = 1.74 segments is _____.
(A) 3.195 V (B) 3.915 V
(C) 5.391 V (D) 5.915 V

Common Data for Questions 14 and 15:

A 450 V series motor runs at 500 rpm taking a current of 40 A. Total resistance of the armature and field circuits is $0.8\ \Omega$, assuming flux is proportional to field current and that the load is reduced so that motor draws a current of 30 A.

14. The percentage change in torque is
(A) 38.75% (B) 41.75%
(C) 39.75% (D) 43.75%

15. Speed is _____ rpm
 (A) 627.94
 (B) 697.4
 (C) 679.42
 (D) 649.72
16. A cumulative compounded long shunt motor is driving a load at rated torque and rated speed. If the series field is shunted by a resistance of the series field, keeping the torque constant then
 (A) the armature current increases
 (B) the armature current decreases
 (C) the series field current remains the same
 (D) the motor fails to run and comes to a standstill
17. A differentially compounded DC motor with interpoles and with brushes on the neutral axis is to be driven as a generator in the direction with the same polarity of the terminal voltage. It will then
 (A) be a cumulatively compounded generator but interpole coil connections are to be reversed
 (B) be a differentially compounded generator but the interpole coil connections are to be reversed
 (C) be a cumulatively compounded generator without reversing the interpole coil connections
 (D) be a differentially compounded generator without reversing the interpole coil connections
18. In case of an armature controlled separately excited DC motor drive with closed loop speed control, an inner current loop is useful because it
 (A) limits the peak current of the motor to the permissible value.
 (B) helps in improving energy efficiency of the drive.
 (C) limits the speed of the motor to a safe value.
 (D) reduces the steady state speed error.
19. The compensating winding in a DC machine
 (A) is located on interpoles for improving commutation
 (B) is a separate winding in armature slots for compensation of the armature reaction
 (C) is located on pole shoes for avoiding the flashover at the commutator surface
 (D) is located on pole shoes to avoid the sparking at the brushes
20. The DC motor which can provide zero speed regulation at full load without any controller is
 (A) shunt
 (B) series
 (C) differential compound
 (D) cumulative compound

21. In relation to DC machines, match the following and choose the correct combination:

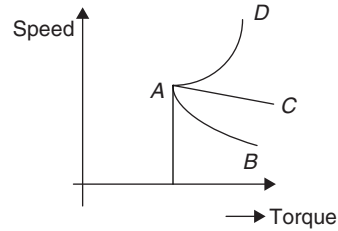
List-I (performance variables)	List-II (proportional to)
P. Armature emf (E)	1. Flux (ϕ), speed (ω), armature current (I_a)
Q. Developed power (P)	2. I_a only
R. Developed torque (T)	3. ϕ and I_a only
	4. ϕ and ω only
	5. I_a and ω only

- (A) P – 4, Q – 3, R – 1
 (B) P – 2, Q – 3, R – 4
 (C) P – 3, Q – 4, R – 1
 (D) P – 2, Q – 5, R – 4
22. A DC series motor driving an electric train faces a constant power load. It is running at rated speed and rated voltage. If the speed has to be brought down to 0.5 p.u., the supply voltage has to be approximately brought down to
 (A) 0.25 p.u. (B) 0.5 p.u.
 (C) 0.707 p.u. (D) 0.8 p.u.
23. An electric motor whose torque-speed characteristic in the form of a rectangular hyperbola will have
 (A) constant speed
 (B) constant torque
 (C) constant output power
 (D) none of the above
24. The armature resistance of permanent magnet DC motor is $0.5\ \Omega$ at no load the motor draws 1.2 A from a supply voltage of 25 V and runs at 1500 rpm. The efficiency of the motor while it is operating on load at 1500 rpm drawing a current of 3 A from the same source will be
 (A) 48.12%. (B) 54.96%.
 (C) 62.28%. (D) 74.56%.
25. A 220 V DC series motor takes 30 A when giving its rated output at 1500 rpm. Its resistance is $0.25\ \Omega$. The value of resistance which must be added to obtain rated torque at 1000 rpm is
 (A) $1.26\ \Omega$. (B) $2.36\ \Omega$.
 (C) $3.26\ \Omega$. (D) $4.26\ \Omega$.
26. A 4-pole lap-wound DC generator has a developed power of P W and voltage of E volt. Two adjacent brushes of the machine are removed as they are worn out. If the machine operates with the remaining brushes, the developed voltage and power that can be obtained from the machine are

(A) $\frac{E}{4}, \frac{P}{4}$ (B) $\frac{E}{2}, \frac{P}{2}$

(C) $E, \frac{P}{4}$ (D) $E, \frac{P}{2}$

27. An electric motor with 'constant output power' will have a torque speed characteristic in the form of a
 (A) straight line parallel to speed axis.
 (B) straight line through the origin.
 (C) straight line parallel to torque axis.
 (D) rectangular hyperbola.
28. A DC series motor fed from rated supply voltage is overloaded and its magnetic circuit is saturated. The torque-speed characteristic of this motor will be approximately represented by which curve of figure.

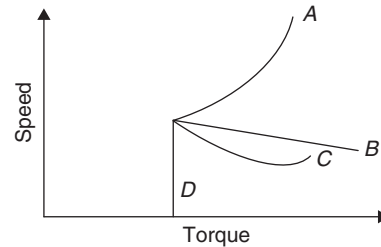


- (A) curve A (B) curve B
 (C) curve C (D) curve D
29. A permanent magnet DC commutator motor has a no-load speed of 6000 rpm when connected to 120 V DC supply. The armature resistance is 2.5Ω and other losses may be neglected. Supply voltage of 60 V developing a torque of 0.5 Nm, is
 (A) 2326 rpm (B) 2673 rpm
 (C) 2836 rpm (D) 5346 rpm

Practice Problems 2

Directions for questions 1 to 50: Select the correct alternative from the given choices.

- The core losses of a certain magnetic material, operated at constant flux density are 2200 and 3500 W at frequencies 50 Hz and 75 Hz respectively. Determine the hysteresis and eddy current loss at a frequency of 100 Hz
 (A) 1067 W, 3867 W
 (B) 3867 W, 1067 W
 (C) 2167 W, 1867 W
 (D) 1867 W, 2167 W
- A 4-pole lap-wound DC armature with 600 conductors draws 60 A from the mains. Its armature reaction per pole is
 (A) 775 AT peak sinusoidal
 (B) 625 AT peak triangular
 (C) 500 AT peak triangular
 (D) 500 AT peak sinusoidal
- In a DC motor torque developed in (Nm) is
 (A) $\frac{\omega_n}{E_b I_a}$ in the direction of ω_n (speed in rad/s)
 (B) $\frac{\omega_n}{E_b I_a}$ opposite to the direction of ω_n
 (C) $\frac{E_b I_a}{\omega_n}$ in the direction of ω_n
 (D) $\frac{E_b I_a}{\omega_n}$ opposite to the direction of ω_n
- A DC series motor is overloaded and its magnetic circuit is saturated. The torque speed characteristic of this motor will be



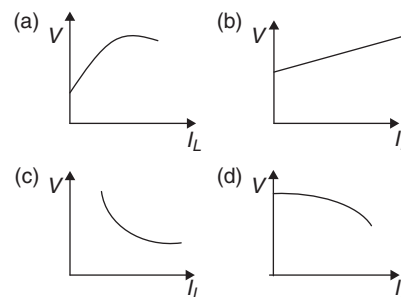
- (A) A (B) B
 (C) C (D) D
- A 6-pole DC generator with 36 coils has a two layer lap winding. What can be the pole pitch?
 (A) 6 (B) 12
 (C) 24 (D) 48
 - A separately excited DC generator has ratings of 240 V, 2500 rpm and 10 A. $R_a = 10 \Omega$. Armature and field winding is excited with rated DC voltage. If armature draws 5 A, torque developed is
 (A) 1.63 Nm (B) 3.63 Nm
 (C) 2.225 Nm (D) 0.25 Nm
 - A 240 V DC series motor, running at 1500 rpm takes 20 A when giving its rated output. Total series resistance = 0.5Ω . To obtain rated torque at 1000 rpm, value of resistance that must be added is
 (A) 3.7Ω (B) 5.7Ω
 (C) 2.2Ω (D) 3.2Ω
 - A permanent magnet DC motor when connected to a 100 V DC supply runs at a no load speed of 4000 rpm. $R_a = 2 \Omega$ and other losses may be neglected. When supplied with a voltage of 50 V and developing a torque of 0.5 Nm speed of motor is
 (A) 2000 rpm (B) 1832 rpm
 (C) 2836 rpm (D) 4346 rpm

9. A cumulatively compounded DC motor runs at 1500 rpm at no load. At full load the combined drop in the armature and field is 8% and the flux increases by 12%. Neglecting magnetic saturation, full-load speed is equal to
 (A) 1632 rpm (B) 1432 rpm
 (C) 1232 rpm (D) 1032 rpm
10. The prime mover of an over compounded DC generator, supplying power to infinite bus is cut off. Which among the following is true?
 (A) Machine stops running
 (B) Machine runs as cumulatively compounded motor in reverse direction
 (C) Machine runs in reverse direction as a differentially compounded motor
 (D) Machine runs in same direction as a differentially compounded motor
11. The armature resistance of a 4-pole lap connected DC generator with 480 conductors is $0.08\ \Omega$. The conductors are reconnected to form a wave winding, other things remaining the same. The new value of armature resistance will be
 (A) $0.02\ \Omega$ (B) $0.08\ \Omega$
 (C) $0.64\ \Omega$ (D) $0.32\ \Omega$
12. A DC series motor with 2-poles runs at 500 rpm with its two field coils connected in series. Now if the field coils are connected in parallel. The new speed will be (Assume magnetic circuit is unsaturated and torque is constant)
 (A) 250 rpm (B) 500 rpm
 (C) $250\sqrt{2}$ rpm (D) 1000 rpm
13. A DC generator on load has its brushes on geometric neutral axis. The magnetic neutral axis gets shifted in the direction of rotation. If a 90° lead is given to the brushes,
 (A) The magnetic neutral axis coincides with geometric neutral axis
 (B) Magnetic neutral axis is shifted forward by 90°
 (C) Magnetic neutral axis is shifted forward by more than 90°
 (D) Magnetic neutral axis is shifted forward by less than 90°
14. A 230 V DC series motor has $R_a = 0.4\ \Omega$ and $R_f = 0.2\ \Omega$. It draws a line current of 20 A and runs at a speed of 1500 rpm. Assuming that flux at 10 A line current is 65% of flux at 20 A, what is the speed of the motor at a line current of 10 A and 230 V?
 (A) 1913 rpm (B) 2371 rpm
 (C) 2331 rpm (D) 2531 rpm
15. The output of a 220 V DC shunt motor with an input of 15 kW is 13 kW. $R_f = 100\ \Omega$ and $R_a = 0.08\ \Omega$. The efficiency is maximum when armature copper losses are
 (A) 1670 W (B) 2000 W
 (C) 3000 W (D) 2670 W
16. A full pitch coil in a 4-pole machine has a mechanical angle span of
 (A) 30° (B) 45° (C) 90° (D) 180°
17. The armature circuit resistance and field circuit resistance of a 200 V DC shunt motor are $0.2\ \Omega$, and $200\ \Omega$ respectively. It takes 4 A at no load. The efficiency when motor takes an input current of 20 A is
 (A) 51.73% (B) 81.17%
 (C) 78.2% (D) 73.62%

Common Data Questions 18 and 19:

A 18 kW, 200 V DC shunt motor has an armature resistance of $0.3\ \Omega$ and field resistance of $100\ \Omega$. Speed at rated load is 1500 rpm and full-load efficiency is 75%

18. To limit starting current to 300% of rated current, starting resistance is
 (A) $0.12\ \Omega$ (B) $0.26\ \Omega$
 (C) $0.46\ \Omega$ (D) $1\ \Omega$
19. If the starting resistance in Q18 remains as it is on full load, the speed at full load will then be
 (A) 1134 rpm (B) 1434 rpm
 (C) 1500 rpm (D) 1534 rpm
20. A 4-pole, 230 V, 100 kW DC shunt generator has 600 lap-wound conductors. Two-degree brush lead is given from geometric neutral. The cross and demagnetizing turns per pole will respectively be
 (A) 7790 A and 362 A
 (B) 362 A and 7790 A
 (C) 6680 A and 532 A
 (D) 532 A and 6650 A
21. The armature of a 6-pole lap-wound DC machine has 600 conductors. Ratio of pole arc to pole pitch is 0.7. The compensating winding carries full armature current. The no. of conductors on each pole of this compensating winding will be
 (A) 6 (B) 12 (C) 24 (D) 30
22. At maximum power output, the efficiency of machine is
 (A) 0% (B) 75% (C) 50% (D) 100%
23. The terminal voltage vs. load current characteristics of a DC shunt generator is depicted by which among the following curves



24. In a DC generator, polarity of interpole is
 (A) always south pole
 (B) always north pole
 (C) same as the polarity of main pole that follows in the direction of rotation
 (D) same as the polarity of main pole that precedes in the direction of rotation
25. The external characteristics of a DC generator is
 (A) Magnetisation characteristics + Armature reaction
 (B) Magnetisation characteristics – Armature reaction
 (C) Magnetisation characteristics – Armature reaction – ohmic drop
 (D) Magnetisation characteristics + Armature reaction – ohmic drop
26. Lap-wound DC generators use equalizer rings to
 (A) avoid unequal distribution of current at brushes giving sparkless commutation
 (B) to provide mechanical balance
 (C) to counter armature reaction by producing equalizing voltages in widening
 (D) to reduce the load as the bearings
27. The torque developed by a DC series motor at 8 A is 25 Nm. If the current is increased to 16 A, the torque developed is
 (A) 50 Nm (B) 25 Nm
 (C) 100 Nm (D) 200 Nm
28. Armature reaction in a DC machine is
 (A) in same direction as main poles
 (B) to make an angle with main pole axis depending on load current
 (C) in direct opposition to main poles
 (D) to make an angle of 90° with main pole axis
29. Open slots are used in DC machine armature as
 (A) it reduces coil reactance and aids in commutation
 (B) it increases induced emf per coil
 (C) it is easy to place the winding in the slots
 (D) it reduces armature voltage drop
- Common Data Questions 30 and 31:**
 A separately excited DC motor runs at 1200 rpm at no load and 250 V applied to armature. The field is given rated excitation. When it delivers a load of 8 Nm, speed is 1100 rpm. Rotational losses and armature reaction can be neglected.
30. Armature resistance of motor is
 (A) 3 Ω (B) 5.2 Ω
 (C) 4.4 Ω (D) 7.7 Ω
31. For motor to deliver a torque of 4 Nm at 1100 rpm armature voltage to be applied is
 (A) 200 V (B) 239.6 V
 (C) 250 V (D) 225.2 V
32. A motor has zero speed regulation at full load without the use of any controller. The motor is
 (A) DC series motor
 (B) DC shunt motor
 (C) DC cumulatively compounded motor
 (D) DC differentially compounded motor
33. A DC machine supplies 25 A at 250 V as a generator $R_a = 0.2 \Omega$. The machine operated as a motor with 10% increased flux but terminal voltage and current remaining same, then ratio of motor speed to generator speed is
 (A) 0.87 (B) 0.95
 (C) 0.96 (D) 1.06
34. With regard to a DC machine, the function of compensating winding and interpole is
 (A) neutralize armature reaction and produce residual flux, respectively
 (B) neutralize armature reaction and improve commutation, respectively
 (C) improve commutation and produce residual flux, respectively
 (D) improve commutation and neutralize armature reaction, respectively
35. In relation to DC machines, match the following and choose correct combination
- | | |
|-----------------------------|---|
| P. Developed Power (P) | 1. Flux (ϕ), speed (ω) and armature current (I_a) |
| Q. Developed Torque (T) | 2. I_a and ω only |
| R. Armature EMF (E) | 3. ϕ and ω only |
| | 4. ϕ and I_a only |
| | 5. I_a only |
- (A) P – 1, Q – 4, R – 2
 (B) P – 2, Q – 5, R – 3
 (C) P – 1, Q – 4, R – 3
 (D) P – 2, Q – 5, R – 1
36. A 30 kW DC shunt motor is drawing rated current at a given speed. When driven
 (1) at half-rated speed by armature voltage control
 (2) 2 times rated speed by field control Output power delivered by the motor are approximately
 (A) 15 kW in (1) 45 kW in (2)
 (B) 15 kW in (1) 30 kW in (2)
 (C) 30 kW in (1) 45 kW in (2)
 (D) 30 kW in (1) 30 kW in (2)
37. A DC series motor driving an electric train faces a constant power load at rated speed and rated voltage. For the speed to be 0.5 p.u., the supply voltage should be
 (A) 0.8 p.u. (B) 0.5 p.u.
 (C) 0.7 p.u. (D) 0.6 p.u.

38. Match the following in Group I and Group II with regard to a DC motor

Group I	Group II
P. Field Control	1. Below base speed
Q. Armature Control	2. Above base speed
	3. Above base torque
	4. Below base torque
(A) P – 1, Q – 4	(B) P – 2, Q – 3
(C) P – 1, Q – 3	(D) P – 2, Q – 1

Common Data Questions 39 and 40:

A 7.5 kW, 200 V, 50 A, 1500 rpm DC series motor has following full-load losses expressed in percentage of motor input

Losses in armature resistance and brush = 2.6%

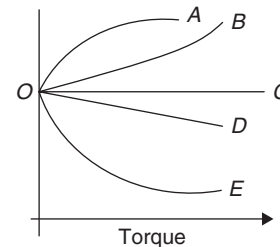
Losses in field circuit = 2.4 %

Friction and windage losses (Rotational loss) = 2.2%

Assume rotational loss to be constant.

39. When motor draws one half rated current at rated voltage Speed in rpm is
 (A) 2057 rpm
 (B) 2889 rpm
 (C) 3079 rpm
 (D) 1500 rpm
40. Shaft power output will be
 (A) 2925 W
 (B) 3925 W
 (C) 4925 W
 (D) 5925 W
41. In an electromechanical energy conversion system, the developed electromagnetic force/torque at in a direction that tends to
 (A) decrease stored energy at constant flux
 (B) increase stored energy at constant mmf
 (C) decrease stored energy at constant mmf
 (D) decrease co-energy at constant mmf
42. The hysteresis loop of a magnetic material has an area of 10 cm². The scales are marked as 1 cm = 4AT and 1 cm = 100 mWb Total hysteresis loss at 50 Hz is
 (A) 50 W
 (B) 100 W
 (C) 200 W
 (D) 400 W
43. A DC shunt motor is started with an open circuited field. Then
 (A) The motor picks up fast and acquires full speed while drawing large current
 (B) Motor does not pick up speed but draws a large current
 (C) The motor does not pick up speed but draws only a small current
 (D) The motor picks up speed fast and acquires full speed while drawing only a small current

44. In a DC series motor with linear magnetization and negligible armature resistance, the motor speed is
 (A) inversely proportional to T
 (B) directly proportional to T
 (C) directly proportional to $\sqrt{T}$
 (D) inversely proportional to $\sqrt{T}$
45. A DC shunt generator coupled to a prime move of constant speed and fixed field resistance is shorted across the terminals. Then short-circuit current is
 (A) several times the rated current
 (B) equal to full-load rated current
 (C) less than maximum rated current
 (D) None of the above
46. The speed torque characteristics of a cumulatively compounded DC motor is



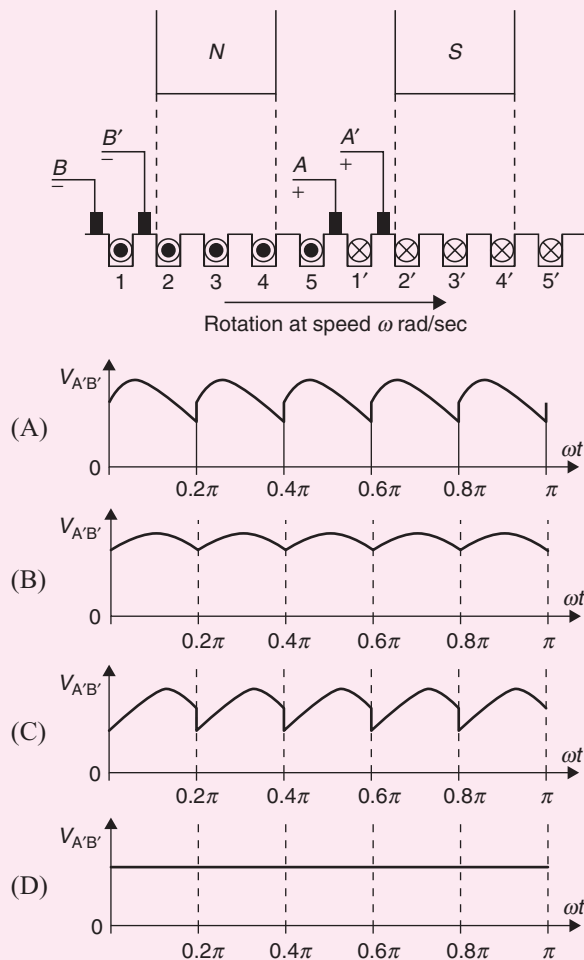
- (A) Curve OA
 (B) Curve OB
 (C) Curve OD
 (D) Curve OE
47. The supply terminals of a DC shunt motor are reversed, then
 (A) it will run as a DC generator
 (B) it will continue to run in same direction
 (C) it will reverse its direction
 (D) it will stop
48. In Ward Leonard system of speed control of DC motor, the lower speed limit is restricted by
 (A) core losses in the motor
 (B) residual magnetism of generator
 (C) mechanical losses of motor and generator together
 (D) All the above
49. A DC machine used as a generator has an efficiency of 80% when output voltage and currents are 200 V and 8 A. If machine is used as a motor and takes 8 A from 200 V supply then efficiency will be
 (A) 80%
 (B) Less than 80%
 (C) More than 80%
 (D) None of these
50. A DC shunt motor is running at 1500 rpm with rated voltage applied to its terminals. Neglecting the saturation of magnetic circuit, if applied voltage is reduced to $\frac{2}{3}$ of original value, motor will run at
 (A) 1005 rpm
 (B) 750 rpm
 (C) 1500 rpm
 (D) 2250 rpm

PREVIOUS YEARS' QUESTIONS

Data for Question 1:

A 240 V, DC shunt motor draws 15 A while supplying the rated load at a speed of 80 rad/s. The armature resistance is 0.5Ω and the field winding resistance is 80Ω

- The external resistance to be added in the armature circuit to limit the armature current to 125% of its rated value is [2008]
(A) 31.1Ω (B) 31.9Ω
(C) 15.1Ω (D) 15.9Ω
- Figure shows the extended view of a 2-pole DC machine with 10 armature conductors. Normal brush positions are shown by A and B , placed at the interpolar axis. If the brushes are now shifted, in the direction of rotation, to A' and B' as shown, the voltage waveform $V_{A'B'}$ will resemble [2009]



- The armature resistance of the motor is, [2010]
(A) 2Ω (B) 3.4Ω
(C) 4.4Ω (D) 7.7Ω
- For the motor to deliver a torque of 2.5 Nm at 1400 rpm, the armature voltage to applied is [2010]
(A) 125.5 V (B) 193.3 V
(C) 200 V (D) 241.7 V

- A 4-point starter is used to start and control the speed of a [2011]
(A) DC shunt motor with armature resistance control
(B) DC shunt motor with field weakening control
(C) DC series motor
(D) DC compound motor
- A 220 V, DC shunt motor is operating at a speed of 1440 rpm. The armature resistance is 1.0Ω and armature current is 10 A. If the excitation of the machine is reduced by 10%, the extra resistance to be put in the armature circuit to maintain the same speed and torque will be [2011]
(A) 1.79Ω (B) 2.1Ω
(C) 3.1Ω (D) 18.9Ω
- A 220 V, 15 kW, 1000 rpm shunt motor with armature resistance of 0.25Ω , has a rated line current of 68 A and a rated field current of 2.2 A. The change in field flux required to obtain a speed of 1600 rpm while drawing a line current of 52.8 A and a field of 1.8 A is [2012]
(A) 18.18 % increase (B) 18.18 % decrease
(C) 36.36 % increase (D) 36.36 % decrease
- A 15 kW, 230 V DC shunt motor has armature circuit resistance of 0.4Ω and field circuit resistance of 230Ω . At no load and rated voltage, the motor runs at 1400 rpm and the line current drawn by the motor is 5 A. At full load, the motor draws a line current of 70 A. Neglect armature reaction. The full load speed of the motor in rpm is _____. [2014]
- A 250 V DC shunt machine has armature circuit resistance of 0.6Ω and field circuit resistance of 125Ω . The machine is connected to 250 V supply mains. The motor is operated as a generator and then as a motor separately. The line current of the machine in both the cases is 50 A. The ratio of the speed as a generator to the speed as a motor is _____. [2014]
- The no-load speed of a 230 V separately excited DC motor is 1400 rpm. The armature resistance drop and the brush drop are neglected. The field current is kept constant at rated value. The torque of the motor in Nm for an armature current of 8 A is _____. [2014]
- A separately excited 300 V DC shunt motor under no load runs at 900 rpm drawing an armature current of 2 A. The armature resistance is 0.5Ω and leakage inductance is 0.01 H. When loaded, the armature current is 15 A. then the speed in rpm is _____. [2014]
- A separately excited DC generator has an armature resistance of 0.1Ω and negligible armature inductance. At rated field current and rated rotor speed, its

open-circuited voltage is 200 V. When this generator is operated at half the rated speed, with half the rated field current, an uncharged 1000 μF capacitor is suddenly connected across the armature terminals. Assume that the speed remains unchanged during the transient. At the time (in microsecond) after the capacitor is connected will the voltage across it reach 25 V? [2015]

- (A) 62.25 (B) 69.3
(C) 73.25 (D) 77.3

13. A separately excited DC motor runs at 1000 rpm on no load when its armature terminals are connected to a 200 V DC source and the rated voltage is applied to the field winding. The armature resistance of this motor is 1 Ω . The no-load armature current is negligible. With the motor developing its full load torque, the armature voltage is set so that the rotor speed is 500 rpm. When the load torque is reduced to 50% of the full load value under the same armature voltage conditions, the speed rises to 520 rpm. Neglecting the rotational losses, the full load armature current (in Ampere) is _____. [2015]

14. A DC motor has the following specifications: 10 hp, 37.5 A, 230 V; flux/pole = 0.01 Wb, number of poles = 4, number of conductors = 666, number of parallel paths = 2. Armature resistance = 0.267 Ω . The armature reaction is negligible and rotational losses are 600 W. The motor operates from a 230 V DC supply. If the motor runs at 1000 rpm, the output torque produced (in Nm) is _____. [2015]

15. A shunt-connected DC motor operates at its rated terminal voltage. Its no-load speed is 200 radian/second. At its torque of 500 Nm, its speed is 180 radian/second. The motor is used to directly drive a load whose load torque T_L depends on its rotational speed ω_r (in radian/second), such that $T_L = 2.78 \times \omega_r$. Neglecting rotational losses, the steady-state speed (in radian/second) of the motor, when it drives this load, is _____. [2015]

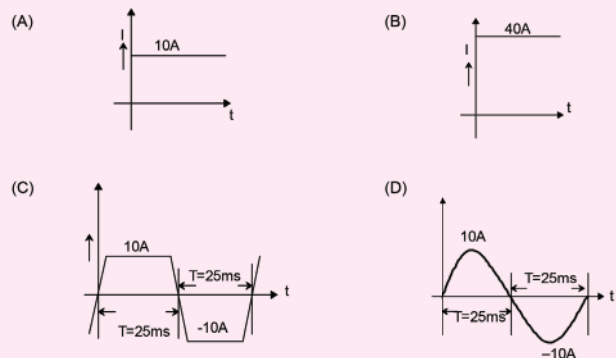
16. A 4-pole, separately excited, wave wound DC machine with negligible armature resistance is rated for 230 V and 5 kW at a speed of 1200 rpm. If the same armature coils are reconnected to form a lap winding, what is the rated voltage (in volts) and power (in kW)

respectively at 1200 rpm of the reconnected machine if the field circuit is left unchanged?

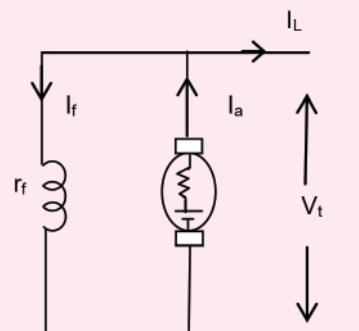
- (A) 230 and 5 (B) 115 and 5
(C) 115 and 2.5 (D) 230 and 2.5

17. With an armature voltage of 100 V and rated field winding voltage, the speed of a separately excited DC motor driving a fan is 1000 rpm, and its armature current is 10 A. The armature resistance is 1 Ω . The load torque of the fan load is proportional to the square of the rotor speed. Neglecting rotational losses, the value of the armature voltage (in Volt) which will reduce the rotor speed to 500 rpm is _____. [2015]

18. A 4-pole, lap-connected, separately excited dc motor is drawing a steady current of 40 A while running at 600 rpm. A good approximation for the wave shape of the current in an armature conductor of the motor is given by [2016]



19. A DC shunt generator delivers 45 A at a terminal voltage of 220 V. The armature and the shunt field resistances are 0.01 Ω and 44 Ω respectively. The stray losses are 375 W. The percentage efficiency of the DC generator is _____. [2016]



ANSWER KEYS**EXERCISES****Practice Problems 1**

1. A	2. B	3. C	4. C	5. D	6. C	7. B	8. C	9. C	10. B
11. A	12. D	13. B	14. D	15. C	16. A	17. C	18. A	19. C	20. C
21. A	22. C	23. C	24. C	25. B	26. C	27. D	28. C	29. B	

Practice Problems 2

1. B	2. A	3. C	4. B	5. A	6. B	7. D	8. B	9. C	10. D
11. D	12. D	13. C	14. B	15. A	16. C	17. C	18. B	19. A	20. A
21. B	22. C	23. D	24. C	25. D	26. A	27. C	28. D	29. A	30. B
31. B	32. D	33. A	34. B	35. C	36. B	37. C	38. D	39. C	40. A
41. A	42. C	43. B	44. D	45. C	46. D	47. B	48. B	49. A	50. C

Previous Years' Questions

1. A	2. A	3. B	4. B	5. B	6. A	7. D	8. 1240.63	9. 1.27	10. 12.5
11. 880	12. B	13. 8	14. 57.78	15. 177 to 183	16. B	17. 47.5 V	18. C	19. 86.84	

Chapter 4

Induction Motors

LEARNING OBJECTIVES

After reading this chapter, you will be able to understand:

- Principle of operation
- Rotor current and power factor
- Power stages in an induction motor
- Synchronous watt
- Rotor torque and breakdown torque
- Equivalent circuit of the rotor
- Starting of induction motors
- Double squirrel cage motor
- Speed control of induction motor
- Revolving field theory of single phase induction motor
- Capacitor split phase motors
- Shaded pole single-phase motor
- Repulsion start induction motor

INTRODUCTION

Three-phase induction motors (IMs) are the widely used AC motors due to their low cost, simple and rugged construction, high reliability, high efficiency. IMs have good speed regulation, less maintenance, simple starting arrangement, reasonable overload capacity. Conversion of electrical power into mechanical power takes place in the rotating part of an electric motor. In DC motors, the electric power is conducted directly to the armature (i.e., rotating part) through brushes and commutator. Hence in this sense, a DC motor can be called a conduction motor. However, in AC motors, the rotor does not receive electric power by conduction but by induction. This is why such motors are known as IMs. An IM can be treated as a rotating transformer. Unlike a transformer due to the presence of air gap, IM has more magnetizing currents, almost lagging 90° the applied voltage and therefore p.f. is low: The no-load current of IM is about 30% to 50% of full-load current, whereas in transformer no-load current is only 2% to 6% of full-load current.

Constructional Details

The IMs essentially consists of two parts:

- (a) A stationary part called stator
- (b) The revolving part, called rotor

Stator

The stator of an IM is made up of a number of stampings, which are slotted on the inner periphery to receive the windings. The stator carries a 3-phase winding which may be star or delta connected and is fed from a 3-phase supply. It is wound for a definite number of poles, the exact number of poles being determined by the requirements of speed.

The number poles P , produced in the rotating field is $P = 2n$, where n is the number of stator slots/pole/phase.

Rotor

Rotor is made up of thin laminations of the same material as stator. These laminations are slotted on the outer periphery. There are two types of rotors.

1. Squirrel-cage rotor: Motors employing this type of rotor are known as squirrel-cage induction motors (SCIMs)
2. Phase wound (or) wound rotor: The motors employing this type of rotor are variously known as 'phase-wound' motors or 'wound' motors or 'slip-ring' motors

Slip-ring Rotor

- Has slotted armature
- Provided with 3-phase double-layer distributed winding housed in the rotor slots

- The rotor windings are connected in star
- The open ends are connected to slip rings
- Brushes resting on the slip rings are further externally connected to a 3-phase star-connected rheostat. This makes possible
 - (a) to increase the starting torque and decrease the starting current and
 - (b) to control the speed of the motor
- A cage motor has a higher efficiency than a slip-ring IM
- A slip-ring IM has a high starting torque and a low starting current
- When 3-phase windings displaced in space by 120° , are fed by three-phase currents displaced in time by 120° , they produce a resultant magnetic flux, which rotates in space. This type of field is known as rotating magnetic field (R.M.F.)
- The resultant flux is of constant value $= \frac{3}{2} \phi_m$, i.e., 1.5 times the maximum values of the flux due to any phase
- The resultant flux rotates around the stator at synchronous speed given by

$$N_s = \frac{120f}{P}$$

- The direction of R.M.F. depends upon the phase sequence

PRINCIPLE OF OPERATION

- When the 3-phase stator windings, are fed by a 3-phase supply, then a magnetic flux of constant magnitude but rotating at synchronous speed, is set up. The flux passes through the air-gap, sweeps past the rotor surface and so cuts the rotor conductors which, as yet, are stationary. Due to the relative speed between the rotating flux and the rotor conductors, an emf is induced in rotor conductors. Since the rotor bars or conductors form a closed circuit, a current is established by the induced emf. The rotor currents generate rotor flux which in turn interacts with the stator flux. The interaction of stator and rotor fluxes produces a torque which makes the rotor to rotate in the direction of R.M.F.
- A 3-phase IM is self-starting. An IM runs at a speed slightly less than the synchronous speed. Hence it is known as an asynchronous motor. It cannot run at synchronous speed.

Slip

The difference between the synchronous speed N_s and the actual speed N of the rotor is known as slip. It is usually expressed as a percentage of the synchronous speed N_s .

$$\text{Slip } s = \frac{N_s - N}{N_s}$$

$N_s - N$ is called slip speed.

The slip speed expresses the speed of the rotor relative to the field.

Frequency of rotor current or emf

$$\text{Rotor frequency } f_2 = sf_1$$

where s = slip and f_1 = supply frequency

When rotor is at standstill, $N = 0$, and $s = 1.0$ and $f_2 = f_1$

- Air-gap flux is rotating synchronously, relative to the stator (or stationary space) but at slip speed relative to the rotor.

Solved Examples

Example 1: A voltmeter gives 120 oscillation per minute when connected to the rotor of an IM.

The stator frequency is 50 Hz. Calculate the slip of the motor?

Solution:

$$f' = sf$$

$$\text{The rotor circuit frequency } f' = \frac{120}{60}$$

$$= 2 \text{ Hz}$$

$$s = \frac{f'}{f} = \frac{2}{50} = 0.04$$

$$\% \text{ slip} = 4\%$$

Rotor Current and Power Factor

If R_2 and X_2 are the resistance and reactance at standstill per phase of rotor and E_2 be the standstill induced emf per phase

$$I_2 = \frac{E_2}{Z_2} = \frac{E_2}{\sqrt{R_2^2 + X_2^2}} \text{ at standstill}$$

$$\text{Rotor power factor} = \frac{R_2}{Z_2} = \frac{R_2}{\sqrt{R_2^2 + X_2^2}}$$

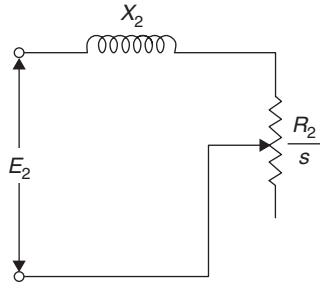
When motor running with slip s , then the rotor current/phase

$$I_2 = \frac{sE_2}{Z_2} = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}} \text{ or } \frac{E_2}{\sqrt{\left(\frac{R_2}{s}\right)^2 + X_2^2}}$$

Now the rotor power factor

$$\cos \phi_2 = \frac{R_2}{\sqrt{R_2^2 + (sX_2)^2}} = \frac{\frac{R_2}{s}}{\sqrt{\left(\frac{R_2}{s}\right)^2 + X_2^2}}$$

This equation gives the rotor equivalent circuit as shown below:



Relation Between Torque and Rotor Power Factor

In case of an IM, the torque is proportional to the product of flux per stator pole and the rotor current. However there is one more factor that has to be taken into account, i.e., the power factor of the rotor.

$$\therefore T_2 \propto \phi I_2 \cos \phi_2 \quad \text{or} \quad T = k \phi I_2 \cos \phi_2$$

where I_2 = Rotor current at standstill

ϕ_2 = Angle between rotor emf and rotor current

k = A constant

Denoting rotor emf at standstill by E_2 , we have that

$$E_2 \propto \phi$$

$$\therefore T \propto E_2 I_2 \cos \phi_2 \quad \text{or} \quad T = k_1 E_2 I_2 \cos \phi_2$$

where k_1 is another constant.

Example 2: The rotor of a 4-pole, 50-Hz slip-ring IM has a resistance of $0.20 \, \Omega$ per phase and runs at 1440 rpm at full load. Calculate the external resistance per phase which must be added to lower the speed to 1300 rpm, the torque being the same as before

Solution: The motor torque is given by

$$T = \frac{KsR_2}{R_2^2 + (sX_2)^2}$$

Since, X_2 is not given,

$$T = \frac{KsR_2}{R_2^2} = \frac{Ks}{R_2}$$

In the first case $T_1 = Ks_1 / 1 \, \Omega$

In the second case, $T_2 = Ks_2 / (R_2 + r)$,

where r is the external resistance per phase, added to the rotor circuit

Since $T_1 = T_2$

$$\therefore Ks_1 / R_2 = Ks_2 / (R_2 + r)$$

$$\text{Or} \quad (R_2 + r) / R_2 = s_2 / s_1$$

Now, $N_s = 120 \times 50 / 4 = 1500 \, \text{rpm}$.

$N_1 = 1440 \, \text{rpm}$, $N_2 = 1300 \, \text{rpm}$.

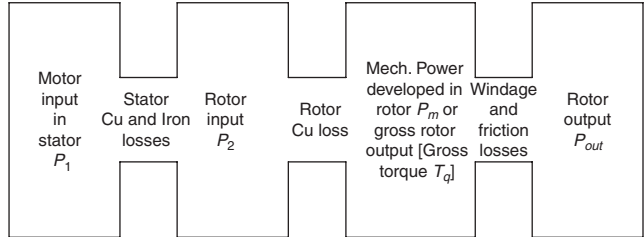
$$\therefore s_1 = \frac{1500 - 1440}{1500} = 0.04$$

$$s_2 = \frac{1500 - 1300}{1500} = 0.133$$

$$\therefore \frac{0.2 + r}{0.2} = \frac{0.133}{0.04}$$

$$\therefore r = 0.465 \, \Omega$$

POWER STAGES IN AN IM



Stator iron loss depends on the supply frequency and the flux density in the iron core. It is practically constant. The iron loss of the rotor is, however, negligible because frequency of rotor currents under normal running conditions is always small.

Torque Developed by an IM

An IM develops gross torque T_g due to gross rotor output P_m . Its value can be expressed either in terms of rotor input P_2 or rotor gross output P_m as given below:

$$T_g = \frac{P_2}{\omega_s} = \frac{P_2}{2\pi N_s}$$

$$\text{or} \quad T_g = \frac{P_m}{\omega} = \frac{P_m}{2\pi N} \quad (N \text{ and } N_s \text{ are in rps})$$

in terms of rotor output.

The shaft torque is due to output power P_{out} which is less than P_m because of rotor friction and windage losses.

$$\therefore T_{sh} = \frac{P_{out}}{\omega} = \frac{P_{out}}{2\pi N}$$

The difference between T_g and T_{sh} equals the torque lost due to friction and windage loss in the motor.

Some Important Relations

$$\frac{\text{Rotor Cu loss}}{\text{Rotor input}} = s$$

Rotor gross output = $(1 - s)$ Rotor input

$$\frac{\text{Rotor Cu loss}}{\text{Rotor gross output}} = \frac{s}{1 - s}$$

$$\text{Rotor efficiency, } \frac{P_m}{P_2} = \frac{N}{N_s}$$

If some power P_2 is delivered to a rotor, then a part sP_2 is lost in the rotor itself as copper loss and the remaining $(1-s)P_2$ appears as gross mechanical power P_m (including friction and windage losses)

$$\therefore P_2 : P_m :: I^2 R : 1 : (1-s) : s$$

IM Torque Equation

The gross torque T_g developed by an IM is given by

$$\begin{aligned} T_g &= \frac{P_2}{2\pi N_s} - N_s \text{ in rps} \\ &= \frac{60P_2}{2\pi N_s} = \frac{9.55P_2}{N_s} - N_s \text{ in rpm} \end{aligned}$$

Now P_2 = rotor Cu loss/s

$$= 3I_2^2 R_2 / s$$

$$I_2 = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}} = \frac{sKE_1}{\sqrt{R_2^2 + (sX_2)^2}}$$

where K is rotor/stator turn ratio per phase

$$\therefore P_2 = 3 \times \frac{s^2 E_2^2 R_2}{R_2^2 + (sX_2)^2} \times \frac{1}{s}$$

$$\therefore T_g = \frac{P_2}{2\pi N_s} = \frac{3}{2\pi N_s} \times \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2}$$

Synchronous Watt (T_{sw})

Synchronous watt is that torque which, at the synchronous speed of the machine under consideration, would develop a power of 1 watt.

$$\text{Rotor input} = T_{sw} \times 2\pi N_s$$

$$\begin{aligned} \therefore T_{sw} &= \frac{\text{Rotor input, } P_2}{2\pi \times N_s} \\ &= \frac{1}{\omega_s} \cdot \frac{N_s}{N} \cdot P_g = \frac{1}{\omega_s} \cdot \frac{N_s}{N} \cdot P_m \end{aligned}$$

Synchronous wattage of an IM equals the power transferred across the air-gap to the rotor.

$\therefore$ Torque in synchronous watt

$$\begin{aligned} &= \text{Rotor input} = \frac{\text{Rotor Cu loss}}{s} \\ &= \frac{\text{Gross output power, } P_m}{(1-s)} \end{aligned}$$

Starting torque, T_{st} :

$$T_{st} = \frac{3}{2\pi N_s} \cdot \frac{E_2^2 R_2}{R_2^2 + X_2^2}$$

Example 3: A 3- ϕ , 50 Hz IM has a full load speed of 1440 rpm. What is slip and rotor frequency of the IM?

- (A) 6%, 4 Hz (B) 10%, 46 Hz
(C) 8%, 6 Hz (D) 4%, 2 Hz

Solution: (D)

$$N_s = \frac{120 \times 50}{4} = 1500 \text{ rpm}$$

$$N = N_s(1 - S) \Rightarrow S = 4\%$$

$$\Rightarrow f_r = S \times f_s \Rightarrow f_r = 2 \text{ Hz}$$

Example 4: An 8-pole, 50 Hz, 3- ϕ slip-ring IM has effective rotor resistance of 0.8 Ω /ph. Stalling speed is 650 rpm. The resistance required in the rotor phase to obtain the maximum torque at starting?

- (A) 0.42 Ω (B) 0.52 Ω
(C) 0.48 Ω (D) 0.58 Ω

Solution: (B)

$$N_s = 750 \text{ rpm}$$

$$S_b = 13.33\%$$

$$\Rightarrow S_b = \frac{R^2}{X^2} = 0.6 \Omega$$

$$\frac{T_{st}}{T_{\max}} = \frac{2a}{1+a^2}, T_{st} = T_{\max}$$

Here $a = 1$

$$\Rightarrow a = \frac{R_2 + r}{X_2}$$

$$\Rightarrow r = .52 \Omega$$

Example 5: The rotor power output of 3- ϕ IM is 15 kW and the corresponding slip is 4%. The rotor ohmic loss will be

- (A) 600 W (B) 625 W
(C) 650 W (D) 700 W

Solution: (B)

$$\text{Rotor loss} = P_g \times \left(\frac{S}{1-S} \right)$$

$$= 15 \times \frac{0.04}{1-0.04} = 625 \text{ W}$$

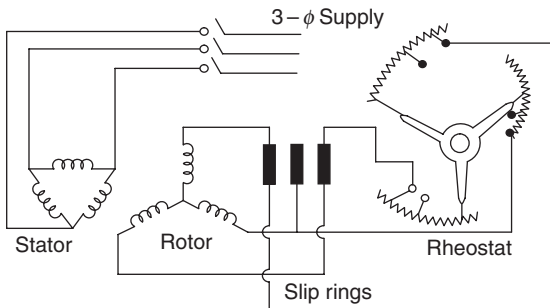
Starting Torque of a SCIM

The resistance of a SCIM is fixed and small compared to its reactance which is very large at start because at standstill, frequency of the rotor currents equals the supply frequency. Hence, the starting current I_2 of the rotor though very large in magnitude, lags by an angle nearly equal to 90° , which results in poor starting torque. It is roughly 1.5 times the full-load torque, although the starting current is 5 to 7 times the full-load current. Hence SCIMs are not useful where the motor has to start against heavy loads.

Starting Torque of a Slip-ring Motor

The starting torque of such a motor is increased by improving its power factor by adding external resistance in the rotor circuit from the star-connected rheostat, the rheostat being progressively cut out as the motor gathers speed. Addition of external resistance, however, increases the rotor impedance and so reduces the rotor current. At first, the effect of improved power factor predominates the current-decreasing effect of impedance. Hence, starting torque is increased. But after a certain point, the effect of increased impedance predominates the effect of improved power factor so the torque starts decreasing.

Starting Torque of a Slip-ring Motor Starting Arrangement



Condition for Maximum Starting Torque

Starting torque is maximum when rotor resistance equals rotor reactance.

$$\text{i.e., } R_2 = X_2$$

Effect of change in supply voltage on starting torque

$$T_{st} = \frac{K_1 E_2^2 R_2}{R_2^2 + X_2^2}$$

Now $E_2 \propto$ supply voltage V

$$\therefore T_{st} = \frac{KV^2 R_2}{R_2^2 + X_2^2},$$

where k is yet another constant.

$$\therefore T_{st} \propto V^2$$

The torque is very sensitive to any changes in supply voltage
Torque under running conditions:

$$T = \frac{3}{2\pi N_s} \cdot \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2}$$

Condition for Maximum Torque (or Breakdown Torque) under Running Conditions

Breakdown torque of a motor is the maximum torque; a motor will develop under increasing load conduction without an abrupt drop in speed and power. It is also sometimes known as pullout torque.

The torque under running conditions is maximum at that value of slip ' s ', which makes rotor reactance per phase equal to rotor resistance per phase.

$$\text{Slip corresponding to maximum torque is 's' = } \frac{R_2}{X_2}$$

$$T_{\max} = \frac{3}{2\pi N_s} \cdot \frac{E_2^2}{2X_2} \text{ Nm}$$

1. The maximum torque is independent of rotor resistance.
2. The speed or slip at which maximum torque occurs is determined by the rotor resistance. Torque becomes maximum when rotor reactance equals its resistance. Hence, by varying rotor resistance (possible only with slip-ring motors) maximum torque can be made to occur at any desired slip.
3. Maximum torque varies inversely as standstill reactance. Hence, it should be kept as small as possible.
4. Maximum torque varies directly as the square of the applied voltage.
5. For obtaining maximum torque at starting ($s = 1$), rotor resistance must be equal to rotor reactance.

Rotor Torque and Breakdown Torque (Pull Out Torque)

The rotor torque T at any slip s can be expressed in terms of the maximum (or breakdown) torque T_b as

$$T = T_b \left[\frac{2}{\left(\frac{s_b}{s}\right) + \left(\frac{s}{s_b}\right)} \right]$$

where T_b = Breakdown or maximum torque
 s_b = Breakdown or pull-out slip

Torque/speed or Torque/Slip Curves

$$T \propto \phi I_2 \cos \phi_2 \text{ or}$$

$$T = K\phi \frac{sE_2 R_2}{R_2^2 + (sX_2)^2}$$

At normal speeds, close to synchronous speed, the term sX_2 is small and hence negligible compared to R_2 .

$$\therefore T \propto \frac{s}{R_2} \text{ or}$$

$T \propto s$ if R_2 is constant. Hence, for low values of slip (or close to N_s) the torque/slip (torque/speed) curve is approximately a straight line. As slip increases (i.e., motor speed falls due to increasing load on the motor), the torque also increases and becomes maximum when $s = \frac{R_2}{X_2}$.

As the slip further increases (i.e., motor speed falls) with further increase in motor load, then R_2 becomes negligible as compared to sX_2 .

Therefore, for large values of slip

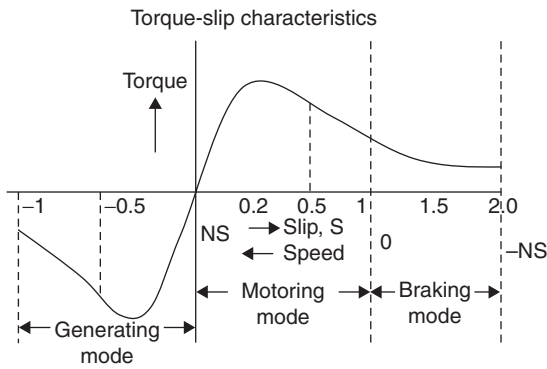
$$T \propto \frac{s}{(sX_2)^2} \propto \frac{1}{s}$$

Hence, the torque/slip curve is a rectangular hyperbola. Beyond the point of maximum torque, any further increase in motor load results in decrease of torque developed by motor. The result is that motor slows down and operation of the motor lies between the values of $s = 0$ and that corresponding to maximum torque.

The maximum torque does not depend on R_2 , but the exact location of $T_{\max}$ is dependent on it. Greater the R_2 , greater is the value of slip at which the maximum torque occurs.

Torque-slip (or) Torque-speed Characteristics

Torque-slip (or) Torque-speed characteristics of three-phase IM under different modes as shown below.



When speed varies from 0 to N_s and slip varies from 0 to 1 then the IM operates under motoring mode. If the slip goes to negative value and speed greater than synchronous speed. Then the Induction machine operates under generating mode. Similarly if the speed is negative and slip is greater than unity then the motor operates under Braking mode.

Effect of Change in Supply Voltage on Torque and Speed

$$T = \frac{K\phi s E_2 R_2}{R_2^2 + (sX_2)^2}$$

As $E_2 \propto \phi \propto V$, where V is the supply voltage

$$\therefore T \propto sV^2$$

Torque at any speed is proportional to the square of the applied voltage. Changes in supply voltage affect both starting torque and running torque.

If supply voltage V decreases, then torque T also decreases. Hence, for maintaining the same torque, slip increases, i.e., speed falls.

Let V change to V' , s to s' and T to T'

$$\text{then } \frac{T}{T'} = \frac{sV^2}{s'V'^2}$$

Example 6: The power input to a 415 V, 50 Hz, 6-pole, 3- ϕ IM running at 975 rpm is 40 kW. The stator losses are 1 kW and friction and windage losses are 2 kW. The efficiency of the motor is

- (A) 92.5% (B) 91%
(C) 90.06% (D) 88%

Solution: (C)

$$\Rightarrow S = \frac{N_s - N}{N_s} = 0.025$$

$$\begin{aligned} \text{Rotor Cu loss} &= S \times \text{Rotor Input} \\ &= 0.025 \times (40 - 1) \end{aligned}$$

$$\text{Total loss} = 3975 \text{ W}$$

$$\text{Output} = 36,025 \text{ W}$$

$$\begin{aligned} \% \text{ Efficiency} &= \frac{\text{Output}}{\text{Input}} = 90.06\% \\ &= \frac{36025}{4000} \times 100 = 90.06\% \end{aligned}$$

Important Relations

(a) Ratio of full-load torque and maximum torque

$$\frac{T_{fl}}{T_{\max}} = \frac{2as_f}{a^2 + s_f^2},$$

$$\text{where } a = \frac{R_2}{X_2}$$

s_f – full-load slip

In fact $a = s_m$, slip corresponding to maximum torque.

In that case the ratio $\frac{T_{fl}}{T_{\max}}$ is given by

$$\frac{T_{fl}}{T_{\max}} = \frac{2s_ms_f^2}{s_m^2 + s_f^2}$$

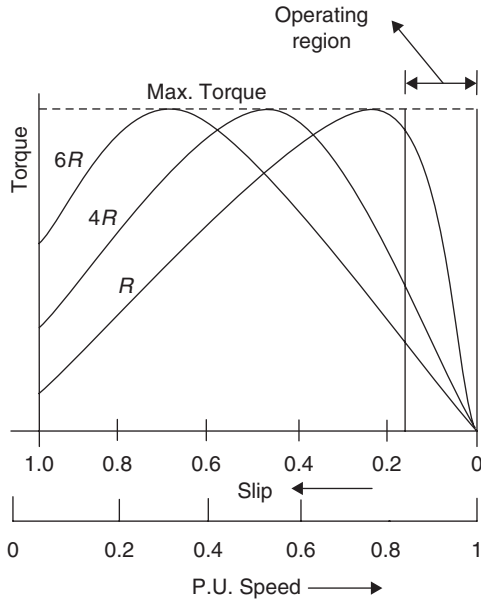
In general,

$$\frac{\text{Operating torque at any slips}}{\text{Maximum torque}} = \frac{2as}{a + s^2}$$

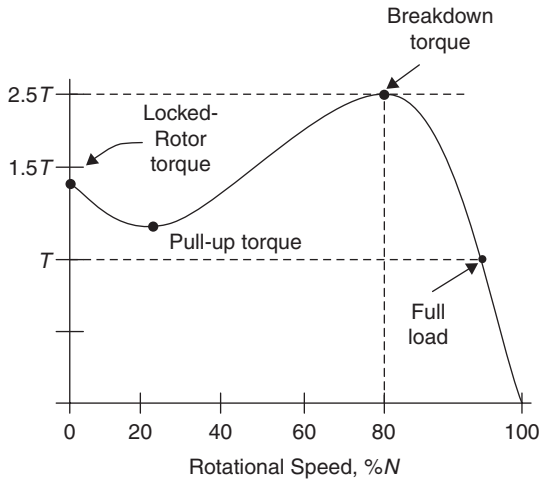
(b) Ratio of starting torque and maximum torque

$$\frac{T_{st}}{T_{\max}} = \frac{2a}{1 + a^2}$$

$$\text{where } a = \frac{R_2}{X_2}$$



Torque/Speed Curve



Example 7: A 12-pole 3-phase, 600-V 50-Hz, star-connected, IM has rotor-resistance and standstill reactance of 0.02 and 0.4 Ω per phase, respectively. Calculate:

- Speed corresponding to maximum torque.
- Ratio of full-load torque to maximum torque, if the speed is 495 rpm.

Solution: $N_s = \frac{120 \times 50}{12} = 500$ rpm

For $r = 0.02$ and $x = 0.4 \Omega$, the slip corresponding to maximum torque is related is

$$S_{MT} = a = r_2/x_2 = \frac{0.02}{0.4} = 0.05$$

- Corresponding speed
 $= 500 (1 - S_{MT}) = 500 (1 - 0.05)$
 $= 475$ rpm

- Full-load speed = 495 rpm

$$S_f = 0.01$$

$$\frac{\text{Full-load torque}}{\text{Maximum torque}} = \frac{2as_f}{a^2 + s_f^2}$$

$$= \frac{2 \times 0.05 \times 0.01}{0.05^2 + 0.01^2} = 0.38$$

Example 8: An 8-pole 50-Hz, 3-phase, slip-ring IM has effective rotor resistance of 0.08 Ω /ph. Stalling speed is 650 rpm. How much resistance must be inserted in the rotor phase to obtain the maximum torque at starting? Ignore magnetizing current and stator leakage impedance.

Solution: The stalling speed corresponding to stalling (or maximum torque) and to maximum slip under running conditions.

$$N_s = 120 \times 50/8 = 750 \text{ rpm}$$

Stalling speed is = 650 rpm

s_b = slip corresponding to max torque.

$$s_b = \frac{750 - 650}{750} = \frac{2}{15}$$

$$= 0.133 \text{ or } 13.33\%$$

$$s_b = R_2/X_2$$

$$\therefore X_2 = 0.08 \times 15/2 = 0.6 \Omega$$

$$\frac{T_{st}}{T_{\max}} = \frac{2a}{1 + a^2}$$

Since $T_{st} = T_{\max}$

$$\therefore 1 = \frac{2a}{1 + a^2} \quad \text{or} \quad a = 1$$

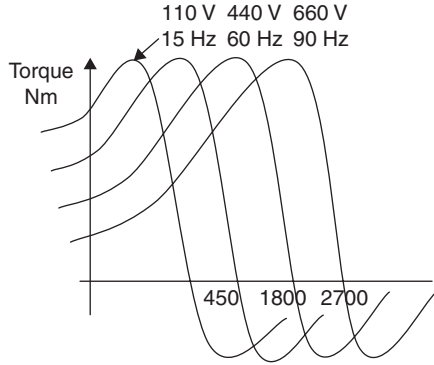
Let r be the external resistance per phase added to the rotor circuit, then

$$A = \frac{R_2 + r}{x_2} \quad \text{or} \quad 1 = \frac{0.08 + r}{0.6}$$

$$\therefore r = 0.52 \Omega/\text{phase}$$

Shape of Torque/Speed Curve

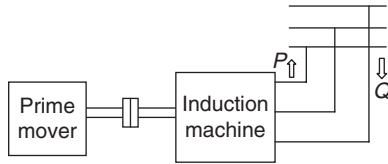
For a SCIM shape of its torque/speed curve depends on the voltage and frequency applied to its stator. In practice, supply voltage and frequency are varied in the same proportion in order to maintain a constant flux in the air gap. Under those conditions, shape of the torque/speed curve remains the same but its position along the X -axis (i.e., speed axis) shifts with frequency.



Above fig shows torque/speed curve of an 11 kW, 440 V, 60 Hz, 3-phase SCIM. Since the shape of the torque/speed curve remains same at all frequencies, it follows that torque developed by an SCIM is the same whenever slip-speed is the same.

Induction Generator

When run faster than its synchronous speed, an IM runs as a generator called an induction generator. To run the IM faster than its synchronous speed a prime mover is needed.



For creating its own magnetic field, it absorbs reactive power ' Q ' from the line and delivers active power ' P ' to the 3-phase line. The active power is directly proportional to the slip above the synchronous speed.

EQUIVALENT CIRCUIT OF THE ROTOR

When motor is loaded, the rotor current I_2 is given by

$$I_2 = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}}$$

From the above relation it appears that the rotor circuit which actually consists of a fixed resistance R_2 and a variable reactance sX_2 (proportional to slip) connected across $E_r = sE_2$ (figure 1(a)) can be looked upon as equivalent to a rotor circuit having a fixed reactance X_2 connected in series with a variable resistance $\frac{R_2}{s}$ (inversely proportional to slip) and supplied with constant voltage E_2 as shown in fig (b).

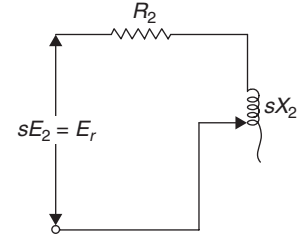


Figure 1(a)

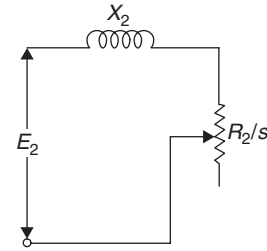


Figure 1(b)

Now, the reactance $\frac{R_2}{s} = R_2 + R_2 \left(\frac{1}{s} - 1 \right)$

It consists of two parts:

1. The first part R_2 is the rotor resistance itself and represents the rotor Cu loss.
2. The second part is $R_2 \left(\frac{1}{s} - 1 \right)$. This is known as the

load resistance R_L and is the electrical equivalent of the mechanical load on the motor

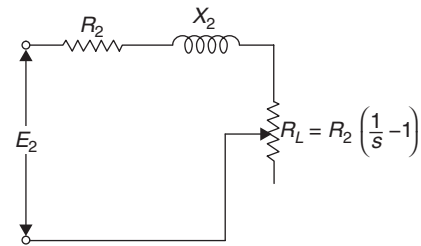
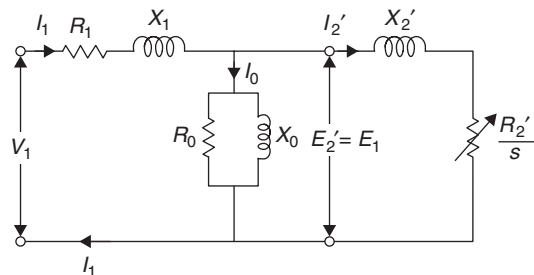
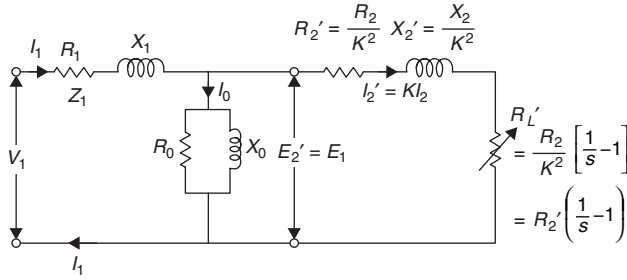


Figure 1(c)

Equivalent Circuit of an IM

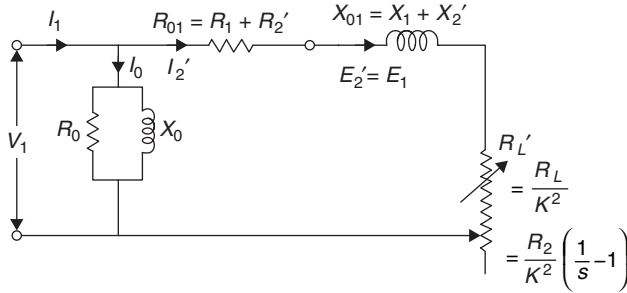
The equivalent circuit of an IM where all values have been referred to primary, i.e., stator is shown in the below figure.





Here, 'K' refers to the transformation ratio of the given machine. Since IMs can be visualized as three-phase transformers with a rotating secondary; 'K' is the effective turns ratio.

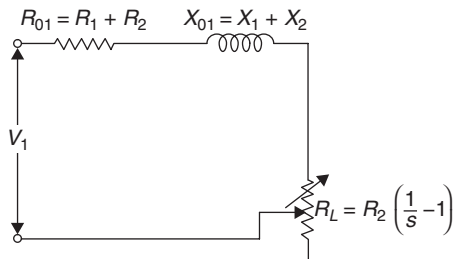
The exciting circuit may be transferred to left, to make calculations simple, thereby we get approximate equivalent circuit of the IM.



Condition for Maximum Power Output

Approximate equivalent circuit of an IM with the following assumptions is shown in figure.

1. Exciting circuit is omitted, i.e., I_0 is neglected and
2. K is assumed unity



With the above assumptions, the power output is maximum when the equivalent load resistance is equal to the standstill leakage impedance of the motor.

i.e., $R_L = Z_{01}$

Corresponding slip, $s = \frac{R_2}{R_2 + Z_{01}}$

Maximum gross power output,

$$P_{gmax} = \frac{3V_1^2}{2(R_{01} + Z_{01})}$$

where V_1 is voltage/phase of the motor and K has been taken as unity.

STARTING OF IMS

An IM at rest is like a transformer with a short-circuited secondary. Therefore, if normal supply voltage is applied to the stationary motor, a very large initial current, i.e., 5 to 7 times the full-load current is taken by the primary and develop only 1.5 to 2.5 times their full-load torque.

- The wound-rotor IMs are started by introducing external resistance across the slip-rings which also improves the starting torque.
- SCIMs are started by applying a reduced voltage at starting and then increasing to the full line voltage as the motor picks up speed.

Direct-switching or Line Starting of IMs

- No effort is made to reduce the starting current. Motors of small capacity can be started by direct-switching.

$$\frac{T_{st}}{T_{fl}} = \left(\frac{I_{sc}}{I_{fl}} \right)^2 S_{fl}$$

Starting of SCIMs

A reduction in the stator applied voltage can be accomplished in three ways

- (a) Stator resistance (or) reactor starting
- (b) Auto-transformer starting
- (c) Star-delta starting

(a) Stator resistance or reactor starting:

- A resistor or a reactor is inserted in between motor and supply terminals. Their purpose is to drop some voltage and hence reduce the initial current drawn by the motor.
- If the stator voltage is reduced to the fraction 'x' (of the rated voltage) the short-circuit current also reduced to the fraction 'x'

$$\text{i.e., } I_{st} = \frac{xV_1}{Z_1} = xI_{sc}$$

$$\begin{aligned} \frac{T_{st}}{T_{fl}} &= \left[\frac{I_{st}}{I_{fl}} \right]^2 \cdot S_{fl} \\ &= \left(\frac{xI_{sc}}{I_{fl}} \right)^2 \cdot S_{fl} \\ &= x^2 \left(\frac{I_{sc}}{I_{fl}} \right)^2 \cdot S_{fl} \end{aligned}$$

Starting Torque with Reactor Starting

$$\text{Starting torque with direct switching} = \left(\frac{xV_1}{V_1} \right)^2 = x^2$$

Auto-transformer Starting

The reduction in the stator impressed voltage is accomplished by means of auto-transformer

$$I_{st} = \frac{xV_1}{Z_1} = xI_{sc}$$

Neglecting the no-load current of the auto-transformer, primary VA = secondary VA.

i.e., $I_L V_1 = xV_1 I_{st}$, where I_L is line (primary) current

$$\therefore I_{st} = \frac{I_L}{x} = xI_{sc}$$

$$\therefore I_L = x^2 I_{sc}$$

$$\frac{T_{st}}{T_{fl}} = \left[\frac{X I_{sc}}{I_{fl}} \right]^2 \cdot s_{fl} = x^2 \left[\frac{I_{sc}}{I_{fl}} \right]^2 \cdot s_{fl}$$

Star-delta Starting

This method of starting is used for motors designed to operate normally in delta. Initially, the stator winding is connected in star, so that the voltage applied to the stator is

$\left(\frac{1}{\sqrt{3}} \right)$ times the voltage. After the motor attains a speed

nearly equal to the steady-state speed, the stator winding are connected in delta.

Let V_L be the line voltage with stator winding star-connected, then starting current/phase

$$= \left(\frac{V_L}{\sqrt{3}} \right) \frac{1}{Z_{sc}} = (I_L)_Y$$

with stator connected in delta the starting current/phase

$$(I_{st})_D = V_L \left(\frac{1}{Z_{sc}} \right)$$

and the line current

$$(I_{st})_{L\Delta} = \sqrt{3} (I_{st})_{\Delta}$$

$$\therefore (I_L)_Y = (I_{st})_Y = \frac{V_L}{\sqrt{3} Z_{sc}} = \frac{1}{\sqrt{3}} (I_{st})_{\Delta}$$

Starting line current with

$$\therefore \frac{\text{star-connected stator}}{\text{Starting line current with delta-connected stator}} = \left(\frac{1}{\sqrt{3}} \right) \left(\frac{1}{\sqrt{3}} \right) = \frac{1}{3}$$

$$(I_{st})_Y = \frac{1}{\sqrt{3}} (I_{sc})_D$$

$$\frac{\text{Starting torque with stator}}{\text{Starting torque in direct switching}}$$

$$= \frac{\left(\frac{V_L}{\sqrt{3}} \right)^2}{V_L^2} = \frac{1}{3}$$

$$\frac{\text{Starting torque with star } - \Delta \text{ starter, } T_{st}}{\text{Full load torque with stator winding in } \Delta, (T_{FL})_{\Delta}}$$

$$\begin{aligned} &= \frac{\frac{1}{\omega_s} (I_{st})_Y^2 \frac{r_2}{1}}{\frac{1}{\omega_s} (I_{fl})_{\Delta}^2 \frac{r_2}{s_{fl}}} \\ &= \frac{1}{3} \left[\frac{(I_{sc})_{\Delta}}{(I_{fl})_{\Delta}} \right]^2 \cdot s_{fl} \end{aligned}$$

Common Data for Examples 9 to 11:

An SCIM has a starting current of 6 times the full-load current at a slip of 0.04. Calculate the line current and starting torque in PU of full-load values of the following methods.

Example 9: Direct switching

Solution: In Direct switching method,
Line current = Starting current

$$\text{Starting current, } I_{st} = 6I_{fl}$$

$$\begin{aligned} \frac{T_{st}}{T_{fl}} &= \left(\frac{I_{st}}{I_{fl}} \right)^2 \times s_{fl} = (6)^2 \times 0.04 \\ &= 1.44 \text{ PU} \end{aligned}$$

Example 10: Auto-transformer starting with motor current limited to 2.0 PU

Solution: Line current $= x \cdot I_{st} = x^2 I_s = \frac{1}{9} 6I_{fl} = \frac{2}{3} I_{fl}$

$$I_{sc} = 6I_{fl}$$

$$\therefore x = \frac{I_{st}}{I_{sc}} = \frac{1}{3}$$

$$\frac{T_{st}}{T_{fl}} = \left(\frac{xI_{sc}}{I_{fl}} \right)^2 \times s_{fl} = x^2 \left(\frac{I_{sc}}{I_{fl}} \right)^2 \times s_{fl}$$

$$= \frac{1}{9} \times 36 \times 0.04$$

$$= 0.16 \text{ PU}$$

Example 11: Star-delta starting.

During starting period the motor is star-connected

Solution: I_{st} per phase = $\frac{1}{\sqrt{3}} I_{sc}$ per phase

$$= \frac{1}{\sqrt{3}} \times 6 I_{FL} = 3.45 \text{ PU}$$

$$\frac{T_{st}}{T_{fl}} = \left[\frac{I_{st}}{I_{fl}} \right]^2 s_{fl}$$

$$= \left[\frac{I_{sc}}{\sqrt{3} I_F} \right]^2 s_{fl}$$

$$= \frac{1}{3} \left[\frac{I_{sc}}{I_{FL}} \right]^2 s_{fl}$$

$$= \frac{1}{3} \times (6)^2 \times 0.04 = 0.48 \text{ PU}$$

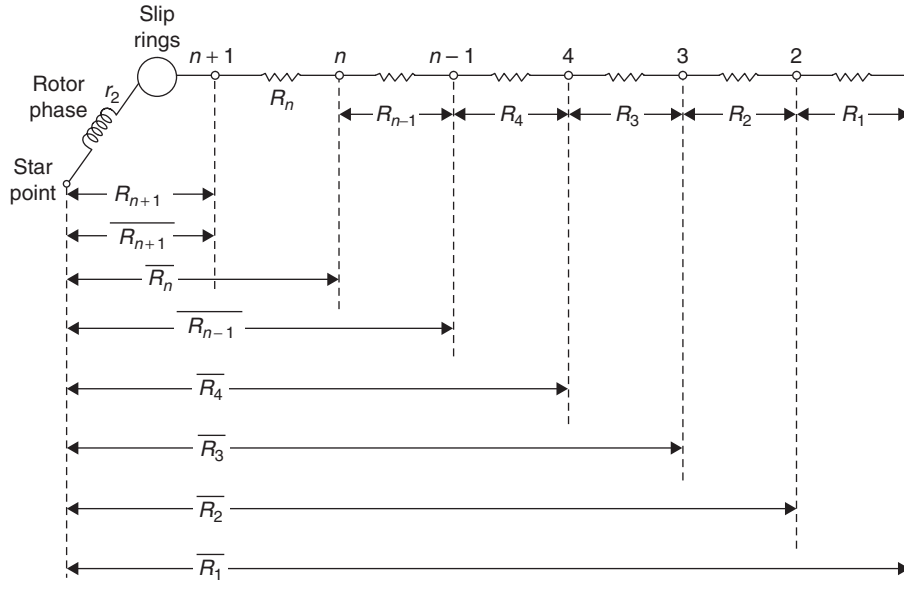
Starting of Wound-rotor Motors

All the starting methods that are used for SCIM are applicable to wound rotor IMs also. However, in a wound rotor IM, the starting current is limited for a safe value by increasing the impedance of the rotor circuit at the time of starting. This is possible since an external resistance can be included in the rotor circuit at the time of starting the motor. It is gradually cut out as the motor comes up to speed.

Rotor resistance starting offers the following advantages

1. It is simple and cheap
2. It limits starting current to a safe value
3. It can increase the starting torque even to a value equal to the pull-out torque
4. The rotor p.f. (and hence the starting p.f.) is improved

Calculation of Starter Resistance Steps



Assumptions

1. During the start-up period, the load torque remains constant
2. No-load current of the motor is neglected
3. The input current varies between a maximum of I_{lmax} and a minimum of I_{lmin}

$$\frac{\bar{R}_2}{\bar{R}_1} = \frac{\bar{R}_3}{\bar{R}_2} = \dots = \frac{\bar{R}_n}{\bar{R}_{n-1}}$$

$$= \frac{\bar{R}_{n+1}}{\bar{R}_n} = \frac{r_2}{\bar{R}_n} = \gamma (\text{say})$$

$$\text{Then } \bar{R}_2 = \bar{R}_1 \gamma, \bar{R}_3 = \bar{R}_2 \gamma = \bar{R}_1 \gamma^2; \dots \bar{R}_{n+1} = \bar{R}_1 \gamma^n = r_2$$

$\gamma = \sqrt[n]{s_m}$, where s_m is the slip under normal operating conditions.

$$R_1 = \bar{R}_1 - \bar{R}_2 = (1 - \gamma) \bar{R}_1$$

$$R_2 = \bar{R}_2 - \bar{R}_3 = (1 - \gamma) \bar{R}_2$$

$$= (1 - \gamma) \gamma \bar{R}_1 = \gamma R_1$$

$$R_3 = \gamma^2 R_1; \dots \dots R_n = \gamma^{n-1} R_1$$

Crawling

This problem is found, particularly in SCIM. They sometimes exhibit a tendency to run stably at speeds as low as one-seventh of their synchronous speed, N_s . This phenomenon is known as crawling of an IM.

This is due to 7th harmonic currents in the three stator windings which have a phase difference of $7 \times 120^\circ = 2 \times 360^\circ + 120^\circ = 120^\circ$.

They set up a forward speed equal to $\frac{1}{7}$ th of the synchronous speed of the fundamental torque.

Cogging or Magnetic Locking

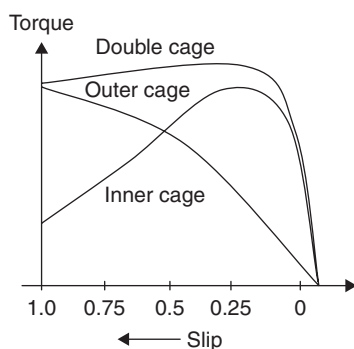
The rotor of a SCIM sometimes refuses to start at all, particularly when the voltage is low

This happens when the number of stator teeth S_1 is equal to number of rotor teeth S_2 and is due to the magnetic locking between the stator and rotor teeth. This is because of the minimum reluctance of magnetic path when the stator and rotor teeth face each other rather than when the teeth of one element are opposite to the slots on the other. It is in such positions of minimum reluctance that the rotor tends to remain fixed and thus cause serious trouble during starting. Cogging of SCIMs can be easily overcome by making the number of rotor slots prime to the number of stator slots.

Double SCIM

The main disadvantage of SCIM is its poor starting torque, which can be overcome by having a cage of high resistance. The motor has two independent cages on the same rotor, one inside the other.

The outer cage has high resistance and low ratio of reactance to resistance whereas the inner cage has low resistance but, being situated deep in the rotor has a large ratio of reactance-to-resistance. Hence, the outer cage develops maximum torque at starting, while the inner cage does so at about 15% slip.



SPEED CONTROL OF IM

The speed control techniques of an IM can be mainly classified as

1. Control from stator side
 - (a) by changing the applied voltage or line voltage control
 - (b) by changing the applied frequency
 - (c) by changing the number of stator poles
2. Control from rotor side
 - (d) rotor rheostat control
 - (e) by operating two motors in concatenation or cascade
 - (f) by injecting an emf in the rotor circuit (secondary foreign voltage control)

Voltage control, rotor-resistance control and secondary foreign voltage control methods are commonly known as slip control methods.

Changing the Applied Voltage

This method is the cheapest and the easiest, but is rarely used because

1. A large change in voltage is required for a relatively small change in speed.
2. This large change in voltage will result in a large change in the flux density thereby seriously disturbing the magnetic conditions of the motor.

Changing the Applied Frequency

This method is also used very rarely.

$$N_s = \frac{120f}{P}$$

Changing the Number of Stator Poles

Change of number of poles is achieved by having two or more entirely independent stator windings in the same slots. Each winding gives a different number of poles and hence different speeds. This method is easily applicable to SCIMs because the squirrel-cage rotor adopts itself to any reasonable number of stator poles.

Rotor Rheostat Control

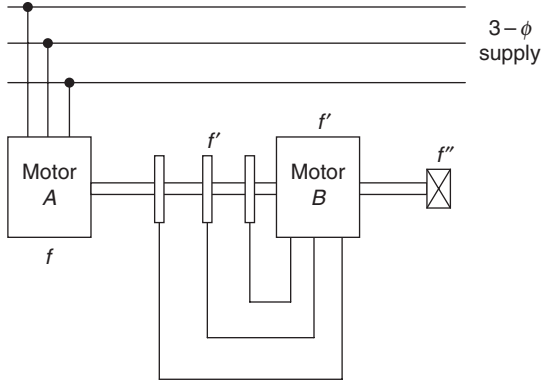
This method is applicable to slip-ring motors alone. The motor speed is reduced by introducing an external resistance in the rotor circuit.

Drawbacks of Rotor Rheostat Control

1. With increase in rotor resistance, I^2R losses also increase which decreases the operating efficiency of the motor. The losses are directly proportional to the reduction in speed.
2. Double dependence of speed, not only on R_2 but on load as well.

Cascade or Concatenation or Tandem Connection

In this method, two motors are used and are ordinarily mounted on the same shaft, so that both run at the same speed.



The stator winding of the main motor *A* is connected to the mains, while that of the auxiliary motor *B* is fed from rotor circuit of motor *A*.

There are at least three ways in which the combination may be run.

1. Main motor *A* may be run separately from the supply.

In that case, the synchronous speed is $N_{sa} = \frac{120f}{P_a}$, where P_a = number of stator poles of motor *A*.

2. Auxiliary motor *B* may be run separately from the mains (with motor *A* being disconnected). In that case, synchronous speed is $N_{sb} = \frac{120f}{P_b}$, where P_b =

Number of stator poles of motor *B*.

3. The combination may be connected in cumulative cascade, i.e., in such a way that the phase rotation of the stator fields of both motors is in the same direction. The synchronous speed of the cascaded set, in this case, is $N_{sc} = \frac{120f}{(P_a + P_b)}$.

4. The fourth possible connection is the differential cascade. In this method, the phase rotation of stator field of motor *B* is opposite to that of motor *A*. This reversal of phase rotation of stator of motor *B* is obtained by interchanging any of its two leads. The synchronous speed of the set is $N_{sc} = \frac{120f}{(P_a - P_b)}$ (for $P_a \neq P_b$).

As differentially cascaded set has a very small or zero starting torque, this method is rarely used. Moreover, the above expression for synchronous speed becomes meaningless for $P_a = P_b$.

5. V/f control.

We know that Torque (τ)

$$= \frac{\text{power}}{\omega} = \frac{\eta \sqrt{3} V 1 \cos \phi}{N}$$

(N is the speed)

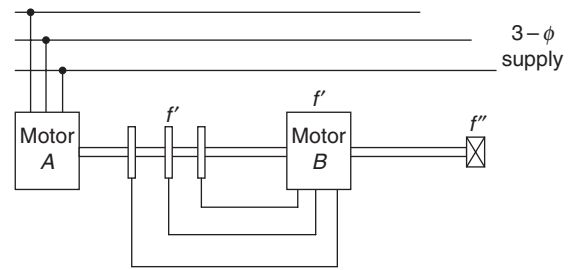
$$\text{But; } N = \frac{120f}{P} (1 - S)$$

Now, if voltage and frequency are constant, torque stays constant and if torque stays constant and V/f = constant; slip remains the same.

Example 12: A 4-pole IM and a 6-pole IM are connected in cumulative cascade. The frequency in the secondary circuit of the 6-pole motor is observed to be 2.0 Hz. Determine the slip in each machine and combined speed of the set. (supply frequency = 50 Hz)

Solution: Speed of cascaded set

$$N_{sc} = 120 \times 50 / (4 + 6) = 600 \text{ rpm}$$



$$s = \frac{f''}{f} = \frac{2}{50} = 0.04$$

Let N be the actual speed of concatenated set,

$$\therefore 0.04 = \frac{600 - N}{600} \text{ or } N = 576 \text{ rpm}$$

$$N_{sa} = 120 \times 50 / 4 = 1500 \text{ rpm}$$

$$s_a = \frac{1500 - 576}{1500} = 0.616 \text{ or } 61.6\%$$

$$\begin{aligned} f' &= s_a f \\ &= 0.616 \times 50 \\ &= 30.8 \text{ Hz} \end{aligned}$$

$$\begin{aligned} N' &= \text{Synchronous speed of 6-pole motor with frequency } f' \\ &= 120 \times 30.8 / 6 = 616 \text{ rpm} \end{aligned}$$

$$\begin{aligned} s_b &= \frac{N' - N}{N} = \frac{616 - 576}{616} \\ &= 0.064 \end{aligned}$$

or 6.4%

Injecting an emf in the Rotor Circuit

In this method, speed of an IM is controlled by injecting a voltage in the rotor circuit. The injected voltage must have the same frequency as the slip frequency. There is no restriction to phase of the injected emf. In this method,

1. Any speed, within the working range can be obtained and smooth speed control is possible.
2. By adjusting the phase and magnitude of the injected emf, the power factor of the system can be improved.

REVOLVING FIELD THEORY OF SINGLE-PHASE IM

A single-phase IM has a stator winding and a squirrel-cage rotor. When an alternating quantity given, it can be represented two oppositely rotating vectors of half magnitude.

$$F = 1/2 F_m \cos(\theta - \omega t) + 1/2 F_m \cos(\theta + \omega t)$$

In the above equation first one is the forward rotating field with half of the maximum value of the pulsating mmf along the axis of the winding then the slip of the rotor with respect to the forward rotating field.

$$s_f = \frac{(N - N_s)}{N_s} = s$$

The second term is the backward rotating field with half the magnitude. The slip corresponding one is $s_b = 2 - s$

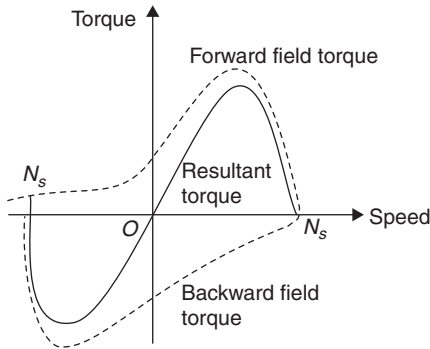


Figure 2 Torque speed characteristics of single winding single-phase motor

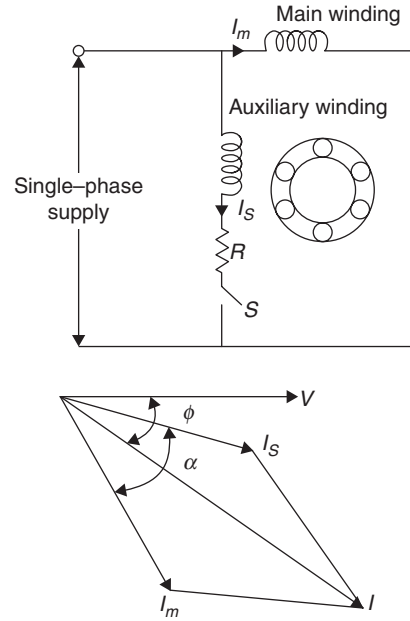
Single-phase Induction Motors

A single-phase IM is not self-starting. To make the motor self-starting, it is temporarily converted into a two-phase motor during starting period. For this purpose, in addition to the main winding, the stator of a single-phase motor is provided with an extra winding, known as starting (or auxiliary winding). The currents flowing through these windings should have a phase difference of 90° so that the motor behaves like a two-phase motor. These two currents produce a rotating magnetic field and hence make the motor self-starting.

The following are the methods by which necessary phase-difference between the two currents can be obtained.

Split Phase Machine

The main winding has low resistance but high reactance whereas the starting winding has a high resistance but low reactance. Hence the current I_s drawn by the starting winding lags behind the applied voltage V by a small angle whereas current I_m taken by the main winding lags behind V by a very large angle.



The phase angle between ' I_s ' and ' I_m ' is made as large as possible (i.e., near to 90°) because the starting torque of a split-phase motor is proportional to $\sin \alpha$. A centrifugal switch ' S ' is connected in series with the starting winding and is located inside the motor to automatically isolate the starting winding from the supply when the motor speed has reached 70% to 80% of steady state speed.

The value of resistance of auxiliary winding, r_a for the production of maximum torque at starting is

$$r_a = \left(\frac{x_a}{x_m} \right) (r_m + Z_m)$$

$$\text{or } r_a = \left(\frac{N_a}{N_m} \right)^2 (r_m + z_m)$$

($\because$ reactance $\propto$ square of number of turns)

Capacitor Split-phase Motors

In these motors, the phase displacement between the main winding current I_m and auxiliary winding current I_a is produced by connecting a capacitor in series with the auxiliary winding.

The starting torque $\propto I_a \cdot I_m \cdot \sin \beta$

where β = angle between I_a and I_m . By choosing suitable value of capacitance, the angle β can be made 90° at standstill, which results in highest possible starting torque but best compromise with the running torque is to make β slightly less than 90° . This type of starting gives the highest starting torque.

Capacitor split-phase motors can be further classified as

1. Capacitor start induction run
2. Capacitor start capacitor run

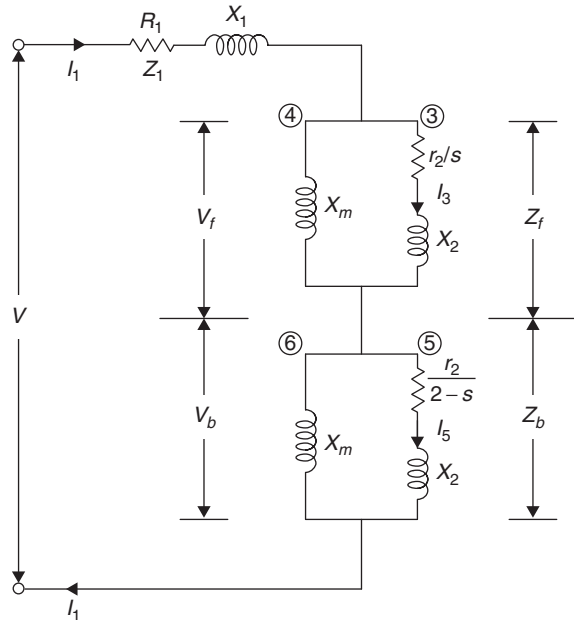
The value of capacitive reactance to be inserted in series with the auxiliary winding to get maximum starting torque is

$$X_c = X_a + \frac{r_a \cdot r_m}{Z_m + X_m}$$

$\therefore$ The value of capacitor, $C = \left(\frac{1}{\omega X_c} \right)$ farad

Equivalent Circuit of a Single-phase IM

A single-phase IM may be viewed as consisting of two motors, having a common stator winding, but with their respective rotors revolving in opposite directions. The equivalent circuit of such a motor based on double field revolving theory is shown in fig.



The stator impedance is $Z = R_1 + jX_1$. The impedance of each rotor is $(r_2 + jx_2)$, where r_2 and x_2 represent half the actual rotor values in stator terms (i.e., x_2 stands for half the standstill reactance of the rotor, as referred to stator). Since iron loss has been neglected, the magnetizing branch

is shown consisting of magnetizing reactance only. Each rotor has been assigned half the magnetizing reactance (i.e., x_m represents half the actual reactance). The impedance of the 'forward running' rotor is

$$Z_f = \frac{jx_m \left(\frac{r_2}{s} + jx_2 \right)}{\frac{r_2}{s} + j(x_m + x_2)}$$

and it runs with a slip of s .

The impedance of backward running rotor is

$$Z_b = \frac{jx_m \left(\frac{r_2}{2-s} + jx_2 \right)}{\frac{r_2}{2-s} + j(x_m + x_2)}$$

and it runs with a slip of $(2-s)$. Under standstill conditions $V_f = V_b$, but under running conditions V_f is almost 90% to 95% of applied voltage.

The forward torque in synchronous watt is

$$T_f = \frac{I_3^2 r_2}{s}$$

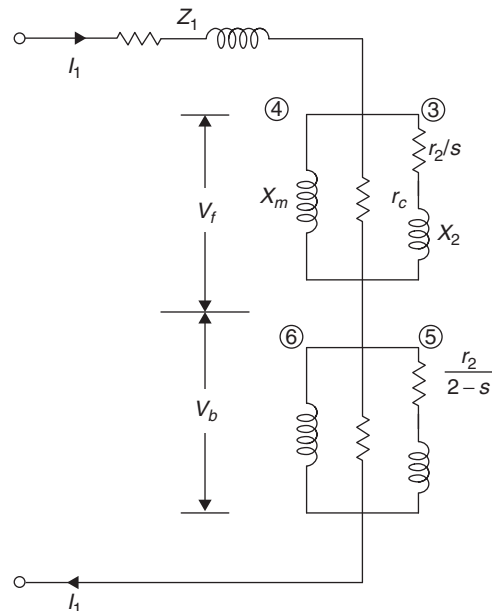
Similarly backward torque is

$$T_b = \frac{I_5^2 r_2}{2-s}$$

The total torque is $T = T_f - T_b$

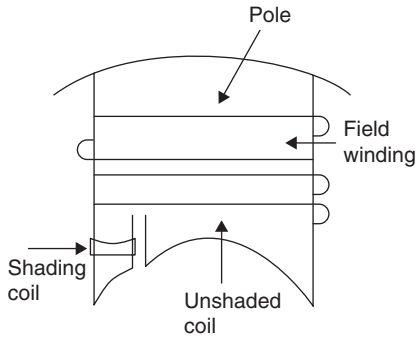
Equivalent Circuit with Core Loss

The core loss can be represented by an equivalent resistance which may be connected in parallel as shown in fig.

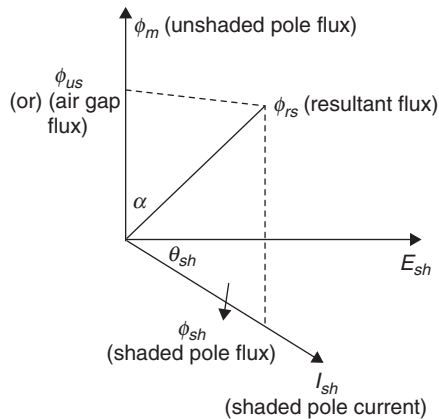


Shaded-pole Single-phase Motor

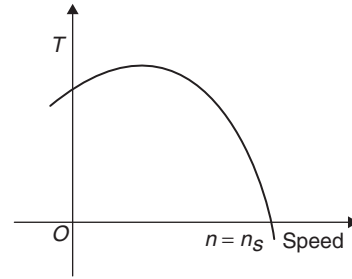
In this type motor, the small portion of each pole is covered with a short-circuited copper coil known as shading coil (or) shaded portion of the pole. The other part of the pole is known as unshaded part as shown in the figure. When an alternating current passed through the coil, the motor rotates in the direction of unshaded portion to shaded portion.



Below phasor diagram shows the flux and induced emf of the shaded-pole IM.



The Torque speed characteristics of the shaded-pole motor show below:



Repulsion Start IM

In Repulsion motors the rotor and stator windings are inductively coupled. In these motors the stator has a distributed winding and the rotor is similar to the DC motor armature, i.e., the windings are connected to commutator. The Repulsion start IM start as a repulsion motor by the transformer action and the motor run as an IM by short-circuiting the commutator segment using a centrifugal switch.

Reluctance Motor

The reluctance motor is just similar to unexcited salient-pole synchronous machine. In these motor the stator produces a rotating field and the reluctance of the magnetic path offered by the rotor to the rotating field is a function of the space angle. Then the rotor aligns in the minimum reluctance path with respect to the synchronous rotating flux. The motor is self-starting by the induction principle by providing short-circuited copper bars in the rotor.

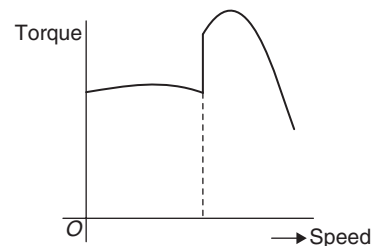


Figure 3 Reluctance motor torque–speed characteristics

EXERCISES

Practice Problems I

Directions for questions 1 to 27: Select the correct alternative from the given choices.

- A 3- ϕ , 50 Hz IM has a full load speed of 1440 rpm. What is slip and rotor frequency of the IM?
(A) 6%, 4 Hz (B) 10%, 46 Hz
(C) 8%, 6 Hz (D) 4%, 2 Hz
- An 8-pole, 50 Hz, 3- ϕ slip-ring IM has effective rotor resistance of $0.8 \Omega/\text{ph}$. Stalling speed is 650 rpm. The resistance required in the rotor phase to obtain the maximum torque at starting?
(A) 0.42Ω (B) 0.52Ω
(C) 0.48Ω (D) 0.58Ω
- The rotor power output of 3- ϕ IM is 15 kW and the corresponding slip is 4%. The rotor ohmic loss will be
(A) 600 W (B) 625 W
(C) 650 W (D) 700 W
- In case of 3- ϕ IM, shaft power is 2700 W and mechanical losses are 180 W. At a slip of 4%, the rotor ohmic losses are
(A) 115.2 W (B) 120 W
(C) 108 W (D) 105 W
- A 3- ϕ IM having a star-connected rotor has an induced emf of 80 V between slip-rings at stand still on open circuit. Find the current and power factor when the slip-rings are connected to $3 \Omega/\text{phase}$ rheostat?

- (A) 8.85 A and 0.748 Lag
(B) 8.16 A and 0.707 Lag
(C) 9.5 A and 0.8 Lag
(D) 9.5 A and 0.6 Lag
6. A 3- ϕ IM is driving the full-load torque which is independent of speed. If the line voltage drops to 85% of the rated value, the increase in motor Cu losses is _____
(A) 123% (B) 33%
(C) 38.4% (D) 33.6%
7. If the supply voltage of a 3- ϕ SCIM is reduced by 20% then the maximum torque will _____
(A) increase by 20%
(B) decrease by 20%
(C) increase by 36%
(D) decrease by 36%
8. A cascaded set of IM, consists of two motors A and B with 4 poles and 6 poles, respectively. The motor A is connected to a 50 Hz supply. What is the speed of the set _____?
(A) 3000 rpm (B) 600 rpm
(C) 1500 rpm (D) 1000 rpm
9. A 3- ϕ IM has a full-load slip of 3% at normal voltage. Which one of the following will be the value of the slip of the motor if it develops the same torque theoretically while operating at 110% of its normal voltage?
(A) 2.48% (B) 0.248
(C) 0.483% (D) 4.83%
10. A 6-pole, 3- ϕ , 60 Hz IM runs at 1000 rpm, developing maximum torque. Rotor resistance per phase is 1.2 Ω . Neglecting stator impedance, then for developing maximum torque, the external resistance to be connected in series with the each rotor phase will be
(A) 7.2 Ω (B) 6 Ω
(C) 1.44 Ω (D) 1.2 Ω

Common Data for Questions 11 and 12:

A 400 V, 4-pole 3- ϕ 50 Hz, IM has a rotor resistance and reactance per phase of 0.02 Ω and 0.2 Ω , respectively. Stator to rotor turns ratio is 4.

11. The $T_{\max}$ and corresponding slip of motor.
(A) 320 Nm, 10% (B) 320 Nm, 20%
(C) 159.23 Nm, 10% (D) 159.23 Nm, 20%
12. The full-load slip and power output, if the maximum torque is twice the full-load torque, are _____?
(A) .027, 12.16 kW (B) 0.373, 12.16 kW
(C) 0.27, 16.18 kW (D) 0.373, 14.3 kW

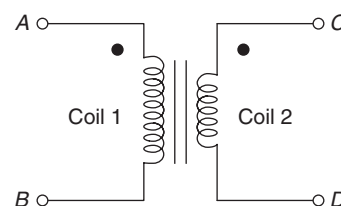
Common Data for Questions 13 and 14:

A 3- ϕ , star-connected 400 V, 50 Hz, 4-pole IM has the following per phase constants in ohm referred to stator and $Nr = 1440$ rpm.

$$R_1 = .15 \Omega, x_1 = .45 \Omega, r_2 = 0.12 \Omega$$

$$x_2 = 0.45 \Omega, x_m = 28.5 \Omega$$

13. The stator current I_1 is _____?
(A) 71.06 $\angle -21.47^\circ$ (B) 122.97 $\angle -21.47^\circ$
(C) 71.06 $\angle -36.47^\circ$ (D) 122.97 $\angle -36.47^\circ$
14. The mechanical power developed is _____
(A) 87.73 kW (B) 6.45 kW
(C) 41.73 kW (D) 100.49 kW
15. The power input to a 415 V, 50 Hz, 6-pole, 3- ϕ IM running at 975 rpm is 40 kW. The stator losses are 1 kW and friction and windage losses are 2 kW. The efficiency of the motor is
(A) 92.5% (B) 91%
(C) 90.06% (D) 88%

Common Data for Question 16:

The figure above shows coils 1 and 2, with dot markings as shown, having 2000 and 3000 turns, respectively. Both the coils have a rated current of 25 A. Coil-1 is excited with single phase, 400 V, 50 Hz supply.

16. In the auto-transformer obtained in above question the current in each coil is
(A) coil-1 is 15 A and coil-2 is 10 A.
(B) coil-1 is 10 A and coil-2 is 25 A.
(C) coil-1 is 10 A and coil-2 is 15 A.
(D) coil-1 is 25 A and coil-2 is 10 A.
17. A three-phase slip-ring IM is fed from the rotor side with stator winding short-circuited. The frequency of the currents flowing in the short-circuited stator winding is
(A) zero.
(B) supply frequency.
(C) slip frequency.
(D) frequency corresponding to rotor speed.
18. Skew is used in IMs in order to reduce torque due to
(A) space harmonics.
(B) slot harmonics.
(C) time harmonics.
(D) reverse rotating fields.
19. An IM is fed from a balanced three – phase supply at rated voltage and frequency through a bank of three single-phase transformers connected in delta – delta. One unit of the bank develops fault and is removed. Then
(A) single phasing will not occur but the motor terminal voltages will become unbalanced and the machine can be loaded to the extent of 57.7% of its rating.
(B) the machine can be loaded to the extent of 57.7% of its rating with balanced supply at its terminals.

- (C) the machine can be loaded to the extent of 66.66% with balanced supply at its terminals.
 (D) single phasing will occur and the machine fails to start.
20. A 3-phase, delta-connected IM fed from unbalanced supply will have
 (A) zero sequence currents.
 (B) negative sequence component current.
 (C) less heating of the rotor.
 (D) All of these.
21. In case of a split-phase motor, the phase shift between currents in the two windings is around
 (A) 120° . (B) 90° .
 (C) 80° . (D) 30° .
22. In an IM, if the air-gap is increased
 (A) speed will reduce.
 (B) breakdown torque will reduce.
 (C) efficiency will improve.
 (D) power factor will be lowered.
23. A three-phase SCIM has a full load efficiency of 0.85 and a maximum efficiency of 0.9. It is operated at a slip of 0.65 by applying a reduced voltage. The efficiency of the motor at this operating point is
 (A) is less than 0.5.
 (B) greater than 0.6.
 (C) in the range of 0.85 ± 0.05 .
 (D) None of the above
24. A single-phase IM with identical main and auxiliary windings can be made self-starting by connecting
 (A) a capacitor in series with the main winding.
 (B) a capacitor in series with the auxiliary winding.
 (C) a capacitor across the supply terminals.
 (D) the main and auxiliary windings in series.
25. Which of the following starting method for an IM is inferior in view of the poor starting torque per ampere of the line current drawn?
 (A) Star-Delta method
 (B) Direct on line starting
 (C) Auto-transformer method of starting
 (D) Series inductor method of starting
26. The type of single-phase IM having the highest power factor at full load is
 (A) Split-phase type
 (B) Shaded-pole type
 (C) Capacitor-start type
 (D) Capacitor-run type
27. A plot of input power versus voltage was drawn from the no – load test readings (at different supply voltages) of a 3-phase IM. This curve was extrapolated to intersect the Y-axis. The intersection point yields
 (A) stray load loss.
 (B) stator copper loss.
 (C) core loss.
 (D) friction and windage loss.

Practice Problems 2

Directions for questions 1 to 50: Select the correction alternative from the given choices.

1. The speed of the rotor field of an IM is (s is the slip and N_s , the synchronous speed)
 (A) $s \times N_s$ in the same direction as stator field
 (B) N_s in the same direction as stator field
 (C) $s \times N_s$ in the opposite direction as stator field
 (D) N_s in the opposite direction as that of stator field
2. The rotor of a three-phase IM draws a current which is lagging at a power factor of 0.707. The spatial displacement between stator and rotor magnetic field is
 (A) 45° (B) 90°
 (C) 135° (D) 180°
3. A synchronous motor and an IM are connected to a common bus. For the power factor to be unity at the bus,
 (A) synchronous motor should be over excited
 (B) synchronous motor should be under excited
 (C) the speed of synchronous motor and synchronous speed of IM should be same
 (D) synchronous motor is normally excited
4. The starting current of an IM is 6 times the full-load current. If the full-load slip is 3%, ratio of starting torque to full-load torque is
 (A) 1.2 (B) 1.08
 (C) 0.8 (D) 0.6
5. The rotor power output of a 3- ϕ IM is 12 kW at a slip of 4%. The rotor copper losses would be
 (A) 400 W (B) 500 W
 (C) 600 W (D) 700 W
6. The rotor resistance per phase of a 4-pole 3-phase, 50 Hz IM is 1.4Ω . It develops maximum torque at 1250 rpm. For developing maximum torque at starting, the resistance to be connected in series with each rotor phase will be
 (A) 8.4 ohm (B) 1.4 ohm
 (C) 7.2 ohm (D) 7 ohm
7. A 3- ϕ IM operating at rated voltage has a full-load slip of 4%. It is now operated at 110% of rated voltage. What should be the value of slip theoretically, if it develops the same torque?
 (A) 3.30% (B) 0.330%
 (C) 1.483% (D) 4.83%

8. A 3- ϕ supply is given to the rotor of a 3- ϕ slip-ring IM with short-circuited stator. The frequency of current in the stator is
 - (A) supply frequency
 - (B) frequency corresponding to rotor speed
 - (C) slip frequency
 - (D) zero
9. A 3- ϕ IM running at 4% slip takes an input power of 10 kW. If stator resistance and core losses are assumed negligible, then the torque developed in synchronous watts is
 - (A) 12 kW
 - (B) 9 kW
 - (C) 10 kW
 - (D) 0.4 kW
10. A 3- ϕ , 6-pole, 50 Hz IM is delivering 3500 W at a slip of 0.04. Stray losses are found to be 100 W. The gross torque is
 - (A) 3750 synchronous watts
 - (B) 35.81 Nm
 - (C) Either A or B
 - (D) 41.58 kgm
11. A 3- ϕ 4-pole 400 V 50 Hz Y-connected IM has rotor resistance of 0.1Ω and stand still reactance of 0.3Ω . The effective stator to rotor turns ratio is 4. If the motor is to drive a constant-load torque of 100 Nm, the minimum resistance to be added in series in rotor circuit for the motor to start on load
 - (A) 0.01Ω
 - (B) 0.11Ω
 - (C) 0.02Ω
 - (D) 0.22Ω
12. A 3- ϕ IM has a rotor resistance of 0.02Ω /phase. If the resistance is increased to 0.04Ω /phase, the maximum torque
 - (A) increase by 50%
 - (B) becomes two times
 - (C) remains unaltered
 - (D) will reduce to half
13. When the stator of a 3- ϕ IM is excited with a 3- ϕ balanced supply, the rotor of the IM runs in same direction as the rotating stator magnetic field. The law that is obeyed here is
 - (A) Faraday's law of electromagnetic induction
 - (B) Lenz's law
 - (C) Fleming's left hand rule
 - (D) Newton's law of motion
14. An SCIM having a rated slip of 2.5% on full load has a starting torque of 0.9 full-load torque, then starting current is
 - (A) equal to full-load current
 - (B) twice full-load current
 - (C) four times full-load current
 - (D) six times full-load current
15. Maintaining the voltage to frequency ratio of an IM constant, the frequency of voltage applied is increased. Assuming stator impedance is zero, maximum torque
 - (A) increases and occurs at large values of slip
 - (B) remains constant and occurs at smaller values of slip
 - (C) remains constant and occurs at larger values of slip
 - (D) increases and occurs at smaller values of slip
16. In an SCIM, blocked rotor test is done determines its equivalent
 - (A) to determine shunt resistance and reactance as seen from rotor
 - (B) to determine series resistance and reactance as seen from rotor
 - (C) to determine shunt resistance and reactance as seen from the stator
 - (D) to determine series resistance and reactance as seen from the stator
17. An ac IM is used for a speed control application is driven from an inverter. Name plate details of motor $V: 415$, $V_{ph}: 3$, $f: 50$ Hz, $N: 1440$ rpm. Keeping v/f constant, inverter output frequency is set to 40 Hz and motor is operated at half rated slip. The running speed of motor is
 - (A) 1200 rpm
 - (B) 1140 rpm
 - (C) 1176 rpm
 - (D) 1340 rpm
18. Both frequency and voltage of a 3- ϕ IM, driving a constant-load torque are halved. Ignore stator impedance and core losses. In this condition,
 - (A) Flux in the air gap remains constant
 - (A) Stator current remains constant
 - (C) The per unit slip remains constant
 - (D) Difference between synchronous speed and actual speed remains constant

Among the above, correct statements are

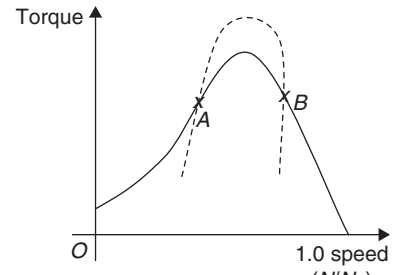
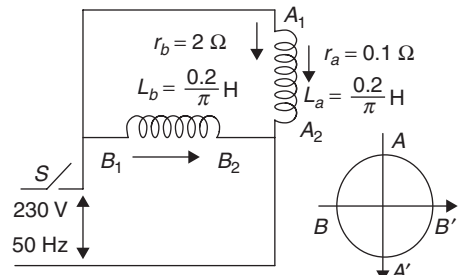
 - (A) A, B, D
 - (B) B, C, D
 - (C) A and D
 - (D) All the above are correct
19. Only the main winding of a 1- ϕ IM is excited. The forward and backward rotating fields
 - (A) are equal
 - (B) are zero
 - (C) forward rotating field is more than backward rotating field
 - (D) forward rotating field is less than backward rotating field
20. The main winding impedance and auxiliary winding impedance of a single-phase 4-pole, capacitor-start IM has the following stand still values.

$$Z_m (\text{main}) = 3 + j4 \Omega$$

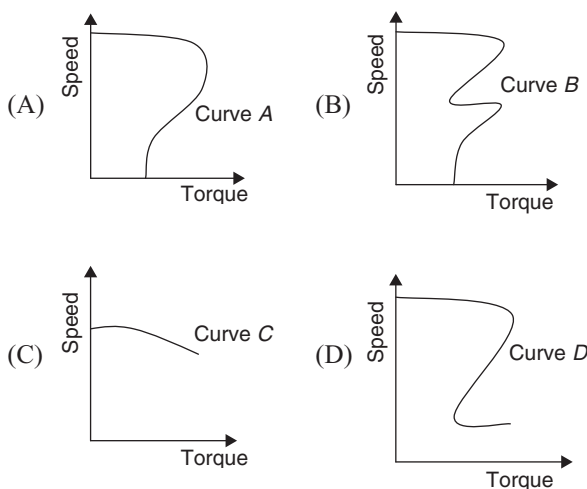
$$Z_a (\text{aux}) = 12 + j10 \Omega$$

Value of capacitor required to produce 90° phase difference between currents in main winding and auxiliary winding

 - (A) $167.53 \mu\text{F}$
 - (B) $197.53 \mu\text{F}$
 - (C) $267.53 \mu\text{F}$
 - (D) $287.53 \mu\text{F}$

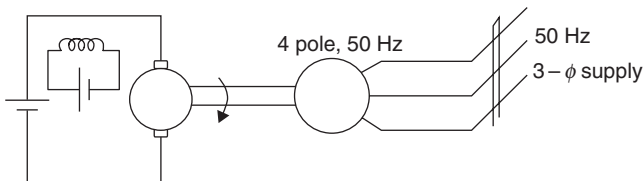
21. A 400 V, 10 kW, 6-pole 50 Hz Y-connected IM has full-load slip of 5%. The output torque of machine at full load is
 (A) 2.9 Nm (B) 92.5 Nm
 (C) 100.51 Nm (D) 248.73 Nm
22. The single-phase IM that has the greatest power factor at full load is
 (A) split-phase type
 (B) shaded-pole type
 (C) capacitor-start type
 (D) capacitor-run type
23. A 3- ϕ IM runs in anticlockwise direction when supplied with a phase sequence of R – B – Y. For a clockwise rotation, the phase sequence should be
 (A) Y – R – B (B) B – Y – R
 (C) B – R – Y (D) either (A) or (B)
24. For an IM ratio of rotor power output to air gap power is
 (A) $(1-s)^2$ (B) $(1-s)$
 (C) $\sqrt{1-s}$ (D) $(1-\sqrt{s})$
25. The voltage applied to a 3- ϕ IM, running at no load is reduced to one half from the rated value. Then the speed and stator current
 (A) decreases and increases, respectively
 (B) both decreases
 (C) both remain practically constant
 (D) remain same and decreases, respectively
26. A 3- ϕ IM is started DOL at the rated voltage. If starting current is 4 times full-load current, and full-load slip is 5% ratio of starting torque to full-load torque is
 (A) 0.2 (B) 0.4
 (C) 0.8 (D) 1
27. The rotor resistance of a 1- ϕ IM, driving a fan load is made high in order to achieve
 (A) high efficiency
 (B) low starting torque
 (C) quick acceleration
 (D) reduced size
28. The equivalent circuit parameters of a 3-phase 6-pole 400 V, 50 Hz Y-connected IM are as follows.
 $r_1 = 0.9 \Omega$, $r_2' = 0.7 \Omega$, $x_1 = x_2' = 1.4 \Omega$, $x_m = 40 \Omega$
 The starting torque when motor is started direct on line
 (A) 105.06 Nm (B) 195.06 Nm
 (C) 235.06 Nm (D) 285.06 Nm
29. The no load and blocked rotor test data for a 3- ϕ , 10 kW, 400 V, 4-pole, 50 Hz star-connected IM drawing 20 A on full load is as given
 No load test: 400 V 5 A 1120 W
 Blocked rotor: 70 V 16 A 692 W
 Estimate the full load efficiency of motor
 (A) 77% (B) 82%
 (C) 83.4% (D) 86%
30. A 400 V, 50 Hz, 25 hp, 3- ϕ IM is drawing 25 A current at 0.8 p.f. lag. Stator and rotor Cu losses are 10 kW and 700 W, respectively. Friction and windage losses are 950 W and core loss is 1100 W. The air gap power of the motor will be
 (A) 10.806 kW (B) 12.85 kW
 (C) 16.32 kW (D) 11.95 kW
31. A 3- ϕ , 400 V, 50 Hz, 4-pole, 1400 rpm, Y-connected SCIM is controlled from a 3- ϕ voltage source inverter with constant v/f control. The motor has the following parameters referred to the stator. $R_r' = 1.0 \Omega$, $X_s = X_r' = 20 \Omega$. Neglecting stator resistance and rotational losses, the stator line to line voltage and frequency to obtain maximum torque at starting will be
 (A) 323 V, 40.3 Hz
 (B) 266.6 V, 33.3 Hz
 (C) 100 V, 12.5 Hz
 (D) 200 V, 25 Hz
32. Figure below shows the speed torque curves of a 3- ϕ SCIM and its load. Points A and B represents the operating conditions. Which among them is stable
- 
- (A) A (B) B
 (C) both are stable (D) none are stable
33. AA' and BB' are the axes of the main winding A_1A_2 and auxiliary winding B_1B_2 , respectively, of a 230 V, 50 Hz 1- ϕ IM. The direction of axes shows the direction of flux when currents in winding are as shown in the figure. When switch S is closed,
- 
- (A) motor rotates clockwise
 (B) motor rotates anticlockwise
 (C) motor rotates momentarily and comes to a halt
 (D) motor does not rotate
34. In a shaded-pole IM, the rotor runs from
 (A) shaded portion to the unshaded portion of the pole while flux in shaded pole leads that of unshaded pole

- (B) shaded portion to the unshaded portion of the pole while flux in the shaded pole lags that of unshaded pole
 (C) unshaded portion to shaded portion and flux in the unshaded pole leads that of shaded pole
 (D) unshaded portion to the shaded portion, and flux in unshaded pole lags that of shaded pole
35. A 3- ϕ 50 Hz IM has a full load speed of 960 rpm. For this motor, the speed of stator field with respect to revolving rotor field is
 (A) 1000 rpm (B) 0 rpm
 (C) 40 rpm (D) 960 rpm
36. Two slip-ring IMs having number of poles 6 and 2, respectively, are connected in cascade. If supply frequency is 50 Hz, speed of the main motor is
 (A) 1500 rpm (B) 3000 rpm
 (C) 750 rpm (D) 1000 rpm
37. The torque speed characteristics of a repulsion start induction run motor is



- (A) Curve A (B) Curve B
 (C) Curve C (D) Curve D
38. The power output of an induction machine is minimum when
 (A) when equivalent resistance is equal to equivalent reactance of the motor
 (B) equivalent resistance equal to slip multiplied by equivalent reactance
 (C) when slip is equal to zero
 (D) when slip is equal to one
39. The main winding and auxiliary winding current of a 2 winding single-phase IM are $I_m = 15$ A and $I_a = 7.5$ A at stand still. The auxiliary winding current leads the main winding current by $\alpha = 45^\circ$ electrical. The two windings are in space quadrature and the effective number of turns are $N_m = 150$ and $N_a = 175$. Amplitude of forward rotating stator mmf is
 (A) 1508 AT (B) 1237 AT
 (C) 1045 AT (D) 653 AT

40. A separately excited DC machine is coupled to a 50 Hz, 3-phase 4-pole induction machine as shown in the figure. The DC machine is energized first and machines rotate at 1600 rpm. The induction machine is also connected to a 50 Hz 3- ϕ source, phase sequence being consistent with the direction of rotation. In steady state,



- (A) both machine acts as generators
 (B) DC machine acts as generator and induction machine motor
 (C) DC machine acts as motor and induction machine generator
 (D) both machines acts as motor
41. Which type of motor is most suitable for a computer printer drive?
 (A) Reluctance motor
 (B) Hysteresis motor
 (C) Shaded-pole motor
 (D) Stepper motor
42. By adding resistance in the rotor circuit of a slip-ring IM, the starting current and torque, compared to DOL
 (A) both are reduced
 (B) both are increased
 (C) starting current is reduced but starting torque is increased
 (D) starting current is increased and starting torque is reduced
43. An IM is connected to a 3- ϕ supply constituting three 1- ϕ transformers connected in Delta-Delta configuration. If one transformer is removed, then
 (A) single phasing occurs and machine does not start
 (B) single phasing does not occur and the terminal voltages become unbalanced and machine can be loaded to 57.7% of its rated capacity
 (C) machine can be loaded to 57.7% of rated capacity with balanced supply at the terminals
 (D) machine can be loaded to 66.67% of its rated power with balanced supply at terminals
44. The skewing of rotor slots in a 3- ϕ IM reduces
 (A) harmonic torques and noise but increases torque
 (B) harmonic torques and noise but increases starting torque
 (C) noise but increases breakdown torque and harmonic torque
 (D) harmonic torque, noise and breakdown torque and starting torque
45. If the air gap of an IM is increased, then
 (A) pull out torque is reduced
 (B) power factor is lowered

- (C) efficiency is improved
(D) speed is reduced

Common Data for Questions 46 and 47:

The starting current of a 3- ϕ SCIM is 8 times the full-load current and the full-load slip is 5%.

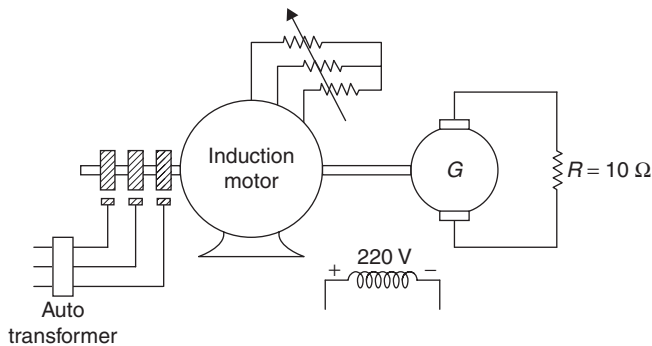
46. If an auto transformer is used to reduce voltage at starting, the auto transformer ratio to provide 2 PU starting torque
(A) 79% (B) 57.7%
(C) 70.7% (D) 82%
47. If a starting torque of 0.5 PU is required, then starting current in PU should be
(A) 2.16 (B) 4.16
(C) 3.75 (D) 3.16

Common Data for Questions 48 and 49:

A slip-ring IM has the following name plate details

$$V : 400 \text{ V}, f : 50 \text{ Hz, Poles } 6, V_{ph} : 3$$

It is fed from rotor side through an auto transformer and stator is connected to a variable resistance.



The motor is coupled to a 220 V, separately excited DC generator feeding power to fixed resistance of 20Ω . By adjusting the variable resistance, motor is run at 960 rpm. Two watt meters used to measure input power gives the following readings $\rightarrow W_1 = 2000 \text{ W}$, $W_2 = -400 \text{ W}$.

48. The speed of rotation of stator magnetic field with respect to rotor structure is
(A) 40 rpm in direction of rotation
(B) 40 rpm opposite to direction of rotation
(C) 1000 rpm in direction of rotation
(D) 1000 rpm opposite to direction of rotation
49. Neglecting all losses of both the machines, generator power output and current through resistance
(A) 64 W, 1.79 A
(B) 1236 W, 7.86 A
(C) 1536 W, 8.76 A
(D) 1880 W, 9.7 A

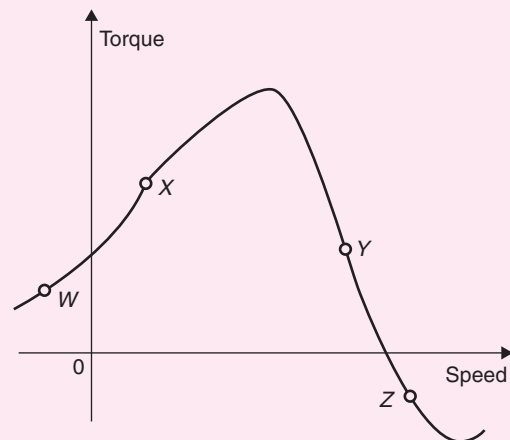
Common Data for Question 50:

Consists of two statements: one is Assertion (A) and the other is Reason (R). You have to examine these two statements and select the answer using the code given below.

- (A) Both A and R are individually correct and R is the correct explanation of A
(B) Both A and R are individually correct, but R is not the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true
50. A: The maximum value of developed torque in a 3- ϕ IM remains a constant for various speeds in the sub synchronous region in the v/f control.
R: The ratio of voltage and frequency over the considered speed range is maintained a constant and hence flux in the machine is almost constant.

PREVIOUS YEARS' QUESTIONS

1. A 400 V, 15 kW, 4-pole, 50 Hz, Y-connected IM has full-load slip of 4%. The output torque of the machine at full load is [2004]
(A) 1.66 Nm (B) 95.50 Nm
(C) 99.47 Nm (D) 624.73 Nm
2. On the torque/speed curve of IM shown in the below figure, four points of operation are marked as W, X, Y and Z. Which one of them represents the operation at a slip greater than 1? [2005]



- (A) W (B) X
(C) Y (D) Z

3. For an IM, operating at a slip s , the ratio of gross power output to air gap power is equal to: [2005]
 (A) $(1-s)^2$ (B) $(1-s)$
 (C) $\sqrt{1-s}$ (D) $(1-\sqrt{s})$
4. Under no load condition, if the applied voltage to an IM is reduced from the rated voltage to half the rated value, [2005]
 (A) the speed decreases and the stator current increases
 (B) both the speed and the stator current decrease
 (C) the speed and the stator current remain practically constant
 (D) there is negligible change in the speed but the stator current decreases
5. A three-phase cage IM is started by direct-on-line (DOL) switching at the rated voltage. If the starting current drawn is 6 times the full-load current, and the full-load slip is 4%, the ratio of the starting developed torque to the full-load torque is approximately equal to [2005]
 (A) 0.24 (B) 1.44
 (C) 2.40 (D) 6.00
6. In a single-phase IM driving a fan load, the reason for having a high resistance rotor is to achieve [2005]
 (A) low starting torque
 (B) quick acceleration
 (C) high efficiency
 (D) reduced size
7. Determine the correctness of otherwise of the following **Assertion [a]** and the **Reason [r]**.
Assertion: Under V/f control of IM, the maximum value of the developed torque remains constant over a wide range of speed in the sub-synchronous region.
Reason: The magnetic flux is maintained almost constant at the rated value by keeping the ratio V/f constant over the considered speed range. [2005]
 (A) Both [a] and [r] are true and [r] is the correct reason for [a]
 (B) Both [a] and [r] are true but [r] is not the correct reason for [a]
 (C) Both [a] and [r] are false
 (D) [a] is true but [r] is false
8. For a single-phase capacitor start IM which of the following statements is valid? [2006]
 (A) The capacitor is used for power factor improvement
 (B) The direction of rotation can be changed by reversing the main winding terminals
 (C) The direction of rotation cannot be changed
 (D) The direction of rotation can be changed by interchanging the supply terminals
9. The speed of a 4-pole IM is controlled by varying the supply frequency while maintaining the ratio of supply frequency (V/f) constant. At rated frequency of 50 Hz and rated voltage of 400 V its speed is 1440 rpm. Find the speed at 30 Hz, if the load torque is constant. [2006]
 (A) 882 rpm (B) 864 rpm
 (C) 840 rpm (D) 828 rpm
10. A 3-phase, 4-pole, 400 V, 50 Hz, star-connected IM has following circuit parameters $r_1 = 1.0 \Omega$, $r_2' = 0.5 \Omega$, $x_1 = x_2' = 1.2 \Omega$, $x_m = 35 \Omega$ The starting torque when the motor is started direct-on-line is (use approximate equivalent circuit model) [2006]
 (A) 63.6 Nm (B) 74.3 Nm
 (C) 190.8 Nm (D) 222.9 Nm
11. A 3-phase, 10 kW, 400 V, 4-pole, 50 Hz, star-connected IM draws 20 A on full load. Its no load and blocked rotor test data are given below:
 No Load test: 400 V 6A 1002 W
 Blocked Rotor test: 90 V 15 A 762 W
 Neglecting copper loss in no Load test and core loss in Blocked Rotor test, estimate motor's full load efficiency. [2006]
 (A) 76% (B) 81%
 (C) 82.4% (D) 85%
12. A three-phase SCIM has a starting torque of 150% and a maximum torque of 300% with respect to rated torque at rated voltage and rate frequency. Neglect the stator resistance and rotational losses. The value of slip for maximum torque is [2007]
 (A) 13.48% (B) 16.24%
 (C) 18.92% (D) 26.79%
- Common Data for Questions 13 to 15:**
 A three-phase SCIM has a starting current of seven times the full-load current and full-load slip of 5%
13. If an autotransformer is used for reduced voltage starting to provide 1.5 per unit starting torque, the autotransformer ratio (%) should be [2007]
 (A) 57.77% (B) 72.56%
 (C) 78.25% (D) 81.33%
14. If a star-delta starter is used to start this IM, the per unit starting torque will be [2007]
 (A) 0.607 (B) 0.816
 (C) 1.225 (D) 1.616
15. If a starting torque of 0.5 per unit is required then. The per unit starting current should be [2007]
 (A) 4.65 (B) 3.75
 (C) 3.16 (D) 2.13
16. A 230 V, 50 Hz, 4-pole, single-phase IM is rotating in the clockwise (forward) direction at a speed of 1425 rpm. If the rotor resistance at standstill is 7.8Ω

then the effective rotor resistance in the backward branch of the equivalent circuit will be [2008]

- (A) $2\ \Omega$ (B) $4\ \Omega$
(C) $78\ \Omega$ (D) $156\ \Omega$

17. A 400 V, 50 Hz, 30 hp, three-phase IM is drawing 50 A current at 0.8 power factor lagging. The stator and rotor copper losses are 1.5 kW and 900 W, respectively. The friction and windage losses are 1050 W and the core losses are 1200 W. The air-gap power of the motor will be [2008]

- (A) 23.06 kW (B) 24.11 kW
(C) 25.01 kW (D) 26.21 kW

18. A 400 V, 50 Hz, 4-pole, 1400 rpm, star-connected SCIM has the following parameters referred to the stator.

$$R'_r = 1.0\ \Omega, X'_s = X'_r = 1.5\ \Omega.$$

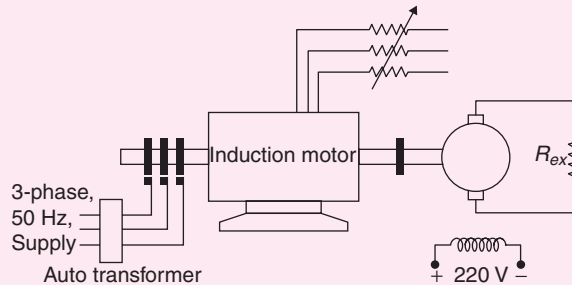
Neglect stator resistance and core and rotational losses of the motor.

The motor is controlled from a 3-phase voltage source inverter with constant V/f control. The stator line-to-line voltage (rms) and frequency to obtain the maximum torque at starting will be [2008]

- (A) 20.6 V, 2.7 Hz
(B) 133.3 V, 16.7 Hz
(C) 266.6 V, 33.3 Hz
(D) 323.3 V, 40.3 Hz

Common Data for Questions 19 and 20:

A 3-phase, 440 V, 50 Hz, 4-pole, slip-ring IM is fed from the rotor side through an auto-transformer and the stator is connected to a variable resistance as shown in the figure.



The motor is coupled to a 220 V, separately excited DC generator feeding power to fixed resistance of $10\ \Omega$. Two-wattmeter method is used to measure the input power to IM. The variable resistance is adjusted such that the motor runs at 1410 rpm and the following readings were recorded.

$$W_1 = 1800\ \text{W}, W_2 = -200\ \text{W}$$

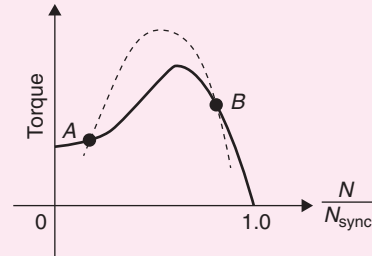
19. The speed of rotation of stator magnetic field with respect to rotor structure will be [2008]

- (A) 90 rpm in the direction of rotation
(B) 90 rpm in the opposite direction of rotation
(C) 1500 rpm in the direction of rotation
(D) 1500 rpm in the opposite direction rotation.

20. Neglecting all losses of both the machines, the DC generator power output and the current through resistance (R_{ex}) will, respectively, be [2008]

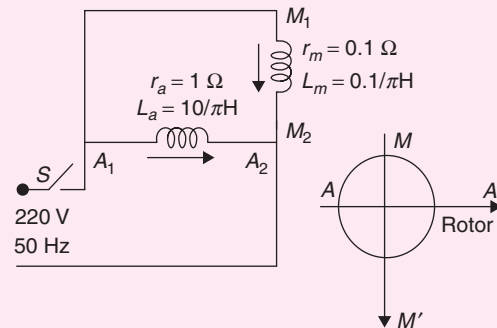
- (A) 96 W, 3.10 A (B) 120 W, 3.46 A
(C) 1504 W, 12.26 A (D) 1880 W, 13.71 A

21. A 3-phase SCIM supplied from a balanced 3-phase source drives a mechanical load. The torque-speed characteristics of the motor (solid curve) and of the load (dotted curve) are shown. Of the two equilibrium points A and B , which of the following options correctly describes the stability of A and B ? [2009]



- (A) A is stable B is unstable
(B) A is unstable B is stable
(C) Both are stable
(D) Both are unstable

22. A 220 V, 50 Hz, single-phase IM has the following connection diagram and winding orientations shown. MM' is the axis of the main stator winding (M_1M_2) and AA' is that of the auxiliary winding (A_1A_2). Directions of the winding axes indicate direction of flux when currents in the windings are in the directions shown. Parameters of each winding are indicated. When switch S is closed, the motor [2009]



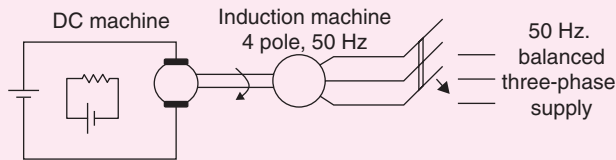
- (A) rotates clockwise
(B) rotates anticlockwise
(C) does not rotate
(D) rotates momentarily and comes to a halt

23. A balanced three-phase voltage is applied to a star-connected IM, the phase to neutral voltage being V . The stator resistance, rotor resistance referred to the stator, stator leakage reactance, rotor leakage reactance referred to the stator, and the magnetizing

reactance are denoted by r_s , r_r , x_s , x_r and X_m , respectively. The magnitude of the starting current of the motor is given by [2010]

- (A) $\frac{V}{\sqrt{(r_s + r_r)^2 + (x_s + x_r)^2}}$
 (B) $\frac{V}{\sqrt{r_s^2 + (x_s + X_m)^2}}$
 (C) $\frac{V}{\sqrt{(r_s + r_r)^2 + (X_m + x_r)^2}}$
 (D) $\frac{V}{\sqrt{r_s^2 + (X_m + x_r)^2}}$

24. A separately excited DC machine is coupled to a 50 Hz, three-phase, 4-pole induction machine as shown in the figure. The DC machine is energized first and the machines rotate at 1600 rpm. Subsequently the induction machine is also connected to a 50 Hz, three-phase source, the phase sequence being consistent with the direction of rotation. In steady state, [2010]



- (A) Both machines act as generators
 (B) The DC machine acts as a generator, and the induction machine acts as a motor
 (C) The DC machine acts as a motor, and the induction machine acts as a generator
 (D) Both machines act as motors
25. A three-phase 440 V, 6-pole, 50 Hz, SCIM is running at a slip of 5%. The speed of stator magnetic field with respect to rotor magnetic field and speed of rotor with respect to stator magnetic field are. [2011]
- (A) zero, -50 rpm (B) zero, 950 rpm
 (C) 1000 rpm, -5 rpm (D) 1000 rpm, 950 rpm
26. The slip of an IM normally does not depend on [2012]

- (A) rotor speed
 (B) synchronous speed
 (C) shaft torque
 (D) core-loss component
27. The locked rotor current in a 3-phase, star-connected 15 kW, 4-pole, 230 V, 50 Hz IM at rated conditions is 50 A. Neglecting losses and magnetizing current, the approximate locked rotor line current drawn when the motor is connected to a 236 V, 57 Hz supply is [2012]
- (A) 58.5 A (B) 45.0 A
 (C) 42.7 A (D) 55.6 A

28. Leakage flux in an IM is [2013]
- (A) flux that leaks through the machine
 (B) flux that links both stator and rotor windings
 (C) flux that links none of the windings
 (D) flux that links the stator winding or the rotor winding but not both

29. A 4-pole IM, supplied by a slightly unbalanced three-phase 50 Hz source, is rotating at 1440 rpm. The electrical frequency in Hz of the induced negative sequence current in the rotor is [2013]
- (A) 100 (B) 98
 (C) 52 (D) 48

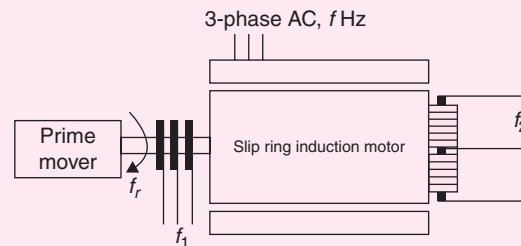
30. An 8-pole, 3-phase, 50 Hz IM is operating at a speed of 700 rpm. The frequency of the rotor current of the motor in Hz is [2014]

31. A 3-phase, 50 Hz, 6-pole IM has a rotor resistance of 0.1Ω and reactance of 0.92Ω . Neglect the voltage drop in stator and assume that the rotor resistance is constant. Given that the full-load slip is 3%, the ratio of maximum torque to full-load torque is [2014]
- (A) 1.567 (B) 1.712
 (C) 1.948 (D) 2.134

32. A three-phase, 4-pole, self-excited induction generator is feeding power to a load at a frequency f_1 . If the load is partially removed, the frequency becomes f_2 . If the speed of the generator is maintained at 1500 rpm in both the cases, then [2014]
- (A) $f_1, f_2 > 50$ Hz and $f_1 > f_2$
 (B) $f_1 < 50$ Hz and $f_2 > 50$ Hz
 (C) $f_1, f_2 < 50$ Hz and $f_2 > f_1$
 (D) $f_1 > 50$ Hz and $f_2 < 50$ Hz

33. A single-phase IM draws 12 MW power at 0.6 lagging power. A capacitor is connected in parallel to the motor to improve the power factor of the combination of motor and capacitor to 0.8 lagging. Assuming that the real and reactive power drawn by the motor remains same as before, the reactive power delivered by the capacitor in MVAR is [2014]

34. A three-phase slip-ring IM, provided with a commutator winding, is shown in the figure. The motor rotates in clockwise direction when the rotor windings are closed.

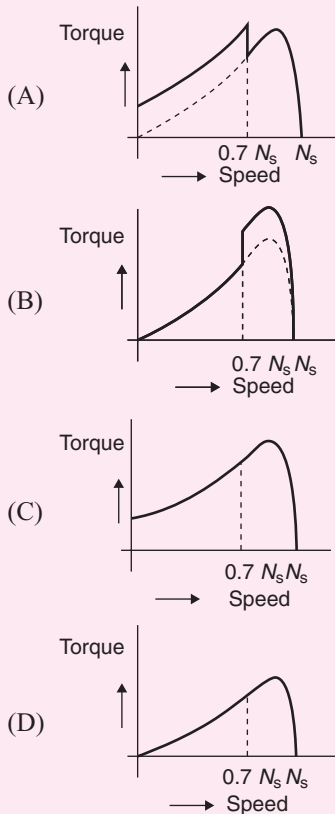


If the rotor winding is open-circuited and the system is made to run at rotational speed f_r with the help of prime-mover in anticlockwise direction, then the frequency of

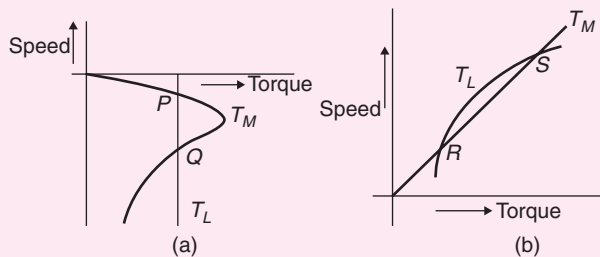
voltage across slip rings is f_1 and frequency of voltage across commutation brushes is f_2 . The values of f_1 and f_2 , respectively, are [2014]

- (A) $f + f_r$ and f (B) $f - f_r$ and f
(C) $f - f_r$ and $f + f_r$ (D) $f + f_r$ and $f - f_r$

35. A single-phase IM is provided with capacitor and centrifugal switch in series with auxiliary winding. The switch is expected to operate at a speed of $0.7 N_s$, but due to malfunctioning the switch fails to operate. The torque-speed characteristic of the motor is represented by [2014]



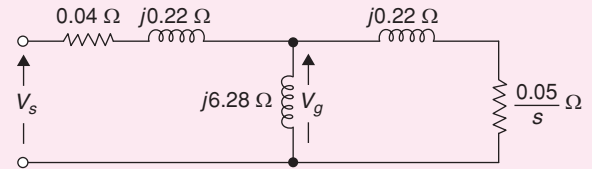
36. The torque-speed characteristics of motor T_M and load (T_L) for two cases are shown in the figures (a) and (b). The load torque is equal to motor torque at points P, Q, R and S



The stable operating points are [2014]

- (A) P and R (B) P and S
(C) Q and R (D) Q and S

37. The figure shows the per-phase equivalent circuit of a two-pole three-phase induction motor operating at 50 Hz. The “air-gap” voltage, V_g across the magnetizing inductance, is 210 V rms, and the slip, s , is 0.05. The torque (in Nm) produced by the motor is _____. [2015]



38. A 220 V, 3-phase, 4-pole, 50 Hz induction motor of wound rotor type is supplied at rated voltage and frequency. The stator resistance, magnetizing reactance, and core loss are negligible. The maximum torque produced by the rotor is 225% of full load torque and it occurs at 15% slip. The actual rotor resistance is 0.03 Ω /phase. The value of external resistance (in Ohm) which must be inserted in a rotor phase if the maximum torque is to occur at start is _____. [2015]

39. The direction of rotation of a single-phase capacitor run induction motor is reversed by [2016]

- (A) Interchanging the terminals of the AC supply
(B) Interchanging the terminals of the capacitor
(C) Interchanging the terminals of the auxiliary winding
(D) Interchanging the terminals of both the windings.

40. The starting line current of a 415V, 3-phase, delta connected induction motor is 120A, when the rated voltage is applied to its stator winding. The starting line current at a reduced voltage of 110V, in ampere is _____. [2016]

ANSWER KEYS

EXERCISES

Practice Problems 1

1. D	2. B	3. B	4. B	5. B	6. C	7. D	8. B	9. A	10. B
11. C	12. A	13. A	14. C	15. C	16. A	17. C	18. B	19. B	20. B
21. D	22. D	23. D	24. B	25. C	26. D	27. D			

Practice Problems 2

1. B	2. C	3. A	4. B	5. B	6. D	7. A	8. C	9. C	10. C
11. B	12. C	13. B	14. D	15. B	16. D	17. C	18. A	19. C	20. A
21. C	22. D	23. C	24. B	25. B	26. C	27. C	28. C	29. B	30. B
31. C	32. B	33. D	34. C	35. B	36. C	37. D	38. C	39. A	40. C
41. D	42. C	43. C	44. D	45. B	46. A	47. D	48. A	49. C	50. A

Previous Years' Questions

1. C	2. A	3. B	4. D	5. B	6. B	7. A	8. B	9. C	10. A
11. B	12. D	13. C	14. B	15. C	16. A	17. C	18. B	19. D	20. C
21. A	22. B	23. A	24. C	25. A	26. D	27. B	28. D	29. B	30. 3.35
31. C	32. C	33. 7	34. A	35. C	36. B	37. 400 to 403		38. 0.16 to 0.18	
39. C	40. 31.8								

Chapter 5

Synchronous Machines

LEARNING OBJECTIVES

After reading this chapter, you will be able to understand:

- Armature windings
- Damper windings
- EMF equation
- Flux and MMF phasors in synchronous machines
- Operation of a salient pole synchronous machine
- Slip test
- Parallel operation of alternators
- Synchronizing power and synchronizing torque
- Synchronous motors
- Servomotor
- Stepper motor

INTRODUCTION

A synchronous machine is a doubly-excited AC machine, because its field winding is energized from a DC source and its armature winding is connected to an AC source. When operating as a motor, the synchronous machine absorbs active power from an AC source. During its operation as a generator, a synchronous machine delivers AC power. However, the field winding of a synchronous machine always absorbs power from a DC source.

Since a synchronous generator delivers AC output, it is also known as an alternator. Alternators operate on the same fundamental principles of electromagnetic induction as DC generators. In large synchronous machines, the field winding is always provided on the rotor because of the following advantages:

1. The output current can be led directly from fixed terminals on the stator to the load circuit.
2. Insulation of stationary armature winding for high AC voltage is easier compared to rotating armature.
3. The sliding contacts i.e. slip rings are transferred to the low-voltage, low power DC field circuit which can, therefore, be easily insulated.
4. The armature conductors can be made easily braced to prevent any deformation which could be produced by the high centrifugal forces or the mechanical stresses set up as a result of short-circuited currents.

- Small AC generators and of low voltage rating are usually made of rotating armature. In such generators the required magnetic field is produced by DC electromagnet placed on the stator.

In large alternators placed on the excitation is usually provided from a small DC shunt or compound generator mounted on the shaft of alternator itself.

- Synchronous machines are of two types depending upon the geometrical structure of the rotor, viz., (i) salient pole or projected pole and (ii) cylindrical-rotor, round-rotor or non-salient pole type.
- The cylindrical construction is used for two or four-pole steam-turbine generators. That is why these are called turbo-generators. A cylindrical-rotor synchronous machine is characterized by long core length and small diameter so as to limit the centrifugal forces developed in the high-speed rotor.
- The salient-pole construction is the most suitable for multipolar slow-speed water-turbine generators. That is why salient-pole synchronous generators are called hydro-generators. A salient-pole synchronous machine has small core length and large diameter so that large number of field poles can be accommodated on the rotor periphery.

The frequency of the induced emf is given by $f = \text{No. of cycles/revolution} \times \text{No. of revolutions/second}$

$$= \frac{P}{2} \times \frac{N}{60} = \frac{PN}{120} \text{ Hz}$$

Armature Windings

The armature windings in DC machines are closed winding but in case of alternator, windings are open, in the sense that there is no closed path for the armature currents in the winding itself. Hence there are no parallel paths for an AC winding.

The two types of armature windings commonly used for 3-phase alternators are

1. Single-layer winding
2. Double-layer winding

There are two types of interconnection of the 3-phase windings. One Y and other Δ . If similar terminals are joined, we get a star configuration and for delta connection dissimilar ends are connected together.

Damper Windings

Damper windings consist of low-resistance bars embedded in slots in the pole-faces of salient-pole machines. The projecting ends of the bars are short-circuited at both ends by strips of the same material as used for bars. Damper windings are not used on turbo-generators. But the solid-steel rotor cores of such machines provide path for eddy currents, especially in the quadrature axis, where the iron may form an equivalent circuit, thus producing the same effects as those of damper bars. These damper bars are useful in preventing the hunting (monetary speed fluctuations) in generators and are needed in synchronous motors to provide the starting torque. Under normal running conditions, damper bars do not carry any current because rotor runs at synchronous speed.

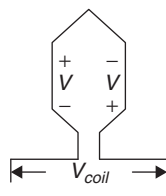
Short-pitch winding: Pitch factor/chording factor.

Short-pitched coils are used owing to the following advantages.

1. Saving of copper due to reduced length of end connections.
2. Waveform of generated emf is improved as the distorting harmonics are reduced or totally eliminated.
3. Due to the elimination of high frequency harmonics, eddy currents and hysteresis losses are reduced thereby increasing the efficiency.

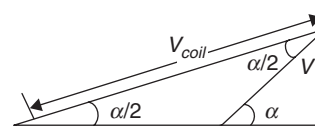
Disadvantages

Use of short-pitched coil results in decrease of induced emf. In full pitched coils, emf's induced by two coils sides are exactly 180° out of phase:



Hence $V_{coil} = 2V$

But, in case of short-pitched coils, this angle is $180 - \alpha$ $\therefore$ (say)



$$\therefore \text{Hence } V_{coil} = 2V \cos \frac{\alpha}{2}$$

$$\text{Pitch factor} = \frac{\text{Resultant emf of short pitched coil}}{\text{Resultant emf of full coil}}$$

$$= \frac{2V \cos \alpha/2}{2V} = \cos \alpha/2 \quad (\alpha \text{ in electrical degrees})$$

Distribution or Breadth Factor or Winding Factor or Spread Factor

In practice, not all coil slots are used. The winding is neither fully distributed in all slots, nor fully concentrated under one pole. This is indicated by distribution factor:

The distribution factor (K_d) is defined as

$$K_d = \frac{\text{e.m.f with distributed winding}}{\text{e.m.f with concentrated winding}} \quad (\text{or})$$

$$\frac{\text{Vector sum of coil e.m.f. s}}{\text{Arithmetic sum of coil e.m.f. s}}$$

$$K_d = \frac{\sin m\beta/2}{m \sin \beta/2}$$

where β is slot angle

$$\beta = \frac{180^\circ}{\text{No. of slots/pole}}$$

m = No. of slots/pole/phase

$m\beta$ is phase spread angle.

Solved Examples

Example 1: A 3-phase, 16-pole alternator has 144 slots. What is the distribution factor?

Solution: Distribution factor, $K_d = \frac{\sin(m\beta/2)}{m \sin \beta/2}$

n = no. of slots per pole

m = no. of slots per pole/phase

β = slot angle

$$n = \frac{144}{16} = 9; \quad m = \frac{144}{3 \times 16} = 3$$

$$\beta = \frac{180}{9} = 20^\circ$$

$$K_d = \frac{\sin(3 \times 20/2)}{3 \sin \frac{20}{2}} = 0.96$$

Example 2: What is the distribution factor for a uniformly distributed 3-phase winding having a phase spread of 60° ?

Solution: For a uniformly distributed winding, distribution factor is given by

$$k_d = \frac{\sin \sigma / 2}{\sigma / 2}$$

where σ = Phase spread in rad.

For $\sigma = 60^\circ$

$$k_d = \frac{\sin 30^\circ}{30 \times \frac{\pi}{180}} = 0.955$$

EMF EQUATION

The induced emf (RMS value) of an alternator is given by

$$E/\text{Ph} = 4.44 K_p K_d f \phi T \dots\dots V$$

where K_p = Pitch factor

K_d = Distribution factor

ϕ = Flux per pole in webers

T = Number of turns per phase

Effect of Harmonics on Pitch and Distribution Factors

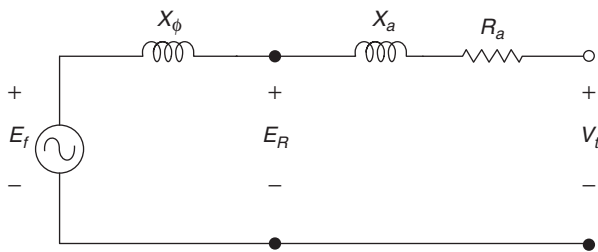
For an n^{th} -order harmonic

$$K_c = \cos \frac{n\alpha}{2} \text{ and}$$

$$K_d = \frac{\sin nm\beta/2}{m \sin n\beta/2}$$

FLUX AND MMF PHASORS IN SYNCHRONOUS MACHINES

Synchronous machine equivalent circuit:



where $X_s = X_\phi + X_a$
 X_a = Armature Leakage Reactance
 X_ϕ = Reactance of air gap
 R_a = Armature Resistance
 E_f = emf generated
 V_t = Terminal voltage

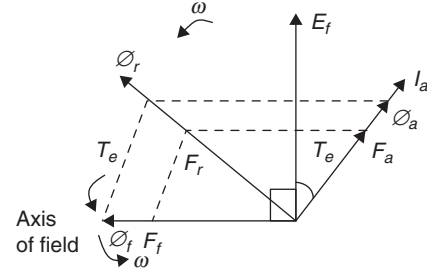


Figure 1 Axis of phase a

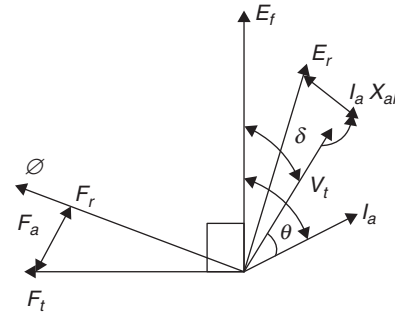
Combined space and time phasor diagram of a cylindrical rotor alternator for a lagging p.f. load.

In an alternator

1. If I_a lags the excitation emf, E_f by 90° the nature of F_a is demagnetizing.
2. If I_a leads the excitation emf, E_f by 90° , the nature of armature F_a is magnetizing.

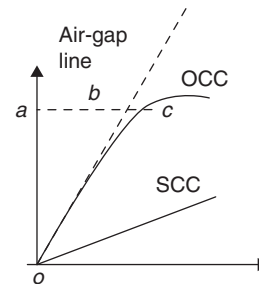
Phasor Diagram of a Cylindrical Rotor Alternator

The phasor diagram is drawn assuming that the per phase quantities, i.e. V_t the terminal voltage, I_a the armature current, r_a the armature resistance, X_{al} the armature leakage reactance and $\cos \theta$ power factor, are known.



Characteristics of an Alternator Open-circuit Characteristic (OCC)

For obtaining OCC, the alternator is driven at constant rated speed and the open-circuit terminal voltage is noted as the field current is increased from zero. Thus the OCC is a graph between the field current I_f or field mmf F_f and the induced voltage. It is just like the B-H curve.



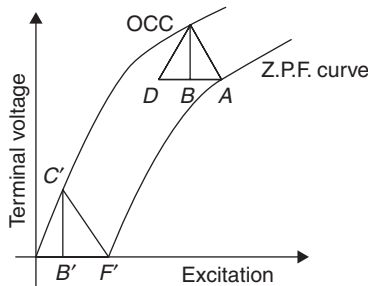
Short-circuit Characteristics (SCCs)

SCC is obtained by short-circuiting the armature (i.e. stator) windings through a low-resistance ammeter. The field current is gradually increased from zero, until the short-circuit armature current has not reached its maximum safe value, equal to 125% to 150% of the rated current. During this test, the speed which is not necessarily synchronous is kept constant.

Zero Power-factor Characteristic and Potier Triangle

Zero power factor characteristic (z.p.f.c.) of an alternator is a plot between the armature terminal voltage and its field current for constant values of armature current and speed. z.p.f.c. in conjunction with o.c.c. is useful in obtaining the armature leakage reactance X_{al} and armature reaction mmf F_a . For an alternator z.p.f.c. is obtained as follows:

1. The synchronous machine is run at rated synchronous speed by the prime mover.
2. A purely inductive load (z.p.f. load) is connected across the armature terminals and field current is increased till full-load armature current is flowing.
3. The load is varied in steps and the field current at each step is adjusted to maintain armature current. The plot of armature terminal voltage and field current recorded at each step gives the zero-power factor characteristic at full-load armature current.



The triangle ABC shown above is known as Potier triangle. If the armature resistance is assumed zero and the armature current is kept constant, then the size of Potier triangle ABC remains constant and can be shifted parallel to itself with its corner remaining on the OCC and its corner A, tracing the z.p.f.c. Thus the z.p.f.c. has the same shape as the O.C.C. and is shifted vertically down-ward by an amount equal to $F_a X_{al}$ (i.e. leakage reactance drop) and horizontally to the right by an amount equal to the armature reaction mmf F_a or the field current equivalent to the armature reaction mmf, i.e. $BC = I_a X_{al}$ and $AB = F_a$. For determining X_{al} and F_a experimentally, entire z.p.f.c. is not necessary, but only two points A and F' are sufficient. The point A (PA = rated voltage) is obtained actually loading the over-excited alternator by an under-excited synchronous motor, so that rated armature current flows in the alternator. The point F' on the z.p.f.c. corresponds to the zero terminal voltage and

can, therefore, be obtained by performing short-circuit test. So here OF' is the field current required to circulate short-circuit current equal to the armature current at which point A is determined in the zero power factor test.

Now draw a horizontal line AD, parallel and equal to F'O. Through point D, draw a straight line parallel to the air-gap line, intersecting the O.C.C. at C. Draw CB perpendicular to AD. Then from the Potier triangle ABC

$$BC = I_a X_{al} \quad \text{and} \quad AB = F_a$$

Since I_a at which point A is obtained is known, X_{al} can be calculated.

Voltage Regulation of an Alternator

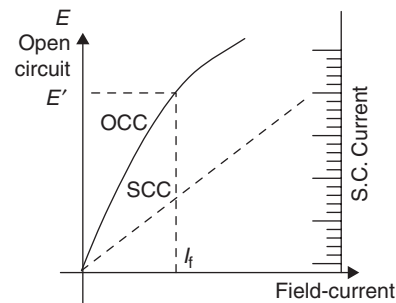
It is defined as the 'rise in voltage when full-load is removed (field excitation and speed remaining the same)', expressed as a percentage (or p.u.) of the rated voltage.

$$\begin{aligned} \therefore \text{Voltage regulation} &= \frac{E_f - V_t}{V_t} \text{ in p.u.} \\ &= \frac{E_f - V_t}{V_t} \times 100 \text{ in percentage.} \end{aligned}$$

The following are the methods to determine voltage regulation.

Synchronous Impedance or EMF Method

In this method, OCC and SCC of given alternator are obtained and are drawn on a common field current base.



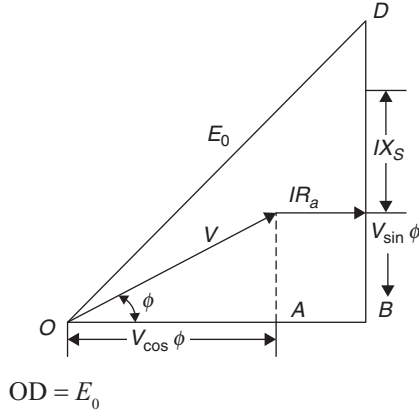
Consider a field current I_f . The OC voltage corresponding to this field current is E_1 . When the armature winding is short-circuited, whole of this voltage E_1 is used to circulate the armature short-circuit current I_1 , against the synchronous impedance Z_s

$$E_1 = I_1 Z_s; \quad \therefore Z_s = \frac{E_1 (\text{open-circuit})}{I_1 (\text{short-circuit})}$$

After determining R_a , X_s can be calculated as

$$X_s = \sqrt{Z_s^2 - R_a^2}$$

Knowing R_a and X_s , the phasor diagram can be drawn for any load and any power factor.



$$OD = E_0$$

$$\therefore E = \sqrt{OB^2 + BD^2}$$

$$E_0 = \sqrt{(V \cos \phi + IR_a)^2 + (V \sin \phi + IX_s)^2}$$

$$\text{and \% regulation} = \frac{E_0 - V}{V} \times 100$$

For leading p.f., ϕ is taken -ve and +ve for lagging p.f.

The synchronous impedance Z_s calculated by this method is higher than actual value. Hence the value of regulation so obtained is always more than that found from an actual test. That is why it is called pessimistic method.

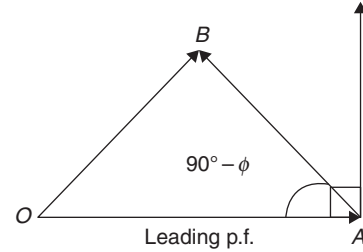
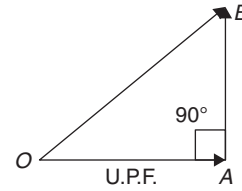
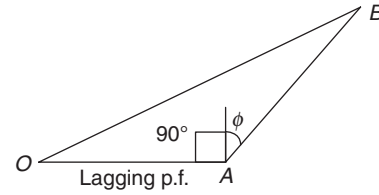
Ampere-turn or MMF Method

- In this method, OC and SC data are used.
- The voltage is attributed entirely to armature reaction (i.e. armature leakage reaction drop, which is rather small, is clubbed with armature reaction). In other words, it is assumed that the change in terminal voltage on load is entirely due to armature reaction. Normally R_a is neglected.
- Since R_a is neglected and is small for the low voltage applied on short circuit, the p.f. may be assumed to be zero lagging.
- Therefore, the field mmf is used to overcome the demagnetizing effect of armature reaction.
- The excitation required to overcome the demagnetizing effect of armature reaction is determined on the unsaturated portion of the OCC.
- This method is optimistic since the regulation calculated by this method is less than the actual value.

Determination of Voltage Regulation, by MMF Method

1. The field current I_{f1} corresponding to rated voltage is obtained from the OCC.
2. The field current I_{f2} , to circulate rated armature current I_a , is obtained from the SCC.

3. $I_f \cos \phi$ is the load p.f., I_{f2} is added vectorially at an angle of $(90 \pm \phi)$ with I_{f1} to get the resultant field current I_f as shown in following figures.



$$OA = I_{f1}, AB = I_{f2}$$

$$OB = I_f, \phi = \text{Power factor angle}$$

The emf E , corresponding to I_f is read from the OCC

$$\% \text{ Regulation} = \frac{E - V}{V} \times 100$$

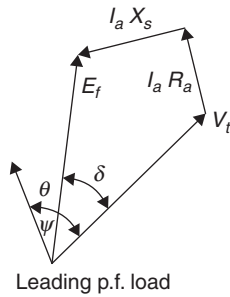
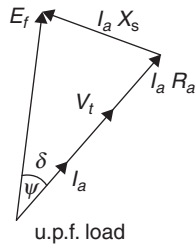
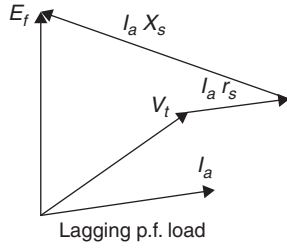
Zero Power Factor or Potier Triangle Method

In the EMF and MMF methods, the effects of armature leakage reactance and armature reaction are clubbed. ZPF method is based on the separation of armature leakage reactance drop and the armature reaction effects. Hence the regulation obtained by this method is more accurate. In addition to no-load curve, full-load ZPF curve is also required.

New A.S.A. (American Standard Association) Method

This method is essentially a modification of the MMF method and gives satisfactory results both for cylindrical and salient-pole machines. New A.S.A. methods require O.C.C. and z.p.f.c.

Voltage Phasor Diagram for an Alternator



For lagging p.f. load

$$E = \sqrt{(V_t \cos \theta + I_a r_a)^2 + (V_t \sin \theta + I_a X_s)^2}$$

For u.p.f. load

$$E = \sqrt{(V_t + I_a r_a)^2 + (I_a X_s)^2}$$

For leading p.f. load

$$E = \sqrt{(V_t \cos \theta + I_a r_a)^2 + (V_t \sin \theta - I_a X_s)^2}$$

Example 3: A three-phase, Y-connected, 1000 kVA, 2000 V, 50 Hz alternator has the following readings:

I_F (A)	10	20	25	30	40	50
OC (V_{LL})	800	1500	1760	2000	2350	2600
SCI _a (A)		200	250	300		
OC V_p	462	866	1016	1155	1357	1501

$$R_a = 0.2 \Omega / \text{phase}$$

Find the full-load voltage regulation at 0.8 p.f. leading.

Solution: Full-load phase voltage = $\frac{2000}{\sqrt{3}} = 1155 \text{ V}$

$$\text{KVA} = \frac{\sqrt{3} V_{LL} I_{fl}}{1000}$$

$$\Rightarrow 1000 \times 1000 = \sqrt{3} \times 2000 \times i_{fl}$$

$$\Rightarrow I_{fc} = 288.7 = I_a$$

at 0.8 leading

$$E = V_p + I_a R_a = 1155 + (288.7 \times 0.2 \angle \cos^{-1}(0.8))$$

$$= 1155 + 46.2 + j34.44 = 1201.7 \angle 1.6^\circ$$

$$I_{F1} = I_F \angle (90 + \alpha) = 32 \angle (90 + 1.6^\circ)$$

(Ampere-turns method)

$$= -0.89 + j 31.98 \text{ A}$$

$$I_{f2} = 29 \angle (180 + 36.87^\circ) = 29 \angle 216.87^\circ \text{ A}$$

$$I_f = I_{f1} + I_{f2} = 28.15 \angle 131.18^\circ$$

at this I_f , $E = 1098 \text{ V}$

$$\Rightarrow V_R = \frac{10998 - 1155}{1155}$$

$$= -5.02\%$$

Example 4: A 3-phase alternator delivers power to a balanced 3-phase load of power factor 0.8 lagging. It is observed that the open-circuit emf phasor leads the corresponding terminal voltage phasor by 15° . Neglecting the effect of harmonics, what is the angle between the axis of the main field mmf and the axis of the armature mmf?

Solution: Angle between E_f and $\phi_{ar} = \delta + \phi$

$$= 15 + \cos^{-1}(0.8) = 51.86^\circ$$

[where δ is load angle and ϕ is power factor angle]

Example 5: A 3-phase, 400 V, delta-connected alternator has a synchronous impedance of $(0 + j20)$ ohm per phase. If it delivers a load of 12 kVA at zero power factor leading, then what is the value of percentage voltage regulation.

Solution: Load = $12 \times 10^3 \text{ VA} = \sqrt{3} \times V_L I_L$

$$\text{Line current } I_L = \frac{12 \times 10^3}{\sqrt{3} \times 400}$$

$$\text{Phase current} = \frac{I_L}{\sqrt{3}} = \frac{12 \times 10^3}{3 \times 400} = 10 \text{ A}$$

$$\% \text{ Regulation} = \frac{IR \cos \phi - I X \sin \phi}{V}$$

$$\therefore \theta = 90^\circ$$

$$\% \text{ Regulation} = \frac{-I X \sin 90}{V}$$

$$= \frac{10 \times 20}{400} = -50\%$$

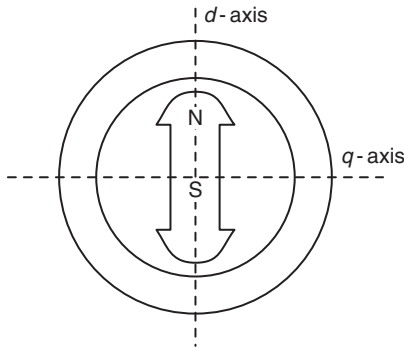
Example 6: A single-phase alternator has a synchronous reactance of 2 ohm and negligible resistance. If it supplies 15 A to a purely capacitive load at 200 V, then what is the value of generated emf?

Solution: Generated emf $E_f = \bar{V}_t + \bar{I}_a X_s$
 $= 200 + (j15)(j2)$
 $= 170 \text{ V}$

OPERATION OF A SALIENT-POLE SYNCHRONOUS MACHINE

Non-salient-pole synchronous machines have a uniform air-gap, because of which its reactance remains the same irrespective of the spatial position of rotor. However a synchronous machine with salient or projected poles has non-uniform air gap due to which its reactance varies with rotor position. A salient-pole machine possesses two-axes of geometric symmetry.

1. Fixed-pole axis called direct axis or d -axis
2. Axis passing through the centre of the inter polar space, called the quadratic axis or q -axis.



To analyse the operation of a salient-pole machine two-reaction theory is used, according to which the armature mmf, F_a is resolved into two sinusoidal components, one F_{ad} along the d -axis and the other F_{aq} along the q -axis. So two mmf act on the d -axis, i.e. field mmf and armature mmf (F_{ad}) whereas only one mmf armature mmf (F_{aq}) acts on the q -axis. The magnetic reluctance is low along the poles and high between the poles.

According to two-reaction theory,

1. armature current I_a can be resolved into two components, i.e. I_d perpendicular to E_0 and I_q along E_0 as shown in figure.

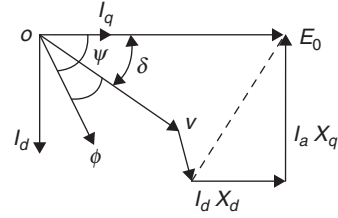
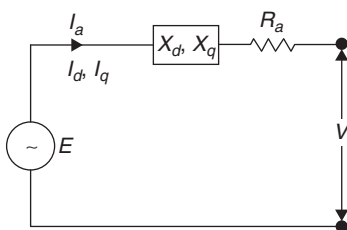


Figure 2 Phasor diagram for a salient-pole alternator

2. armature reactance has two components, i.e. q -axis armature reactance X_{ad} associated with I_d and d -axis armature reactance X_{aq} associated with I_q . If we include the armature leakage reactance X_l , which is the same on both axes, we get

$$X_d = X_{ad} + X_l$$

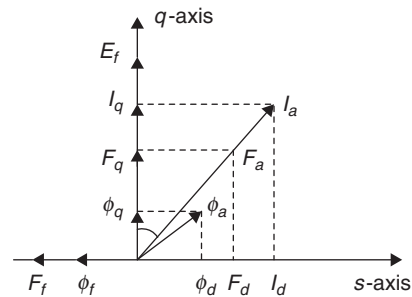
$$\text{and } X_q = X_{aq} + X_l$$

Since reluctance is higher on the q -axis, because of the large air-gap, we have

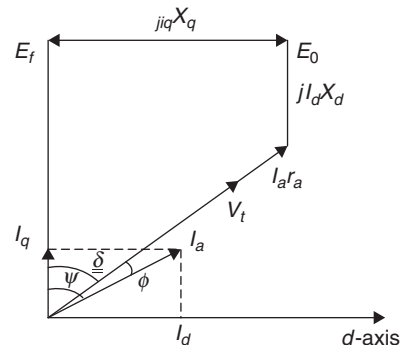
$$X_{aq} < X_{ad}, X_q < X_d \text{ or } X_d > X_q$$

TWO-REACTION THEORY

In a cylindrical rotor synchronous machine, the permeance offered to a mmf wave is Independent on rotor position of the wave axis with respect to field-pole axis. But in salient-pole synchronous the permeance offered is different because the rotor alignment is different therefore the mmf vector also different. The mmf vector and induced emf in salient-pole synchronous machine are as shown below. The reactance of direct and quadrature axis are X_d and X_q , respectively, generally $X_d > X_q$.



The phasor diagram of the generated voltage and terminal voltage of salient-pole synchronous machine is as shown in the following figure.



The direct and quadrature axis currents are

$$I_d = I_a \sin \Psi$$

$$I_q = I_a \cos \Psi$$

The generated emf in salient-pole synchronous machine is

$$E_f = V_t + I_a R_a + jX_d I_d + jX_q I_q$$

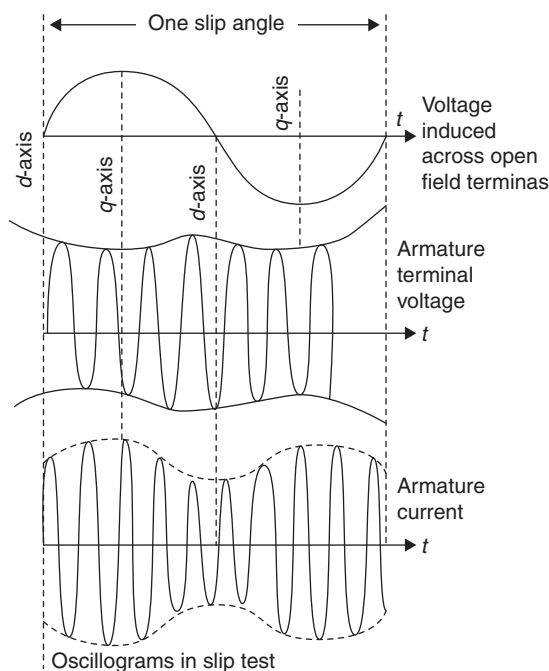
Slip Test

From this test X_d and X_q can be determined. The synchronous machine is driven by a separate prime-mover (or motor) at a speed slightly different from synchronous speed. The field winding is left open, and positive sequence balanced voltages of reduced magnitude (around 25% of rated value) and of rated frequency are impressed across the armature terminals. Under these conditions, the relative velocity between the field poles and the rotating armature mmf wave is equal to the difference between synchronous speed and the rotor speed, i.e. slip speed. A small AC voltage across the open field winding indicates that the field poles and rotating mmf wave are revolving in the same direction - and that is what is required in slip test. If field poles revolve in a direction opposite to the rotating mmf wave, negative sequence reactance would be measured.

The field winding is open-circuited and rotor is rotated at synchronous speed N_s . As the excitation is absent, armature would draw heavy current from terminals. Now, the armature current would vary from minimum (d -axis) and maximum (at the q -axis position).

It would vary at the rate of twice the corresponding slip frequency.

The voltage at the terminal would vary from minimum at (q -axis) to maximum (at d -axis). Oscillations of voltage and current are shown in following figure.



$$= \frac{[\text{Maximum armature terminal voltage per phase}]}{[\text{Minimum armature current per phase}]}$$

and

$$X_q = \frac{[\text{Minimum armature terminal voltage per phase}]}{[\text{Maximum armature current per phase}]}$$

Power Developed by a Synchronous Generator

The per phase power developed by an alternator is given by

$$P_d = \frac{E_0 V}{X_d} \sin \delta + \frac{1}{2} V^2 \left[\frac{1}{X_q} - \frac{1}{X_d} \right] \sin 2\delta$$

$$= \frac{E_0 V}{X_d} \sin \delta + \frac{V^2 (X_d - X_q)}{2 X_d X_q} \sin 2\delta$$

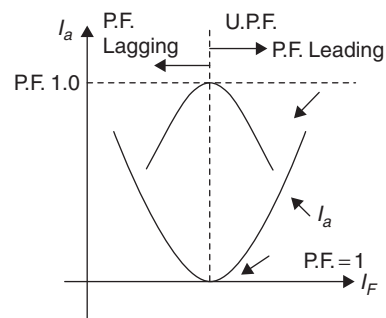
The above expression for P_d consists of two components, the first term represents the power due to field excitation and the second term gives the reluctance power, i.e. power due to saliency.

If $X_d = X_q$, i.e. the machine has a cylindrical rotor, then the reluctance power is zero and the power is given by the first term only. If on the other hand, the field has not excited, i.e. $E_0 = 0$, then the first term in the above expression becomes zero and the power developed is given by second term. The value of δ is positive for a generator and negative for a motor.

Power-factor Control of Alternators

An under excited alternator operates at a leading power factor and absorbs reactive power from infinite bus.

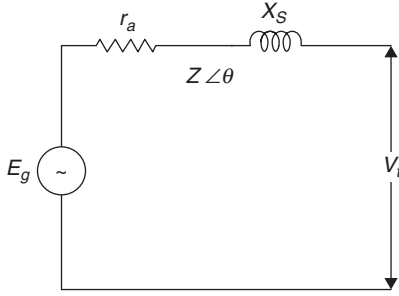
The over excited alternator operates at a lagging p.f. and delivers reactive power to infinite bus.



Effect of field current on an alternator connected to infinite bus is given by the V-curve and inverted V-curve shown in above figure.

Power Angle Characteristics of Synchronous Machine

The power flow in a cylindrical rotor synchronous generator is analysed by



In the above circuit E_g represent the generated voltage in generator and V_t represent the terminal voltage across the generator terminals after subtracting the armature and friction drop from the generated voltage. Therefore the power input (i.e., power developed) in generator

$$P_{in} = \frac{E_g V_t}{Z_s} \sin(\delta - \alpha) + \frac{E_g^2}{Z_s^2} r_a$$

δ – is the angle between generated voltage and terminal voltage.

$$\therefore \theta = \text{impedance angle} = \tan^{-1} \left(\frac{X}{R} \right)$$

$$\alpha = 90^\circ - \theta$$

Maximum power input occurs at $\frac{dP_{in}}{d\delta} = 0$

$$\frac{E_g V_t}{Z_s} \cos(\delta - \alpha) = 0$$

$$\delta = 90^\circ + \alpha = 180^\circ - \theta$$

$$\therefore \text{Maximum input power is } P_{in}(\text{max}) = \frac{E_g V_t}{Z_s} + \frac{E_g^2}{Z_s^2} r_a$$

Power output equation of generator

$$P_{out} = \frac{E_g V_t}{Z_s} \sin(\delta + \alpha) - \frac{V_t^2}{Z_s^2} r_a$$

Maximum real power output obtained at $\frac{dP_{out}}{d\delta} = 0$

$$\frac{E_g V_t}{Z_s} \cos(\delta + \alpha) = 0$$

$$\delta + \alpha = 90^\circ$$

$$\delta = 90^\circ - \alpha = \theta$$

$\therefore$ The corresponding maximum output power is

$$P_{out(\text{max})} = \frac{E_g V_t}{Z_s} + \frac{V_t^2}{Z_s^2} r_a$$

Similarly Reactive power output at generator terminals

$$Q_{out} = V_t \left[\frac{E_g}{Z_s} \sin(\theta - \delta) - \frac{V_t}{Z_s} \sin \theta \right] \text{ (or)}$$

$$Q_{out} = \frac{V_t E_g}{Z_s} \cos(\delta - \alpha) - \frac{V_t^2}{Z_s} X_s$$

Maximum output reactive power obtained at

$$\frac{dQ_{out}}{d\delta} = \frac{V_t E_g}{Z_s} \sin(\delta + \alpha) = 0$$

$$\Rightarrow \delta + \alpha = 0$$

$$\begin{aligned} \therefore Q_{out(\text{max})} &= \frac{V_t}{Z_s} \left[E_g - \frac{V_t}{Z_s} \cdot X_s \right] \\ &= \frac{V_t E_g}{Z_s} = \frac{V_t^2}{Z_s} X_s \end{aligned}$$

If $r_a = 0$

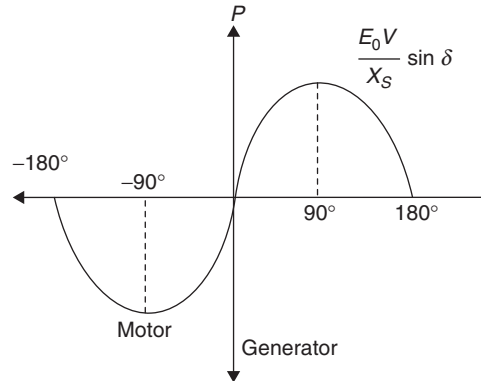
$$\begin{aligned} Q_{out} &= \frac{V_t E_g}{X_s} \cos \delta - \frac{V_t^2}{X_s} \\ &= \frac{V_t}{X_s} [E_g \cos \delta - V_t] \end{aligned}$$

1. when $E_g \cos \delta > V_t$, then Q_{out} positive. Therefore the generator delivers the reactive power.
2. when $E_g \cos \delta = V_t$, then Q_{out} is zero. Therefore the generator neither absorbs nor delivers reactive power.
3. when $E_g \cos \delta < V_t$, then Q_{out} is negative therefore the generator absorbs the reactive power from bus bar.

For cylindrical rotor synchronous machine

$$P = \frac{E_0 V}{X_s} \sin \delta$$

The power vs. load angle curve has sinusoidal shape and is called power angle characteristic.



The Power-angle Curves of Synchronous Generator

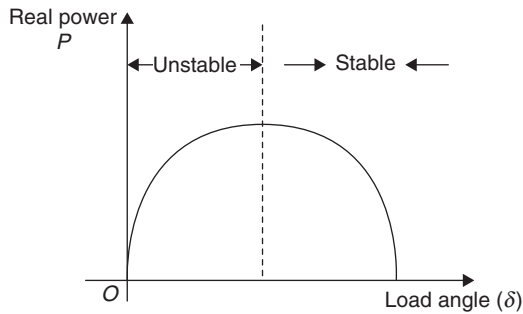


Figure 3 Real power versus load angle characteristic

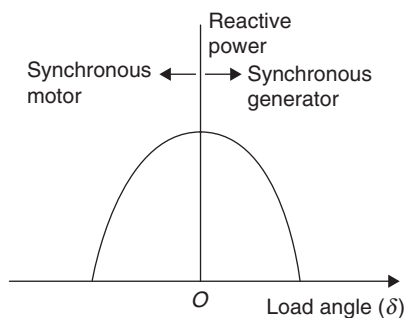
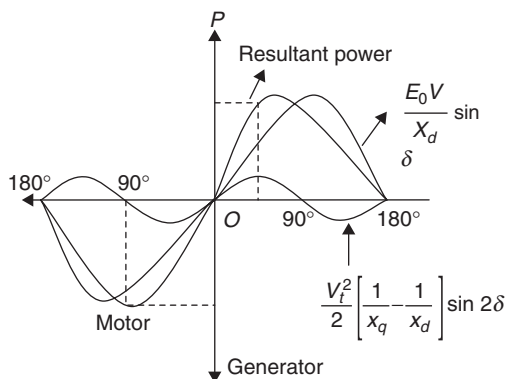


Figure 4 Reactive power versus load angle characteristic

For salient-pole synchronous machine

$$P = \underbrace{\frac{E_0 V}{X_d} \sin \delta}_{\text{Fundamental component}} + \underbrace{\frac{1}{2} V^2 \left[\frac{1}{X_d} - \frac{1}{X_q} \right] \sin 2\delta}_{\text{Second harmonic component}}$$



Common data for Example 7:

A 400 V, 4-pole 3- ϕ 50 Hz, induction motor has a rotor resistance and reactance per phase of 0.02 Ω and 0.2 Ω , respectively. Stator-to-rotor turns ratio is 4.

Example 7: A synchronous generator with a synchronous reactance of 1.2 p.u. is connected to an infinite bus whose voltage is 1 p.u., through an equivalent reactance of 0.3 p.u. What should be the value of emf to give a maximum output of 1.2 p.u.?

Solution: $P = \frac{E_f V_t}{x} \sin \delta$

$$P_{\max} = \frac{E_f V_t}{X}$$

$$E_f = \frac{X P_{\max}}{V_t} = \frac{(1.2 + 0.3) \times 1.2}{1.0} = 1.8 \text{ p.u.}$$

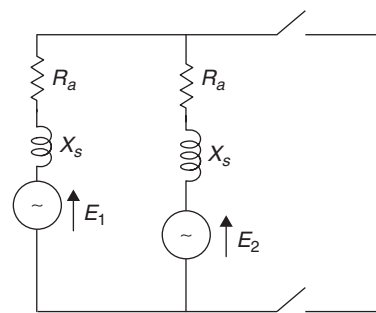
PARALLEL OPERATION OF ALTERNATORS

Connecting an alternator in parallel with another alternator or with common bus-bars is known as synchronizing. Conditions to be satisfied for synchronization of alternators are:

1. The terminal voltage of the incoming alternator must be the same as bus-bar voltage
2. The speed of the incoming machine must be such that its frequency $\left(f = \frac{NP}{120} \right)$ equals bus-bar frequency
3. The phase of the alternators must be same as that of bus-bar voltage, it means that the synchronizing switch must be closed at (or very near) the instant the two voltages have correct phase relationship

Synchronizing Current and Synchronizing Power

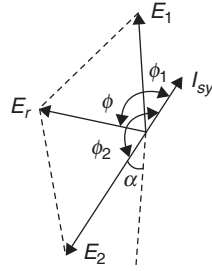
Once synchronized properly, two alternators continue to run in synchronism. Any tendency on the part of one to drop out of synchronism is immediately counteracted by the production of a synchronizing torque, which brings it back to synchronism.



When in exact synchronism, the two alternators have equal terminal voltages and are in exact phase opposition.



But now suppose that E_2 falls back by a phase angle of α electrical degrees as shown in following figure.



Neglecting the resistance R_a , then $\theta = 90^\circ$ and

$$E_r = 2E \sin \alpha/2$$

since α is small (α is in electrical radians)

$$I_{sy} = \frac{E_r}{2X_s} = \frac{\alpha E}{2X_s}$$

From the above phasor diagram, it can be seen that machine no. 1 is generating and supplying synchronizing power whereas machine no. 2 acts a motor and receives the synchronizing power. Hence I_{sy} is generating current with respect to machine no. 1 and motoring current with respect to machine no. 2

$$\begin{aligned} P_{sy} &= E_1 I_{sy} \cos \phi_1 = E I_{sy} \cos (90^\circ - \theta) \\ &= E I_{sy} \sin \theta = E I_{sy} \quad (\because R_a \text{ is neglected and } \theta = 90^\circ) \end{aligned}$$

Substituting the value of I_{sy}

$$\begin{aligned} P_{sy} &= E \cdot \alpha E / 2X_s \\ &= \alpha E^2 / 2X_s \text{ per phase} \end{aligned}$$

$$\left(\text{More accurately, } P_{sy} = \frac{\alpha E^2 \sin \theta}{2X_s} \right)$$

This is the value of the synchronizing power when two alternators are connected in parallel and are on no-load.

Alternators Connected to Infinite Bus-bars

In this case impedance (or reactance) of only one alternator has to be considered

$$\begin{aligned} \therefore I_{sy} &= \frac{\alpha E}{X_s} \\ P_{sy} &= \frac{\alpha E^2}{X_s} \\ \left[\text{or } P_{sy} &= \frac{\alpha E^2 \sin \theta}{X_s} \right] \end{aligned}$$

Synchronizing torque (T_{sy}) can be calculated from the value of synchronizing power by using relation

$$\begin{aligned} \omega \times T_{sy} &= P_{sy} \\ \text{i.e. } T_{sy} &= \frac{P_{sy}}{\omega} \end{aligned}$$

SYNCHRONIZING POWER AND SYNCHRONIZING TORQUE

The rate at which synchronous power, ' P ' varies with δ is called the synchronizing-power coefficient P_{sy} . It is also known as stiffness of coupling, rigidity factor or stability factor.

The coefficient P_{sy} is equal to the slope of the power angle curve.

For a cylindrical rotor machine

$$P_{sy} = \frac{dP}{d\delta} = \frac{EV}{X_s} \cos \delta$$

$$P_{sy} = \frac{dP}{d\delta} = \frac{EV}{X_d} \cos \delta + V^2 \left[\frac{1}{X_q} - \frac{1}{X_d} \right] \cos 2\delta$$

From the above equations it is obvious that P_{sy} is inversely proportional to X_s or X_d and is directly proportional to E , the excitation voltage.

Hence, an over excited synchronous machine is more rigidly coupled than the one which is under excited. Large air gap decreases the value of X_s or X_d , thus a synchronous machine with longer air gap is more stiffer than the one with smaller air-gap units of synchronizing power coefficient are watt per electrical radian.

The variation of synchronous power with the change of load angle is called the synchronizing power. It exists only during the transient state, i.e. whenever there is a sudden disturbance in load (or steady-state operating conditions). Once the steady-state is reached, the synchronizing power reduces to zero.

The synchronizing power flows from or to the bus in order to maintain the relative velocity between interacting stator and rotor field, zero, once the equality is reached, the synchronizing power vanishes.

Hunting

The operation of a synchronous machine is satisfactory, if the mechanical speed of the rotor is equal to the stator field speed i.e. if the relative speed between the rotor and stator field is zero. Any departure from these conditions gives rise to synchronizing forces which tend to maintain this equality. With a sudden change in the load, the rotor swings or oscillates first to one side and then to other side of the new equilibrium position of load angle. This phenomenon involving the oscillations of the rotor about its final equilibrium position, is called hunting. During the rotor oscillation or hunting, the orientation of phasor, E_f changes relative to terminal voltage, V_t and because of this reason, hunting is also called phase-swinging.

Common data for Example 8:

A 3- ϕ , star-connected 400 V, 50 Hz, 4-pole induction motor has the following per phase constants in ohm referred to stator and $N_r = 1440$ rpm.

$$R_1 = .15 \, \Omega, x_1 = .45 \, \Omega, r_2 = 0.12 \, \Omega$$

$$x_2 = 0.45 \, \Omega, x_m = 28.5 \, \Omega$$

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Example 8: A 3000 kVA, 6-pole star-connected alternator runs at 1000 rpm in parallel with other machines on 3300 V bus-bars. The synchronous reactance is 25%. Calculate the synchronizing power for 2° (mech) displacement and the corresponding synchronizing torque.

Solution: Alternator is working in parallel with many alternators, hence it may be considered to be connected to infinite bus-bars.

$$\text{Voltage/phase} = 3300 / \sqrt{3} = 1905 \text{ V}$$

$$\text{Full-load current, } I = 3 \times 10^6 / \sqrt{3} \times 3300 = 525 \text{ A}$$

$$\text{Now, } IX_s = 25\% \text{ of } 1905$$

$$\therefore X_s = 0.25 \times 1905 / 525 \\ = 0.9075 \Omega$$

$$P_{sy} = 3 \times \alpha E^2 / X_s$$

$$\alpha = 2^\circ \text{ mech } \alpha_{elect} = 2 \times \frac{6}{2} = 6^\circ$$

$$\alpha = 6 \times \frac{\pi}{180} = \frac{\pi}{30} \text{ elect rad}$$

$$\therefore P_{sy} = \frac{3 \times \pi \times 1905^2}{30 \times 0.9075 \times 1000} \\ = 314.2 \text{ kW}$$

$$T_{sy} = \frac{60 \cdot P_{sy}}{2\pi N_s} = 9.55 \frac{P_{sy}}{N_s} \\ = 9.55 \times \frac{314.2 \times 10^3}{1000} \\ = 3000 \text{ Nm}$$

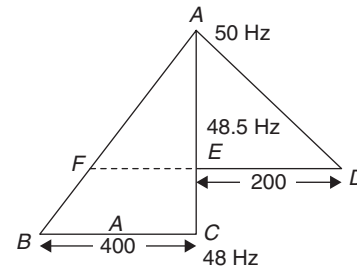
Load Sharing of Alternators Connected in Parallel

- Effect of change in excitation: The change in excitation merely changes the p.f. at which the load is delivered without affecting the load so long as the input to prime mover remains unchanged, i.e. kVA component of its output is changed (indirect proportion with excitation) but not its kW output.
- Effect of change in the power input to prime mover: The amount of load taken up by an alternator running, in parallel with other machines, directly depends upon its driving torque or in other words upon the power input (or the angular advance of its rotor). The increase in the power input to prime mover does not disturb the division of reactive power.

Example 9: An alternator has a per unit impedance of 0.9 p.u. to a base of 20 MVA, 33 kV, find the p.u. impedance to the base of 50 MVA and 11 kV.

$$\text{Solution: } Z_{new} = Z_{old} \frac{MVA_{new}}{MVA_{old}} \times \left(\frac{kV_{old}}{kV_{new}} \right)^2 \\ = 0.9 \times \frac{50}{20} \times \left(\frac{33}{11} \right)^2 = 20.25$$

Example 10: An alternator of 400 kW is driven by a prime mover of speed regulation 4% and another alternator of 200 kW is driven by a prime mover of speed regulation 3%. Find the total load they can take, without over loading either of the machines



Solution: For getting maximum load, DE is extended to cut AB at F.

$$\text{Max load} = DF$$

$$\frac{FE}{BC} = \frac{AE}{AC}$$

$$\frac{FE}{400} = \frac{(50 - 48.5)}{(50 - 48)}$$

$$FE = \frac{400 \times 1.5}{2} = 300 \text{ kW}$$

$$\text{Maximum load } FD = FE + ED \\ = 300 + 200 = 500 \text{ kW}$$

Short-circuit Ratio (SCR)

Field current required to produce

$$\bullet \text{ SCR} = \frac{\text{rated voltage on open circuit}}{\left(\text{Field current required to produce rated current on 3-}\phi \text{ short circuit} \right)}$$

- A lower value of SCR means a greater change in field current to maintain a constant terminal voltage and lower value of steady-state stability limit.
- Lesser the SCR, lesser is the size, weight and cost of machine.
- If there were no magnetic saturation, SCR is the reciprocal of the p.u. value of saturated synchronous reactance.
- Modern alternators have SCR between 0.5 and 1.5.

SYNCHRONOUS MOTORS

A synchronous motor construction is identical with an alternator. It has following features.

1. It runs either at synchronous speed or not at all, i.e. while running it maintains a constant speed.

$$N_s = \frac{120f}{P}$$

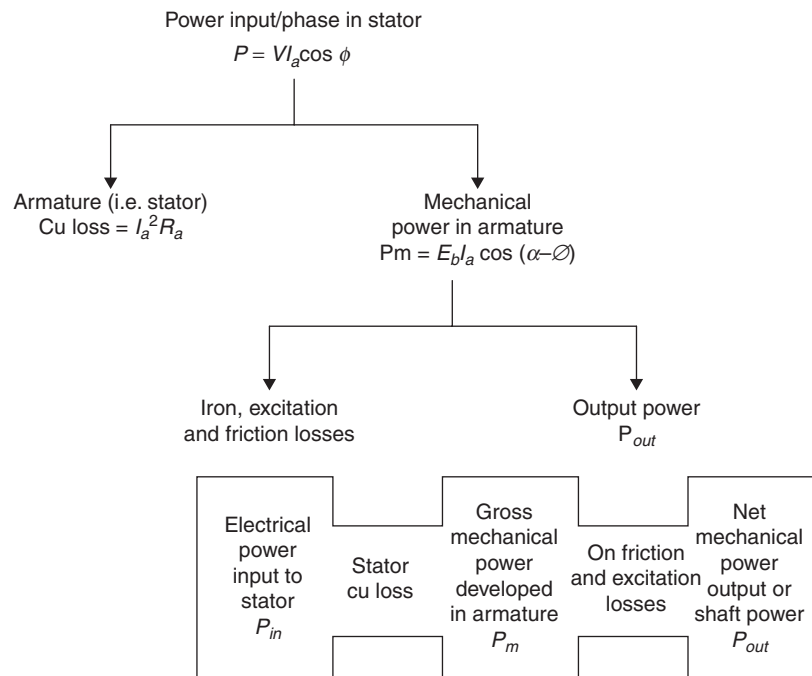
2. It is not inherently self-starting. It has to be run up to synchronous (or near synchronous) speed by some means (known as starting methods) before it is connected to the supply.
3. A synchronous motor can be operated under a wide range of power factors both lagging and leading. A range of zpf lag to zpf lead through unity can be obtained by adjusting the excitation. Hence it can be used for power factor improvement purposes, in addition to supplying driving torque to drive loads.

Principle of Operation

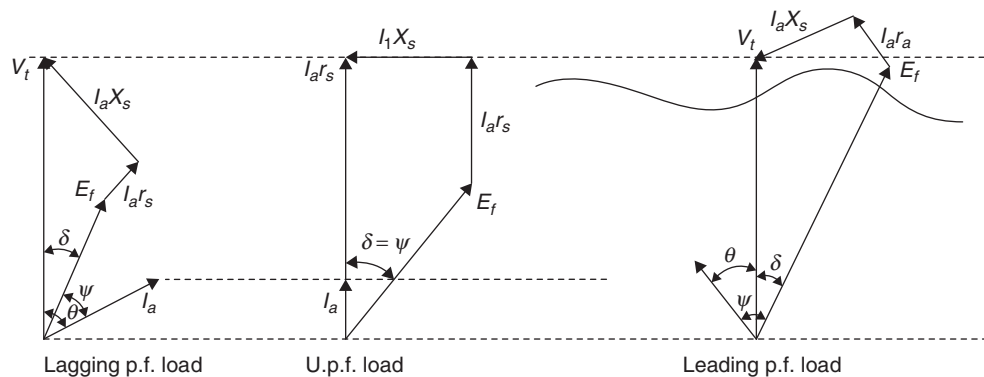
A synchronous motor works on the principle of magnetic locking between rotating air-gap flux and rotor poles.

Method of Starting

The rotor (which is as yet unexcited) is speeded up to synchronous/near synchronous speed by some arrangement and then field circuit is excited. The moment this synchronously (or near) rotating rotor is excited, and it is magnetically locked into position with the stator poles. A synchronous motor may be started with the help of auxiliary motor connected to the same shaft or by means of induction torque obtained by damper bars.

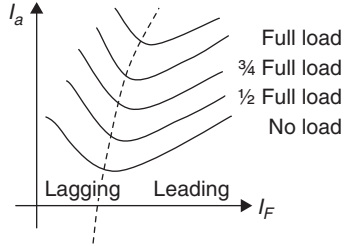


Synchronous Motor Phasor Diagram

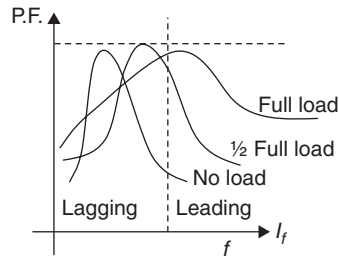


V- and Inverted V-curves

The V-curves of a synchronous motor show how armature current varies with its field current when motor input is kept constant and are so called because of their shape. Minimum armature current corresponds to unity power factor.



The inverted V-curves of a synchronous motor show how power factor varies with excitation (or I_f) when motor input is constant. The curve for p.f. looks like inverted 'V' curve.



In case of synchronous motor

1. If armature current I_a lags the excitation emf, E_f 90° , the nature of armature MMF or armature reaction MMF F_a is magnetizing.
2. If armature current I_a leads E_f by 90° , the nature of armature MMF, or armature reaction MMF F_a is demagnetizing.

Effect of Increased Load with Constant Excitation

1. As load on the motor increases the armature current, I_a increases regardless of level of excitation (i.e. normal, under, or over-excited conditions)
2. For under-and over-excited motors p.f. tend to approach unity with increase in load
3. Both with under-and over-excited, change in p.f. is greater than in I_a with increase in load
4. For normal excited case, when load is increased, change in I_a is greater than in p.f. which tends to become increasingly lagging.

Effect of Changing Excitation on Constant Load

The variations in the excitation of a synchronous motor running with a given load produce variations in its load angle only. When excitation is decreased the load angle increases and vice-versa.

Example 11: A 3-phase, induction motor draws 1000 kVA at a p.f. of 0.8 lag. A synchronous condenser is connected in parallel to draw an additional 750 kVA at 0.6 p.f. lead. What is the p.f. of the total load supplied by the mains?

Solution: Reactive power drawn by induction motor

$$Q_m = 1000 \sin 36.86$$

$$j 600 \text{ kVAR lagging}$$

Reactive power supplied by condenser

$$Q_c = 750 \sin(\cos^{-1} 0.6)$$

$$= -j 600 \text{ kVA lead}$$

Total reactive power $Q = Q_m + Q_c$

$$Q = j600 + -j600 = 0$$

Power factor of total load

$$\cos \left[\tan^{-1} \left[\frac{Q}{P} \right] \right] = 1$$

Example 12: A 2300 V, 3-phase synchronous motor driving a pump is provided with a line ammeter and a field rheostat. When the rheostat is adjusted such that the AC line current is minimum, the ammeter reads 8.8 A. What is the power being delivered to the pump, neglecting losses? How should the rheostat be adjusted so that the motor operates at 0.8 leading power factor? What is the reactive power supplied by the motor at this new power factor?

Solution: Line current will be minimum at unity power factor

$$\text{Power delivered} = \sqrt{3} \times 2300 \times 8.8 \times 1 = 35.05 \text{ kW}$$

For operating motor at 0.8 leading power factor field current should be increased.

As the load is unaltered $I \cos \phi$ remains same.

$$\text{So } I_{a2} \cos \phi_2 = I_{a1} \cos \phi_1$$

$$I_{a2} = \frac{8.8 \times 1}{0.8} = 11 \text{ A}$$

Leading kVAR supplied

$$= \frac{\sqrt{3} \times 2300 \times 11 \times 0.6}{1000} = 26.3$$

Power Developed by a Synchronous Motor

$$P_m = \frac{E_b V}{Z_s} \cos(\theta - \alpha) - \frac{E_b^2}{Z_s} \cos \theta$$

where α = Load angle

θ = Internal or Machine angle

Since R_a is generally negligible, $Z_s = X_s$ so that $\theta \cong 90^\circ$. Hence, mechanical power developed,

$$P_m = \frac{E_b V}{X_s} \cos(90^\circ - \alpha) = \frac{E_b V}{X_s} \sin \alpha$$

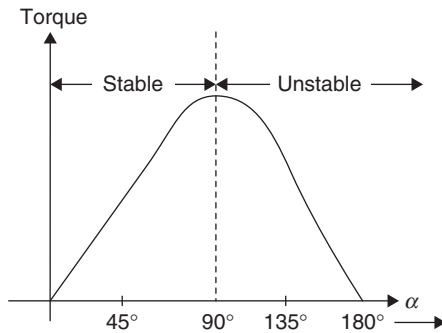
Condition for Maximum Power Developed

Maximum power can be developed when $\theta = \alpha$

$\therefore$ Value of maximum power $(P_m)_{\max}$

$$= \frac{E_b V}{Z_s} - \frac{E_b^2}{Z_s} \cos \alpha \quad \text{or} \quad (P_m)_{\max} = \frac{E_b V}{Z_s} - \frac{E_b^2}{Z_s} \cos \theta$$

The above expression shows that maximum power and hence torque ($\because$ speed is constant) depends on V and E_b . Maximum value of θ (and hence α) is 90° . The variation of torque (or power) with load angle α is shown in following figure.



If R_a is neglected then $Z_s \cong X_s$ and $\theta = 90^\circ$.

$\therefore \cos \theta = 0$

$$(P_m)_{\max} = \frac{E_b V}{X_s}$$

This corresponds to pull-out or maximum torque.

Special Motors

Servomotor

The motors that are used in automatic control systems are called servomotors. When the objective of the system is to control the position of an object, then the system is called servomechanism. The servomotors are used to convert an electrical signal (control voltage) applied to them into an angular displacement of the shaft. The suitability of a motor

for a particular application depends on the characteristics of the system, the purpose of the system and its operating conditions.

A servomotor should have the following features.

1. Linear relationship between the speed and electric control signal
2. Steady-state stability
3. Wide range of speed control
4. Linearity of mechanical characteristics throughout the entire speed range
5. Low mechanical and electrical inertia and
6. Fast response.

A two-phase induction motor with certain special design features can be used as servomotors, they are

1. The rotor of servomotor is built with high resistance, so that its $\frac{X}{R}$ (Inductive reactance/resistance) ratio is small which results in linear speed-torque characteristics.
2. The excitation voltage applied to two stator windings should have a phase difference of 90° .

Stepper Motor

A stepper motor transforms electrical pulses into equal increments of rotor shaft motion called steps. A one-to-one correspondence exists between the electrical pulses and the motor steps. The number of teeth or poles on the rotor and the number of poles on the stator determine the size of the step.

$$\text{Step angle} = \frac{360}{\left[\begin{array}{c} \text{No. of rotor poles} \\ \text{or teeth} \end{array} \right] \times \left[\begin{array}{c} \text{No. of stator pole} \\ \text{pairs} \end{array} \right]}$$

The frequency and sequence of pulses given to the stator windings is controlled by switching devices. Stepper motors are used in computer peripherals, X-Y plotters, robots and machine tools.

EXERCISES

Practice Problems I

Directions for questions 1 to 26: Select the correct alternative from the given choices.

1. Per unit synchronous impedance of synchronous machine is 1.10. Its SCR is _____?
(A) 1.20 (B) 0.417
(C) 0.909 (D) 0.833
2. A synchronous generator with a synchronous reactance of 1.3 p.u. is connected to an infinite bus whose voltage is 1 p.u., through an equivalent reactance of 0.2 p.u., for maximum output of 1.2 p.u., the alternator emf must be
(A) 1.5 p.u. (B) 1.56 p.u.
(C) 1.8 p.u. (D) 2.5 p.u.
3. A single-phase alternator has a synchronous reactance of 2Ω and negligible resistance. If it supplies 10 A to a purely capacitive load at 200 V, then the generated emf will be _____
(A) 240 V (B) 220 V
(C) 200 V (D) 180 V
4. An alternator has 18 slots/pole and the first coil lies in slots 1 and 16. What is the pitch factor for 5th harmonic _____?
(A) 0.96 (B) 0.259
(C) 0.289 (D) 0.829
5. An alternator on open circuit generates 400 V and 60 Hz. When the field current is 3.6 A. Neglecting saturation,

determine the open-circuit emf. When the frequency is 40 Hz and the field current is 3 A _____

- (A) 160 V (B) 200 V
(C) 220 V (D) 240 V

6. A salient-pole alternator has $X_d = 1.0$ p.u., $X_q = 0.6$ p.u. and $r = 0$. If this alternator delivers a power of 0.9 p.u. to an infinite bus at rated voltage and at p.f. 0.8 lag. Its load angle is _____

- (A) 42.048° (B) 21.024°
(C) 60.234° (D) 36.06°

7. A synchronous generator is running over excited with $E_f = 1.40$ p.u. This machine with a synchronous reactance of 1.20 p.u. is delivering a synchronous power of 0.5 p.u. to the bus. If the prime mover torque is increased by 1%, by how much will the P and Q change?

- (A) P increases by 1% but the Q decreases by 0.475%
(B) P decreases by 1% but the Q increases by 0.475%
(C) P and Q both increases by 1%
(D) P and Q both decreases by 1%

8. A 3-phase, 415 V, 6-pole, 50 Hz, star-connected synchronous motor develops a torque of 220 Nm and the stator winding has a synchronous reactance of 2 ohm/phase. Its load angle is _____. Its emf is 520 V (L-L)

- (A) 24.66° (B) 36.80°
(C) 18.40° (D) 12.33°

9. A 4-pole, 3-phase, 50 Hz, star-connected alternator has 60 slots with 4 conductors per slot. Coils are short-pitched by 3 slots. If the phase spread is 60° , find the line voltage induced for a flux per pole of 0.943 Wb distributed in space.

- (A) 7.613 kV (B) 5.234 kV
(C) 13.185 kV (D) 12.42 kV

10. A 3- ϕ , star-connected alternator supplies a load of 10 MW of P.F of 0.85 lag and at 11 kV, terminal voltages, its resistance is 0.1 ohm per/phase and synchronous reactance 0.66 ohm per phase. The line value of emf generated is _____

- (A) 6.625 kV (B) 13.21 kV
(C) 7.22 kV (D) 11.4863 kV

11. A 3-phase star-connected alternator is rated at 1600 kVA, 13.5 kV. The armature resistance drop and synchronous reactance drops are 3% and 10%, respectively. Find the regulation of alternator on full load at a power factor of 0.8 leading?

- (A) -2.1% (B) -3.1%
(C) +3.1% (D) 2.1%

Common Data for Questions 12 and 13:

A 3- ϕ star-connected 400 V synchronous motor takes a power input of 6 kW at rated voltage. Its synchronous reactance is 10 Ω per phase and resistance is negligible. If the excitation voltage is adjusted equal to the rated voltage of 400 V then

12. The load angle is _____

- (A) 20.0° (B) 24.02°
(C) 22.02° (D) 29.82°

13. The armature current is _____

- (A) 8.82 A (B) 7.86 A
(C) 6.43 A (D) 4.4 A

Common Data for Questions 14 and 15:

A 500 kVA, 11 KV, 3- ϕ star-connected alternator has the following data:

Friction and windage losses = 1000 W

core loss = 1500 W

Field winding resistance at 75°C is 180 Ω

Effective armature resistance per phase = 4 Ω

Field copper loss = 1000 W

For the above data

14. The load at which maximum efficiency occurs is _____.

- (A) 65% of full load
(B) 58% of full load
(C) 70% of full load
(D) 80% of full load

15. The maximum efficiency at 0.8 p.f lagging is _____.

- (A) 96.85% (B) 86.89%
(C) 78.79% (D) 70.23%

16. List-I lists different applications and List-II lists motors for these applications. Match the application with the most suitable motor.

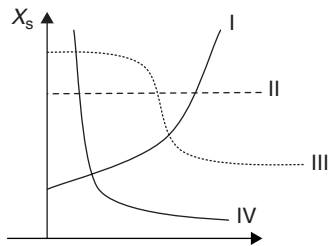
List-I	List-II
A. Escalator	1. Single-phase induction motor
B. Food mixer	2. Three-phase induction motor
C. Cassette tape recorder	3. Permanent magnet DC motor
D. Domestic water pump	4. Universal motor
	5. DC series motor
	6. Stepper motor

	A	B	C	D
(A)	1	4	3	2
(B)	1	4	2	3
(C)	5	4	3	6
(D)	1	4	3	5

17. A rotating electrical machine having its self-inductance of both the stator and the rotor windings independent of the rotor position will be definitely not develop

- (A) synchronizing torque.
(B) starting torque.
(C) reluctance torque.
(D) hysteresis torque.

18. A synchronous motor on load draws a current at a leading power factor angle ϕ . If the internal power factor angle which is the phase angle between the excitation emf and the current in time phasor diagram, is Ψ , then the air gap excitation m. m. f. lags the armature mmf by
- (A) $\pi/2 + \Psi$ (B) Ψ
 (C) $\Psi - \phi$ (D) $\frac{\pi}{2} - \Psi$
19. The developed electromagnetic force and/or torque in electro-mechanical energy conversion system act in a direction that tends
- (A) to decrease the co-energy at constant mmf.
 (B) to decrease the stored energy at constant mmf.
 (C) to decrease the stored energy at constant flux.
 (D) to increase the stored energy at constant mmf.
20. The following motor definitely has a permanent magnet rotor
- (A) Brushless DC motor.
 (B) DC commutator motor.
 (C) Universal motor.
 (D) Stepper motor.
21. In the figure the characteristic that corresponds to the variation of synchronous reactance of synchronous motor with field current is



- (A) curve I (B) curve II
 (C) curve III (D) curve IV
22. X_d , X_d' and X_d'' are steady-state d -axis synchronous reactance, transient d -axis reactance and sub-transient d -axis reactance of synchronous machine, respectively. Which of the following relation is true?
- (A) $X_d'' > X_d' > X_d$
 (B) $X_d' > X_d'' > X_d$

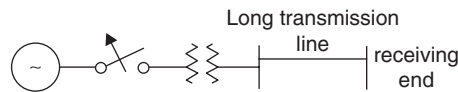
- (C) $X_d > X_d' > X_d''$
 (D) $X_d > X_d'' > X_d'$
23. An isolated engine driven synchronous generator is feeding a partly inductive load. A capacitor is now connected across the load to completely compensate the inductive current. For this operating condition
- (A) the field current has to be increased and fuel input left unaltered.
 (B) the field current has to be reduced and fuel input left unaltered.
 (C) the field current and fuel input have to be reduced.
 (D) the field current and fuel input have to be increased.
24. In relation to the synchronous machines, which one of the following statements is false?
- (A) the damper bars help the synchronous motor self-start.
 (B) in salient-pole machines, the quadrature axis synchronous reactance is less than the direct axis synchronous reactance.
 (C) the V curve of a synchronous motor represents the variation in the armature current with field excitation, at a given output power.
 (D) the short-circuit ratio is the ratio of the field current required to produce the rated voltage on open circuit to the rated armature current.
25. A three-phase synchronous motor connected to AC mains is running at full-load unity power factor. If the load is reduced by half, with excitation unaltered, its new power factor will be
- (A) leading
 (B) lagging
 (C) unity
 (D) dependent on machine parameters
26. A 4-pole, 3-phase, double layer winding is housed in a 36 slot stator for an AC machine with a phase spread of 60° . The coil span is 7 slot pitches. Number of slots in which top and bottom layers belonging to same phases is
- (A) 12 (B) 14
 (C) 24 (D) 28

Practice Problems 2

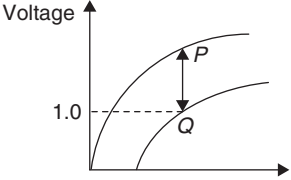
Directions for questions 1 to 100: Select the correct alternative from the given choices.

1. Two similar alternators are working in parallel. They have identical field excitations and supply a common load. Now the steam supply to one of the alternator is increased keeping the field excitation constant, its active power component
- (A) remains the same, but the reactive power component will increase
 (B) decreases while reactive power increases
 (C) increases and reactive power also increases
 (D) increases while reactive power contribution remains unchanged
2. Two mechanically coupled alternators deliver power at 50 Hz and 60 Hz, respectively. The highest speed of the alternator is
- (A) 3000 rpm (B) 3600 rpm
 (C) 500 rpm (D) 600 rpm

3. The phase groups of machines A and B are 60° and 120° , respectively. If the breadth factor of machine A is x times that of B, then value of x is
 (A) 1 (B) <1
 (C) >1 (D) None of these
4. For the same power rating an alternator at lower voltage will be
 (A) larger in size (B) smaller in size
 (C) less noisy (D) more efficient
5. When an alternator, designed for operation at 60 Hz is operated at 50 Hz
 (A) kVA rating increases by 1.2
 (B) operating voltage reduces by $\frac{5}{6}$
 (C) operating voltage increases by 1.2
 (D) operating voltage reduces by $\left(\frac{5}{6}\right)^2$
6. When a balanced 3- ϕ distributed type armature winding is carrying 3- ϕ balanced currents, the strength of rotating magnetic field is
 (A) three times the amplitude of each constituent pulsating magnetic field
 (B) equal to the amplitude of each constituent pulsating magnetic field
 (C) half the amplitude of each pulsating magnetic field
 (D) one and a half times the amplitude of each constituent pulsating magnetic field
7. The armature of a 1- ϕ alternator is completely wound with T single turn coils distributed uniformly, the induced voltage in each turn is 2 V (RMS). The emf of the whole winding is
 (A) 2 T V (B) 1.27 T V
 (C) 1.11 T V (D) 1.414 T V
8. A 3- ϕ , 4-pole alternator has 48 stator slots carrying the 3- ϕ distributed winding. Each coil of the winding is short-circuited by one slot pitch. The winding factor is
 (A) $\frac{1}{16} \cot 7.5^\circ$ (B) $\frac{1}{8} \cot 7.5^\circ$
 (C) $\frac{1}{16 \sin 7.5^\circ}$ (D) $\cos 7.5^\circ$
9. A 50 kW, 3- ϕ Y-connected, 50 Hz, 415 V cylindrical rotor synchronous generator operates at rated excitation with 0.8 p.f. leading. The motor efficiency excluding field and stator losses is 90% and $X_s = 2.5 \Omega$. The power angle of the machine is approximately
 (A) 27° (B) 54°
 (C) -27° (D) -54°
10. A stationary alternator is connected across line bus bars. It
 (A) will get short-circuited
 (B) will run as a synchronous motor
 (C) will decrease bus bar voltage through momentarily
 (D) will distinct generated emf's of other alternators connected in parallel
11. Two alternators P and Q operate in parallel and supply a load of 20 MW at 0.8 p.f. lag. By adjusting the steam input of P its power output is adjusted to 12 MW and by changing its excitation its p.f. is adjusted to 0.9 lag. Find p.f. of alternator Q
 (A) 0.557 lag (B) 0.657 lag
 (C) 0.557 lead (D) 0.657 lead
12. A 50 Hz alternator is initially connected to a long loss-less transmission line which is open circuited at the receiving end. With field voltage held constant, generator is disconnected from transmission line. Which of the following can be said about steady-state terminal voltage and field current of the generator?



- (A) magnitude of terminal voltage decreases and magnitude of field current does not change
 (B) magnitude of terminal voltage increases and field current does not change
 (C) magnitude of terminal voltage increases and field current increases
 (D) magnitude of terminal voltage does not change and field current decreases
13. A 250 MW, 21 kV, 50 Hz, 3- ϕ , 4-pole alternator having a rated p.f. = 0.8 has moment of inertia $30 \times 10^3 \text{ kg m}^2$. The inertia constant H will be
 (A) 1.18 s (B) 1.91 s
 (C) 2.68 s (D) 4.82 s
14. A 80 kVA, 400 V, Y-connected synchronous machine generates rated open-circuit voltage of 400 V at a field current of 16 A. The short-circuit armature current at a field current of 10 A is equal to rated armature current. The per unit saturated synchronous reactance is
 (A) 0.571 (B) 1.25
 (C) 0.625 (D) 1.63
15. In relation to synchronous machines, which one of the following statements is false?
 (A) In salient-pole machines, direct axis synchronous reactance is greater than quadrature axis synchronous reactance
 (B) Damper winding helps synchronous motor self-start
 (C) Short-circuit ratio is the ratio of field current required to produce, the rated voltage on O.C. to rated armature current
 (D) The V curve of a synchronous motor represent variation in armature current with field excitation at a given output power

16. A 50 Hz, 6-pole, 100 MVA, 22 kV turbo generator is delivering rated MVA at 0.8 p.f. Suddenly a fault occurs reducing electric output by 30%. Neglect losses and assume constant power input to the shaft. The accelerating torque of generator in KNm at time of fault is
 (A) 534.76 (B) 356.27
 (C) 296.35 (D) 178.13
17. Two 3- ϕ , Y-connected alternators are to be paralleled to a set of common bus bars. The armature has a per phase synchronous reactance of 1.5Ω and negligible armature resistance. The line voltage of first machine is adjusted to 3300 V and of second machine is 3200 V. The machine voltages are in phase at the instant they are paralleled. Under this condition, the synchronizing current per phase will be
 (A) 19.1 A (B) 33.08 A
 (C) 38.2 A (D) 66.16 A
18. A 4-pole, 3- ϕ , double-layer winding is housed in a 36 slot stator for an AC machine with 60° phase speed. Coil span is 7 slot pitches. No. of slots in which top and bottom layers belong to different phases is
 (A) 24 (B) 18
 (C) 12 (D) 0
19. A standalone engine drives synchronous generator is feeding a partly inductive load. A capacitor is now connected across the load to completely nullify the inductive current. For this operating condition
 (A) field current and fuel input has to be reduced
 (B) field current and fuel input has to be increased
 (C) field current has to be increased and fuel input left unaltered
 (D) field current has to be reduced and fuel input left unaltered
20. Curves X and Y denote O.C. and full-load ZPF characteristics of a synchronous generator. Q is a point on ZPF characteristics at 1 p.u. voltage. The vertical distance PQ gives voltage drop across.

 (A) synchronous reactance
 (B) magnetizing reactance
 (C) Potier reactance
 (D) leakage reactance
21. A round rotor generator with internal voltage $E_1 = 2$ p.u. and $X = 0.9$ p.u. is connected to a round rotor synchronous motor with internal voltage $E_2 = 1.2$ p.u. and $x = 1.1$ p.u. The reactance of line connecting generator and motor is 0.5 p.u. When generator supplies 0.4 p.u. power, rotor angle difference between machines is
 (A) 30° (B) 24°
 (C) 18° (D) 45°
22. The phase sequence of a 3- ϕ alternator can be reversed by
 (A) reversing the field current and doubling the number of poles
 (B) doubling the number of poles without reversing the field current
 (C) reversing the field current keeping direction of rotation same
 (D) reversing the direction of rotation keeping the field current same
23. A 5000 kVA, 10,000 V 1500 rpm, 50 Hz alternator is connected with infinite bus bar. Its synchronous reactance is 20%. The no-load synchronizing torque when there is a displacement of 0.5° mechanical is
 (A) 672 kW (B) 772 kW
 (C) 872 kW (D) 972 kW
24. The terminals of a 3- ϕ unloaded alternator is shorted. The role of damper windings here is to
 (A) establish flux through the direct axis magnetic circuit of the machine
 (B) repel the armature flux and confine it to leakage flux path in the air gap
 (C) allow only partial linkage of armature flux with main field winding
 (D) confine the armature flux to completely link the damper winding
25. Synchronous impedance method is used to calculate the regulation of a synchronous generator. This is
 (A) lower than actual because of saturation of magnetic circuit
 (B) higher than actual due to saturation of magnetic circuit
 (C) nearly accurate as generator is normally operated with unsaturated magnetic circuit
 (D) nearly accurate as it takes into account the magnetic saturation
26. The armature current of an alternator lags behind the excitation emf by an angle ϕ . The electrical angle between field axis and the axis of armature reaction field will be
 (A) ϕ (B) 90°
 (C) $90^\circ - \phi$ (D) $90^\circ + \phi$
27. In an alternator, the stator field,
 (A) leads the rotor field and electromagnetic torque is developed in a direction opposite to direction of rotation of rotor
 (B) leads the rotor field and electromagnetic torque is developed in the direction of rotation of rotor
 (C) lags the rotor field and electromagnetic torque is developed opposite to direction of rotation of rotor
 (D) lags the rotor field and electromagnetic torque is developed in same direction of rotation of rotor

28. A 50 Hz 4-pole turbo generator rated 100 MVA, 11 kV has been given a mechanical input of 80 MW suddenly for an electrical load of 40 MW, then speed of rotor after 10 cycles will be, if machine inertia constant is $8 \mu\text{J/MVA}$
- (A) 1515 rpm (B) 1468 rpm
(C) 496 rpm (D) None of these
29. The maximum electrical output of a synchronous generator is
- (A) $\frac{X_s}{V_t C_f}$ (B) $\frac{V_t^2}{X_s}$
(C) $\frac{C_f^2}{X_s}$ (D) $\frac{V_t C_f}{X_s}$
30. An alternator is synchronized with the infinite bus bar. If the input to prime mover is cut off with field supply remaining constant
- (A) it gets burnt
(B) it behaves as an induction motor but run in reverse direction
(C) it behaves as a synchronous motor and will rotate in same direction
(D) it behaves as a synchronous motor and rotate in reverse direction
31. High frequency on aircraft alternators is used to
- (A) free system from external disturbance
(B) compensate for high speed
(C) compensate for high altitudes
(D) reduce bulk
- Common Data for Questions 32 and 33:**
A 4-pole, 50 Hz, synchronous generator has 36 slots in which a double layer winding is housed. Each coil has 10 turns and is short pitched by an angle of 30° electrical. The fundamental flux per pole is 0.025 Wb.
32. The line to line induced emf for a 3- ϕ Y connection is approximately
- (A) 969 V (B) 869 V
(C) 1369 V (D) 1069 V
33. The line to line induced emf for a 2- ϕ connection is approximately
- (A) 873 V (B) 973 V
(C) 1269 V (D) 1069 V
34. The fifth-harmonic component of phase emf for the 3- ϕ Y connection
- (A) 0 (B) 269
(C) 281 (D) 808
35. The induced emf (line to line) is close to
- (A) 11.24 kV (B) 13.54 kV
(C) 16.24 kV (D) 14.24 kV
36. The power or torque angle is close to
- (A) 13.9° (B) 16.3°
(C) 12.9° (D) 30.3°
37. Distribution factor K_d is also known as
- (A) Length factor
(B) Breadth factor
(C) Winding Factor
(D) None of the above
38. Which losses are reduced, by having laminated poles and pole shoes?
- (A) Eddy current losses
(B) Hysteresis losses
(C) Iron losses
(D) Copper losses
39. Two 3-phase star-connected alternators, whose armature has a per phase synchronous reactance of 0.85Ω and negligible armature resistance, are to be paralleled to a set of common bus bars. The line voltage of the first machine is adjusted to 330 V and that of second machine is adjusted to 320 V the synchronizing current per phase would be
- (A) 16.97 A (B) 15.88 A
(C) 3.39 A (D) 1.697 A
40. Which among the following is not correct about distributed winding?
- (A) Improves the waveform of induced emf
(B) Provides better utilization of core
(C) Enhances armature reaction
(D) Facilitates cooling
41. A 3.3 kV single-phase alternator delivering a current of 100 A at 0.8 p.f. lagging, produces a full-load current of 100 A on short circuit by a field excitation of 5 A and produces an emf of 250 V on open circuit by the same field current. The armature resistance is 0.65Ω . The percentage regulation of the alternator is
- (A) -6% (B) -8.9%
(C) +8.9% (D) +6%
42. For a uniformly distributed 3-phase winding the value of distribution factor would be
- (A) 0.955 (B) 0.8
(C) 0.755 (D) 0.675
43. A 6.6 MVA, 3-phase star-connected, 6.6 kV, 2-pole turbo generator has a synchronous impedance of $(0.2 + j1.4)\Omega$ per phase. The generator losses are as given below
- (a) S.C load loss = 110 kW
(b) O.C core loss at rated voltage = 45 kW
(c) Friction and windage losses = 25 kW
(d) Field winding resistance = 3Ω
(e) Field current = 100 A

Common Data for Questions 35 and 36:

A 500 kVA, 13.2 kV, 3- ϕ Y-connected cylindrical-pole synchronous generator has a synchronous reactance of 80Ω . Neglect armature resistance and consider operation at full load and u.p.f.

Neglecting the change in field current compute the efficiency at rated load 0.8 p.f. leading

- (A) 94.65% (B) 95.46%
(C) 96% (D) 92%

44. The power angle at which a synchronous generator will develop maximum power will be

(A) $\delta = \theta$ and $\theta = \tan^{-1} \left(\frac{X_s}{R_e} \right)$

(B) $\delta = \theta$ and $\theta = \tan^{-1} \left(\frac{R_e}{X_s} \right)$

- (C) $\delta = \theta$ only
(D) $\delta = \theta$ and $\theta = \tan^{-1}(X_s)$

45. A 380 V, 100 kVA, 0.8 p.f. leading delta-connected, 60 Hz synchronous machine has a synchronous reactance of 2Ω and negligible armature resistance. The friction and windage losses are 1 kW and the core loss is 1.6 kW. Calculate the power supplied to the load if the load is drawing a line current of 21 A at 0.8 p.f. leading
(A) 9 kW (B) 9.5 kW
(C) 8.5 kW (D) 11.785 kW

46. Given below are the test results that were conducted on an alternator for a field excitation of 20 A. Test Results
S.C. test: Armature current = 200 A

O.C. test: Terminal voltage = 1000 V

The magnitude of internal voltage drop within the machine for a load current of 25 A is

- (A) 1000 V (B) 500 V
(C) 250 V (D) 125 V

47. A synchronous machine with large air gap has higher synchronizing power due to
(A) Larger air gap offers large reluctance to the path of flux produce by armature mmf
(B) Large air gap reduces the effect of armature reaction
(C) Large air gap results in small value of synchronous reactance
(D) All of the above

48. The methods employed to reduce hunting in alternators are by

- (A) Employing heavy flywheels
(B) Putting dash-pots on the engine governors
(C) Using squirrel cage windings on the surface of the rotor
(D) All of the above

49. Synchronous generator with a synchronous reactance of 1.2 p.u. is connected to an infinite bus whose voltage is 1 p.u. through an equivalent reactance of 0.3 p.u. What should be the value of emf to give a maximum output of 1.2 p.u.?

- (A) 0.8 p.u. (B) 1.8 p.u.
(C) 2.8 p.u. (D) 3.65 p.u.

50. Power angle δ depends upon which of the following factors:

- (i) Supply voltage
(ii) Armature current
(iii) Power factor
(A) (i) and (ii) (B) (ii) and (iii)
(C) (i) and (iii) (D) (i), (ii) and (iii)

51. The increase in field current causes an increase in armature current in a synchronous motor, fed from an infinite bus and is delivering half full-load. Then the motor

- (A) delivers reactive power to the bus and absorb active power from the bus
(B) Absorbs reactive power from the bus and delivers active power to the bus
(C) Absorbs active power and reactive power from the bus
(D) deliver reactive power and active power to the bus

52. The field current of a synchronous motor operating at 0.8 p.f. lag is continuously increased. Then

- (A) the armature current increases up to a certain value of field current, and thereafter decreases
(B) the armature current decreases up to a certain value of field current and then increases
(C) the power factor increases up to a certain value of field current and then decreases
(D) the power factor decreases up to a certain value of field current and then increases

53. During hunting of synchronous motor

- (A) negative phase sequence currents are generated
(B) harmonics are developed in the armature circuit
(C) damper bar develops torque
(D) field excitation increases

54. A 11 kV synchronous motor is connected to a load taking 0.8 p.f. lagging current. The armature reaction is

- (A) cross-magnetizing (B) demagnetizing
(C) non-effective (D) magnetizing

55. A 3- ϕ synchronous motor connected to AC mains is running at full load and unity p.f. If the shaft load is reduced by half, with field current held constant, new power factor will be

- (A) unity
(B) leading
(C) lagging
(D) dependent on m/c parameters

56. During hunting of a synchronous motor

- (A) negative phase sequence current are generated
(B) harmonics are developed in armature circuit
(C) damper bar develops torque
(D) field excitation increases

57. A synchronous condenser is used to compensate a 2400 kVA 0.65 p.f. load so that the p.f. becomes unity. The kVA rating of condenser is

- (A) 27 kVAr (B) 1560 kVAr
(C) 1825 kVAr (D) 840 kVAr
58. If the excitation of a salient-pole synchronous motor running under normal excitation is reduced to zero, then
(A) it becomes an induction motor
(B) it becomes a reluctance motor
(C) it remains a synchronous motor
(D) motor stops
59. Inverted V curves for a synchronous motor show variation of
(A) power factor and supply voltage when motor is hunting
(B) field current and supply voltage when excitation is constant
(C) power factor with **DC** field current when load on motor is constant
(D) None of these
60. A 3- ϕ synchronous motor is connected to an infinite bus. It is operating at half load with normal excitation. When load on the machine is suddenly increased
(A) speed decreases and then becomes synchronous
(B) speed first increases and then becomes synchronous
(C) speed will fluctuate around synchronous speed and then becomes
(D) speed remains unchanged
61. Damper windings are provided
(A) on pole faces
(B) on armature
(C) on rotor shaft
(D) on stator frame
62. The torque angle in a synchronous machine is the angle between
(A) rotating stator flux and rotor poles
(B) magnetizing current and back emf
(C) supply voltage and back emf
(D) none of the above
- Common Data for Questions 63 and 64:**
A synchronous motor connected to an infinite bus at 1 p.u. voltage draws 0.8 p.u. current at u.p.f. Its synchronous reactance is 0.9 p.u. and resistance is negligible.
63. The excitation voltage (E) and load angle δ will be
(A) 0.616 and 35.7 lag
(B) 0.616 and 35.7 lead
(C) 1.23 and 35.7 lag
(D) 1.23 and 35.7 lead
64. Keeping the excitation voltage same, the load on the motor is increased such that motor current increases by 20%. The operating power factor becomes
(A) 0.812 lag (B) 0.812 lead
(C) 0.749 lag (D) 0.749 lead
65. A synchronous motor runs at N_s rpm at full load. The speed of the synchronous motor at half full-load will be
(A) Half the rated rpm
(B) $1/4^{\text{th}}$ the rated rpm
(C) Remains the same
(D) Double the rated rpm
66. A 3-phase synchronous motor driving a constant load torque draws power from infinite bus bar at leading power factor. With increase in excitation the power angle and power factor will
(A) Both power angle and p.f. decreases
(B) Power angle increases, p.f. decreases
(C) Power angle decreases, p.f. increases
(D) Both power angle and p.f. would increase
67. What should be the nature of excitation for a synchronous motor to operate as synchronous condenser?
(A) Under excited
(B) Over Excited
(C) Normally excited
(D) cannot be determined
68. The maximum value of torque that a motor can develop without losing synchronism is called
(A) Starting torque (B) Running torque
(C) Pull in torque (D) Pull out torque
69. For synchronous motors v-curves are plotted between the values of
(A) Armature current and power factor
(B) Armature current and Field current
(C) Field current and power factor
(D) Terminal voltage and Field current
70. A 1000 hp, 3.75 kV, 3-phase, star-connected synchronous motor has a synchronous impedance of $(5 + j12) \Omega$ per phase. It is excited to develop an O.C emf of 2.5 kV. The ratio of current due to supply voltage and current due to induced emf is
(A) 1:2 (B) 2:3
(C) 3:2 (D) 2:1
71. Which of the following is false with regards to applications of a synchronous motor?
(A) Power factor correction equipment
(B) Voltage regulation equipments
(C) Constant speed drives
(D) Variable load drives
72. A synchronous motor is operated on no-load u.p.f. What should be done to the field current for the power factor to be leading?
(A) should not be changed
(B) should be increased
(C) should be decreased
(D) none of the above
73. If the excitation of a synchronous motor on infinite mains at no-load is increased, the current drawn by it will be of

- (A) Unity p.f.
(B) Zero p.f. lagging
(C) It will not draw any current
(D) Zero p.f. leading
74. What would happen to load angle of a synchronous motor which is supplied power from a common bus bar and the bus voltage and frequency are reduced while field current and load torque are held constant?
(A) Load angle would decrease
(B) Load angle would increase
(C) Load angle would remain the same
(D) the motor will lose synchronism
75. If the phase angle between the excitation emf and current of a synchronous motor is given by θ_1 . The synchronous motor on load draws a current at lagging p.f. θ_2 . Then the armature mmf leads the excitation mmf by
(A) θ_2 (B) θ_1
(C) $\theta_1 - \theta_2$ (D) $\theta_1 + \theta_2$
76. What may be the causes if the synchronous motor fails to attain synchronous speed?
(A) Low supply voltage (B) Excessive load
(C) Field excited (D) All of the above
77. In what ways magnitude of armature current and power factor of a synchronous motor would vary if it is connected to an infinite bus, supplying constant torque, operating at u.p.f.
(A) Armature current will increase and p.f. will decrease
(B) Armature current will decrease and p.f. will increase
(C) Both armature current and p.f. will decrease
(D) Both armature current and p.f. will increase
78. A 440 V, 3- ϕ , 50 Hz circuit takes 10 A at 0.8 lagging p.f. A synchronous motor is used to raise the p.f. to unity. If the motor is driving a mechanical load of 10 kW and the motor has an efficiency of 80%, calculate the kVa input given to the motor and its p.f.
(A) 13.63, 0.64 lead
(B) 12.75, 0.875 lead
(C) 15.231, 0.75 lead
(D) 13.3, 0.94 lead
79. Which among the following is INCORRECT regarding a servomotor?
(A) They have non-linear relationship between the speed and electric control signal
(B) It has steady-state stability
(C) Wide range of speed control
(D) Low mechanical and electrical inertia
80. When a synchronous motor operates unexcited it is a
(A) AC series motor
(B) Universal motor
(C) Reluctance motor
(D) Repulsion motor
81. The maximum value of power developed in the synchronous motor is dependent on which of the following factors
(A) Maximum value of δ
(B) Rotor excitation
(C) Supply voltage
(D) All of the above
82. The purpose of damper winding in a synchronous motor is to
(A) reduce noise
(B) minimize circulating currents
(C) provide starting torque
(D) prevent hunting and provide starting torque.
83. Back emf set up in the stator of a synchronous motor will depend on
(A) Rotor speed, excitation and load angle
(B) Rotor excitation only
(C) Rotor speed only
(D) Rotor excitation and rotor speed
84. If the field windings of a synchronous motor are shorted on themselves and the motor is switched on then it will
(A) not start
(B) start and continue to run as an induction motor
(C) start as an induction motor and then run as synchronous motor
(D) Burn immediately
85. A 2300 V, 3-phase synchronous motor driving a pump is provided with a line ammeter and a field rheostat. When the rheostat is adjusted such that the AC line current is minimum, the ammeter reads 8.8 A. What is the reactive power supplied by the motor, if the motor was operating at 0.8 p.f. leading?
(A) 26.3 kVAR
(B) 36.3 kVAR
(C) 62.3 kVAR
(D) None of the above
86. A 3- ϕ star-connected 440 V, 20 A, 50 Hz, 500 rpm, develops an internal power of 6.7 kW. Calculate the effective resistance of the motor winding when operated at rated current and 0.8 leading p.f.
(A) 1.5 Ω (B) 2.5 Ω
(C) 3.5 Ω (D) 4.5 Ω
87. A 3- ϕ , 2.3 kV, star-connected synchronous motor has an effective resistance and synchronous reactance of 0.45 Ω and 4.5 Ω , respectively. If ' θ ' was the internal angle of the motor then what would happen to the value of θ if the value of R was increased by 0.9 Ω
(A) remain the same (B) 2θ
(C) 3θ (D) 4θ
88. A salient-pole synchronous motor has $X'_q = 0.35$ p.u. and $X'_d = 0.55$ p.u.. If the excitation is adjusted to 1.2 p.u. and the motor is connected to a bus bar of

- 1.0 p.u. voltage. The maximum power that the motor can deliver without loss of synchronism will be
 (A) 0.958 p.u. (B) 1.58 p.u.
 (C) 2.58 p.u. (D) 0.858 p.u.
89. The stepping angle of a stepper motor be increased by
 (A) Decreasing the number of phases
 (B) Reducing the number of poles
 (C) Depends on the type of motor
 (D) Both (A) and (B)
90. The number of stacks for a stepper motor of 6 poles and a stepping angle of 15° would be
 (A) Single (B) Double
 (C) Three (D) Four
91. Which of the following statements are true regarding servo motors
 (i) The motor output torque is proportional to the voltage applied
 (ii) The direction of torque developed depends on the instantaneous polarity of control voltage
 (iii) The motor output torque is inversely proportional to the voltage applied
 (iv) The direction of torque developed is independent of the polarity of control voltage
 (A) (i) and (ii) (B) (ii) and (iii)
 (C) (i) and (iv) (D) (iii) and (iv)
92. In a 3-phase synchronous motor the air gap flux will
 (A) lag behind the armature mmf
 (B) lead the armature mmf
 (C) in phase with armature mmf
 (D) either lead or lag based on load
93. For a 3- ϕ synchronous motor if
 E_0 was O.C. emf/phase
 V was Terminal voltage/phase
 θ was the internal angle with lagging p.f. The relationship among the quantities will be
 (A) $E_0 = [V^2 + E_R^2 + 2VE_R \cos(\theta + \phi)]^{\frac{1}{2}}$
 (B) $E_0 = [V^2 + E_R^2 - 2VE_R \cos(\theta + \phi)]^{\frac{1}{2}}$
 (C) $E_0 = [V^2 + E_R^2 + 2VE_R \cos(\theta - \phi)]^{\frac{1}{2}}$
 (D) $E_0 = [V^2 + E_R^2 - 2VE_R \cos(\theta - \phi)]^{\frac{1}{2}}$
94. A 440 V, 1- ϕ synchronous motor gives a net output mechanical power of 6.5 kW and operates at 0.8 lagging p.f. If the effective resistance is 0.9Ω and mechanical losses are 350 W then the estimated armature current is
 (A) 15A (B) 25A
 (C) 20 A (D) 30 A
95. The 'pull-out torque' of a synchronous motor is x times the 'full-load torque'. x can vary from
 (A) 0.5 to 1.5
 (B) 1.5 to 2.5
 (C) 1.25 to 3.5
 (D) It is a constant value 0.6
96. A 440 V, 50 Hz, 3-phase circuit takes a 10 A at lagging p.f. of 0.8 A synchronous motor is used to raise the p.f. to unity. Calculate the efficiency of the motor when driving a mechanical load of 8.5 kW, supplied with an input of 13.63 kVa at a power factor of 0.6472
 (A) 60% (B) 66%
 (C) 71.25% (D) 76.66%
97. A 4-pole, 3-phase, 50 Hz, star-connected synchronous motor has a negligible armature resistance and armature reactance of $8 \Omega/\text{phase}$. The excitation is such as to give an E_{OC} of 13.2 kV. If the motor is connected to a 11.5 kV, 50 Hz supply. The maximum load that the motor can supply during synchronism is
 (A) 18.97 kW
 (B) 18.97 MW
 (C) 17.89 kW
 (D) 17.89 MW
98. Which among the following is the expression for maximum mechanical power developed by a synchronous motor if the synchronous impedance was purely reactive and armature resistance is negligible
 (A) $(P_{\text{mech}})_{\text{max}} = 3 \left[\frac{EV}{Z_s} - \frac{E^2}{Z_s} \cos \theta \right]$
 (B) $(P_{\text{mech}})_{\text{max}} = \frac{3EV}{Z_s}$
 (C) $(P_{\text{mech}})_{\text{max}} = 3 \left[\frac{EV}{Z_s} - \frac{E^2}{Z_s} \right]$
 (D) None of the above
- Common Data for Questions 99 and 100:**
 A 3- ϕ star-connected 400 V synchronous motor takes a power input of 6 kW at rated voltage. Its synchronous reactance is 10Ω per phase and resistance is negligible. If the excitation voltage is adjusted equal to the rated voltage of 400 V, then
99. The load angle is _____
 (A) 20° (B) 24.02°
 (C) 22.02° (D) 29.82°
100. The armature current is _____
 (A) 8.82 A (B) 7.80 A
 (C) 6.43 A (D) 4.4 A

PREVIOUS YEARS' QUESTIONS

- For a 1.8° , 2-phase bipolar stepper motor, the stepping rate is 100 steps/second. The rotational speed of the motor in rpm is [2004]
(A) 15 (B) 30
(C) 60 (D) 90
- A 400 V, 50 kVA, 0.8 p.f. leading Δ -connected, 50 Hz synchronous machine has a synchronous reactance of $2\ \Omega$ and negligible armature resistance. The friction and windage losses are 2 kW and the core loss is 0.8 kW. The shaft is supplying 9 kW load at a power factor of 0.8 leading. The line current drawn is [2004]
(A) 12.29 A (B) 16.24 A
(C) 21.29 A (D) 36.88 A
- A 500 MW 3-phase Y-connected synchronous generator has a rated voltage of 21.5 kV at 0.85 p.f. The line current when operating at full-load rated conditions will be [2004]
(A) 13.43 kA (B) 15.79 kA
(C) 23.25 kA (D) 27.36 kA

Common Data for Questions 4a and 4b:

A 1000 kVA, 6.6 kV, 3-phase star-connected cylindrical-pole synchronous generator has a synchronous reactance of $20\ \Omega$. Neglect the armature resistance and consider operation at full load and unity power factor.

- (a) The induced emf (line-to-line) is close to [2005]
(A) 5.5 kV (B) 7.2 kV
(C) 9.6 kV (D) 12.5 kV
(b) The power (or torque) angle is close to
(A) 13.9° (B) 18.3°
(C) 24.6° (D) 33.0°
- In relation to the synchronous machines, which one of the following statements is false? [2005]
(A) In salient-pole machines, the direct-axis synchronous reactance is greater than the quadrature-axis synchronous reactance
(B) The damper bars help the synchronous motor self-start
(C) Short-circuit ratio is the ratio of the field current required to produce the rated voltage on open circuit to the rated armature current
(D) The V-curve of a synchronous motor represents the variation in the armature current with field excitation, at a given output power
- A synchronous generator is feeding a zero power factor (lagging) load at rated current. The armature reaction is [2006]
(A) magnetizing
(B) demagnetizing
(C) cross-magnetizing
(D) ineffective
- A 3-phase, 400 V, 5 kW, star-connected synchronous motor having an internal reactance of $10\ \Omega$ is operating at 50% load, unity p.f. Now, the excitation is increased by 1%. What will be the new load in per cent, if the power factor is to be kept same? Neglect all losses and consider linear magnetic circuit. [2006]
(A) 67.9% (B) 56.9%
(C) 51% (D) 50%

Common Data for Questions 8 to 10:

A 4-pole, 50 Hz, synchronous generator has 48 slots in which a double layer winding is housed. Each coil has 10 turns and is short pitched by an angle to 36° electrical. The fundamental flux per pole is 0.025 Wb.

- The line-to-line induced emf (in volts), for a three-phase star connection is approximately [2006]
(A) 808 (B) 888
(C) 1400 (D) 1538
- The line-to-line induced emf (in volts), for a three-phase connection is approximately [2006]
(A) 1143 (B) 1332
(C) 1617 (D) 1791
- The fifth harmonic component of phase emf (in volts), for a three-phase star connection is [2006]
(A) 0 (B) 269
(C) 281 (D) 808
- A three-phase synchronous motor connected to AC mains is running at full load and unity power factor. If its shaft load is reduced by half, with field current held constant, its new power factor will be [2007]
(A) unity
(B) leading
(C) lagging
(D) dependent on machine parameters
- A 100 kVA, 415 V (line), star-connected synchronous machine generates rated open-circuit voltage of 415 V at a field current of 15 A. The short-circuit armature current at a field current of 10 A is equal to the rated armature current. The per unit saturated synchronous reactance is [2007]
(A) 1.731 (B) 1.5
(C) 0.666 (D) 0.577
- A three-phase, three-stack, variable reluctance step motor has 20 poles on each rotor and stator stack. The step angle of this step motor is [2007]
(A) 3° (B) 6°
(C) 9° (D) 18°
- In a stepper motor, the detent torque means [2008]
(A) minimum of the static torque with the phase winding excited.

- (B) maximum of the static torque with the phase winding excited.
- (C) minimum of the static torque with the phase winding unexcited.
- (D) maximum of the static torque with the phase winding unexcited.

Common Data for Questions 15 and 16:

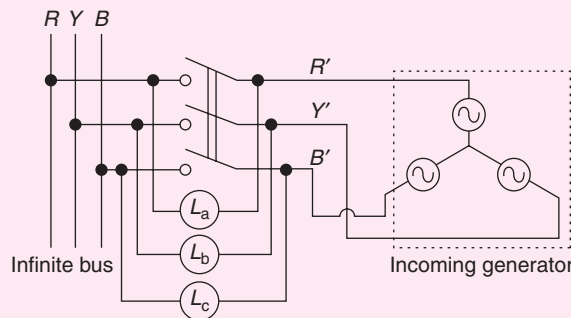
A synchronous motor is connected to an infinite bus at 1.0 p.u. voltage and draws 0.6 p.u. current at unity power factor. Its synchronous reactance is 1.0 p.u. and resistance is negligible.

15. The excitation voltage (E) and load angle (δ) will, respectively, be [2008]
 - (A) 0.8 p.u. and 36.86° lag
 - (B) 0.8 p.u. and 36.86° lead
 - (C) 1.17 p.u. and 30.96° lead
 - (D) 1.17 p.u. and 30.96° lag
16. Keeping the excitation voltage same, load on the motor is increased such that the motor current increases by 20%. The operating power factor will become [2008]
 - (A) 0.995 lagging
 - (B) 0.995 leading
 - (C) 0.791 lagging
 - (D) 0.848 leading
17. A field excitation of 20 A in a certain alternator results in an armature current of 400 A in short circuit and a terminal voltage of 2000 V on open circuit. The magnitude of the internal voltage drop within the machine at a load current of 200 A is [2009]
 - (A) 1 V
 - (B) 10 V
 - (C) 100 V
 - (D) 1000 V
18. The direct axis and quadrature axis reactances of a salient-pole alternator are 1.2 p.u. and 1.0, respectively. The armature resistance is negligible. If this alternator is delivering rated kVA at u.p.f. and at rated voltage then its power angle is. [2011]
 - (A) 30°
 - (B) 45°
 - (C) 60°
 - (D) 90°
19. A star-connected 400 V, 50 Hz, 4-pole synchronous machine gave the following open-and short-circuit test results:

Open-circuit test: $V_{oc} = 400$ V (RMS, line-to-line) at field current, $I_f = 2.3$ A

Short-circuit test: $I_{sc} = 10$ A (RMS, phase) at field current, $I_f = 1.5$ A

The value of per phase synchronous impedance in Ω at rated voltage is _____. [2014]
20. A three-phase synchronous generator is to be connected to the infinite bus. The lamps are connected as shown in the figure for the synchronization. The phase sequence of bus voltage is R-Y-B and that of incoming generator voltage is R'-Y'-B'.



It was found that the lamps are becoming dark in the sequence $L_a-L_b-L_c$. It means that the phase sequence of incoming generator is [2014]

- (A) opposite to infinite bus and its frequency is more than infinite bus
 - (B) opposite to infinite bus but its frequency is less than infinite bus
 - (C) same as infinite bus and its frequency is more than infinite bus
 - (D) same as infinite bus and its frequency is less than infinite bus
21. A 20-pole alternator is having 180 identical stator slots with 6 conductors in each slot. All the coils of a phase are in series. If the coils are connected to realize single-phase winding, the generated voltage is V_1 . If the coils are reconnected to realize three-phase star-connected winding, the generated phase voltage is V_2 . Assuming full pitch, single-layer winding, the ratio V_1/V_2 is [2014]
 - (A) $\frac{1}{\sqrt{3}}$
 - (B) $\frac{1}{2}$
 - (C) $\sqrt{3}$
 - (D) 2
 22. In a synchronous machine, hunting is predominantly damped by [2014]
 - (A) mechanical losses in the rotor
 - (B) iron losses in the rotor
 - (C) copper losses in the stator
 - (D) copper losses in the rotor
 23. A non-salient-pole synchronous generator having synchronous reactance of 0.8 p.u. is supplying 1 p.u. power to a unity power factor load at a terminal voltage of 1.1 p.u. Neglecting the armature resistance, the angle of the voltage behind the synchronous reactance with respect to the angle of the terminal voltage in degrees is _____. [2014]
 24. A three-phase, salient-pole synchronous motor is connected to an infinite bus. It is operated at no load at normal excitation. The field excitation of the motor is first reduced to zero and then increased in the reverse direction gradually. Then the armature current.
 - (A) increases continuously
 - (B) first increases and then decreases steeply

- (C) first decreases and then increases steeply
(D) remains constant
25. A three-phase, 11 kV, 50 Hz, 2 pole, stars connected, cylindrical rotor synchronous motor is connected to an 11 kV, 50 Hz source, Its synchronous reactance is 50Ω per phase, and its stator resistance is negligible. The motor has a constant field excitation. At a particular load torque, its stator current is 100 A at unity power factor. If the load torque is increased so that the stator current is 120 A, then the load angle (in degrees) at this load is _____. [2015]
26. Consider a system consisting of a synchronous generator working at a lagging power factor, a synchronous motor working at an overexcited condition and a directly grid-connected induction generator. Consider capacitive VAR to be a source and inductive VAR to be a sink of reactive power. Which one of the following statements is TRUE? [2016]
- (A) Synchronous motor and synchronous generator are sources and induction generator is a sink of reactive power.
(B) Synchronous motor and induction generator are sources and synchronous generator is a sink of reactive power.
(C) Synchronous motor is a source and induction generator and synchronous generator are sinks of reactive power.
(D) All are sources of reactive power.
27. A three-phase, 50 Hz salient-pole synchronous motor has a per-phase direct-axis reactance (X_d) of 0.8 pu and a per-phase quadrature-axis reactance (X_q) of 0.6 pu. Resistance of the machine is negligible. It is drawing full-load current at 0.8 pf (leading). When the terminal voltage is 1 pu, per-phase induced voltage, in pu, is _____. [2016]

ANSWER KEYS

EXERCISES

Practice Problems 1

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. C | 3. D | 4. B | 5. B | 6. B | 7. A | 8. D | 9. C | 10. D |
| 11. B | 12. C | 13. A | 14. A | 15. C | 16. A | 17. C | 18. D | 19. C | 20. A |
| 21. C | 22. C | 23. B | 24. D | 25. B | 26. A | | | | |

Practice Problems 2

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|
| 1. D | 2. D | 3. B | 4. A | 5. B | 6. D | 7. B | 8. B | 9. C | 10. A |
| 11. B | 12. D | 13. A | 14. C | 15. C | 16. A | 17. A | 18. C | 19. D | 20. D |
| 21. A | 22. D | 23. C | 24. B | 25. B | 26. C | 27. C | 28. A | 29. D | 30. C |
| 31. D | 32. D | 33. A | 34. A | 35. B | 36. C | 37. B | 38. A | 39. C | 40. C |
| 41. D | 42. A | 43. C | 44. A | 45. C | 46. D | 47. D | 48. D | 49. B | 50. D |
| 51. A | 52. B | 53. C | 54. D | 55. B | 56. C | 57. C | 58. B | 59. C | 60. C |
| 61. A | 62. A | 63. D | 64. D | 65. C | 66. A | 67. B | 68. D | 69. B | 70. C |
| 71. D | 72. B | 73. D | 74. B | 75. C | 76. B | 77. A | 78. D | 79. A | 80. C |
| 81. D | 82. D | 83. B | 84. C | 85. A | 86. D | 87. C | 88. A | 89. D | 90. D |
| 91. A | 92. A | 93. B | 94. C | 95. C | 96. B | 97. B | 98. B | 99. C | 100. A |

Previous Years' Questions

- | | | | | | | | | |
|-------|-------|-------|----------------|-------|--------------------------|-------|-----------|-----------|
| 1. B | 2. C | 3. B | 4. (a) B (b) C | 5. C | 6. B | 7. A | 8. C | 9. C |
| 10. A | 11. B | 12. C | 13. B | 14. D | 15. D | 16. A | 17. D | 18. B |
| 20. A | 21. D | 22. D | 23. 33.61 | 24. B | 25. -48 to -46° | 26. A | 27. 1.606 | 19. 15.06 |

TEST Electrical Machines

Ref: ELE1431435

Time: 60 min.

Directions for questions 1 to 25: Select the correct alternative from the given choices.

1. An auto transformer supplies a load of 3 KW at 150 V at a unity power factor.

If the applied voltage is 230 V. The power transferred to the load inductively is

- (A) 1.043 KW (B) 1.956 KW
(C) 3 KW (D) 1.5 KW

2. Neglecting the effects of armature reaction, if the excitation of a synchronous motor running with a constant load is decreased from its normal value it leads to

- (A) increase in p.f.
(B) increase in I_a
(C) decrease in p.f.
(D) increase in I_a and decrease in p.f.

3. No load test on a 3-phase induction motor was conducted at different supply voltage and a plot of input power versus voltage was drawn. This curve is extrapolated to intersect the Y axis. This intersection point yields

- (A) friction and windage loss
(B) core loss
(C) stator copper loss
(D) stray load loss

4. In a 3-phase delta/star transformer, the phase displacement of secondary line voltage with respect to corresponding primary line voltages will be _____.

- (A) zero (B) 30° lag
(C) 30° lead (D) 180°

5. If a 230 V d.c. series motor is connected to a 230 V a.c. supply,

- (A) the motor will vibrate violently.
(B) the motor will run with less efficiency and more sparking.
(C) the motor will not run.
(D) the fuse will be blown.

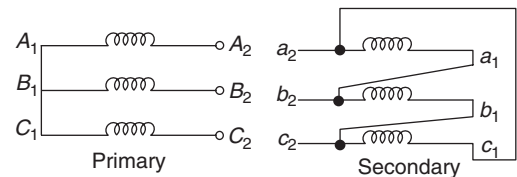
6. A 3-phase 20-pole, synchronous generator has 180 stator slots with single-layer full pitch coils. There are 6 conductors per slot and all coils per phase are connected in series. The rotor is driven at 300 rpm and the flux pole (sinusoidally distributed) is 25 mWb. The voltage induced per phase is nearest to

- (A) 500 V
(B) 1000 V
(C) 1500 V
(D) 2000 V

7. The phase sequence of an alternator reverses if

- (A) The field current is reversed keeping the direction of rotation same
(B) The field current remains the same but the direction of rotation is reversed
(C) Field current is reversed and number of poles doubled
(D) Number of poles doubled and field current remain same

8. Three single phase transformer are connected to form a 3 phase transformer bank. The transformer are connected as in figure. The transformer connection will be represented by



- (A) Yd0 (B) Yd1 (C) Yd6 (D) Yd11

9. The primary current in a current transformer is dictated by

- (A) The secondary burden
(B) The core of the transformer
(C) The load current
(D) None of these

10. A 100 MVA synchronous generator operates on full load UPF at a frequency of 50 Hz. The load is suddenly reduced to 50 MW. Due to time lag in governor system, the system valve begins to close after 0.4 seconds. The frequency at the end of 0.4 seconds is _____. (Given $H = 5$ KW – seconds/KVA of generator capacity)

- (A) 51 Hz (B) 52 Hz (C) 53 Hz (D) 49 Hz

11. Which of the following conditions are necessary for parallel operation of alternators?

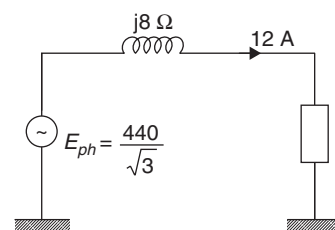
1. Equal terminal voltage
2. Same frequency
3. Same phase sequence
4. Same KVA rating
5. Same $\frac{X}{R}$ ratio

- (A) 1, 2 and 3 (B) 2, 3 and 5
(C) 1, 2 and 4 (D) 2, 3 and 4

12. A 400 V, 20 kW, 4 pole, 50 Hz Y-connected induction motor has full load slip of 3.4%. The output torque of the machine at full load is

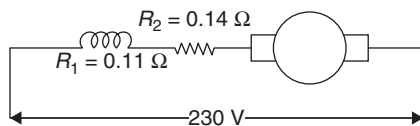
- (A) 250 Nm (B) 141 Nm (C) 120 Nm (D) 131.8 Nm

13. The per phase synchronous reactance of a star connected 440 V, 50 Hz alternator is 8Ω . It supplies a balanced capacitive load current of 12 A, as shown in figure. If zero voltage regulation is desirable, the load power factor should be

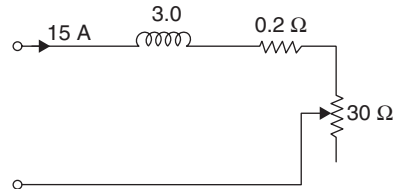


- (A) 0.923 (B) 0.783 (C) 0.85 (D) 0.65

14. If the applied voltage of a certain transformer is increased by 50% and the frequency is reduced to 50% (assuming that the magnetic circuit remains unsaturated), the maximum core flux density will _____.
 (A) change to three times the original value
 (B) change to 1.5 times the original value
 (C) change to 0.5 times as the original value
 (D) remain the same as the original value.
15. A 20 kVA 400V/200V single phase transformer with a percentage resistance of 4% and percentage reactance of 8% is supplying a current of 80 A to a resistive load. The value of load voltage is
 (A) 193.6 V (B) 387.2 V
 (C) 190.4 V (D) 197.8 V
16. A DC series motor is rated 230 V, 1000 rpm, 80 A. The series field resistance is 0.11Ω and the armature resistance is 0.14Ω . If the flux at an armature current of 20 A is 0.4 times of that under rated condition, calculate the speed at the reduced armature current of 20 A.



- (A) 2168 rpm
 (B) 876 rpm
 (C) 2678 rpm
 (D) 1980 rpm
17. A separately excited dc machine, having an armature resistance of 2 ohms was working on a 220 V supply and drawing 10 A armature current from the source. When the supply voltage suddenly dropped to 200 V, assuming that the field circuit source voltage remained unaffected, how will the armature current of the machine react to the change?
 (A) It will initially rise to 11 A and then settle down to 10 A.
 (B) It will fall momentarily to 9.09 A and then slowly attain 10 A.
 (C) It will reduce to zero first and then settle back to 10 A.
 (D) It will remain unaffected by the change and continue to be 10 A.
18. A 50 KW synchronous motor is tested by driving it by another motor. When the excitation is not switched on, driving motor takes 800 W. When the armature is short circuited and rated armature current of 10 A is passed through it, the driving motor requires 2500 W. On open circuiting the armature with rated excitation, driving motor takes 1800 W. Neglecting the losses in the driving motor, the efficiency of the synchronous motor is
 (A) 87.71%
 (B) 91.83%
 (C) 83.05%
 (D) 85.32%
19. A 3 ϕ , 50 Hz, 6 pole induction motor has an equivalent diagram as shown in the figure. The torque developed in the motor is



- (A) 113.4 Nm
 (B) 92.8 Nm
 (C) 185.6 Nm
 (D) 193.4 Nm
20. Two transformers having equal voltage ratio are connected in parallel. They have equal impedance and equal $\frac{X}{R}$ ratio. The ratio of full load KVA delivered to the sum of their individual KVA ratings is
 (A) 0.5 (B) 1 (C) 0.6 (D) 0.75
21. A 400 V, 50 Hz, 100 KVA, 0.9 pf leading, delta connected synchronous machine has synchronous reactance of 3Ω . The armature resistance can be neglected. The core loss of the machine is 1.2 KW and the friction and windage losses are 2.8 KW. If the shaft is supplying 20 KW at 0.9 leading, then the line current drawn is
 (A) 36.88 A
 (B) 52.16 A
 (C) 18.44 A
 (D) 26.08 A
22. A 200 V dc series motor runs at 500 rpm when taking a current of 25 A. The resistance of the armature is 0.5Ω and that of field is 0.3Ω . If the current remains constant, the resistance necessary to reduce the speed to 250 rpm is
 (A) 1.8Ω (B) 3.6Ω (C) 5.4Ω (D) 7.2Ω
23. A 3 – ϕ squirrel cage induction motor has a starting torque of 140% and a maximum torque of 280% with respect to rated torque at rated voltage and rated frequency. Neglect the stator resistance and rotor losses. The value of slip for maximum torque is
 (A) 300%
 (B) 26.79%
 (C) 2.67%
 (D) 267.9%

Common Data for Questions 24 and 25:

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A 250 V separately excited DC motor has armature resistance of 2.5 ohm. When driving a load at 600 rpm with constant torque, the armature takes 20 A. The motor is controlled by a DC chopper operating with a frequency of 400 Hz and an input voltage of 250 V D.C.

24. What should be the value of duty ratio, if it is desired to reduce the speed from 600 rpm to 400 rpm?
(A) 0.63 (B) 0.55

(C) 0.73 (D) 0.85

25. The motor speed at rated current and a duty ratio of 0.5, if the motor is regenerating is?
(A) 225 rpm (B) 525 rpm
(C) 325 rpm (D) 425 rpm

ANSWER KEYS

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. A | 2. D | 3. A | 4. C | 5. B | 6. B | 7. B | 8. B | 9. C | 10. A |
| 11. A | 12. D | 13. A | 14. A | 15. A | 16. C | 17. C | 18. B | 19. D | 20. B |
| 21. A | 22. B | 23. B | 24. C | 25. B | | | | | |

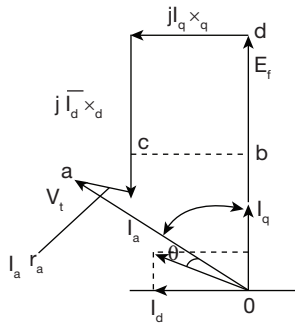
ELECTRICAL MACHINES TEST I

Number of Questions: 35

Section Marks: 90

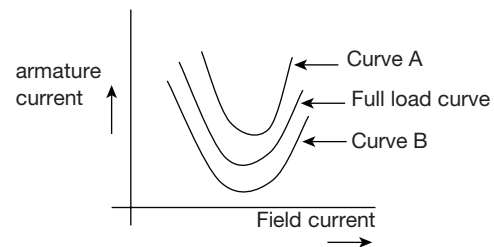
Directions for questions 1 to 35: Select the correct alternative from the given choices.

1. The phasor diagram representation of salient pole synchronous motor as shown in the figure. Then the synchronous motor operated is in _____ condition.



- (A) Over excitation (B) Under excitation
(C) Normal excitation (D) No excitation
2. In which of the following methods is the regulation is more than the actual regulation value.
(A) EMF method
(B) Ampere – turn method
(C) ZPF method
(D) None of the above
3. The Buchholz relay is generally placed
(A) Inside the transformers tank
(B) Near to bushings of the transformer
(C) Between transformer tank and conservator
(D) None of the above
4. A 500V, 10 pole DC Shunt motor has rated armature current of 220 A with simplex wave winding. Find the current per path of the armature if the armature resistance is 5Ω .
(A) 20A (B) 100A
(C) 40A (D) 110A
5. A transformer has copper loss of 3% and reactance drop of 4% when tested at full load. The full – load regulation at U. P. F is
(A) 3% (B) 5%
(C) 4% (D) 8%
6. Which of the following torque has maximum value.
(A) Locked Rotor torque
(B) Break down torque
(C) Pull – up torque
(D) Full load torque
7. The conditions to be satisfied for proper synchronization of alternators with the bus bar are.
(1) The terminal voltages are same
(2) The frequencies are same

- (3) Phase sequences are same
(A) 1 and 2 (B) 2 and 3
(C) 1 and 3 (D) 1, 2 and 3
8. The highest possible speed of 60 Hz alternator is
(A) 3600 rpm (B) 1500 rpm
(C) 3000 rpm (D) Insufficient data
9. Which of the following methods are suitable for reversing the direction of rotation of a D. C. motor
(1) By reversing the current direction in the armature
(2) By reversing the field current direction
(3) By reversing Both armature and field current direction.
(A) 1 and 2 (B) 1 and 3
(C) 2 and 3 (D) 1, 2, and 3
10. A long shunt generator running at 1,000 rpm. Supplies 30 kW at a terminal voltage of 200 V. the resistance of armature, shunt field and series field are 0.06, 120 and 0.08 ohms respectively. The total copper loss of the machine is
(A) 3554 watts (B) 1380 watts
(C) 1714 watts (D) 2174 watts
11. The stepping angle for a 3 – stack; 18 – tooth variable reluctance motor is
(A) 7.56° (B) 6.67°
(C) 1.56° (D) 4.57°
12. A 230 V, 50 Hz, 6 – pole, 1 – phase, induction motor takes input current of 3 A and output power is 500 watts. calculate the efficiency of motor at 0.8 pf
(A) 85.42% (B) 88.96%
(C) 90.57% (D) Insufficient data
13. The V – Curves of synchronous motor as shown in the figure. In that curve – B Represents.



- (A) Half load curve (B) $\frac{1}{4}$ Full load curve
(C) $\frac{1}{2}$ Full load curve (D) $\frac{3}{2}$ Full load curve
14. A Δ/Y connected 3 – phase transformer has a voltage ratio of (Δ) 22 kV/330 kV(Y) a (line to line). The transformer is feeding 500 MW and 200 MVAR to the grid (330 kV) The VA rating of the transformer is

- (A) 500 MVA (B) 700 MVA
(C) 200 MVA (D) 538 MVA
15. A 3 – ϕ 400 V, 50 Hz, 6 pole induction motor has a stator winding of 36 slots and rotates at a speed of 900 rpm. The fifth harmonic belt factor of a stator winding is
(A) 0.965 (B) 0.707
(C) 0.854 (D) 0.258
16. A five alternators are working in parallel with their terminal voltage equal to the rated value. In that one of the machine, which has a synchronous reactance of 40% and resistance 2% delivers a power output in kW equal to 60% of its rated KVA. If the emf of this unit equals 1.2 times the terminal voltage. The power factor of the machine is
(A) 0.5 lagging (B) 0.643 lagging
(C) 0.8 lagging (D) Unity
17. A dc series motor running a fan at 1500 rpm takes 25 A from 250 V. the armature plus field resistance is 0.8Ω if an additional resistance of 5Ω is inserted in series with the armature circuit. Then find the corresponding speed of the machine.
(A) 1000 rpm (B) 1160 rpm
(C) 1500 rpm (D) 1480 rpm
18. A 20 kW, 400 V, 3 – phase, 4 – pole, 50 Hz delta connected squirrel cage induction motor gave the following data. For blocked rotor test: 220 V, 25A, 10 kW Calculate the starting torque at a rated voltage and frequency for a 500 watts of stator core loss at rated voltage and frequency. [The d. c. resistance measured between any two stator terminals is 0.8Ω]
(A) 210 Nm (B) 200 Nm
(C) 150 Nm (D) 174.16 Nm
19. A 3–phase 11 kV, 20 MW, star connected synchronous generator has a synchronous impedance of $0.6 + j8$ ohms per phase, if the excitation is such that the open circuit voltage is 15 kV. Calculate the power factor at the maximum power output of the machine.
(A) 0.79 (B) 0.64
(C) 0.82 (D) 0.54
20. A single phase transformer with a ratio of 400/100 V takes a no load current of 5A at 0.3 power factor lagging. If the secondary supplies a current of 100 A at a p. f 0.6 lagging. The current taken by the primary is
(A) 25 A (B) 29.76 A
(C) 30.94 A (D) 27.84 A
21. A 3–phase, star connected, 50 Hz synchronous generator has direct – axis synchronous reactance of 0.8PU and quadrature – axis synchronous reactance of 0.5PU. the generator delivers rated KVA at rated voltage Calculate the voltage regulation at 0.8p. f. lagging when Resistive drop at full load is 0.02PU
(A) 31% (B) 42%
(C) 38% (D) 61%
22. A 3–phase, 15 kV, 20 MW, Y– Connected synchronous generator has synchronous impedance of $0.6 + j5.0$ ohm per phase. Find the maximum power output of generator for open circuit voltage of 18 KV
(A) 54 MW (B) 18 MW
(C) 21.6 MW (D) 64.8 MW
23. A 50 KVA, 1–phase transformer 2000 volts to 200 volts has primary resistance of 1.2Ω and a secondary resistance of 0.02Ω . Calculate the full load efficiency at 0.85 p.f if the iron loss of the transformer is 90% of the full load loss.
(A) 95.50% (B) 95.93%
(C) 97.32% (D) 91.79%
24. A 500 KVA transformer has an efficiency of 95% at full load if the maximum efficiency occurs at two third of full load. Calculate the efficiency at three quarters of full load and 0.8 p.f at all loads.
(A) 94.82% (B) 95.34%
(C) 96.72% (D) 97.82%
25. A 3 – phase 400 V, 1500 rpm slip ring induction motor is operating with 4% slip and taking a stator current of 50 A. speed of the motor is reduced at constant torque to 800 rpm. Using stator voltage control calculate the new value of stator current.
(A) 123.58A (B) 171.38A
(C) 173.54A (D) 23.5A
26. A 200 watts split phase motor draws its starting winding current of 5 A lagging the supply voltage by 20° electrical and its running winding current is 8 A lagging by 60° electrical. The power factor of the motor at steady state condition is
(A) 0.173 lagging (B) 0.70 lagging
(C) 0.766 lagging (D) 0.45 lagging
27. A 20 MVA, 10 kV, 4 – pole, 50 Hz alternator is connected to infinite bus – bars. The short circuit current is 3 times the normal full load current and the moment of inertia of the rotating system is $20,000 \text{ kg} - \text{m}^2$. The period of oscillation of alternator is
(A) 1.01 sec (B) 1.01 m sec
(C) 4.47 m sec (D) 1.01 μsec
28. The self and mutual inductance of a two winding transformer are
 $L_1 = 5 \text{ mH}$, $L_2 = 8 \text{ mH}$, $M_{12} = M_{21} = 2 \text{ mH}$
The transformer is connected to 100 volts, 50 Hz supply. Calculate the primary winding current when the transformer secondary is connected to 0.4mH load.
(A) 70.37 A (B) 35.18 A
(C) 140.74 A (D) 17.59 A
29. In a Hopkinson's test on two 200 V, 200 kW generators, the circulating current is equal to the full load current and in addition 100 A are taken from the supply. The efficiency of the machine is
(A) 91.34% (B) 95.34%
(C) 98.35% (D) 90.12%

30. A 5 kW, 200 V, 50 Hz 4 – pole 3 – ϕ star connected induction motor has the following data
No load: Line voltage 200 V, line current 5A; total input 300 W
Blocked rotor: Line voltage 100 V, line current 25 A, total input 1800 W calculate the blocked rotor input when the normal voltage is applied.
(A) 1800 W (B) 5600 W
(C) 2100 W (D) 3600 W
31. A 300 W, single phase 50 Hz, 350 V universal motor runs at 1500 rpm and takes a current 2 A when connected to 350 V AC Supply. Calculate the back emf of the motor. Assume $R_a = 25\Omega$ and $L_a = 0.5H$
(A) 0 V (B) 104 V
(C) 110 V (D) 100 V

32 and 33 Common data Question:

A 50 kW, 3-phase slip – ring induction motor of negligible stator impedance runs at a speed of 0.95 times synchronous speed at rated torque. The slip at maximum torque is 5 times

the full load value. [If the rotor resistance of the motor is increased by 6 times]

32. The rotor copper loss at rated torque
(A) 36.84 kW (B) 15.78 kW
(C) 18.42 kW (D) 11.052 kW
33. The speed corresponding to maximum torque
(A) $1.5N_s$ (B) $0.5N_s$
(C) $-1.5N_s$ (D) $-0.5N_s$

Linked answer questions 34 and 35:

A 250 V d. c series motor runs at 600 rpm. When taking a line current of 30 A. the resistance of the armature is 0.4Ω and that of the series field 0.3Ω if the motor develops, the same torque when armature diverter of 20Ω used [Assume a straight line magnetization curve]

34. The armature current of motor is
(A) 17.08 A (B) 5.94 A
(C) 8.04 A (D) 2.97 A
35. The speed of the motor is
(A) 600 rpm (B) 1200 rpm
(C) 1152 rpm (D) 1054 rpm

ANSWER KEYS

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. A | 2. B | 3. C | 4. D | 5. A | 6. B | 7. D | 8. A | 9. A | 10. A |
| 11. B | 12. C | 13. A | 14. D | 15. D | 16. B | 17. B | 18. D | 19. A | 20. B |
| 21. D | 22. A | 23. D | 24. B | 25. B | 26. B | 27. A | 28. A | 29. B | 30. B |
| 31. B | 32. B | 33. D | 34. B | 35. C | | | | | |

HINTS AND EXPLANATIONS

1. Form the given phasor diagram the salient pole synchronous motor operated in leading power factor that means the motor operated in over excited condition.
Choice (A)
2. Choice (B)
3. The relay arranged in the pipe – line between the transformers tank and separate oil conservator.
Choice (C)
4. Rated armature current $I_a = 220$ A
Number of poles $P = 10$
With simplex wave winding
Number of parallel paths $A = 2$
Current per path $= I_C = \frac{I_a}{A} = \frac{220}{2} = 110$ A
Choice (D)
5. Given data
 P . U. resistance = 0.03
 P . U. reactance = 0.04
Voltage regulation at unity p. f full load = 3%
Choice (A)
6. By observing the Torque – speed characteristic of induction motor, Breakdown torque has maximum value.
Choice (B)

7. Choice (D)

$$8. \text{Speed} = \frac{120f}{P}$$

$$= \frac{120 \times 60}{2} = 3600 \text{ rpm} \quad \text{Choice (A)}$$

9. The rotation of D. C. motor is reversed either By reversing the field current (or) armature current direction.
Choice (A)

10. Power output = 30,000 watts
Terminal voltage = 200 volts
Output current $= \frac{\text{Power output}}{V} = \frac{30000}{200} = 150$ A

$$\text{Shunt field current } i_{sh} = \frac{V}{R_{sh}} = \frac{200}{120} = 1.67 \text{ A}$$

$$\text{Armature current} = 150 + 1.67 = 151.67 \text{ A}$$

$$\text{Armature copper loss} = (151.67)^2 \times 0.06 = 1380.22 \text{ watts}$$

$$\text{Series field copper loss} = (151.67)^2 \times 0.08 = 1840.30 \text{ watts}$$

$$\text{Shunt field copper loss} = 200 \times 1.67 = 334 \text{ watts}$$

$$\text{Total copper loss} = 3554.52 \text{ watts}$$

Choice (A)

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11. Number of stacks (or) phases $n = 3$
 Number of teeth (or) poles $P = 18$
 Stepping angle $\alpha = \frac{360}{np} = \frac{360}{3 \times 18} = 6.67^\circ$ Choice (B)
12. Power output = 500 watts
 Power input $P_n = 230 \times 3 \times 0.8 = 552$ watts
 Motor efficiency $= \frac{P_o}{P_{in}} = \frac{500}{552} \times 100 = 90.57\%$
 Choice (C)
13. As observing the V –Curves of synchronous motor the curve B is below the full load curve. So. Only option (A) load is below the full load. Choice (A)
14. Load MVA $\bar{S} = 500 + j200$
 $= 538$ MVA
 Choice (D)
15. Belt factor $K_{dn} = \frac{\sin \frac{nmr}{2}}{m \sin \frac{n\gamma}{2}}$
 Fifth harmonic $n = 5$
 Slots per pole per phase $= \frac{36}{3 \times 6} = 2$
 Slot angular pitch $\gamma = \frac{P \times 180}{S} = \frac{6 \times 180}{36} = 30$
 $K_{as} = \frac{\sin\left(\frac{5 \times 2 \times 30}{2}\right)}{2 \sin\left(\frac{5 \times 30}{2}\right)} = \frac{\sin 150}{2 \sin 75} = 0.258$ Choice (D)
16. given $\frac{I_a r_a}{V_t} = 2\%$ & $\frac{I_a X_s}{V_t} = 40\%$
 $E_f^2 = (V_t \cos \theta + I_a r_a)^2 + (V_t \sin \theta + I_a X_s)^2$
 $= V_t^2 \left[\left(\cos \theta + \frac{I_a r_a}{V_t} \right)^2 + \left(\sin \theta + \frac{I_a X_s}{V_t} \right)^2 \right]$
 $I_a = 0.6 I_{\text{rated}}$
 $(1.2 V_t)^2 = V_t^2 [(\cos \theta + 0.02 \times 0.6)^2 + (\sin \theta + 0.4 \times 0.6)^2]$
 $1.44 = 1.0577 + 0.024 \cos \theta + 0.48 \sin \theta$
 $0.3823 = 0.024 \cos \theta + 0.48 \sin \theta$
 $\times 0.48 \sin(\theta + 2.862) = 0.3823$
 $\times \sin(\theta + 2.862) = 0.7964$
 $\theta + 2.862 = 52.79$
 $\theta = 49.93$
 Power factor $= \cos \theta = 0.643$ lagging Choice (B)
17. $E_{a1} = 250 - 25 - 0.8 = 230$ V
 $E_{a2} = 250 - \frac{n_2}{60} (5 + 0.8)$
 $= 250 - \frac{n_2}{60} \times 5.8$

- $$\frac{E_{a2}}{E_{a1}} = \frac{N_2 \phi_2}{N_1 \phi_1} \Rightarrow \frac{250 - 0.097 n_2}{230} = \frac{n_2 \cdot \frac{n_2}{60}}{(1500)(25)}$$
- $$\Rightarrow \frac{230}{60} n_2^2 - (250)(1500)(25) + (0.097)(1500)(25) n_2 = 0$$
- $$\Rightarrow 3.83 n_2^2 + 3637.5 n_2 + 9,37,5000 = 0$$
- $$n_2 = 1160.14 \text{ rpm}$$
- Choice (B)
18. With the stator winding in delta, the resistance per phase R is obtained from the relation.
- $$\frac{R \times 2R}{3R} = 0.8 \Rightarrow R = \frac{3}{2} \times 0.8 = 1.2 \Omega$$
- effective resistance per phase $= 1.2 \times 1.2 = 1.44$
 Power input at rated voltage during blocked rotor test
 $= 10 \times \left(\frac{400}{220} \right)^2 = 33.05$ KW
 Stator current at rated voltage during blocked rotor test
 $= 33.05 \times \frac{400}{220} = 60.06$ A
 Air gap power at rotated voltage and frequency
 $P_g = 33057 - 3 \left(\frac{60.06}{\sqrt{3}} \right)^2 \times 1.44 - 500 = 27,357.43$ W
 Synchronous speed $=$
 $\omega_s = \frac{4\pi f}{P} = \frac{4\pi \times 50}{4} = 50\pi \text{ rad/sec}$
 Starting torque $= \frac{27,357}{50\pi} = 174.16$ N – m
 Choice (D)
19. Current corresponding to maximum output
- $$I_{\max} = \frac{\sqrt{V^2 + E^2 - 2VE \cos \theta}}{Z_s}$$
- $$= \frac{\sqrt{\left(\frac{11000}{\sqrt{3}} \right)^2 + \left(\frac{15000}{\sqrt{3}} \right)^2 - 2 \left(\frac{11000}{\sqrt{3}} \times \frac{15000}{\sqrt{3}} \right) \cos(85.71)}}{8.02}$$
- $$< 1290.41 \text{ A}$$
- Power factor $= \frac{P_{\max}}{v I_{\max}}$
- Maximum power output/phase $= \frac{EV}{Z_s} - \frac{V^2}{Z_s} \cos \theta$
- $$= \frac{\frac{15000}{\sqrt{3}} \times \frac{11000}{\sqrt{3}}}{8.02} - \frac{\left(\frac{11000}{\sqrt{3}} \right)^2}{8.02} \cos 85.71$$
- $$= 6.48 \text{ MW} = 6,48,1655$$
- $$\cos \phi = \frac{6,48,1655}{\frac{11000}{\sqrt{3}} \times 1290.41} = 0.79$$
- Choice (A)

$$\begin{aligned} 20. \cos\phi_2 = 0.6 &\Rightarrow \phi_2 = \cos^{-1}(0.6) = 53.13^\circ \\ \cos\phi_0 = 0.3 &\Rightarrow \phi_0 = \cos^{-1}(0.3) = 72.54^\circ \\ K &= \frac{100}{400} = \frac{1}{4} \end{aligned}$$

$$I_2 = KI_1 = 100 \times \frac{1}{4} = 25 \text{ A}$$

$$I_0 = 5 \text{ A}$$

$$\text{Angle between } I_0 \text{ and } I_2 = 72.54 - 53.13 = 19.41^\circ$$

$$\begin{aligned} I_1 &= \sqrt{5^2 + 25^2 + 2 \times 25 \times 5 \times \cos 19.41^\circ} \\ &= 29.76 \text{ A} \end{aligned} \quad \text{Choice (B)}$$

$$\begin{aligned} 21. I_a &= 1 \text{ P.U.}, V = 1 \text{ P.U.}, X_d = 0.8 \text{ P.U.} \\ X_q &= 0.5 \text{ P.U.}; R_a = 0.02 \text{ P.U.} \\ \tan\Psi &= \frac{V \sin\phi + I_a X_q}{V \cos\phi + I_a R_a} = \frac{1 \times 0.6 + 1 \times 0.5}{1 \times 0.8 + 1 \times 0.02} = 1.341 \end{aligned}$$

$$\Psi = 53.29^\circ$$

$$\delta = \Psi - \phi = 53.29 - 36.86^\circ = 16.43^\circ$$

$$I_d = I_a \sin\Psi = 1 \times \sin(53.29) = 0.80 \text{ A}$$

$$I_q = I_a \cos\Psi = 1 \times \cos(53.29) = 0.59 \text{ A}$$

$$E_0 = V \cos\delta + I_q R_a + I_d X_d = 1.61 \text{ V}$$

$$\% \text{ regulation} = \frac{1.61 - 1}{1} \times 100 = 61\% \quad \text{Choice (D)}$$

$$22. P_{\max} \text{ per phase} = \frac{EV}{X_s}$$

Where V is the terminal voltage (or bus – bar. Voltage on general) and E is the emf of the machine

$$P_{\max} = \frac{\left(\frac{15,000}{\sqrt{3}}\right)\left(\frac{18,000}{\sqrt{3}}\right)}{5} = 18,000 \text{ KW/phase}$$

$$P_{\max} = 3 \times 18,000 = 54,000 \text{ KW} = 54 \text{ MW} \quad \text{Choice (A)}$$

$$\begin{aligned} 23. R_{02} &= R_2 + KR_1 \\ &= 0.02 + \frac{1.2}{100} = 0.032\Omega \end{aligned}$$

$$\text{Full load current} = \frac{50,000}{200} = 250 \text{ A}$$

$$\begin{aligned} \text{Full load copper loss} &= I_2^2 R_{02} = 250^2 \times 0.032 \\ &= 2000 \text{ watts} \end{aligned}$$

$$\text{Iron loss} = 0.9 \times 2000 = 1800 \text{ watts}$$

$$\text{Total loss} = 1800 + 2000 = 3800 \text{ watts}$$

$$\text{Full load output} = 50 \times 0.85 = 42.5 \text{ kW} = 42,500 \text{ watts}$$

$$\text{Full load } \eta = \frac{42,500}{42,500 + 3800} = 91.79\% \quad \text{Choice (D)}$$

$$\begin{aligned} 24. \text{ Full load output at } 0.8 \text{ p.f.} &= 500 \times 0.8 = 400 \text{ kW} \\ \text{Full load efficiency } \eta &= 95\% \text{ (or) } 0.95 \end{aligned}$$

$$\text{Full load input} = \frac{\text{Output}}{\eta} = \frac{400}{0.95} = 421.05 \text{ kW}$$

$$\begin{aligned} \text{Total losses on full load} &= P_i + P_c = \text{Input} - \text{Output} \\ &= 421.05 - 400 = 21.05 \text{ kW} \end{aligned} \quad \text{—————(1)}$$

Maximum efficiency occurs at two third of full load

$$\left(\frac{2}{3}\right)^2 P_c = P_i \Rightarrow \frac{4}{9} P_c = P_i \quad \text{—————(2)}$$

$$\text{From (1) and (2) } P_c + \frac{4}{9} P_c = 21.05$$

$$\frac{13}{9} P_c = 21.05$$

$$\text{Copper loss at Full load is } P_c = 14.57 \text{ kW}$$

$$\text{Iron loss } P_i = \frac{4}{9} \times 14.57 = 6.47 \text{ kW}$$

At three quarter of full load and 0.8 p.f

$$\text{Output} = \frac{3}{4} \times 500 \times 0.8 = 300 \text{ kW}$$

$$\text{Total losses} = 6.47 + \left(\frac{3}{4}\right)^2 \times 14.57 = 14.66 \text{ kW}$$

$$\text{Efficiency } \eta = \frac{\text{Output}}{\text{Input}} = \frac{300}{300 + 14.66} = 0.9534$$

$$= 95.34\%$$

Choice (B)

$$25. \text{ Supply voltage } V = 400 \text{ V}$$

$$\text{Slip } S = 0.04$$

$$\text{Stator current } I_1 = 50 \text{ A}$$

$$\text{Synchronous speed} = 1500 \text{ rpm}$$

$$\text{Reduced speed } N = 800 \text{ rpm}$$

$$\text{Slip} = \frac{1500 - 800}{1500} = \frac{700}{1500} = 0.47$$

$$\text{Torque developed } T \propto SV^2$$

$$\begin{aligned} V a \sqrt{\frac{1}{S}} &\Rightarrow V^1 = V \sqrt{\frac{S}{S^1}} \\ &= 400 \sqrt{\frac{0.04}{0.47}} = 116.69 \text{ V} \end{aligned}$$

$$\text{Stator current } I_1 \propto SV$$

$$I_1^1 = I^1 \times \frac{S^1 V^1}{SV} = \frac{50 \times 0.47 \times 116.69}{0.04 \times 400}$$

$$I_1^1 = 171.38 \text{ A}$$

Choice (B)

$$26. \text{ Starting winding current } i_s = 5 \angle -20^\circ \text{ A}$$

$$\text{Running winding current } I_m = 8 \angle -60^\circ \text{ A}$$

$$\begin{aligned} \text{Total current } I_L &= I_s + I_m \\ &= 5 \angle -20^\circ + 8 \angle -60^\circ \\ &= 12.25 \angle -44.80^\circ \end{aligned}$$

$$\text{Power factor } \cos(-44.80) = 0.70 \text{ lagging} \quad \text{Choice (B)}$$

$$27. \text{ Moment of inertia of the rotating system } J = 20,000 \text{ kg} - \text{m}^2.$$

$$\text{Phase voltage } E = \frac{10,000}{\sqrt{3}} = 5773.50 \text{ volts}$$

$$F = 50 \text{ Hz}$$

$$\text{Synchronous speed } N_s = \frac{2f}{P} = \frac{2 \times 50}{4} = 25 \text{ rps}$$

$$\text{Full load current } I = \frac{20,000 \times 1000}{\sqrt{3} \times 19,000} = \frac{2000}{\sqrt{3}} \text{ A}$$

$$\text{Short circuit current } I_{sc} = 3I = \frac{6000}{\sqrt{3}}$$

$$\begin{aligned} \text{Time period of oscillation } T &= 9.1n_s \sqrt{\frac{J}{EI_s f}} \\ &= 9.1 \times 25 \sqrt{\frac{20,000}{10,000 \times \frac{6000}{\sqrt{3}} \times 50}} = 1.01 \text{ sec} \quad \text{Choice (A)} \end{aligned}$$

28. Consider all the given parameters are referred to primary
 $a = 1$, $L_1 - aM = 5 - 2 = 3 \text{ mH}$
 $aM = 2 \text{ mH}$
 $a^2 L_2 - aM = 8 - 2 = 6 \text{ mH}$
load inductance L referred to primary $a^2 L = 0.4 \text{ mH}$
total inductance seen by the primary applied voltage

$$\begin{aligned} &= (L_1 - aM) + \frac{(aM)(a^2 L_2 - aM + a^2 L)}{aM + a^2 L_2 - aM + a^2 L} \\ &= 3 \times 10^{-3} + \frac{(2 \times 10^{-3})(6 + 0.4) \times 10^{-3}}{(2 + 6 + 0.4) \times 10^{-3}} \\ &= 3 \times 10^{-3} + \frac{2 \times 6.4 \times 10^{-3}}{8.4} \\ &= 3 \times 10^{-3} + 1.52 \times 10^{-3} = 4.523 \times 10^{-3} \text{ H} \\ \text{Total reactance at the primary terminals} \\ &= 2\pi \times 50 \times 4.523 \times 10^{-3} = 1.420 \Omega \\ \text{Current primary winding} &= \frac{100}{1.420} = 70.37 \text{ A} \end{aligned}$$

Choice (A)

29. Output current of the generator

$$I_1 = \frac{20000}{200} = 100 \text{ A}, I_2 = 100 \text{ A}$$

$$\eta = \sqrt{\frac{I_1}{I_1 + I_2}} = \sqrt{\frac{1000}{1100}} = 95.34\% \quad \text{Choice (B)}$$

30. Short circuit current with normal voltage

$$= 25 \times \frac{200}{100} = 50 \text{ A}$$

$$\begin{aligned} \text{Blocked rotor input with normal voltage} &= 1800 \times \left(\frac{50}{25}\right)^2 \\ &= 1800 \times 4 = 5600 \quad \text{Choice (B)} \end{aligned}$$

31. $X_a = 2\pi f L = 2\pi \times 50 \times 0.5 = 157.07 \Omega$

$$\begin{aligned} V^2 &= (E_{bac} + I_a R_a)^2 + (I_a X_a)^2 \\ E_{bac} &= -I_a R_a + \sqrt{V^2 - (I_a X_a)^2} \\ &= -2 \times 25 + \sqrt{(350)^2 - (2 \times 157.07)^2} \\ &= -50 + 154.32 = 104.32 \quad \text{Choice (B)} \end{aligned}$$

32. Motor output $= P_{out} = 50 \text{ kW}$

$$\text{Speed at rated torque } N = 0.95 N_s$$

$$\text{Full load slip } S_f = \frac{N_s - N}{N_s} = \frac{N_s - 0.95 N_s}{N_s} = 0.05$$

$$\text{Slip corresponding to maximum torque } S_{max} = 5S_f$$

$$S_{max} = 0.05 \times 5 = 0.25$$

$$\begin{aligned} \Rightarrow \frac{\text{Full load torque}}{\text{maximum torque}} &= \frac{T_f}{T_{max}} = \frac{2S_{max} S_f}{S_{max}^2 + S_f^2} \\ &= \frac{2 \times 0.05 \times 0.25}{(0.25)^2 + (0.05)^2} = 0.384 \end{aligned}$$

When the rotor circuit resistance is increased 6 times, the magnitude of maximum torque will remain unchanged because it is independent of load but the slip. Corresponding to the maximum torque will change. Let the new slip corresponding to maximum torque be S'_{max}

$\Rightarrow$ Since slip corresponding to maximum torque is proportional to rotor resistance provided its stand-still reactance is fixed so

$$S'_{max} = S_{max} \times \frac{R^1}{R} = S_{max} \times 6 = 1.5$$

$$\text{Now } \frac{T_f}{T_{max}} = \frac{2 \times S_f^1 S'_{max}}{(S_f^1)^2 + (S'_{max})^2}$$

Where S_f^1 is the new full load slip.

$$0.384 = \frac{2 \times S_f^1 \times 1.5}{(S_f^1)^2 + (1.5)^2}$$

$$\Rightarrow S_f^1 - 7.8125 S_f^1 + 2.25 = 0$$

$$\Rightarrow S_f^1 = 0.3, 7.51$$

So new speed at full load

$$N^1 = (1 - S_f^1) N_s = (1 - 0.3) N_s = 0.7 N_s$$

$$\text{Gross torque at full load} = \frac{50 \times 1000}{2\pi \times \frac{N}{60}} = \frac{50 \times 1000}{2\pi \times \frac{0.7 N_s}{60}}$$

New power output at full load

$$= \text{Gross torque at full load} \times \frac{2\pi N^1}{60}$$

$$= \frac{50 \times 1000}{2\pi \times \frac{0.95 N_s}{60}} \times \frac{2\pi N^1}{60}$$

$$= \frac{50,000 \times 0.7 N_s}{0.95 N_s} = 36,842.10 = 36.84 \text{ kW}$$

Rotor copper loss at rated torque

$$= \frac{\text{Power Output}}{1 - S} \times S = \frac{36.842}{0.7} \times 0.3$$

$$= 15.78 \text{ kW}$$

Choice (B)

33. Speed corresponding to maximum torque $= N_s(1 - S_{\max})$
 $= N_s(1 - 45) = -0.5N_s$ Choice (D)

34. Resistance of the motor $= 0.4 + 0.8 = 1.2\Omega$

$$E_{b1} = 250 - 30 = 1.2 = 214V$$

Let the motor input current be I_2

$$\text{Series field voltage drop} = 0.8I_2$$

$$\text{Potential difference at brushes} = 250 - 0.8I_2$$

$$\therefore \text{Armature divertor current} = \left[\frac{250 - 0.8I_2}{20} \right] A$$

$$\therefore \text{Armature current } I_{a2} = \left(\frac{20.8I_2 - 250}{20} \right)$$

As torque in both cases same

$$\therefore \phi_1 I_{a1} = \phi_2 I_{a2}$$

$$30 \times 30 = I_2 \times \left(\frac{20.8I_2 - 250}{20} \right)$$

$$\Rightarrow 18000 = 20.8I_2^2 - 250I_2$$

$$\Rightarrow 20.8I_2^2 - 250I_2 - 1800 = 0$$

$$I_2 = 17.08A$$

Potential difference at brushes

$$= 250 - (17.08 \times 0.8) = 236.33V$$

$$I_{a2} = \frac{20.8 \times 17.08 - 236.33}{20} = 5.94A \quad \text{Choice (B)}$$

35. $E_{b2} = 236.33 - 5.94 \times 0.4 = 233.954V$

$$\Rightarrow N_2 = \frac{233.95}{214} \times \frac{30}{17.08} \times 600 = 1152.10 \text{ rpm}$$

Choice (C)

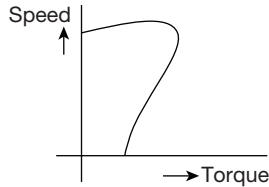
ELECTRICAL MACHINES TEST 2

Number of Questions: 35

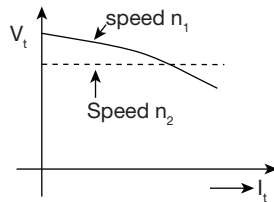
Section Marks: 90

Directions for questions 1 to 35: Select the correct alternative from the given choices.

1. Which machine has the following torque speed characteristics.



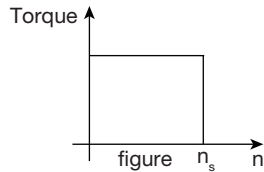
- (A) DC – shunt motor
(B) DC – Compound motor
(C) Induction motor
(D) Servo motor
2. A 200 V, 50 A, 4-pole DC-motor has commutator with a diameter of 60 cm rotates at 1500 rpm and has a brush width of 2 cm. The commutation time of DC motor is.
(A) 0.212 m sec (B) 0.424 m sec
(C) 1.2 m sec (D) 0.824 m sec
3. The DC shunt machine external characteristics as shown in the below figure.



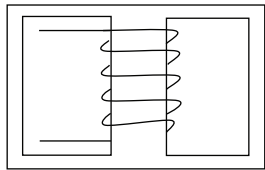
- (A) $n_2 < n_1$ (B) $n_2 > n_1$
(C) $n_1 = n_2$ (D) any of the above
4. The stored magnetic energy can be calculated by equation(s)
(1) $\frac{1}{2} \frac{\psi^2}{L}$ (2) $\frac{1}{2} HB$
(A) Only 1 (B) Only 2
(C) Both 1 and 2 (D) None of the above
5. The magnitude of electromagnetic torque in all rotating machines can be expressed as
(A) $T_e \propto (\text{Stator field strength}) (\text{Rotor field strength}) \sin \delta$
(B) $T_e \propto (\text{Stator field strength}) (\text{Rotor field strength}) \cos \delta$
(C) $T_e \propto (\text{Stator field strength}) (\text{Rotor field strength})$
(D) None of the above
6. Which of the following statements are correct when three phase induction motor fails to start.
(1) One or more fuses may be blown
(2) Voltage may be too low
(3) The starting load may be too heavy

- (A) Only 1 (B) Both (1) and (2)
(C) Both (2) and (3) (D) All of the above
7. Two 1 – ϕ transformers are connected in parallel at no load one has a turn ratio of 4000/440 V and a rating of 250 KVA, the other has a ratio of 4,000/500 V and a rating of 300 KVA. The leakage reactance of each transformer is 3%. The no load circulating current is
(A) 568.18 A (B) 1250 A
(C) 1168.18 A (D) 31.81 A
8. A 500 KVA, 50 Hz single phase, 6000/5000 V auto transformer has an emf per turn 10 V and maximum flux density 1.2 wb/m². The core area of the transformer is
(A) 0.0375 mm² (B) 0.0375 cm²
(C) 375 cm² (D) 375 mm²
9. A 100 KVA, 200 V, 50 Hz, 1 – ϕ alternator has an armature resistance of 0.02 ohm and an armature leakage reactance of 0.08 ohm. Calculate the voltage induced in the armature when the alternator is delivering rated current at a load power factor of 0.8 leading.
(A) 187.88 V (B) 212.11 V
(C) 233.44 V (D) 170.00 V
10. A 3-phase, Y-connected synchronous generator supplies a current of 20 A having phase angle of 30° lagging at 400 V. Calculate the direct axis current (Assume $X_d = 20$ ohm and $X_q = 8$ ohm)
(A) 5.54 A (B) 14.41 A
(C) 7.05 A (D) 10 A
11. A stepper motor has a step angle of 3° and stepping frequency is 3000 rps. The shaft speed of motor is.
(A) 3000 rpm (B) 25 rpm
(C) 3000 rps (D) 25 rps
12. The hysteresis loop of a magnetic material has an area of 10 cm² with the scales given as 1 cm = 2AT and 1cm = 25 mwb at 60 Hz. The total hysteresis loss is
(A) 60 W (B) 30 W
(C) 25 W (D) 50 W
13. The reactance voltage in dc machines can be expressed as
(A) Coefficient of self Inductance $\times$ Rate of change of current
(B) $L \times \frac{2I}{T_c}$
(C) Both (A) and (B)
(D) None of the above
14. The maximum power developed in synchronous motor when.
(A) $\theta = \alpha$ (B) $\theta = \alpha + 90$
(C) $\theta = \alpha - 90$ (D) $\theta + \alpha = 0$

15. Which motor has the following torque speed characteristic as shown in the figure



- (A) Universal motor (B) Stepper motor
(C) Hysteresis motor (D) Reluctance motor
16. The transformer has core area of 2 cm^2 and lengths 10 cm in each outer leg and an area of 3 cm^2 and a length of 5 cm in the central leg as shown in the figure. A coil of 1500 turns carrying 20 mA is placed around the central leg. Calculate B in the centre leg ($\mu = 0.005 \text{ H/m}$)



- (A) 0.8 T (B) 1.2 T
(C) 0.6 T (D) 1.4 T
17. A 6 – pole, DC machine has 40 slots and 8 conductors per slot. The total flux per pole is 25 mwb. The machine is rotated at a speed of 1500 rpm. The generated emf of the machine when it is lap connected winding is.
(A) 200 V (B) 600 V
(C) 400 V (D) 300 V
18. A 3-phase, 8-pole, 50 Hz star connected cylindrical rotor alternator develops an open circuit emf of 415 V. The armature has 48 slots with two layer winding full pitched with 2 turns each. The alternator has mean air gap diameter 0.2 m, core length 0.5 m. The peak value of fundamental flux density wave in air gap.
(A) 0.05 T (B) 2.38 T
(C) 1.2 T (D) 1.82 T
19. Two 1 – ϕ transformers, one is 200 KVA and the other is 100 KVA are connected in parallel to the same bus – bar on the primary side, their no – load secondary voltages being 2000 V and 1800 respectively. The resistance and reactance are 2.0Ω , 3.0Ω and 6Ω , 4Ω respectively. The no – load circulating current of transformers is
(A) 40 A (B) 13.34 A
(C) 15.44 A (D) 17.88 A
20. The speed of separately excited d. c. motor is controlled by a chopper. The supply voltage is 100 V, armature circuit resistance = 0.2 ohm, armature circuit inductance = 10 mH and motor constant = 0.02 V/rpm. The motor drives a constant load torque requiring an average current of 15 A. calculate the range of speed control for continuous motor current

- (A) 0 to 4850 rpm (B) 0 to 9700 rpm
(C) 0 to 1500 rpm (D) 0 to 3000 rpm

21. In a double cage induction motor, if the outer cage has an impedance at stand still $(3 + j1.5) \text{ ohm}$. Calculate the slip at which the two cages develop equal torques, if the inner Cage has an impedance of $(0.5 + j3) \text{ ohm}$ at stand still.
(A) 38.06% (B) 20%
(C) 5.02% (D) 4.08%
22. A 20 kW, 4-pole, 50 Hz, 3-phase induction motor has friction and windage losses 2% of the output. The full load copper loss at a slip of 5% is
(A) 1073 watts (B) 1020 watts
(C) 1473 watts (D) 1420 watts
23. A 200 V, d. c shunt machine has an armature circuit resistance of 0.4Ω and field circuit resistance of 100 Ω . If this machine is connected to 200 V supply mains and line current of 50 A. The ratio of e. m. f generated when the machine is working as a generator and motor is
(A) 1.18 (B) 1.22
(C) 0.81 (D) 1.02
24. A 3000 V star connected synchronous motor has synchronous impedance of $0.2 + j3 \Omega$ per phase if an excitation e.m.f of 400 V and input power of 1200 kW at rated voltage is given to the motor. The armature current is.
(A) 769.83 A (B) 576.19 A
(C) 192.33 A (D) 285.56 A
25. A 200 KVA lighting transformer has a full load loss of 5 kW, the losses being equally divided between iron and copper. During a day, the transformer operates on full load for 5 hours, on half load for 8 hours, the output being negligible for the remainder of the day. The all day efficiency of the transformer is
(A) 72.07% (B) 94.75%
(C) 92.75% (D) 97.56%
26. The equivalent circuit of a 400V, 3-phase, 1500 rpm induction motor with a stator connected winding has the following impedances per phase referred to the stator at stand still.
Stator: $(0.2 + j2) \text{ ohm}$; Rotor : $(0.8 + j4) \text{ ohm}$
Magnetising branch : $(20 + j40) \text{ ohm}$
The maximum torque developed in the induction motor is.
(A) 155.17 Nm (B) 165.53 Nm
(C) 122.016 Nm (D) 141.38 Nm
27. Two alternators A and B operate in parallel and supply a load of 20 MW at 0.8 p.f lagging. By adjusting steam supply and excitation of machine A, its power output and p.f are 8000 kW and 0.9 lagging respectively. The p.f of alternator B is

- (A) 0.732 leading (B) 0.732 lagging
(C) 0.447 leading (D) 0.447 lagging
28. A 2,000 V, 3-phase star-connected, synchronous, motor has a resistance of 0.4 ohm per phase and a synchronous reactance of 3 ohm per phase. The motor is operating at 0.6 power factor leading with a line current of 150 A. The generated e.m.f per phase is.
(A) 1154 volts (B) 1511 volts
(C) 1607 volts (D) 700 volts
29. A 50 Hz, 6-pole induction motor has full load slip of 8% the rotor resistance /phase = 0.02 ohm and stand still reactance = 0.4 ohm. Calculate the speed at which maximum torque occurs.
(A) 1000 rpm (B) 1050 rpm
(C) 900 rpm (D) 950 rpm
30. A load of 800 KVA at 0.85 power factor lagging is to be shared by two three phase transformers A and B of equal ratings. If the equivalent delta impedances as referred to secondary are $(3 + j6)\Omega$ for A and $(2 + j4)\Omega$ for B. The load supplied by transformer A is
(A) 1195.13 KVA (B) 597.56 KVA
(C) 268.57 KVA (D) 531.43 KVA
31. A 250 V, 2000 rpm, 15 A, separately excited DC motor has an armature resistance of 3 Ω . Rated DC voltage is applied to both the armature and field winding of the motor. If the armature draws 10 A from the source, the torque developed by the motor is
(A) 14.68 Nm (B) 4.89 Nm
(C) 17.90 Nm (D) 8.95 Nm

Common Data Questions for 32 and 33:

A 3-phase induction motor having a 4 – pole, Star connected stator winding runs on 415 V, 50 Hz. The rotor resistance and stand still reactance are 0.15 ohm and 0.8 ohm per phase. The ratio of stator to rotor turns is 1.5, full load slip is 4%

32. The developed full load torque is
(A) 124.48 Nm (B) 304.54 Nm
(C) 248.56 Nm (D) 314.46 Nm
33. The maximum torque of the machine is
(A) 248.56 Nm (B) 304.54 Nm
(C) 320.46 Nm (D) 504.56 Nm

Linked Answer Questions 34 and 35:

A 200 W, 230 V, 50 Hz capacitor start motor has the following constants, for the main and auxiliary windings;

Main winding $Z_m = (4 + j3.5)$ ohm

auxiliary winding $Z_a = (9 + j3)$ ohm

Assume the main and auxiliary winding currents are in quadrature at starting.

34. The phase angle between auxiliary winding current and supply voltage is
(A) 41.18 lead (B) 71.57 lead
(C) 18.43 lead (D) 48.82 lead
35. The value of starting capacitor is
(A) 239.81 μ F (B) 13.28 μ F
(C) 10.28 μ F (D) 102.81 μ F

ANSWER KEYS

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. B | 3. A | 4. C | 5. A | 6. D | 7. B | 8. C | 9. A | 10. B |
| 11. D | 12. B | 13. C | 14. A | 15. C | 16. B | 17. A | 18. B | 19. D | 20. A |
| 21. A | 22. A | 23. B | 24. D | 25. B | 26. B | 27. B | 28. B | 29. D | 30. D |
| 31. A | 32. A | 33. B | 34. D | 35. A | | | | | |

HINTS AND EXPLANATIONS

1. The torque – speed characteristics of Induction motor same as the given figure. Choice (C)
2. Peripheral velocity of diameter

$$v = \pi D n = \pi \times 60 \times \frac{1500}{60} \text{ cm/s}$$
 But $v \times \text{time of commutation} = \text{Brush width}$
 Time of commutation = $\frac{2}{\pi \times 1500} = 0.424 \text{ m sec}$
 Choice (B)
3. As the speed of the DC-machine decreases, the terminal voltage also decreases up to rated current. Choice (A)
4. Choice (C)

5. Choice (A)

6. Choice (D)

7. Reactances seen from the secondary side are

$$X_{1(\Omega)} = \frac{3}{100} \times \frac{440 \times 440}{250 \times 10^3} = 0.023 \Omega$$

$$X_{2(\Omega)} = \frac{3}{100} \times \frac{500 \times 500}{300 \times 10^3} = 0.025 \Omega$$

The difference of induced voltage is 60V

$$\text{The circulating current is } = I_c = \frac{60}{0.023 + 0.025}$$

$$= 1,250 \text{ A}$$

Choice (B)

8. $E = 4.44 \phi_m f N$ volt

$$\phi_m = \frac{\frac{E}{N}}{4.44 f} = \frac{10}{4.44 \times 50} = 0.045$$

$$\text{Core area} = \frac{0.045}{1.2} = 0.03753 = 375 \text{ cm}^2$$

Choice (C)

9. Full load rated current = $\frac{100 \times 1000}{200} = 500 \text{ A}$

$$IR_a = 500 \times 0.02 = 10 \text{ V}$$

$$IX_L = 500 \times 0.08 = 40 \text{ V}$$

$$\begin{aligned} E &= [(V \cos \phi + IR_a)^2 + (V \sin \phi - IX_L)^2]^{1/2} \\ &= [(200 \times 0.8 + 10)^2 + (200 \times 0.6 - 40)^2]^{1/2} \\ &= 187.88 \text{ V} \end{aligned}$$

Choice (A)

10. $\cos \phi = \cos 30^\circ = 0.866$

$$\sin \phi = \sin 30^\circ = 0.5$$

$$\tan \delta = \frac{I_a X_q \cos \phi}{V + I_a X_q \sin \phi} = \frac{20 \times 8 \times 0.866}{400 + 20 \times 8 \times 0.5} = 0.288$$

$$\delta = 16.10^\circ$$

$$I_a = I_a \sin(\phi + \delta) = 20 \sin(30 + 16.10) = 14.41 \text{ A}$$

Choice (B)

11. $\eta = \frac{\beta \times f}{360^\circ}$

$$= \frac{3 \times 3000}{360} = 25 \text{ rps}$$

Choice (D)

12. Hysteresis loss $P_h = 2 \times 10 \times 25 \times 60 \times 10^{-3}$

$$P_h = 30 \text{ watts}$$

Choice (B)

13. Choice (C)

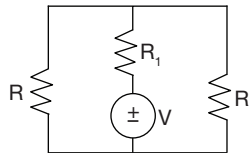
14. Choice (A)

15. By observing the torque – speed characteristics of the given motors, the Hysteresis motor has similar characteristics.

Choice (C)

16. $V_m = m \cdot m \cdot f = NI = 1500 \times 20 \times 10^{-3} = 30 \text{ AT}$

The equivalent electrical circuit for the problem



$$\text{Total resistance} = \frac{R \times R}{R + R} + R_1 = \frac{R}{2} + R_1$$

Hence total reluctance =

$$\begin{aligned} R &= \frac{1}{2} \times \frac{10 \times 10^{-2}}{0.005 \times 2 \times 10^{-4}} + \frac{5 \times 10^{-2}}{0.005 \times 3 \times 10^{-4}} \\ &= 50,000 + 33,333.34 \\ &= 83,333.34 \text{ AT/wb} \end{aligned}$$

$$\phi(\text{total}) = \frac{V_m}{R} = \frac{30}{83,333.34} = 3.6 \times 10^{-4}$$

$$B(\text{in centre leg}) = \frac{\phi}{s} = \frac{3.6 \times 10^{-4}}{3 \times 10^{-4}} = 1.2 \text{ T}$$

Choice (B)

17. For lap winding $A = P = 6$

$$Z = 40 \times 8 = 320$$

$$E = \frac{\phi Z N}{60} \times \frac{P}{A}$$

$$= \frac{25 \times 10^{-3} \times 320 \times 1500}{60} = 200$$

Choice (A)

18. $N_{ph} = \frac{48 \times 2}{3} = 32$

$$\text{Slots per pole phase} = \frac{48}{8 \times 3} = 2$$

$$\text{Slot angular pitch} = \gamma = \frac{180}{8} = 22.5$$

$$K_d = \frac{\sin \frac{2 \times 22.5}{2}}{2 \sin \frac{22.5}{2}} = 0.98$$

$$K_p = 1$$

$$E = 4.44 \phi f N_{ph} \cdot K_w (\because K_w = K_p K_d)$$

$$\phi = \frac{4}{P} B_{lp} I_r$$

$$B_{ip} = \frac{P \phi}{4 I_r} = \frac{8 \times 0.0596}{4 \times 0.1 \times 0.5} = 2.384 \text{ T}$$

Choice (B)

19. Total impedance = $\sqrt{(R_A + R_B)^2 + (X_A + X_B)^2}$

$$= \sqrt{5^2 + 10^2} = 11.18 \Omega$$

$$\text{Circulating current} = \frac{200}{11.18} = 17.88 \text{ A}$$

Choice (D)

20. The minimum speed is zero when $E_b = 0$

$$\begin{aligned} V_t &= E_b + I_a R_a \\ &= 15 \times 0.2 = 3 \text{ V} \end{aligned}$$

Maximum speed corresponds to $\alpha = 1$ when

$$V_t = V = 100 \text{ V}$$

$$E_b = V - I_a R_a = 100 - 15 \times 0.2 = 97 \text{ V}$$

$$N = \frac{E_b}{K_a \phi} = \frac{97}{0.02} = \frac{9700}{2} = 4850 \text{ rpm}$$

Choice (A)

21. S – be the slip at which two cage develop equal torques.

$$Z_1 = \sqrt{\left(\frac{3}{s}\right)^2 + (1.5)^2} \quad \text{and} \quad Z_2 = \sqrt{\left(\frac{0.5}{s}\right)^2 + 3^2}$$

$$\left(\frac{I_1}{I_2}\right)^2 = \left(\frac{Z_2}{Z_1}\right)^2 = \frac{\left(\frac{0.25}{s^2}\right) + 9}{\frac{9}{s^2} + 2.25}$$

Power input to outer cage $P_1 = \frac{I_1^2 R_1}{S}$

$$P_1 = I_1^2 \times \frac{3}{S}$$

$$P_2 = I_2^2 \times \frac{0.5}{S}$$

$$\frac{T_1}{T_2} = \frac{P_1}{P_2} = \left(\frac{I_1}{I_2} \right)^2 \times 6 = \frac{\left(\frac{0.25}{S^2} \right) + 9}{\left(\frac{9}{S^2} \right) + 2.25} \times 6$$

$$\frac{T_1}{T_2} = \frac{P_1}{P_2} = \left(\frac{0.25}{S^2} + 9 \right) 6$$

$$\Rightarrow \frac{9}{S^2} - \frac{1.5}{S^2} = 51.75 \Rightarrow \frac{7.5}{S^2} = 51.75$$

$$\Rightarrow S^2 = 0.145 \Rightarrow S = 0.38 \quad \text{Choice (A)}$$

22. Motor output $P_{\text{out}} = 20,000$ watts
Friction and windage loss $P_w = 2\% \times 20,000$
 $= 0.02 \times 20,000 = 400$ watts
rotor gross output $= 20,000 + 400 = 20,400$
rotor copper loss $= \frac{0.05}{0.95} \times 20,400 = 1073.68$

Choice (A)

23. given line current $I_L = 50$ A

$$I_f = \frac{200}{100} = 2 \text{ A}$$

Armature current $I_a = 50 + 2 = 52$ A

Generated e. m. f in generator $E_{g1} = V_t + I_a R_a$
 $= 200 + 52 \times 0.4 = 220.8$ volts

Generated emf in motor $E_{m1} = 200 - 48 \times 0.4 = 180.8$

$$\text{ratio} = \frac{E_{g1}}{E_{m1}} = \frac{220.8}{180.8} = 1.22 \quad \text{Choice (B)}$$

24. given $V_t = \frac{3000}{\sqrt{3}} = 1732.05$ V

$$E_f = \frac{4000}{\sqrt{3}} = 2309.5 \text{ V}$$

$$Z_s = \sqrt{0.2^2 + 3^2} = 3.006 \Omega$$

$$\alpha_z = \tan^{-1} \left(\frac{0.2}{3} \right) = 3.81^\circ$$

Per phase input power

$$\frac{1200 \times 1000}{3} = \frac{1732.05 \times 2309.5}{3.006} \sin(\delta - \alpha_z) +$$

$$\left(\frac{1732.05}{3.006} \right)^2 \times 0.2$$

$$40,000 = 13330728.36 \cdot \sin(\delta - \alpha_z) + 66400.73$$

$$333599.27 = 1330728.36 \cdot \sin(\delta - \alpha_z)$$

$$0.25 = \sin(\delta - \alpha_z) \Rightarrow \delta - \alpha_z = 14.51$$

$$\Rightarrow \delta = 18.32^\circ$$

$$I_a Z_s = [1732.05^2 + 2309.05^2 - 2 \times 1732.05 \times 2309.5 \cos 18.32^\circ]^{1/2}$$

$$I_a = \frac{858.40}{3} = 285.56 \text{ A} \quad \text{Choice (D)}$$

25. The lighting transformers taken a load of unity p. f

Iron loss for 24 hours $= 2.5 \times 24 = 60$ kWh

Full load copper loss $= 2.5$ kW

Copper loss for 5 hours $= 12.5$ kWh

Copper loss for 8 hours at half load

$$= \frac{2.5}{4} \times 8 = 5 \text{ kWh}$$

Total losses $= 60 + 12.5 + 5 = 77.5$

Total output $= (200 \times 5) + 100 \times 4 = 1400$ kWh

$$\eta_{\text{all-day}} = \frac{1400}{1477.5} \times 100 = 94.75\% \quad \text{Choice (B)}$$

26. Maximum slip $S_m = \frac{R_2^1}{\sqrt{R_1^2 + (X_1 + X_2^1)^2}}$

$$= \frac{0.8}{\sqrt{0.2^2 + 6^2}} = 0.1332 = 13.32\%$$

Maximum value of gross torque developed by rotor

$$T_{g\text{max}} = \frac{P_{g\text{max}}}{2\pi N_s} = \frac{\frac{3I_2^1{}^2 R_2^1}{S_m}}{2\pi N_s} = \frac{60}{60}$$

$$I_2^1 = \frac{V_1}{\sqrt{(R_1 + R_2^1)^2 + (X_1 + X_2^1)^2}}$$

$$= \frac{400 / \sqrt{3}}{\sqrt{(0.2 + 0.8)^2 + (2 + 4)^2}} = 37.96 \text{ A}$$

$$T_{g\text{max}} = \frac{3 \times 37.96^2 \times 0.8 \times 60}{0.133 \times 2\pi \times 1500} = 165.53 \text{ Nm} \quad \text{Choice (B)}$$

27. $\cos \phi = 0.8$, $\phi = 36.86^\circ$

$$\tan \phi = 0.7508$$

$$\cos \phi_A = 0.9 \Rightarrow \phi_A = 25.84^\circ$$

$$\tan \phi_A = 0.4843$$

Load kW $= 20,000$ kW, load KVAR $= 15,016$ (lag)

kW of A $= 8,000$, KVAR of A $= 8,000 \times 0.4843 = 3874.4$

kW of B $= 20,000 - 8000 = 12,000$ kW

KVAR of B $= 15016 - 3874.4 = 11141.6$ KVAR

$$\therefore \text{KVA of B} = 12,000 - j11141.6$$

$$= 16374.83 \angle -42.87$$

$$\cos 42.87 = 0.732 \text{ lagging}$$

Choice (B)

28. $\phi = \cos^{-1}(0.6) = 53.13^\circ$ lead

$$\theta = \tan^{-1}\left(\frac{3}{0.4}\right) = 82.40^\circ$$

$$\theta + \phi = 82.40^\circ + 53.13^\circ = 135.53^\circ$$

$$V = \frac{2000}{\sqrt{3}} = 1154.70 \text{ volts}$$

$$Z_s = \sqrt{0.4^2 + 3^2} = 3.02\Omega$$

$$IZ_s = 150 \times 3.02 = 453$$

$$E_b = \sqrt{V^2 + (IZ_s)^2 - 2V(IZ_s)\cos(\theta + \phi)}$$

$$= \sqrt{(1154.70)^2 + (453)^2 - 2 \times 1154.70 \times 453 \cos 135.53^\circ}$$

$$= 1511.65 \text{ volts} \quad \text{Choice (B)}$$

29. $S_m = \frac{R_2}{X_2} = \frac{0.02}{0.4} = 0.05$

$$S_f = 0.08$$

$$N_s = \frac{120 \times 50}{6} = 1000 \text{ rpm}$$

$$S_m = 0.05$$

$$N = (1 - 0.05) \times 1000 = 950 \text{ rpm} \quad \text{Choice (D)}$$

30. $S_A = S \times \frac{Z_B}{Z_A + Z_B} = S \times \frac{1}{1 + \left(\frac{Z_A}{Z_B}\right)}$

$$S = 800(0.85 - j0.52) = 680 - j416$$

$$Z_A/Z_B = \left(\frac{3 + j6}{2 + j4}\right) = 1.5$$

$$S_A = \frac{(680 - j416)}{1.5} = 531.43 \angle -31.45^\circ \quad \text{Choice (D)}$$

31. Back emf $E_b = V - I_a R_a$

$$= 250 - 15 \times 3$$

$$= 205 \text{ volts}$$

$$E_b I_a = T\omega$$

$$\text{Torque } T = \frac{E_b I_a}{\omega} = \frac{205 \times 15 \times 60}{2\pi \times 2000} = 14.68 \text{ Nm.}$$

Choice (A)

32. $K = \frac{\text{rotor turns / phase}}{\text{stator turns / phase}} = \frac{1}{1.5}$

$$E_2 = KE_1$$

$$= \frac{1}{1.5} \times \frac{415}{\sqrt{3}} = 159.73 \text{ V}$$

$$\text{Slip} = 0.04$$

$$N_s = \frac{120 \times 50}{4} = 1500 \text{ rpm}$$

$$\text{Full load torque} = T_f = \frac{3 \times 60}{2\pi N_s} \times \frac{SE_2^2 R_2}{R_2^2 + (SX_2)^2}$$

$$= \frac{3 \times 60}{2\pi \times 1500} \times \frac{0.04 \times 159.73^2 \times 0.15}{0.15^2 + (0.8 \times 0.04)^2}$$

$$= 124.28 \text{ Nm}$$

Choice (A)

33. For maximum torque $S = \frac{R_2}{X_2} = \frac{0.15}{0.8} = 0.1875$

$$T_{\max} = \text{maximum torque}$$

$$= \frac{3 \times 60}{2\pi \times 1500} \times \frac{0.1875 \times 159.73^2 \times 0.15}{0.15^2 + (0.1875 \times 0.8)^2}$$

$$= 304.54 \text{ Nm}$$

Choice (B)

34. Assume X_C be the reactance of the capacitor connected in the auxiliary winding

$$Z_a = 9 + j3 - jX_C = 9 + j(3 - X_C)$$

$$Z_M = 4 + j3.5 = 5.31 \angle 41.18^\circ$$

$$I_m \text{ lags behind } V \text{ by } 41.18^\circ$$

$$\text{given phase angle between } I_m \text{ and } I_a \text{ has to be } 90^\circ$$

$$I_a \text{ must lead } V \text{ by } = 90^\circ - 41.18^\circ = 48.82^\circ \quad \text{Choice (D)}$$

35. For auxiliary winding $\tan \phi_a = X/R$

$$\Rightarrow \tan 48.82^\circ = \frac{3 - X_C}{9} (X = 3 - X_C)$$

$$\Rightarrow 3 - X_C = 10.28 \Rightarrow X_C = 13.28\Omega$$

$$X_C = 13.28\Omega$$

$$\frac{1}{\omega C} = 13.28$$

$$\Rightarrow C = \frac{1}{314 \times 13.28} = 239.81 \mu\text{F}$$

Choice (A)

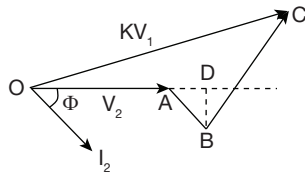
ELECTRICAL MACHINES TEST 3

Number of Questions: 25

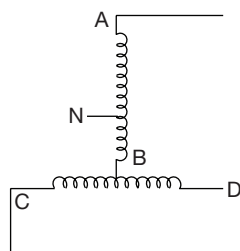
Section Marks: 90

Directions for questions 1 to 25: Select the correct alternative from the given choices.

1. Hysteresis loss depends on
 (A) f (B) f^2
 (C) $f^{1.6}$ (D) f^0
2. A 20KVA, 20000/480V, 60Hz transformer has following results $V_{oc} = 480V$
 $I_{oc} = 1.51A$
 $P_{oc} = 271W$
 $V_{sc} = 1130V$
 $I_{sc} = 1A$
 $P_{sc} = 260W$
 What would be the rating of this transformer, if it is operated at 50Hz?
 (A) 20 KVA, 20000/480V
 (B) 16.67 KVA, 1667/400V
 (C) 24 KVA, 24000/576V
 (D) None of the above
3. In the following phasor diagram k represent turns ratio of the transformer, then AD represents

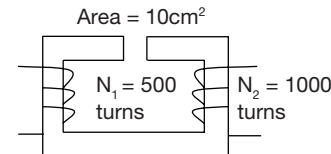


- (A) $I_2 R_{02} \sin \phi$ (B) $I_2 R_{02}$
 (C) $I_2 R_{02} \cos \phi$ (D) $I_2 X_{02} \sin \phi$
4. The total load taken by the V - V bank transformers is
 (A) 2/3 of the total load taken by Δ - Δ bank.
 (B) 57.7% of the total load taken by Δ - Δ bank.
 (C) 86.6% of the total load taken by Δ - Δ bank.
 (D) 42.3% of the total load taken by Δ - Δ bank.
 5. Three 20 MVA transformers are connected in Δ - Δ fashion, when one of the transformer is removed the total load taken by the remaining transformers is _____
 (A) 40 MVA (B) 34.64 MVA
 (C) 51.96 MVA (D) 23.08 MVA
 6. In the following transformer connection, the voltage between AB terminal is 200V. Then the voltage AN is _____



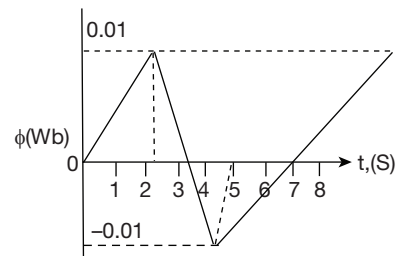
- (A) 100 V (B) 86.6 V
 (C) 133.34 V (D) 66.67 V

7. In no load operation of the transformer, the angle ϕ_o (no load angle) is due to
 (A) no load copper losses in the transformer.
 (B) hysteresis loss in the transformer.
 (C) mechanical displacement of core.
 (D) None of the above.
8. A Δ - Δ bank consisting of three 40-KVA, 2000/200V transformers supplies a load of 90 kVA. If one of the transformer is out of service, calculate the percent of rated load carried by each transformer.
 (A) 130% (B) 70%
 (C) 30% (D) 170%
9. A magnetic material of $\mu_r = 200$ carries two coils on its vertical limbs as shown. Their mutual inductance = ' M '.

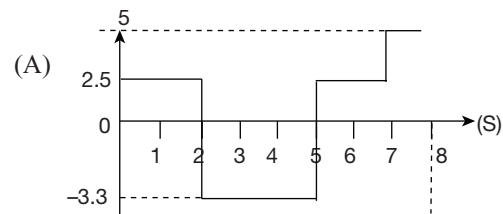


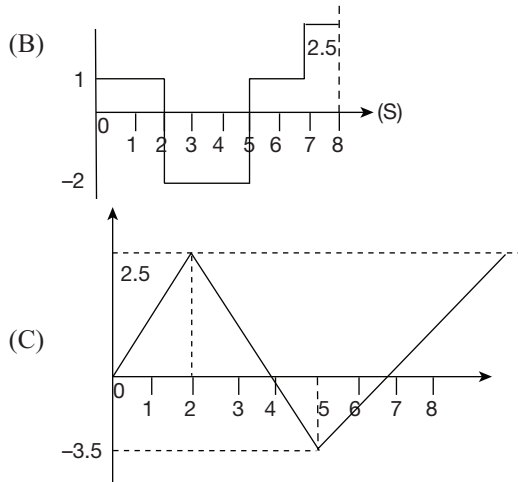
If number of turns of each coil are doubled, what would be the new mutual inductance?

- (A) M (B) $4M$
 (C) $2M$ (D) $M/2$
10. A 50 KVA transformer has 800 turns on the primary and 100 turns on the secondary winding. The primary is connected to 5000V, 50 Hz supply calculate the maximum flux in the core
 (A) 225.22 mwb (B) 22.52 mwb
 (C) 2.81 mwb (D) 28.15 mwb
 - 11.



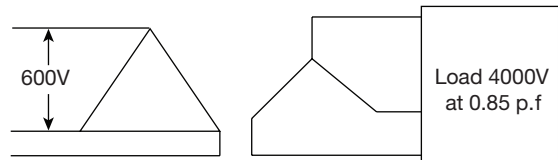
This flux passes through a coil of 500 turns voltage wave form due to this flux will be





(D) None of the above

12. A 500kVA, 3000/300V, 60Hz transformer the open circuit and short circuit test results are as follows:
O.C test: 300V, 5A, 180W
S.C test: 115V, 18A, 400W
 Calculate the power factor on short circuit
 (A) 0.58 lag (B) 0.8 lag
 (C) 0.38 lag (D) 0.62 lag
13. Calculate the regulation of a transformer in which the ohmic loss is 2% of the output and the reactance drop is 6% of the voltage, when the power factor is 0.7 leading.
 (A) 0.02 p.u (B) -0.02 p.u
 (C) 0.05 p.u (D) -0.05 p.u
14. An auto transformer is used to connect 12.6kV distribution line to a 13.8kV distribution line. It must be capable of handling 2000kVA. The three phases are connection *Y - Y*, neutrals solidly grounded. What should be the turns ratio to accomplish this?
 (A) 10 (B) 10.5
 (C) 11 (D) 11.5
15. A 5000VA 480/120V conventional transformer is used as autotransformer to supply 120V to load from a 600V source. Assuming ideal transformer, what are the maximum primary and secondary currents in this condition?
 (A) 10.4A, 52.1A (B) 52.1A, 10.4A
 (C) 1.04A, 5.21A (D) 5.21A, 1.04A
16. A transformer is connected to 4000V, 50Hz. The core loss is 1000 watts, out of which 600 watts are due to hysteresis and the remaining eddy current losses. Calculate the core loss if the supply voltage and frequency are 4800V and 60 Hz respectively
 (A) 720W (B) 1296W
 (C) 1200W (D) 1440W
17. A short circuit when performed on *h.v* side of a 20 kVA, 5000/500V single phase transformer gave the data, 150V, 8A, 400W. If the *L.V* side is delivering full load current at 0.6 p.f and at 500V. Calculate the voltage applied to on *h.v* side.
 (A) 5000V (B) 5500V
 (C) 5180V (D) 5320V
18. A 40 kVA, 4000/400V single phase transformer, the iron and full load copper losses are 400W and 500W respectively. Calculate the efficiency at half load and 0.8 power factor.
 (A) 97.56% (B) 98.48%
 (C) 96.10% (D) 96.85%
19. A 100 kVA, 4000/200V transformer has $R_1 = 3.85 \Omega$, $R_2 = 0.008 \Omega$. The values of reactance's are $X_1 = 5.5 \Omega$ and $X_2 = 0.018 \Omega$. Calculate the total copper loss (Referred to primary) of the transformer.
 (A) 2.40 kW (B) 4.40 kW
 (C) 2 kW (D) 6.40 kW
20. A 500 KVA transformer has a full load loss of 8kW, the losses being equally divided between iron and copper. During a day, the transformer operates on full load 5 hours, one half load for 6 hours and the output being negligible for the remainder of the day. Calculate the All day efficiency.
 (A) 97.04% (B) 96.71%
 (C) 96.89% (D) 98.82%
21. A 3-phase, delta/star connected 11000/400V, 50Hz transformer takes a line current of 8A. When secondary load of 0.6 lagging p.f connected. Calculate the total power output.
 (A) 152.42 kW (B) 50.80 kW
 (C) 30.48 kW (D) 29.33 kW
22. A single phase transformer has 2000 turns on the primary and 400 turns on the secondary. The no load current is 4A at a p.f of 0.25 lagging. Calculate the primary current when the secondary current is 300A at a power factor of 0.85 lagging.
 (A) $62.94 \angle -34.29^\circ$ (B) $60 \angle -31.78^\circ$
 (C) $64.24 \angle -75.52^\circ$ (D) $64 \angle -107.2^\circ$
23. A 50 kVA, Single-phase transformer, 4000V to 400 volts has a primary resistance of 1.2Ω and a secondary resistance of 0.02Ω . Calculate the full load efficiency at 0.9 p.f, if the iron loss of the transformer is 90% of the full load copper loss.
 (A) 97.93% (B) 98.93%
 (C) 96.54% (D) 99.82%
24. A 3-phase transformer has an equivalent resistance of 2% and an equivalent reactance of 8% as shown in figure, when the transformer delivers 4000V at full load current and 0.85 power factor. Calculate transformation ratio of the transformer.



3.122 | Electrical Machines Test 3

- (A) 0.40 (B) 2.45
(C) 0.70 (D) 1.41

25. The results of OC and SC test on a 2000VA, 230/115V transformer are as follows.

$V_{OC} = 230V$	$V_{SC} = 13.2V$
$I_{OC} = 0.45A$	$I_{SC} = 6A$
$P_{OC} = 30W$	$P_{SC} = 20.1W$

What is the transformer efficiency at rated conditions 0.8 p.f. lagging?

- (A) 95%
(B) 94.3%
(C) 91.7%
(D) 90.4%

ANSWER KEYS

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. A | 2. B | 3. C | 4. B | 5. B | 6. C | 7. B | 8. A | 9. B | 10. D |
| 11. A | 12. C | 13. B | 14. B | 15. A | 16. B | 17. C | 18. D | 19. B | 20. A |
| 21. C | 22. A | 23. A | 24. A | 25. C | | | | | |

HINTS AND EXPLANATIONS

1. $W_h = K_h B^{1.6} f V$ watts Choice (A)

2. Ratings $\propto f$;
 $\therefore$ both apparent power and voltage should be multiplied by 50/60

$$\frac{50}{60} \times 20KVA = 16.67KVA$$

$$\frac{50}{60} \times 20000V = 16667V; \frac{50}{60} \times 480 = 400V$$

Choice (B)

3. Choice (C)

4. The $V-V$ bank is taken only 57.7% of the total load taken by $\Delta-\Delta$ bank transformers. Choice (B)

5. When one of the transformer is removed from the $\Delta-\Delta$ bank, the total load taken by the remaining transformers is 86.6% of their own rating.

$$\text{One transformer supplies} = 0.866 \times 20 = 17.32 \text{ MVA}$$

$$\text{Then two transformers supplies total load} = 34.64 \text{ MVA}$$

Choice (B)

6. The diagram given in the question is scott connection therefore the voltage $AN = 2/3 \times$ voltage of $AB = 2/3 \times 200 = 133.34V$ Choice (C)

7. Choice (B)

8. KVA load supplied by each of two transformers

$$= \frac{90}{\sqrt{3}} = 51.96 \text{ kVA}$$

$$\text{Percent of rated load} = \frac{\text{kVA load of each transformer}}{\text{kVA rating of each transformer}}$$

$$= \frac{51.96}{40} \times 100 = 130\%$$

Choice (A)

9. $M = \sqrt{L_1 L_2}$

$$L_1 = \frac{\mu N_1^2 A_1}{l_1}$$

$$L_2 = \frac{\mu N_2^2 A_2}{l_2}$$

$L \propto N^2 \Rightarrow$ Individual inductance become 4 times.

$\therefore M$ also becomes 4 times. Choice (B)

10. Induced e.m.f $E = 4.44 \phi f N$

$$5000 = 4.44 \times \phi \times 50 \times 800$$

$$\phi = 0.02815$$

$$= 28.15 \text{ mwb}$$

Choice (D)

11. $E = -N(d\phi/dt)$

$$0 < t < 2; e = 500 \times \frac{0.01}{2} = 2.5V$$

$$2 < t < 5; e = 500 \times \frac{-0.02}{3} = -3.33V$$

$$7 < t < 5; e = 500 \times \frac{0.01}{2} = 2.5V$$

$$7 < t < 8; e = 500 \times \frac{0.01}{1} = 5V$$

This matches graph (a)

Choice (A)

12. Impedance $Z_{01} = \frac{115}{18} = 6.38\Omega$

$$\text{Resistance } R_{01} = \frac{800}{18^2} = 2.469\Omega$$

$$\text{Power factor on short circuit} = \frac{R_{01}}{Z_{01}}$$

$$= \frac{2.469}{6.38} = 0.38 \text{ lag}$$

Choice (C)

13. $\cos \phi = 0.7 \Rightarrow \phi = 45.57^\circ$

$$\sin \phi = \sin(45.57) = 0.714$$

Per unit regulation at 0.7 leading p.f

$$= 0.02 \times 0.7 - 0.06 \times 0.714 = -0.02$$

Choice (B)

14. $\frac{V_H}{V_L} = \frac{N_1 + N_2}{N_2} = \frac{\sqrt{3} \times 13.8KV}{\sqrt{3} \times 12.6KV}$

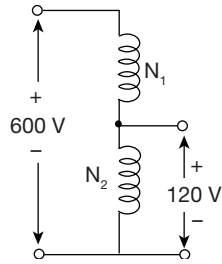
$$\Rightarrow 12.6N_1 + 12.6N_2 = 13.8N_1$$

$$12.6N_2 = 1.2N_1$$

$$N_2/N_1 \approx 10.5$$

Choice (B)

15.



$$\text{New KVA rating} = \left(\frac{480 + 120}{480} \right) \times 5000 \text{ VA} = 6250 \text{ VA}$$

$$\therefore I_p = \frac{S}{V_p} = \frac{6250}{600} = 10.4 \text{ A}$$

$$I_s = \frac{6250}{120} = 52.1 \text{ A}$$

Choice (A)

16. Core loss = hysteresis loss + eddy current loss

$$\text{Hysteresis loss } W_{h1} = 600 \text{ W}$$

$$\text{Eddy current loss } W_{e1} = 400 \text{ W}$$

$$\text{Hysteresis loss} = \frac{W_{h1}}{W_{h2}} = \frac{Af_1}{Af_2}$$

$$\Rightarrow \frac{600}{W_{h2}} = \frac{50}{60}$$

$$\Rightarrow W_{h2} = \frac{600 \times 60}{50} = 720 \text{ watts}$$

$$\text{Eddy current loss} = \frac{W_{e1}}{W_{e2}} = \frac{f_1^2}{Bf_2^2}$$

$$\Rightarrow \frac{400}{W_{e2}} = \frac{50^2}{60^2}$$

$$\Rightarrow W_{e2} = \frac{400 \times 60^2}{50^2} = 576 \text{ watts}$$

$$\text{Total Iron loss} = 1296 \text{ watts}$$

Choice (B)

$$17. Z_{01} = \frac{150}{3} = 50 \Omega$$

$$R_{01} = \frac{400}{3^2} = 44.44 \Omega$$

$$X_{01} = \sqrt{Z_{01}^2 - R_{01}^2} = \sqrt{50^2 - 44.44^2} = 22.91 \Omega$$

$$I_1 = \frac{20000}{5000} = 4 \text{ A}$$

$$\text{Voltage drop on primary side} = I_1(R_{01} \cos \Phi + X_{01} \sin \Phi)$$

$$= 4(44.44 \times 0.6 + 22.91 \times 0.8) = 179.96 \text{ V}$$

$$\therefore \text{Primary voltage has raised to } 5180 \text{ V}$$

Choice (C)

$$18. \text{Copper loss} = 500 \times \frac{1}{4} = 125 \text{ W}$$

$$\text{Iron losses} = 400 \text{ W}$$

$$\text{Output at Half load and 0.8 power factor}$$

$$= 40 \times \frac{1}{2} \times 0.8 = 16 \text{ kW}$$

$$\% \eta = \frac{16}{16 + 0.12 + 0.4} \times 100 = 96.85\%$$

Choice (D)

$$19. I_1 = \frac{100000}{4000} = 25 \text{ A}$$

$$I_2 = \frac{100000}{200} = 500 \text{ A}$$

Total copper loss are whether referred to primary (or) secondary

$$W_{\text{cut}} = 25^2 \times 3.85 + 500^2 \times 0.008$$

$$= 4406.25 \text{ watts}$$

Choice (B)

 20. Iron loss for 24 hours = $4 \times 24 = 96 \text{ kWh}$

$$\text{Full load copper loss} = 4 \text{ kW}$$

$$\text{Full load copper losses for 5 hours} = 4 \times 5 = 20 \text{ kWh}$$

$$\text{Half load copper losses for 6 hours} = 1 \times 6 = 6 \text{ kWh}$$

$$\text{Total losses} = 96 + 20 + 6 = 122 \text{ kWh}$$

$$\text{Total output} = 500 \times 5 + 250 \times 6 = 4000 \text{ kWh}$$

$$\eta_{\text{all day}} = \frac{4000}{4000 + 122} \times 100 = 97.04\%$$

Choice (A)

 21. Primary rating $V_1 = 11000$

$$\text{Secondary rating } V_2 = \frac{400}{\sqrt{3}} = 230.94 \text{ V}$$

$$\text{Primary current } I_1 = \frac{8}{\sqrt{3}} = 4.618 \text{ A}$$

$$\text{Three phase volt - Amp} = 3 \times 11000 \times \frac{8}{\sqrt{3}}$$

$$= 1,52,420.47$$

$$\text{Volt-amp/phase} = 50806.82$$

$$\text{Secondary current } I_2 = \frac{50806.82}{230.94} = 220 \text{ A}$$

$$\text{Total output power} = 50806.82 \times 0.6$$

$$= 30.48 \text{ kW}$$

Choice (C)

$$22. \cos^{-1}(0.85) = 31.78^\circ$$

$$I_2 = 300 \angle -31.78^\circ$$

Secondary load current refer to primary

$$I_2^1 = 60 \angle -31.78^\circ$$

$$\Phi_0 = \cos^{-1}(0.25) = 75.52^\circ$$

$$I_1 = I_0 + I_2^1$$

$$= 4 \angle -75.52^\circ + 60 \angle -31.78^\circ$$

$$= 52 - 35.472i = 62.94 \angle -34.29^\circ$$

Choice (A)

3.124 | Electrical Machines Test 3

23. $K = \frac{400}{4000} = 0.1$

$$R_{02} = R_2 + K^2 R_1 = 0.02 + \frac{1}{100} \times 1.2 = 0.032 \Omega$$

$$\text{Full load } I_2 = \frac{50000}{400} = 125 \text{ A}$$

$$\text{Full load copper loss} = I_2^2 R_{02} = 125^2 \times 0.032 = 500 \text{ W}$$

$$\text{Iron loss} = 90\% \text{ of } 500 = 450 \text{ W}$$

$$\text{Total losses} = 450 + 500 = 950 \text{ W}$$

$$\text{Full load output} = 50 \times 0.9 = 45 \text{ kW}$$

$$\text{Full load } \eta = \frac{45000}{45000 + 950} \times 100$$

$$= 97.93\%$$

Choice (A)

24. Percent regulation = $2 \times 0.85 + 8 \times 0.52 = 5.86\%$

Induced secondary

$$\text{e.m.f} = 4000 + \frac{5.86}{100} \times 4000 = 4234.4 \text{ V}$$

$$\text{Secondary phase voltage} = \frac{4234.4}{\sqrt{3}} = 2444.73 \text{ V}$$

$$\text{Transformation ratio} = K = \frac{2444.73}{6000} = 0.40$$

Choice (A)

25. $P_{\text{out}} = V_s I_s \cos \theta = 115 \text{ V} \times 8.7 \times 0.8 = 800 \text{ W}$

$$(I_s = \frac{1000 \text{ VA}}{115 \text{ V}} = 8.7 \text{ A})$$

$$P_{sc}^1 = \left(\frac{8.7}{6} \right)^2 \times 20.1 = 42.26 \text{ W}$$

$$P_{oc} = \text{constant loss}$$

$$\eta = \frac{800}{800 + 42.26 + 30} = \frac{800}{872.26} = 91.7\%$$

Choice (C)

ELECTRICAL MACHINES TEST 4

Number of Questions: 25

Section Marks: 90

Directions for questions 1 to 25: Select the correct alternative from the given choices.

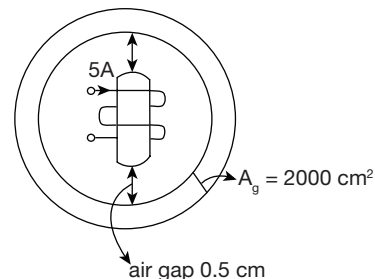
1. Hopkinson's Test on D.C machines is
 (A) No load test. (B) Full load test.
 (C) Break test. (D) None of the above.
2. The compensating winding conductors in D.C machine is (has)
 (A) directly proportional to the square of armature conductors.
 (B) directly proportional to the armature conductors.
 (C) inversely proportional to the armature conductors.
 (D) No relation with the armature conductors.
3. Which of the following generator generally not used for power supply cases
 (A) d.c. shunt generator.
 (B) d.c. series generator.
 (C) d.c. compound generator.
 (D) Both B and C
4. A d.c booster is connected between a station bus bar and a feeder, the characteristics of the booster is straight line through 0 to 40V at 100A and the feeder has a resistance of 0.6 ohm. When the feeder has a current of 100A, calculate the potential difference between the station bus-bar and far end of the feeder
 (A) 60V (B) 40V
 (C) 20V (D) 100V
5. In separately excited D.C Generator load saturation curves is drawn between
 (A) terminal voltage and field current.
 (B) terminal voltage and armature current.
 (C) generated voltage and field current.
 (D) generated voltage and armature current.
6. The overloading of generator is indicated by
 (P) excessive sparking at brushes.
 (Q) overheating of the armature and other parts of the generator.
 (A) only P
 (B) only Q
 (C) both P and Q
 (D) None of the above
7. Match the following

List-I	List-II
P. Swinburne's test	1. Below base speed
Q. Field test	2. Above base speed
R. Field control	3. Constant losses
	4. Stray losses in case of d.c series machines
	5. Temperature rise

- | | | | |
|-----|---|---|---|
| | P | Q | R |
| (A) | 3 | 5 | 1 |
| (C) | 4 | 5 | 2 |

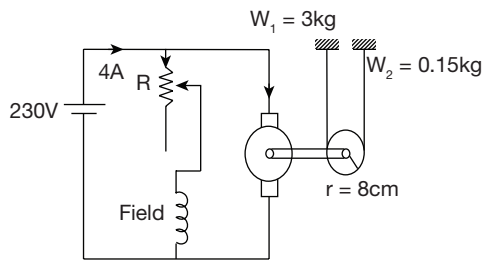
- | | | | |
|-----|---|---|---|
| | P | Q | R |
| (B) | 3 | 4 | 2 |
| (D) | 4 | 5 | 1 |

8. A 4-pole D.C machine has an armature resistance of 0.4Ω and constant losses of the machine 40W. When the machine is working under maximum efficiency, calculate the armature current of the machine.
 (A) 100A (B) 2.5A
 (C) 10A (D) 16A
9. In which of the following motors 3-point starter is not used
 (P) shunt motor. (Q) compound motor.
 (R) series motor.
 (A) Only P (B) Both P and Q
 (C) Both Q and R (D) P, Q and R
10. In a dc generator, the residual magnetism is of the order of
 (A) 2.5% (B) 10%
 (C) 15% (D) 25%
11. A d.c. shunt machine develops an a.c emf of 200V at 1000 rpm. Calculate the torque developed for an armature current of 80A
 (A) 152.8Nm (B) 16 Nm
 (C) 168.8 Nm (D) 136.8 Nm
12. A 250V d.c. shunt motor has an armature resistance of 0.4Ω and runs at 1000 rpm with an armature current of 80A. If the torque is increased four times then the corresponding speed of the motor is _____.
 (A) 560 rpm (B) 440 rpm
 (C) 1560 rpm (D) 1440 rpm
13. Calculate the torque developed in a 8-pole D.C motor having 800 conductors, two paths in parallel, 25 milli-webers of pole flux and the armature current is 100A
 (A) 424 N-m (B) 1272 N-m
 (C) 636 N-m (D) 2544 N-m
14. For the given magnetic structure, assuming infinite permeability with number of turns 1000, what is the magnetic flux?

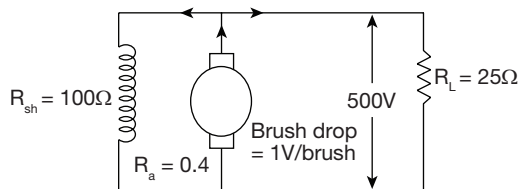


- | | |
|-------------|--------------|
| (A) 0.1256T | (B) 0.2512Wb |
| (C) 0.2512T | (D) 0.1256Wb |

15. A brake test is conducted on D.C shunt motor running at a speed of 1500 rpm as shown in the figure. Calculate efficiency of the motor.



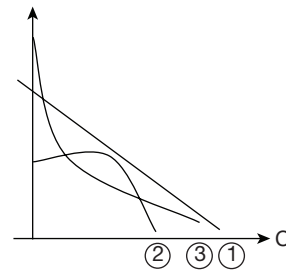
- (A) 38.17% (B) 61.29%
(C) 55.64% (D) 84.25%
16. A 4-pole lap wound d.c generator has 300 coils of single turn each, Resistance of each turn is 0.04Ω . The armature is 40 cm long and 30 cm diameter. Airgap flux density of $0.8T$ is uniform over pole shoe. The mechanical angle of each pole is 30° . For an armature current of 50A, calculate the terminal voltage of machine at a speed of 1500 rpm.
- (A) 412.5V (B) 375V
(C) 373V (D) 337.5V
17. A d.c series motor having a resistance of 2Ω drives a fan for which the torque varies as the square of the speed at 300V, the set run at 500 rpm and takes 30A. The speed is to be raised to 800 rpm by increasing the voltage, when the field is unsaturated. Calculate the terminal voltage of the machine.
- (A) 614.4V (B) 710.4V
(C) 240.0V (D) 204.4V
18. The hysteresis and eddy current losses in a d.c. machine running at 1500 rpm are 500W and 200W respectively. Calculate the total iron loss of the machine at half rated speed of the machine.
- (A) 250W (B) 700W
(C) 300W (D) 254W
19. A 6-pole d.c generator circuit as shown in the figure. Calculate the efficiency of the generator.



- (A) 79.68% (B) 80.85%
(C) 78.12% (D) 79.02%
20. The essential condition for the stable operation of two dc generators in parallel is that they should have the same
- (A) kW rating.
(B) No load and full load speeds.

- (C) Drooping voltage characteristics.
(D) Voltage regulations at full load.

21. A 200V shunt motor has armature and field resistance of 0.2 and 200Ω respectively. The motor drives a torque load, $T_L \propto n^2$. It runs at 1000 rpm, drawing 10A from supply. If an external resistance of 5Ω is given to armature circuit, what would be the new speed of rotation? (Neglect armature reaction & saturation)
- (A) 900 rpm (B) 856 rpm
(C) 841 rpm (D) Data insufficient
22. Given the various torque - speed curves of dc motors 1, 2, 3 may be (respectively)



- (A) Permanent magnet, series, compound
(B) Compound, series, permanent magnet
(C) Series, compound, permanent magnet
(D) Permanent magnet, compound, series
23. Plugging in an motor implies
- (A) rapidly stopping the motor by using mechanical brakes.
(B) rapidly stopping the motor by reversing armature field.
(C) rapidly accelerating the motor by using mechanical means.
(D) rapidly accelerating the motor by reversing armature field.

Common data questions for 24 and 25:

24. A separately excited DC motor is rotated at 1000 rpm, no load test results are as follows

I_f (Amperes)	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8
V_t (Volts)	0	30	60	85	102	115	124	130	134

$V_f = 24V$ and field resistance is adjustable

$R_A = 0.2\Omega$, $V_t = 130V$

The motor drives a load at 1200 rpm.

If $R_f = 60\Omega$, what is the armature voltage?

- (A) 102V (B) 122.4V
(C) 115V (D) 100V
25. If the above motor supplies mechanical load of 4kW at 1450 rpm and rotational losses are 160W, what is the efficiency? (Approximately)
- (A) 96% (B) 91%
(C) 93% (D) 94%

ANSWER KEYS

1. B	2. B	3. B	4. C	5. A	6. C	7. B	8. C	9. C	10. A
11. A	12. A	13. B	14. D	15. A	16. D	17. B	18. C	19. C	20. C
21. C	22. D	23. B	24. B	25. B					

HINTS AND EXPLANATIONS

1. Regenerative (or) Hopkinson's test is a full load test conducted on two similar d.c shunt machines.

Choice (B)

2. Compensating winding Turns $AT_c = \frac{\text{Pole arc}}{\text{Pole pitch}} \times \frac{I_a Z}{2ap}$

$$AT_c \propto Z$$

Choice (B)

3. D.C series generators are not used for power supply because of their rising characteristics.

Choice (B)

4. Voltage drop = $100 \times 0.6 = 60\text{V}$

Booster voltage for 100A current is given as 40V

Net voltage drop = $60 - 40 = 20\text{V}$

Choice (C)

5. Choice (A)

6. Choice (C)

7. Swinburne's test is no load test, the test is conducted to find constant losses and field control method is used to control the speed of the motor above base speed only.

Choice (B)

8. Copper losses = Constant losses $I_a^2 R_a = 40$

$$I_a = \sqrt{\frac{40}{0.4}} = 10\text{A}$$

Choice (C)

9. Choice (C)

10. Choice (A)

11. Mechanical power developed in the arm = $E_b I_a$
 $= 200 \times 80 = 16000 \text{ watts}$

$$T_a = 9.55 \times \frac{E_b I_a}{N}$$

$$= \frac{9.55 \times 200 \times 80}{1000} = \frac{9.55 \times 16000}{1000} = 152.8 \text{ N-m}$$

Choice (A)

12. $T \propto I_a$

$$\Rightarrow \frac{T_{a2}}{T_{a1}} = \frac{I_{a2}}{I_{a1}} \Rightarrow 4 = \frac{I_{a2}}{80} \Rightarrow I_{a2} = 320\text{A}$$

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \Rightarrow \frac{N_2}{1000} = \frac{250 - 320 \times 0.4}{250 - 80 \times 0.4}$$

$$N_2 = 559.63 \text{ r.p.m}$$

$$\Rightarrow N_2 = 560 \text{ rpm}$$

Choice (A)

$$13. T = \frac{0.159 \phi Z I_a P}{A}$$

$$T = \frac{0.159 \times (25 \times 10^{-3}) \times 800 \times 100 \times 8}{2}$$

$$= 1272 \text{ N-m}$$

Choice (B)

$$14. \phi = \frac{NI\mu_0 A_g}{\ell} = \frac{1000 \times 5 \times 4\pi \times 10^{-7} \times 2000 \times 10^{-4}}{2 \times 0.5 \times 10^{-2}}$$

$$= \frac{5000 \times 4\pi \times 10^{-7} \times 0.2}{10^{-2}} = 0.1256 \text{ Wb} \quad \text{Choice (D)}$$

$$15. \text{Net Force on Pulley} = (W_1 - W_2) \times 9.81$$

$$= 2.85 \times 9.81 = 27.95 \text{ N}$$

$$\text{Power output} = \text{Torque} \times \omega$$

$$= 27.95 \times \frac{8}{100} \times \frac{2\pi \times 1500}{60} = 351.23 \text{ watts}$$

$$\% \text{ efficiency} = \frac{\text{Power output}}{\text{Power input}} \times 100$$

$$= \frac{351.23}{230 \times 4} \times 100 = 38.17\% \quad \text{Choice (A)}$$

$$16. \text{electrical angle} = \frac{4}{2} \times 30 = 60^\circ$$

$$\text{Flux per pole} = \frac{2\pi \times 0.15 \times 0.4}{4} \times \frac{60}{180} \times 0.8 = 0.025 \text{ wb}$$

$$\text{Total armature conductors } Z = 300 \times 2 = 600$$

$$\text{Generated e.m.f at no load } E_g = \frac{\phi Z N}{60} \times \frac{P}{A}$$

$$= \frac{0.025 \times 600 \times 1500}{60} \times \frac{4}{4} = 375 \text{ V}$$

$$\text{Number of armature turns per path} = \frac{300}{4} = 75$$

$$\text{Resistance of one path} = 75 \times 0.04 = 3 \Omega$$

$$\text{Resistance of armature circuit} = \frac{3}{4} = 0.75 \Omega$$

$$\text{Terminal voltage } V_t = E_g - I_a R_a$$

$$= 375 - 50 \times 0.75$$

$$= 337.5 \text{ V}$$

Choice (D)

$$17. \text{Torque} \propto I_a^2 \propto N^2$$

$$\Rightarrow I_a \propto N$$

$$\Rightarrow I_{a1} \propto N_1$$

$$\therefore \frac{I_{a2}}{I_{a1}} = \frac{N_2}{N_1} = \frac{800}{500} \Rightarrow I_{a2} = \frac{800}{500} \times 30 = 48 \text{ A}$$

$$E_{b1} = 300 - 30 \times 2 = 240 \text{ V}$$

$$E_{b2} = V - 48 \times 2$$

$$\frac{\phi_1}{\phi_2} = \frac{30}{48} = \frac{5}{8} = \frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_2}$$

$$= \frac{800}{500} = \frac{V - 48 \times 2}{240} \times \frac{5}{8}$$

$$\Rightarrow 614.4 = V - 48 \times 2$$

$$\Rightarrow V = 710.4 \text{ V}$$

Choice (B)

18. $W_i = W_n + W_e = AN + BN^2$

$$\therefore W_n = 500 \text{ W} \Rightarrow A \times \frac{1500}{60} = 500$$

$$\Rightarrow A = 20$$

$$\therefore W_e = 200 \text{ W} \Rightarrow B \times \frac{1500 \times 1500}{60 \times 60} = 200$$

$$\Rightarrow B = 0.32$$

$$\text{New iron loss } W_i = \frac{20 \times 750}{60} + \frac{0.32 \times 750^2}{60^2}$$

$$= 250 + 50 = 300 \text{ W}$$

Choice (C)

19. Terminal voltage = $20 \times 25 = 500 \text{ V}$

$$I_{sh} = \frac{500}{100} = 5 \text{ A}$$

$$I_a = 20 + 5 = 25 \text{ A}$$

$$I_a R_a = 25 \times 0.4 = 10 \text{ V}$$

$$\text{Brush drop} = 2 \text{ V}$$

$$\text{Generated voltage} = 500 + 2 + 10 = 512 \text{ V}$$

$$\eta = \frac{\text{Power output}}{\text{Power generated}} = \frac{500 \times 20}{512 \times 25} \times 100$$

$$= 78.125\%$$

Choice (C)

20. Choice (C)

21. Initially $I_{f1} = 1 \text{ A}$; $I_{a1} = 9 \text{ A}$

$$T_{e1} = K_1 \times 1 \times 9 = T_{L1}$$

$$E_{b1} = K_2 \times 1 \times 1000 = V - I_{a1} R_a = 200 - 9 \times 0.2$$

$$= 198.2 \text{ V}$$

$$\text{Let new steady state armature current} = I_{a2} ;$$

$$\text{Speed} = n_2$$

$$\therefore T_{e2} = K_1 \times 1 \times I_{a2} = T_{L2}$$

$$E_{b2} = K_2 \times 1 \times n_2 = V - I_{a2} (r_a + R_{ext})$$

$$\Rightarrow K_2 n_2 = 200 - I_{a2} \quad (5.2)$$

$$\frac{T_{e2}}{T_{e1}} = \frac{T_{L2}}{T_{L1}} = \frac{K_2 \times 1 \times I_{a2}}{K_2 \times 1 \times 9}$$

$$\Rightarrow \frac{n_2^2}{1000^2} = \frac{I_{a2}}{9} \quad \text{————(1)}$$

$$\text{Similarly } E_{b2}/E_{b1} = \frac{K_1 \times 1 \times n_2}{K_1 \times 1 \times 1000}$$

$$= \frac{200 - I_{a2} \times 5.2}{198.2}$$

$$\Rightarrow \frac{n_2}{1000} = \frac{200 - I_{a2} \times 5.2}{198.2} \quad \text{————(2)}$$

Substituting (2) in (1)

$$\left(\frac{200 - 5.2 I_{a2}}{198.2} \right)^2 = \frac{I_{a2}}{9}$$

$$\Rightarrow I_{a2} \approx 6.36 \text{ A}$$

$$\therefore \frac{n_2}{1000} = \sqrt{\frac{6.36}{9}} \approx 841 \text{ rpm}$$

Choice (C)

22. Choice (D)

23. Choice (B)

24. $E_A = K \phi \omega$

$$V_F = I_F R_F \Rightarrow 24 = I_F \times 60$$

$$\Rightarrow I_F = 0.4 \text{ A}$$

From the table, at 1000rpm, $I_F = 0.4 \text{ A}$ gives

$$V_T = 102 \text{ V}$$

$$\text{at 1200rpm, } E^1 = E_{1000} \times \frac{1200}{1000} = 122.4 \text{ V}$$

Choice (B)

25. $I_A^2 R_A - V_T I_A + P = 0$

$$\Rightarrow I_A^2 (0.2) - V_T I_A + 4160 = 0$$

$$I_A = 33.75 \text{ A}$$

$$E_A = V_T - I_A R_A = 123.25 \text{ V (at 1450rpm)}$$

$$E_A^1 \text{ at 1000rpm} = 85 \text{ V} \Rightarrow I_F = 0.3 \text{ A}$$

$$\text{Copper losses} = I_A^2 R_A + I_F^2 R_F$$

$$\text{Total losses} = I_A^2 R_A + I_F^2 R_F + P_{\text{rotational}} = 395 \text{ W}$$

$$\eta = \frac{4000}{4395} \approx 91\%$$

Choice (B)

ELECTRICAL MACHINES TEST 5

Number of Questions: 25

Section Marks: 90

Directions for questions 1 to 25: Select the correct alternative from the given choices.

1. The torque equation of induction motor under running condition is (N_s = synchronous speed, N_r = rotor speed)

(A) $\frac{3}{2\pi N_r} \left(\frac{SE_2^2 R_2}{R_2^2 + (SX_2)^2} \right)$

(B) $\frac{3}{2\pi N_s} \left(\frac{SE_2^2 R_2}{R_2^2 + (SX_2)^2} \right)$

(C) $\frac{3}{2\pi N_r} \left(\frac{E_2^2 R_2}{\left(\frac{R_2}{S} \right)^2 + X_2^2} \right)$

(D) $\frac{3}{2\pi N_s} \left(\frac{SE_2^2 R_2}{\left(\frac{R_2}{S} \right)^2 + X_2^2} \right)$

2. The frequency of injecting an e.m.f in the Rotor circuit of induction motor is

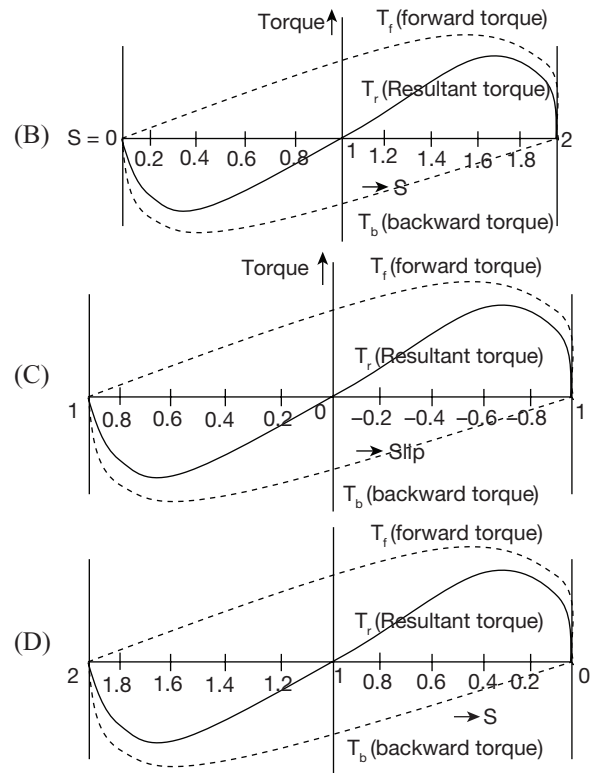
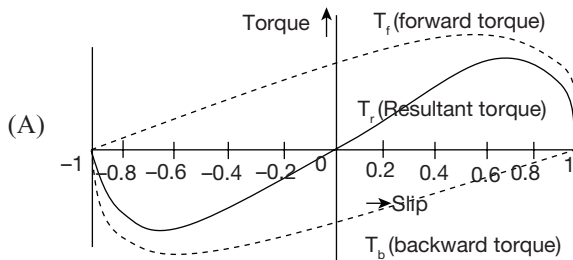
- (A) Same as supply frequency
 (B) Same as rotor frequency
 (C) Half of supply frequency
 (D) Two times of supply frequency

3. Backward torque equation in single phase Induction motor is

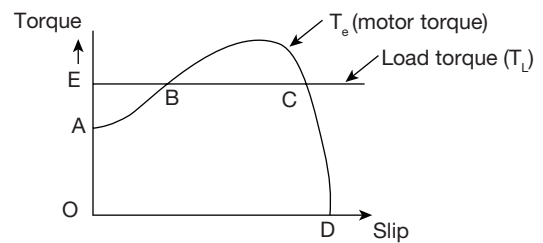
(A) $-k \left(\frac{I_2^2 R_2}{S} \right)$ (B) $-k \left(\frac{I_2^2 R_2}{(1-S)} \right)$

(C) $-k \left(\frac{I_2^2 R_2}{2-S} \right)$ (D) $-k \left(\frac{SI_2^2 R_2}{(2-S)} \right)$

4. Which of the following diagram represents Torque-Slip characteristic of single phase induction motor.

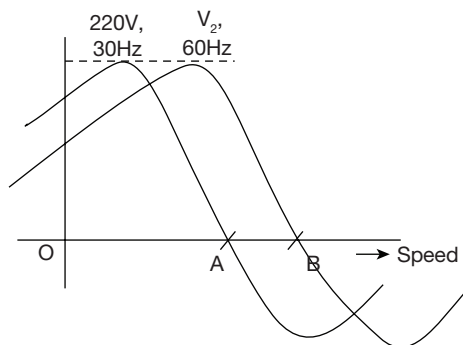


5. Which of the following statements is correct for the below figure



- (A) the operating region is BC.
 (B) the operating region is CD.
 (C) the operating region is AB.
 (D) at starting Load torque is more, so the motor will not start.
6. What is mean by jogging in induction motor?
 (A) The motor move a little at a time by constant starting and stopping.
 (B) The motor runs by 5th (or) 7th harmonic speed.
 (C) The motor start with very high speed.
 (D) None of the above
7. If the rotor resistance of a wound rotor induction motor is doubled, what will happen?

- (A) Induced torque increases, efficiency decreases
(B) efficiency increases, induced torque decreases
(C) Both torque and efficiency increases
(D) Both torque and efficiency decreases
8. A three phase 60Hz two pole SCIM runs at 3580 rotations/min at no load. The slip and electrical frequency at no load are:
(A) 6.5%, 3.9Hz (B) 0.65%, 0.39Hz
(C) 5.6%, 3.3Hz (D) 0.56%, 0.33Hz
9. In a three phase induction machine, if the flux created by each phase takes the maximum value of ϕ_m , total magnitude of rotating flux at any given time is
(A) ϕ_m (B) $2\phi_m$
(C) $1.5\phi_m$ (D) $0.75\phi_m$
10. A 50kW, 440V, 50Hz two pole SCIM has 6% slip at full load. If full load friction and windage losses are 520W and core losses are 500W, what is the output power?
(A) 45.0 kW (B) 50 kW
(C) 52.4 kW (D) 43.2 kW
11. The torque-speed characteristics of 4-pole squirrel cage induction motor as shown in the figure. By varying the voltage and frequency, the flux in the air gap is maintained constant. Then the values of V_2 , A and B are _____



- (A) 440V, 900 rpm, 1800 rpm
(B) 220V, 900 rpm, 1800 rpm
(C) 110V, 1800 rpm, 900 rpm
(D) 110V, 900 rpm, 1800 rpm
12. A 400V, 50Hz, 6-pole, 3 – Φ induction motor runs at a speed of 950 rpm. When connected to 400V line voltage, calculate the speed if the voltage increases to 500V.
(A) 950 rpm (B) 968 rpm
(C) 946 rpm (D) 985 rpm
13. Which of the following are the speed control method on stator side of an induction motor
(P) By changing applied voltage
(Q) By operating two motors in cascade
(R) By changing number of poles
(A) P and Q (B) P and R
(C) Q and R (D) P, Q and R

14. A 3-phase, 4-pole, 50 Hz induction motor takes 50 A at full load of 1450 rpm and develops a torque of 200 N-m. The starting current at rated voltage is 400 A by using star/delta method. Calculate the starting torque of the motor.
(A) 422.4 N-m (B) 52.8 N-m
(C) 211.2 N-m (D) 1267.2 N-m
15. The power input to the rotor of a 400V, 50Hz, 8 pole, 3 phase induction motor is 500 kW. Calculate the total rotor copper loss. When the rotor makes 180 cycles per minute.
(A) 470 kW (B) 3 kW
(C) 30 kW (D) 10 kW
16. The power to a 400V, 50 Hz, 8-pole, 3-phase induction motor running at 700 rpm is 50 kW. The stator losses are 2 kW and the friction and windage losses total 3 kW. Calculate the efficiency of the motor.
(A) 96% (B) 90%
(C) 89.56% (D) 83.56%
17. A 400V, 3-phase, 50Hz, 6-pole star connected induction motor takes a line current of 20A with 0.8 p.f lagging. Its total stator losses are 4% of the input, rotor copper losses are 4% of the input to the rotor and mechanical losses are 2% of the input of the rotor. Then the shaft-torque of the motor is
(A) 110.26 Nw-m (B) 105.85 Nw-m
(C) 99.50 Nw-m (D) 103.73 Nw-m
18. The rotor of a 6-pole, 50Hz slip ring induction motor has a resistance of 0.2 Ω per phase and runs at 950 rpm at full load. Calculate the external resistance per phase which must be added in order to lower the speed to 900 rpm and the torque being 1.5 times as before.
(A) 0.2 Ω (B) 0.66 Ω
(C) 0.266 Ω (D) 0.33 Ω
19. A 3-phase induction motor has a 6-pole, Y – connected stator winding. The motor runs on 60 Hz supply with 250V between lines. The motor resistance and standstill reactance per phase are 0.15 Ω and 0.82 Ω respectively. Calculate the rotor gross output power at a slip of 4% (rotor turns to stator turns ratio is 0.8)
(A) 3278.96W (B) 29510.64W
(C) 9836.88W (D) 8545.55W
20. A 3-phase induction motor having a 4-pole, star connected stator winding runs on 250V, 50Hz supply. The rotor resistance and standstill reactance are 0.2 ohm and 0.8 ohm per phase. The ratio of stator to rotor turns is 1.5. Calculate the developed torque at a full load slip of 5%.
(A) 42.5 N-m (B) 85.00 N-m
(C) 106.26 N-m (D) 58.50 N-m
21. Calculate the power factor of an induction motor at a slip of 5 percent, when the rotor resistance and stand

still reactance of a 3-phase induction motor are 0.02 and 0.08 Ω per phase respectively.

- (A) 0.24 (B) 0.74
(C) 0.85 (D) 0.98

22. Calculate the percentage tapping required on an auto transformer required for a squirrel cage motor to start the motor against 1/5 of full load torque. The short circuit current on normal voltage is 6 times the full load current and the full load slip is 4%.
(A) 62.74% (B) 37.26%
(C) 4.56% (D) 13.41%
23. A 3-phase, 500/300V, Y-Y connected wound rotor induction motor has 0.04 Ω rotor resistance and 0.2 Ω standstill reactance per phase. When the ratio of starting torque to maximum torque is 0.8, then the slip at maximum torque is _____

- (A) 0.4 (B) 0.8
(C) 1 (D) 0.5

24. A 6-pole induction motor and a 8-pole induction motor are connected in cumulative cascade. When the frequency in the secondary circuit of the 8-pole motor is 2Hz and supply frequency is 60Hz. Calculate combined speed of the set.
(A) 870.30 rpm (B) 514.28 rpm
(C) 1160.40 rpm (D) 497.30 rpm
25. A three phase, two pole 50Hz induction motor is observed to be operating at a speed of 2900rpm, with an input power 15.7kW and terminal current of 22.6A. stator winding has a resistance of 0.2 Ω /phase. What is the I^2R power dissipated in the rotor?
(A) 419W (B) 513W
(C) 493W (D) 503W

ANSWER KEYS

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. B | 2. B | 3. C | 4. D | 5. D | 6. A | 7. D | 8. D | 9. C | 10. B |
| 11. A | 12. B | 13. B | 14. A | 15. C | 16. D | 17. C | 18. B | 19. C | 20. A |
| 21. D | 22. B | 23. D | 24. D | 25. B | | | | | |

HINTS AND EXPLANATIONS

1. Choice (B)
2. rotor frequency $f_r = S \times$ supply frequency, therefore the injecting e.m.f has same frequency of rotor frequency with opposite phase sequence. Choice (B)

3. Backward slip = $2 - S$

$$\therefore \text{Backward torque } T_b = -k \left(\frac{I_2^2 R_2}{2 - S} \right)$$

Choice (C)

4. Choice (D)

5. Choice (D)

6. Choice (A)

7. Torque adjusts according to new speed, hence for most motors, it will decrease. Since more I^2R losses happen, it decreases. Choice (D)

$$8. S_{nl} = \frac{n_s - n_{nl}}{n_s} = \frac{3600 - 3580}{3600} \times 100 = 0.56\%$$

$$f = sf = 0.0056 \times 60 = 0.33\text{Hz} \quad \text{Choice (D)}$$

9. Vector addition of three fluxes, 120° electrical apart;
 $\Rightarrow$ gives flux of $1.5\phi_m$ at each instant Choice (C)

10. At full load, rated output is given Choice (B)

11. For constant flux v/f ratio is constant

$$\therefore \frac{V_1}{f_1} = \frac{V_2}{f_2} \Rightarrow v_2 = \frac{220}{30} \times 60 = 440\text{V}$$

$$N_1 = \frac{120 \times f_1}{P} = \frac{120 \times 30}{4} = 900\text{rpm}$$

$$N_2 = \frac{120 f_2}{P} = \frac{120 \times 60}{4} = 1800\text{rpm}$$

Choice (A)

$$12. \text{Synchronous speed } N_s = \frac{120f}{P}$$

$$= \frac{120 \times 50}{6} = 1000\text{rpm}$$

$$\text{Slip } S_1 = \frac{1000 - 950}{1000} = 0.05$$

$$S_2 = S_1 \left(\frac{V_1}{V_2} \right)^2 = 0.05 \times \left(\frac{400}{500} \right)^2 = 0.032$$

$$\text{Speed } N_2 = 1000(1 - 0.032) = 968\text{rpm} \quad \text{Choice (B)}$$

13. By operating two motors in cascade connection mode is rotor side speed control method.

Choice (B)

$$14. \frac{T_{st}}{T_f} = \left[\frac{I_{sc}}{I_f} \right]^2 S_f$$

$$I_{st} = I_{sc} = 400\text{A}$$

$$I_f = 50\text{A}$$

$$N_s = \frac{120 \times 50}{4} = 1500\text{rpm}$$

$$S = \frac{N_s - N}{N_s} = \frac{50}{1500} = 0.033$$

$$T_f = 200\text{N-m}$$

$$T_{st} = 200 \times \left(\frac{400}{50} \right)^2 \times 0.033 = 422.4 \text{ N-m}$$

Choice (A)

15. Synchronous speed $N_s = \frac{120f}{P}$

$$= \frac{120 \times 50}{8} = 750 \text{ rpm}$$

Rotor frequency $f_r = sf_s$

$$\frac{180}{60} = S \times 50 \Rightarrow S = 0.06$$

Rotor speed $N_r = N_s(1 - S) = (1 - 0.06) \times 750$
 $= 705 \text{ rpm}$

Total rotor copper loss = $S \times$ rotor Input
 $= 0.06 \times 500 = 30 \text{ kW}$

Choice (C)

16. Synchronous speed $N_s = \frac{120 \times 50}{8} = 750 \text{ rpm}$

$$\text{Slip} = \frac{N_s - N}{N_s} = \frac{750 - 700}{750} = 0.067$$

Motor input = 50 kW

Output power = rotor Input power – rotor copper loss – friction and windage losses

Rotor input power = Input power – stator loss
 $= 50 - 2 = 48 \text{ kW}$

$$P_{out} = 48 - 0.067 \times 48 - 3 = 41.784$$

$$\% \eta = \frac{P_{out}}{P_{in}} = \frac{41.784}{50} \times 100 = 83.568\%$$

Choice (D)

17. $P_{in} = \sqrt{3} \times 400 \times 20 \times 0.8 = 11085.12 \text{ W}$

Stator losses = $0.04 \times P_{in} = 443.40 \text{ W}$

Rotor input P_r = stator output = $P_{in} -$ Stator losses
 $= 10641.72 \text{ W}$

Rotor copper loss = $0.04 \times P_r = 425.66 \text{ W}$

Mechanical losses = $0.02 \times P_r = 212.83 \text{ W}$

Shaft output $P_{out} = 10641.72 - 425.66 - 212.83$
 $= 10003.23 \text{ watts}$

$$\text{Slip} = \frac{\text{rotor copper loss}}{\text{rotor input}} = 4\%$$

Speed of rotor $N_r = (1 - S)N_s$

$$= (1 - 0.04) \times \frac{120 \times 50}{6} = 960 \text{ rpm}$$

Angular rotor speed $\omega_r = \frac{2\pi N_r}{60}$

$$= \frac{2\pi \times 960}{60} = 100.53 \text{ rad/sec}$$

Shaft torque = $\frac{P_{out}}{\omega_r} = \frac{10003.23}{100.53} = 99.50 \text{ Nw-m}$

Choice (C)

18. Torque $T = \frac{KS R_2}{R_2^2 + (SX_2)^2}$

$$T_2 = 1.5 T_1$$

$$N_s = \frac{120 \times 50}{6} = 1000 \text{ rpm}$$

$$S_1 = \frac{50}{1000} = 0.05$$

$$S_2 = \frac{100}{1000} = 0.1$$

X_2 is not given $T \propto \frac{ks}{R_2}$

$$\frac{T_1}{T_2} = \frac{k S_1}{R_2} \left(\frac{R_2 + r}{k S_2} \right)$$

$$\Rightarrow \frac{T_1}{1.5 T_1} = \frac{0.05}{0.1} \times \frac{R_2 + r}{R_2}$$

$$\Rightarrow 1.33 = \frac{R_2 + r}{R_2} \Rightarrow (1.33 - 1)R_2 = r$$

$$\Rightarrow r = 0.66 \Omega$$

Choice (B)

19. Voltage/phase = $\frac{250}{\sqrt{3}} = 144.33 \text{ V}$

$$K = 0.8$$

Standstill motor e.m.f per phase is

$$E_2 = \frac{250}{\sqrt{3}} \times 0.8 = 115.47 \text{ V}$$

$$S = 0.04$$

Rotor impedance = $\sqrt{0.15^2 + (0.04 \times 0.8)^2} = 0.153 \Omega$

$$I_2 = \frac{SE_2}{Z_r} = \frac{0.04 \times 115.47}{0.153} = 30.18 \text{ A}$$

Total copper loss in rotor = $3I_2^2 R_2$

$$= 3 \times 30.18^2 \times 0.15 = 409.87 \text{ W}$$

Rotor gross output = rotor copper loss $\times \left(\frac{1-S}{S} \right)$

$$= 409.87 \times \left(\frac{1-0.04}{0.04} \right) = 9836.88 \text{ W}$$

Choice (C)

20. $K = \frac{\frac{\text{rotor turns}}{\text{phase}}}{\frac{\text{stator turns}}{\text{phase}}} = \frac{1}{1.5}$

$$E_2 = kE_1 = \frac{1}{1.5} \times \frac{250}{\sqrt{3}} = 96.22 \text{ V}$$

$$S = 0.05$$

$$N_s = \frac{120 \times 50}{4} = 1500 \text{ rpm}$$

$$T_f = \frac{180}{2\pi N_s} \times \frac{SE_2^2 R_2}{R_2^2 + (SX_2)^2}$$

$$= \frac{180}{2\pi \times 1500} \times \frac{0.05 \times 96.22^2 \times 0.2}{0.2^2 + (0.05 \times 0.8)^2} = 42.5 \text{ N-m}$$

Choice (A)

21. Rotor reactance per phase under running condition

$$= SX_2 = 0.05 \times 0.08 = 0.004$$

$$\text{Rotor impedance per phase} = \sqrt{0.02^2 + 0.004^2}$$

$$= 0.0204 \Omega$$

$$\text{Power factor} = \frac{0.02}{0.0204} = 0.98$$

Choice (D)

22. $\frac{T_{st}}{T_f} = \frac{1}{5} = 0.2$

$$\frac{I_{sc}}{I_f} = 6 \Rightarrow \left(\frac{I_{sc}}{I_f} \right)^2 = 36$$

$$S_f = 0.04$$

$$\frac{T_{st}}{T_f} = k^2 \left(\frac{I_{sc}}{I_f} \right)^2 S_f$$

$$\Rightarrow 0.2 = k^2 \times 36 \times 0.04 \Rightarrow k = 0.372$$

Choice (B)

23. $\frac{\text{starting torque}}{\text{maximum torque}} = \frac{2 \times S_m}{1 + S_m^2}$

$$0.8 = \frac{2S_m}{1 + S_m^2}$$

$$\Rightarrow 0.8S_m^2 - 2S_m + 0.8 = 0$$

$$S_m = 2, 0.5$$

Choice (D)

24. $N_s = \frac{120 \times 60}{(8+6)} = 514.28 \text{ rpm}$

$$S = \frac{f^{11}}{f} = \frac{2}{60} = 0.033$$

$$S = \frac{N_s - N}{N_s} \Rightarrow N = (1 - S)N_s$$

$$= (1 - 0.033) \times 514.28$$

$$= 497.30 \text{ rpms}$$

Choice (D)

25. $P_{\text{stator}} = 3I_1^2 R_1 = 3(22.6)^2 0.2 = 306 \text{ W}$

$$P_{\text{gap}} = P_{\text{in}} - P_{\text{stator}} = 15.7 - 0.3 = 15.4 \text{ kW}$$

$$N_s = \frac{120f}{P} = \frac{120}{2} \times 50 = 3000 \text{ rpm}$$

$$\text{Slip} = \frac{3000 - 2900}{3000} = 0.03334$$

$$P_{\text{rotor}} = 0.03334 \times 15.4 \text{ kW} = 513 \text{ W}$$

Choice (B)

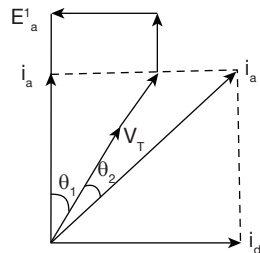
ELECTRICAL MACHINES TEST 6

Number of Questions: 25

Section Marks: 90

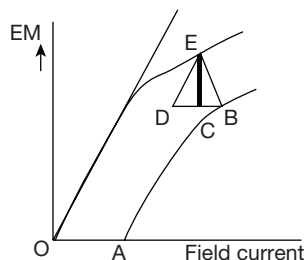
Directions for questions 1 to 25: Select the correct alternative from the given choices.

1. A three phase Y connected 50Hz two pole synchronous machine has a stator with 2000 turns/phase. What rotor flux is required to produce line to line voltage of 5kV?
 (A) $1.2 \times 10^{-2} \text{Wb}$ (B) $1.95 \times 10^{-2} \text{Wb}$
 (C) $6.5 \times 10^{-3} \text{Wb}$ (D) $3.75 \times 10^{-3} \text{Wb}$
2. A location requires 300kW of 60Hz ac power. Power sources are available only at 50Hz. Power is to be generated using synchronous motor - alternator set. How many poles should each of the machines have to convert 50Hz power to 60Hz power?
 (A) 10 pole motor, 12 pole generator
 (B) 12 pole motor, 10 pole generator
 (C) 10 pole motor, 10 pole generator
 (D) 12 pole motor, 12 pole generator
3. If the frequency of an alternator is changed from f_1 to f_2 its synchronous reactance (X_s) becomes _____.
 (A) $(f_1/f_2) X_s$ (B) $(f_2/f_1) X_s$
 (C) X_s (D) $X_s f_2$
4. Given synchronous generator phasor diagram

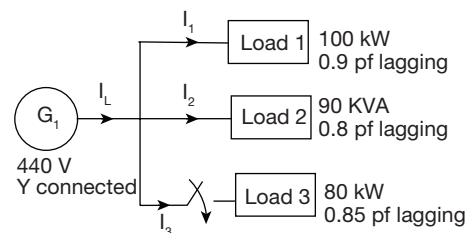


θ_1, θ_2 indicate (respectively)

- (A) torque angle, power factor angle
 - (B) power factor angle, torque angle
 - (C) power factor angle, internal voltage angle
 - (D) internal voltage angle, torque angle
5. _____ torque of synchronous motor can cause the motor to slip and step out of synchronism.
 (A) accelerating (B) Pull - in
 (C) pull - out (D) load
 6. In the following characteristics of an alternator BC represents.



- (A) field current necessary to overcome demagnetizing effect of armature reaction at full load.
 - (B) armature reaction drop at full load.
 - (C) field current necessary to overcome cross magnetizing effect of armature reaction at full load.
 - (D) armature resistance drop at full load.
7. Which of the following equation represent the maximum mechanical power developed in synchronous motor?
 (A) $\frac{E_b V}{Z_s} - \frac{E_b^2}{Z_s}$ (B) $\frac{E_b V}{Z_s} - \frac{E_b^2}{Z_s} \sin \theta$
 (C) $\frac{E_b V}{Z_s} - \frac{E_b^2}{Z_s} \cos \theta$ (D) $\frac{E_b V}{Z_s} \cos \theta$
 8. In an alternator chording angle for fundamental flux wave is α . Then its value for 7th harmonic is
 (A) $\frac{\alpha}{7}$ (B) 7α
 (C) $\frac{6\alpha}{7}$ (D) $\frac{8\alpha}{7}$
 9. Synchronous motors are :
 (A) self starting
 (B) started manually
 (C) started with a squirrel cage
 (D) started with a capacitor
 10. A 40 MVar synchronous condenser operates on a 34.5 kV bus. Synchronous reactance is 150%. Estimate the field excitation required to get 40 MVar power.
 (A) 100% (B) 200%
 (C) 250% (D) 50%
 11. A 480V, 60Hz six pole synchronous motor draws 80A from the line at UPF, full load. Assuming that motor is lossless. Then what is the output torque?
 (A) 629 Nm (B) 429 Nm
 (C) 529 Nm (D) 729 Nm
 12. For the circuit given below, what real and reactive power are provided by the generator when the switch is closed?



- (A) 180 kW, 90kVAr (B) 252 kW, 52.8kVAr
 (C) 252 kW, 152kVAr (D) 108 kW, 52.8kVAr

13. A 100MVA, 11.5kV, 0.75pf lagging, 50Hz two pole Y – connected synchronous generator has a per unit reactance of 0.9 and a per unit armature resistance of 0.012. The magnitude of internal generated voltage at rated conditions is ____ and the torque angle is ____
 (A) 10750, 23° (B) 11556, 37°
 (C) 10750, 37° (D) 11556, 23°
14. A three phase synchronous generator has negligible armature resistance and synchronous reactance $X_d = 1.5\text{P.u.}$ This machine is connected to an infinite bus at voltage $1\angle 0^\circ\text{P.u.}$ what is the internal voltage of the machine when it delivers a current of $1\angle 30^\circ\text{P.u.}$ to the infinite bus
 (A) $1.44\angle 83^\circ$ (B) $1.23\angle 82^\circ$
 (C) $1.32\angle 79^\circ$ (D) None of the above
15. A three phase two pole winding is installed in six slots on a stator. There are 80 turns of wire in each slot of the windings. The three phases are Δ connected, all coils in each phase are connected in series. What is the frequency of voltage produced by this winding? ($N = 3600\text{rm}$)
 (A) 50 Hz (B) 60 Hz
 (C) 80 Hz (D) 48 Hz
16. What may be the highest speed at which two generators mounted on the same shaft be driven, if frequency of one generator is 25 Hz and that of the other is 60 Hz? They cannot have more than 26 poles.
 (A) 120 rpm (B) 360 rpm
 (C) 300 rpm (D) 3000 rpm
17. A 50 kVA, 200 V, 50 Hz, 1 – Φ alternator has an effective armature resistance of 0.02 ohm and an armature leakage of 0.08 ohm. Compute the voltage induced in the armature, when the alternator is delivering rated current at a load power factor of 0.8 leading.
 (A) 216.39 V (B) 192.93 V
 (C) 468.25 kV (D) 372.25 kV
18. A 3-phase alternator has a direct axis synchronous reactance of 0.6 p.u and a quadrature axis synchronous reactance of 0.3 p.u. Calculate the direct axis current when the alternator delivers a full load current at 0.7 p.f lagging.
 (A) 0.71 (B) 0.58
 (C) 0.82 (D) 0.17
19. A 600 kVA, 11 kV, 6-pole, 3 – Φ star connected alternator has percentage resistance and reactance of 2 and 10 respectively. Calculate the synchronizing power per mechanical degree of displacement at full load 0.9 power factor lagging.
 (A) 111.50 W/phase (B) 111.50 kW/phase
 (C) 334.5 W/phase (D) 334.5 kW/phase
20. Two alternators A and B operate in parallel and supply a load of 20 MW at 0.75 p.f lagging. Calculate the power factor of alternator B when alternator A supplies 14,000 kW at 0.9 lagging power factor.
 (A) 0.48 lagging (B) 0.77 lagging
 (C) 0.62 lagging (D) 0.53 lagging
21. Calculate the power angle when a 2000 kVA, 11 kV, 3-phase, Y -connected alternator having a resistance of 0.5 ohm and a reactance of 5 ohm per phase delivers full load current at normal rated voltage and 0.8 p.f lagging.
 (A) 6.60° (B) 40.16°
 (C) 3.30° (D) 36.24°
22. A 2000V star connected synchronous motor has a resistance of 0.4 ohm per phase and a synchronous reactance of 3.0 ohm per phase. The motor is operating at 0.8 power factor leading with a line current of 300 A. Calculate the value of the generated e.m.f per phase.
 (A) 692.89 V/phase
 (B) 1370.19 V/phase
 (C) 1782.36 V/phase
 (D) 1200.12 V/phase
23. A 1500 V, 3-phase, 6-pole, Y -connected synchronous motor runs at 1000 rpm. The excitation is constant and corresponds to an open-circuit terminal voltage of 1500 V. The resistance is negligible as compared with synchronous reactance of 4 Ω per phase. Calculate the torque developed for an armature current of 300 A.
 (A) 5627.44 N-m (B) 5374.21 N-m
 (C) 562.74 N-m (D) 537.42 N-m

Common data questions for 24 and 25:

24. A 480V, 200kW, two pole, three phase, 50 Hz synchronous generators prime mover has the no load speed of 3040rpm and full load speed of 2975rpm. It operates in parallel with a 480 V, 180 kW, four pole, 50 Hz synchronous generator whose prime mover has a no load speed of 1500rpm and full load speed of 1485rpm. Together, they supply 200kW at 0.85PF lagging. Speed drops of generator 1 and generator 2 are respectively. (in percent)
 (A) 2.18, 1.01 (B) 1.01, 2.18
 (C) 0.218, 0.101 (D) 0.101, 0.218
25. What is the power given by generator 2?
 (A) 128 kW (B) 72 kW
 (C) 100 kW (D) 200 kW

ANSWER KEYS

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. A | 3. B | 4. A | 5. C | 6. A | 7. C | 8. B | 9. C | 10. C |
| 11. C | 12. B | 13. D | 14. C | 15. B | 16. C | 17. B | 18. C | 19. B | 20. A |
| 21. C | 22. C | 23. B | 24. A | 25. B | | | | | |

HINTS AND EXPLANATIONS

$$1. V_{\text{phase}} = \frac{5000}{\sqrt{3}} = 2886.8$$

$$E = \sqrt{2} \pi N_c \phi f$$

$$\Rightarrow \phi = \frac{E}{\sqrt{2} \pi N_c f} = \frac{2886.8}{\sqrt{2} \pi \times 2000 \times 50} \approx 6.5 \times 10^{-3} \text{ Wb}$$

Choice (C)

$$2. N_s = \frac{120 f}{p}$$

But since shafts are coupled, mechanical speed of motor and generator should be equal.

$$\text{Hence, } \frac{120 \times 50}{P_1} = \frac{120 \times 60}{P_2}$$

$$\Rightarrow \frac{P_2}{P_1} = \frac{6}{5}$$

Hence 12 pole generator, 10 pole motor will be suitable.

Choice (A)

3. Since armature reaction $\propto f$

$$\therefore X_s \propto f$$

Choice (B)

4. Choice (A)

5. Choice (C)

6. Choice (A)

7. Choice (C)

8. Choice (B)

9. Choice (C)

$$10. V_t = E + j I_a X_s$$

$$1 \angle 0^\circ = E \angle 0^\circ + j I_a (1.5)$$

$$1 - E = 1.5 I_a \angle (90^\circ + 90^\circ)$$

$$\text{At } |I_a| = 1$$

$$1 - E = -1.5$$

$$|E| = 2.5$$

$\therefore$ Field excitation should be 250%

Choice (C)

11. For lossless motor, $P_{\text{OUT}} = P_{\text{IN}}$

$$\tau = \frac{P}{\omega} = \frac{\sqrt{3} V_T I_L \cos \theta}{\left(\frac{120 \times 60}{6} \right) \frac{2\pi}{60}}$$

$$= \frac{66.5 \times 1000}{\left(\frac{1200 \times 2\pi}{60} \right)} \approx 529 \text{ Nm}$$

Choice (C)

$$12. P_3 = 80 \text{ kW}$$

$$Q_3 = P \tan \theta = P \tan(\cos^{-1}(0.85)) = 80 \text{ kW} \times \tan(-31.79^\circ) = -49.6 \text{ KVAR}$$

$$P_2 = S \cos \theta = 90 \text{ KVA} \times 0.8 = 72 \text{ kW}$$

$$Q_2 = S \sin \theta = 90 \text{ KVA} \times 0.6 = 54 \text{ KVAR}$$

$$P_1 = 100 \text{ kW}$$

$$Q_1 = P \tan \theta = 100 \text{ kW} (\tan(\cos^{-1}(0.9)))$$

$$= 100 \tan(25.8) = 48.4 \text{ KVAR}$$

$$\therefore \text{Total real power} = 252 \text{ kW}$$

$$\text{Total reactive power} = 52.8 \text{ KVAR} \quad \text{Choice (B)}$$

$$13. V_{\text{phase}} = \frac{11500}{\sqrt{3}} = 6640 \text{ V}$$

$$Z_{\text{base}} = \frac{3(6640)^2}{100 \times 10^6} = 1.32 \Omega$$

$$R_A = 0.012 \times 1.32 = 0.0158 \Omega$$

$$X_S = 0.9 \times 1.32 = 1.188 \Omega$$

$$\text{Rated armature current} = \frac{S}{\sqrt{3} V_T} = \frac{100 \times 10^6}{\sqrt{3} \times 11.5 \times 10^3} = 5020 \text{ A}$$

$$\text{Pf} = 0.75 \text{ lagging, } \therefore I_A = 5020 \angle -41.41^\circ$$

$$\therefore E_A = V_{\text{phase}} + R_A I_A + j X_S I_A = 6640 \angle 0^\circ + (0.0158)(5020 \angle -41.41^\circ) + j 1.188(5020 \angle -41.41^\circ)$$

$$= 6640 + 79.316 + j[4472.82 - j3936]$$

$$= 6640 + 79.316 + 3936 + j4472.82$$

$$= 10655.316 + j4472.82$$

$$\therefore E_A = 11556.03 \angle 22.73^\circ \approx 11556 \angle 23^\circ \quad \text{Choice (D)}$$

$$14. E \angle \delta = V_a \angle 0^\circ + I_a \angle \theta X_a \angle 90^\circ$$

$$= 1 \angle 0^\circ + 1.5 \angle (90^\circ + 30^\circ)$$

$$= 1 + (-0.75) + j1.3$$

$$= 1.324 \angle 79.1^\circ \approx 1.32 \angle 79^\circ$$

Choice (C)

$$15. f = \frac{np}{120} = \frac{3600 \times 2}{120} = 60 \text{ Hz}$$

Choice (B)

$$16. N = \frac{120 f_1}{P_1} = \frac{120 f_2}{P_2}$$

$$= \frac{120 \times 60}{P_1} = \frac{120 \times 25}{P_2}$$

$$P_1/P_2 = \frac{60}{25}$$

$$P_1, P_2 \leq 26$$

$$\therefore P_1, P_2 = 24, 10$$

$$\therefore N = \frac{120 \times 60}{24} = 300 \text{ rpm}$$

$$= \frac{120 \times 25}{10} = 300 \text{ rpm}$$

Choice (C)

$$17. I = 50000/200 = 250 \text{ A}$$

$$E = [(V \cos \Phi + I R_a)^2 + (V \sin \Phi - I X_L)^2]^{1/2}$$

$$= [(200 \times 0.8 + 250 \times 0.02)^2 +$$

$$(200 \times 0.6 - 250 \times 0.08)^2]^{1/2} = 192.93 \text{ V} \quad \text{Choice (B)}$$

$$18. V = 1 \text{ p.u.}$$

$$X_d = 0.6 \text{ p.u.}$$

$$X_q = 0.3 \text{ p.u.}$$

$$\cos \phi = 0.7 \Rightarrow \phi = \cos^{-1}(0.7) = 45.57^\circ$$

$$\sin \phi = 0.7$$

$$I_a = 1 \text{ p.u.}$$

$$\tan \delta = \frac{I_a X_q \cos \phi}{V + I_a X_q \sin \phi} = \frac{1 \times 0.3 \times 0.7}{1 + 1 \times 0.37 + 0.7} = 0.173$$

$$\delta = 9.81^\circ$$

$$I_d = I_a \sin[\Phi + \delta] = 1 \times \sin(45.57^\circ + 9.81^\circ) = 0.82$$

Choice (C)

$$19. \text{ Full load current } I = \frac{600 \times 10^3}{\sqrt{3} \times 11 \times 10^3} = 31.5 \text{ A}$$

$$V_{ph} = \frac{11000}{\sqrt{3}} = 6350 \text{ V}$$

$$I R_a = 0.02 \times 6350 = 127 \text{ V}$$

$$I X_s = 0.1 \times 6350 = 635 \text{ V} \Rightarrow X_s = 20.15 \Omega$$

$$E = [(V \cos \Phi + I R_a)^2 + (V \sin \Phi + I X_s)^2]^{1/2}$$

$$= [(6350 \times 0.9 + 127)^2 + (6350 \times 0.435 + 635)^2]^{1/2}$$

$$= 6757.97 \text{ V}$$

$$\alpha (\text{mech}) = 1^\circ \Rightarrow \alpha (\text{elect}) = 1 \times \frac{6}{2} = 3^\circ$$

$$\alpha = 3\pi/180 = \pi/60 \text{ elect. radi}$$

$$P_{sy} = \frac{\frac{\pi}{60} \times 6757.97 \times 6350}{20.15} = 111.50 \text{ kW/phase}$$

Choice (B)

$$20. \cos \phi = 0.75 \Rightarrow \phi = 41.40^\circ$$

$$\text{Load kW} = 20000 \text{ kW}$$

$$\text{Load KVAR} = 20000 \tan \Phi$$

$$= 17632.37 \text{ KVAR}$$

$$\text{Load kw supplied by alternator A, } kw_1 = 14000 \text{ kW}$$

$$\text{Load kw supplied by alternator B, } kw_2 = 6000 \text{ kW}$$

$$\text{Load KVAR supplied by alternator A KVAR}_1$$

$$= kw_1 \times \tan \phi_1$$

$$\cos \phi_1 = 0.9 \Rightarrow \phi_1 = 25.84^\circ$$

$$\text{KVAR}_1 = 14000 \times \tan(25.84)$$

$$= 6779.92 \text{ KVAR}$$

$$\text{Load KVAR}_2 \text{ supplied by alternator B KVAR}_2$$

$$= 17632.37 - 6779.92$$

$$= 10852.45 \text{ KVAR}$$

$$\text{KVA}_2 = 6000 - j10852.45$$

$$= 12400 \angle -61.06^\circ$$

$$\cos \phi_2 = 0.48 \text{ lagging}$$

Choice (A)

$$21. \cos \Phi = 0.8 \Rightarrow \Phi = 36.86^\circ$$

$$I = \frac{2000 \times 10^3}{\sqrt{3} \times 11 \times 10^3} = 104.97 \text{ A}$$

$$I R_a = 104.97 \times 0.5 = 52.48 \text{ V}$$

$$I X_s = 104.97 \times 5 = 524.85 \text{ V}$$

$$V/\text{phase} = 11000/\sqrt{3} = 6350.85 \text{ V}$$

$$\tan(\phi + \delta) = \frac{V \sin \phi + I X_s}{V \cos \phi + I R_a} = \frac{6350.85 \times 0.6 + 524.85}{6350.85 \times 0.8 + 52.48}$$

$$\tan(\phi + \delta) = 0.844$$

$$\phi + \delta = 40.16 \Rightarrow \delta = 40.16 - 36.86^\circ$$

$$\Rightarrow \delta = 3.30^\circ$$

Choice (C)

$$22. \phi = \cos^{-1}(0.8) = 36.86^\circ$$

$$\theta = \tan^{-1} \left(\frac{3}{0.4} \right) = 82.40^\circ$$

$$\theta + \phi = 82.40 + 36.86 = 119.26^\circ$$

$$V = \frac{2000}{\sqrt{3}} = 1154.70 \text{ V}$$

$$Z_s = \sqrt{0.4^2 + 3^2} = 3.02 \Omega$$

$$I Z_s = 300 \times 3.02 = 906 \Omega$$

$$E_b = \sqrt{V^2 + E_R^2 - 2V E_R \cos(\theta + \phi)}$$

$$= \sqrt{(1154.70)^2 + (906)^2 - 2 \times 1154.70 \times 906 \times \cos(119.26)}$$

$$= 1782.36 \text{ volt/phase}$$

Choice (C)

$$23. \text{ Voltage/phase} = \frac{1500}{\sqrt{3}} = 866.02 \text{ V}$$

$$\text{Induced e.m.f} = 866.02 \text{ V}$$

$$\text{Impedance drop} = 300 \times 4 = 1200 \text{ V}$$

$$866^2 = 866^2 + 1200^2 - 2 \times 866 \times 1200 \times \cos(90 - \Phi)$$

$$\sin \phi = 0.692 \Rightarrow \phi = 43.78^\circ$$

$$\text{Power input} = \sqrt{3} \times 1500 \times 300 \times \cos(43.78^\circ)$$

$$= 562.74 \text{ kW}$$

$$\text{Torque } T = \frac{9.55 \times 562.74}{1000} = 5374.21 \text{ N-m}$$

Choice (B)

$$24. F_{nl1} = \frac{n_m P}{120} = \frac{3040 \times 2}{120} = 50.67 \text{ Hz}$$

$$F_{fl1} = \frac{2975 \times 2}{120} = 49.58 \text{ Hz}$$

$$F_{nl2} = \frac{1500 \times 4}{120} = 50 \text{ Hz}$$

$$F_{fl2} = \frac{1485 \times 4}{120} = 49.5 \text{ Hz}$$

$$SD_1 = \frac{n_{nl} - n_{fl}}{n_{fl}} \times 100\% = \frac{3040 - 2975}{2975} \times 100 = 2.18\%$$

$$SD_2 = \frac{1500 - 1485}{1485} \times 100 = 1.01\%$$

Choice (A)

$$25. P_2 = S_{p2} (f_{nl2} - f_{sys}) = 360[50 - 49.8] = 72 \text{ kW}$$

$$S_{p2} = \frac{P}{f_{nl} - f_{fl}} = \frac{180 \text{ kW}}{50 - 49.5} = 360 \text{ kW/Hz}$$

$$(S_{pl} = 183.48 \text{ kW/Hz})$$

$$P_{Load} = S_{pl} (f_{nl} - f_{sys}) + S_{p2} (f_{nl} - f_{sys})$$

$$\Rightarrow 200 = 183.48(50.67 - f_{sys}) + 360(50 - f_{sys})$$

$$\Rightarrow f_{sys} = 49.8$$

Choice (B)